



B. Tech. ELECTRICAL ENGINEERING

Syllabus AY: 2024-25



CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Authored By: Charusat



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY



ACADEMIC REGULATIONS & SYLLABUS 2024-25 (Choice Based Credit System)

Faculty of Technology & Engineering
M & V Patel Department of Electrical Engineering
Bachelor of Technology Programme
(Electrical Engineering)

Chandubhai S Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives

M & V Patel Department of Electrical Engineering

Vision

To become an internationally recognized, research intensive education entity, serving as a workforce developer for the nation by providing the best education in close collaboration with society.

Mission

To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.

THE PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	Preparation: To prepare the students with sound academically background, intensive development of knowledge and skills, and with an uncompromising attitude to achieve excellence in the electrical engineering.
PEO 2	Core Competence: To build a broad engineering core competence and to enable the students to solve the problems systematically by providing them exposure to different dimensions of specializations.
PEO 3	Learning Environment: To provide conducive learning environment instilling sound foundation in mathematics, computing techniques, science and humanities along with professional courses for futuristic thinking in an increasingly technology dependent society.
PEO 4	Professionalism: To provide a platform to the students to plan and shape their career in the chosen field and to imbibe ethical values with thorough professionalism and encouragement towards entrepreneurial thinking and traits.

PROGRAM ARTICULATION MATRIX

(-Not Matching, 1-Low, 2-Medium, 3-High)

	PEO 1	PEO 2	PEO 3	PEO 4
To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.	3	3	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO 1	Engineering knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO 4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	The engineer and society: Apply reasoning obtained by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO 1	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.
PSO 2	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Electrical Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-265011, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2024-2025

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING ACADEMIC REGULATIONS

Bachelor of Technology Programmes

Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

Undergraduate programme (B. Tech.)

Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for admissions:

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – I attached

6. Attendance

6.1 All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

6.2 Student's attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, the continuous assessment will be conducted by the respective department /institute.
- 7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these.
- 7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.
- 7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.
- 7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examination

- 7.2.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

- 7.2.2 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grade Point System

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	<80 ≥73	<73 ≥66	<66 ≥60	<60 ≥55	<55 ≥50	<50 ≥45	<45
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9 Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
Distinction	$CGPA \geq 7.5 \text{ \& } \leq 10.0$
First Class	$CGPA \geq 6.0 \text{ \& } < 7.5$
Second Class	$CGPA \geq 5.0 \text{ \& } < 6.0$
Pass Class	$CGPA < 5.0$

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9.1 Maximum duration allowed for Completion of a programme

- Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10 *Detention Criteria*

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- A student will not be allowed to move to third year if he/she has not cleared all the courses of second year.
- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of third year.

11 *Transcript:*

- The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY												
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING												
Sem	Course Code	Course Title	Teaching Scheme					Examination Scheme				
			Contact Hours				Credit	Theory		Practical		Total
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External	
SY Sem-III	XXXXX	University Elective I	0	2	0	2	2			30	70	100
	MA252	Advanced Engineering Mathematics	4	0	0	4	4	30	70	0	0	100
	EE251	Analog and Digital Electronics	3	2	0	5	4	30	70	25	25	150
	EE252.01	Circuit Theory	3	2	0	5	4	30	70	25	25	150
	EE243.01	Electrical Measurement and Industrial Instrumentation	4	2	0	6	5	30	70	25	25	150
	HS121.02A	Creativity, Problem Solving and Innovation		2		2	2	0	0	30	70	100
			14	10	0	24	21	120	280	135	215	750
SY Sem-IV	XXXXX	University Elective II	0	2	0	2	2			30	70	100
	EE254	Electrical Power Generation and Its Economics	3	0	0	3	3	30	70	0	0	100
	EE246	Microprocessor and Microcontrollers	4	2	0	6	5	30	70	25	25	150
	EE247	Transformers and Induction Machines	4	4	0	8	6	30	70	50	50	200
	EE255.01	Control Systems	3	2	0	5	4	30	70	25	25	150
	HS111.03A/ HS113A	Human Values and Ethics/ Indian Knowledge System		2		2	2	0	0	30	70	100
			14	12	0	26	22	120	280	160	240	800

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 3rd Semester Undergraduate Programme
(Effective from Academic year 2024-25)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	BM231: Banking and Insurance	I2IM / FMS
2	CA224: Introduction to Web Designing	CMPICA / FCA
3	CE281.01: Art of Programming	CE / FTE
4	CL281.01: Environmental Sustainability and Climate Change	CL / FTE
5	CL283: SDG Handprint Laboratory	CL / FTE
6	EC281.01: Introduction to MATLAB Programming	EC / FTE
7	EE284: Python Programming	EE/FTE
8	IT283: Web Designing & UI/UX	IT/FTE
9	ME281.01: Engineering Drawing	ME/FTE
10	NR251.01: First Aid & Life Support	NURSING / FMD
11	PD260.01: Basic Laboratory Techniques	PDPIAS/FAS
12	PH233.01: Fundamentals of Packaging	RPCP/FPH
13	PT191.01: Health Promotion and Fitness	ARIP / FMD

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 4th Semester Undergraduate Programme
(Effective from Academic year 2024-25)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	BM241: Health Care Management	I2IM / FMS
2	CA225: Programming the Internet	CMPICA / FCA
3	CE282.01: Web Designing	CE / FTE
4	CL282.01: Basics of Environmental Impact Assessment	CL / FTE
5	EC282.01: Prototyping Electronics with Arduino	EC / FTE
6	EE287: MATLAB Programming	EE/FTE
7	EE288: Maintenance of Household Apparatus	EE/FTE
8	IT284: DATA Visualization	IT/FTE
9	ME282.01: Material Science	ME/FTE
10	NR261.01: Life Style Diseases & Management	NURSING / FMD
11	PH238.01: Cosmetics in daily life	RPCP/FPH
12	PT192.01: Occupational Health & Ergonomics	ARIP / FMD

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – III)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Charotar University of Science and Technology, Gujarat
Re-designed Course Structure of UG in Electrical and Electronics Engineering

Course Code: **EE284**
Course Title: **Python Programming**

UG Engineering Programme	Semester
Electrical Engineering	Third

I RATIONALE

Python programming is an essential subject to learn in today's technological landscape. Python is a high-level, general-purpose programming language that is easy to learn and widely used in various fields, including data science, machine learning, web development, and automation. The language's simplicity and readability make it an ideal choice for beginners, while its extensive libraries and frameworks make it a powerful tool for experts. By learning Python programming, students can acquire valuable skills that can lead to numerous career opportunities, as Python is in high demand in industries such as finance, healthcare, and technology.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Proficiently write, debug and deploy Python code for various applications and industries.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- Write** program for real world problems
- Describe** the Math functions, Strings, List, Tuples and Dictionaries in Python.
- Implement** solution for mathematical problems, processing data, and handling strings.
- Debug** and **Troubleshoot** python code.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme			Total Credits	Examination Scheme (Anna 2021)			
(In Hours)			(L+T+P)	Theory Marks		Practical Marks	
L	T	P	C	ESE	PA	ESE	PA
0	0	2	2	0	0*	70	30

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the Theory PA, out of 30 marks, 10 marks are for micro-project assessment to facilitate integration of COs and the remaining 20 marks is based on lab activities by assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in

integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	2,4	<u>Hands-on</u> practice to use print, input command; variables; operators.	2
2	2,4	<u>Hands-on</u> practice and <u>use cases</u> of if, if-else, elif, nested if-else, for, while statement/loop.	2
3	1,3	<u>Develop</u> Python program to find odd/even year.	1
4	1,3	<u>Develop</u> Python program to compute the GCD of two numbers.	1
5	1,3	<u>Develop</u> Python program to find the square root of a number.	2
6	1,3	<u>Develop</u> Python program to find the exponentiation (Power of a number).	2
7	1,3	<u>Develop</u> Python program to find the first n prime numbers.	2
8	1,3	<u>Develop</u> Python program to multiply two matrices.	2
9	2,4	<u>Hands-on</u> practice to create list and various list methods.	2
10	1,3	<u>Develop</u> Python program to find the maximum of a list of numbers.	2
11	1,3	<u>Develop</u> Python program to search an element in an array.	2
12	1,3	<u>Develop</u> Python program to sort the elements.	2
13	1,3	<u>Hands-on</u> practice to use string and string methods, and file handling.	2
14	1,3	<u>Develop</u> Python program to take command line arguments(word count)	2
15	1,3	<u>Develop</u> Python program to find the most frequent words in a text read from a file.	2
16	2,4	<u>Hands-on</u> practice and <u>use cases</u> for Tuples and Dictionaries.	2
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Software tool: Python v3.0 or higher	1,2,3,4,5, 6,7,8,9,10, 11,12,13, 14,15,16

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Function as a team member.
- 3) Function as a team leader.
- 4) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- 'L' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

IX SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Develop a program that simulates rolling dice. The user should be able to specify the number of dice to roll and the number of sides on each die.
- Develop a program to construct calculator.
- Create a weather app based on python programming.
- Create a chatbot using python programming.

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Think Python	Allen Downey	O'Reilly Media 978-1449330729
2	Python Programming: A Complete Guide for Beginners to Master, Python Programming Language	Brian Draper	Createspace Independent Pub 978-1539434375
3	Python: The Complete Reference	Martin C. Brown	Tata McGraw Hill 978-9387572942

XV SOFTWARE/LEARNING WEBSITES

- <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016>
- <https://www.python.org/about/gettingstarted>
- https://onlinecourses.nptel.ac.in/noc18_cs21/preview

XVI PO-COMPETENCY-CO MAPPING

		POs and PSOs
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Cours e Code	Competency and COs	PO 1 Engine ering knowl edge	PO 2 Proble m analysi s	PO 3 Design /devel opmen t of solutio ns.	PO 4 Condu ct investi gation s of compl ex proble ms	PO 5 Modern tool usage	PO 6 The engine er and society	PO 7 Enviro nment and sustai nabilit y	PO 8 Ethics	PO 9 Individ ual and teamw ork	PO 10 Comm unicati on	PO 11 Projec t manag ement and financ e	PO 12 Life- long learnin g	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent technologies like Artificial intelligence, machine learning and optimization for design, simulation analysis of electrical systems to provide real world solutions to a real world problem.
EE284	Semester IV	Python Programming Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO													
284	Proficiently write, debug and deploy Python code for various applications and industries.	2	2	3	1	3								2	
284.1	Write program for real world problems	1	1	2		3								3	
284.2	Describe the Math functions, Strings, List, Tuples and Dictionaries in Python.	3	2												
284.3	Implement solution for mathematical problems, processing data, and handling strings.	1	1	3	1									2	
284.4	Debug and Troubleshoot python code.		3	1	2	1								1	

XVII NAMES OF RESOURCE PERSONS

a) Jignesh S Patel

b)

Course Code: **MA252**
Course Title: **Advance Engineering Mathematics**

UG Engineering Programme	Semester
Electrical Engineering, Electronics & Communication Engineering	Third

I RATIONALE

Mathematical concepts play a significant role in engineering for comprehending core technical subjects. This course structure is developed to familiarize engineering students with some advanced concepts in mathematics which includes the calculus of vector valued functions, basic transformation such as Laplace Transform, series expansion of periodic functions and curve fitting techniques.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Explain mathematical techniques for the solution of various problems of engineering.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Solve** the complex problems through the concepts of Laplace transforms.
- 2) **Compute** the problems related to the Fourier series.
- 3) **Explain** the vector space, linear span, linear dependence, and basis.
- 4) **Explain** the concepts of vectors, directional derivative, gradient, vector fields, divergence and curl.
- 5) **Solve** the problems through the concepts of Z- transforms.
- 6) **Find** the Fourier Integral and Fourier Transform of various functions.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
4	0	0	4	70	30*	-	-	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes

(UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Laplace Transforms and Its Applications	CO 1. Solve the complex problems through the concepts of Laplace transforms. 1a. <u>Compute</u> the Laplace transform of the <u>given</u> functions 1b. <u>Explain</u> various properties of the Laplace transform 1c. <u>Find</u> the inverse Laplace transform of <u>some</u> functions 1d. <u>Explain</u> various properties of the inverse Laplace transform 1e. <u>Evaluate</u> the inverse Laplace transform of <u>the given</u> functions <u>using</u> convolution theorem 1f. <u>Use</u> the Method of Laplace transforms to <u>solve</u> initial-value problems for linear differential equations	1.1 Laplace transforms as an improper integral and its existence 1.2 Laplace transforms of elementary functions and its properties 1.3 Inverse Laplace transforms and its properties 1.4 First and second shifting theorems. Laplace transforms of derivatives and integrals 1.5 Convolution theorem and its application to obtain inverse Laplace transform. 1.6 Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function). 1.7 Solving differential equations using Laplace transform 1.8 Applications of Laplace transform to ODE: Electric circuits (RLC and RL)
Unit– II Fourier Series	CO 2. Compute the problems related to the Fourier Series. 2a. <u>Explain</u> the existence condition for Fourier series 2b. <u>Compute</u> the Fourier series of <u>the given</u> periodic functions 2c. <u>Find</u> the Fourier series of discontinuous functions 2d. <u>Determine</u> the Half range Fourier sine and cosine series of <u>the given</u> periodic functions	2.1 Periodic functions, Dirichlet's conditions, Trigonometric series 2.2 Euler formulae, Fourier series of periodic functions of period p 2.3 Fourier series of Discontinuous functions 2.4 Fourier series of Even and odd functions; half range Fourier series
Unit– III Linear Algebra	CO 3. Explain the vector space, linear span, linear dependence, and basis. 3a. <u>Verify</u> either a structure forms a vector space or not 3b. <u>Define</u> Linear dependence of set of vectors	3.1 Introduction of Vector Space and its examples 3.2 Linear dependence and independence and its examples 3.3 Basis and its examples

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related POs required by industry)</i>	<i>Topics and Sub-topics</i>
	3c. <u>Explain</u> basis of vector space	
Unit-IV Vector Differential Calculus	CO 4. Explain the concepts of vectors, directional derivative, gradient, vector fields, divergence and curl. 4a. <u>Interpret</u> the concepts of vectors 4b. <u>Evaluate</u> the gradient and directional derivatives of problems 4c. <u>Find</u> the divergence and curl of vector fields 4d. <u>Justify</u> the terms irrotational and solenoidal 4e. <u>Evaluate</u> the scalar potential function of vectors	4.1 Revision of concepts of Vector algebra, Scalar and Vector fields 4.2 Gradient of a scalar functions, Directional derivatives 4.3 Divergence and Curl of a vector field and their properties 4.4 Physical interpretations of gradient, divergence and curl: Irrotational and solenoidal vector fields 4.5 Scalar potential function
Unit – V Z-transforms	CO 5. Solve the problems through the concepts of Z- transforms. 5a. <u>Find</u> the Z-transform of <u>the given</u> sequences 5b. <u>Explain</u> various properties of the Z-transform 5c. <u>Compute</u> the inverse Z-transform of <u>some</u> functions	5.1 Introduction to Z transform. Representation of sequences 5.2 Z-transforms of basic sequences 5.3 Unit impulse and Unit step sequence. Basic operations on sequence 5.4 Properties of Z-transforms: Damping rule and shifting property 5.5 Introduction to inverse Z-transforms. Inverse Z-transform using partial fraction and geometric series
Unit-VI Fourier Transforms	CO 6. Find the Fourier Integral and Fourier Transform of various functions. 6a. <u>Compute</u> the Fourier integral of <u>the given</u> functions 6b. <u>Find</u> the Fourier transform of <u>the given</u> functions 6c. <u>Explain</u> various properties of the Fourier transform 6d. <u>Determine</u> the Fourier sine and cosine transform of <u>some</u> functions	6.1 Introduction to Fourier integral: sine and cosine integrals 6.2 Introduction to Fourier Transform and inverse Fourier Transform 6.3 Properties and examples of Fourier transforms 6.4 Fourier sine and cosine transforms

VI TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Laplace Transforms and Its Applications	15	2	3	4	4	5	0	18
II	Fourier Series	10	1	2	2	3	4	0	12
III	Linear Algebra	07	1	1	2	4	0	0	08
IV	Vector Differential Calculus	08	1	1	2	5	0	0	09
V	Z-transforms	10	1	2	2	3	4	0	12
VI	Fourier Transforms	10	1	1	2	3	4	0	11
Total		60	07	10	14	22	17	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

VII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Laplace Transforms and Its Applications	1	1a. <u>Compute</u> the Laplace transform of the <u>given</u> functions	Convention al, ICT based	Green board, PPTs
	2	1b. <u>Explain</u> various properties of the Laplace transform 1c. <u>Find</u> the inverse Laplace transform of <u>some</u> functions	Convention al, ICT based	Green board, PPTs
	3	1d. <u>Explain</u> various properties of the inverse Laplace transform 1e. <u>Evaluate</u> the inverse Laplace transform of <u>the given</u> functions <u>using</u> convolution theorem	Convention al, ICT based	Green board, PPTs
	4	1f. <u>Use</u> the Method of Laplace transforms to <u>solve</u> initial-value problems for linear differential equations	Convention al, ICT based	Green board, PPTs
Unit– II Fourier Series	5	2a. <u>Choose</u> the relevant electromagnetic relay for <u>the given</u> situation. 2b. <u>Explain</u> the existence condition for Fourier series	Convention al, ICT based	Green board, PPTs
	6	2c. <u>Compute</u> the Fourier series of <u>the given</u> periodic functions 2d. <u>Find</u> the Fourier series of discontinuous functions	Convention al, ICT based	Green board, PPTs
	7	2e. <u>Determine</u> the Half range Fourier sine and cosine series of <u>the given</u> periodic functions	Convention al, ICT based	Green board, PPTs
Unit– III Linear Algebra	8	3a. <u>Verify</u> either a structure forms a vector space or not	Convention al, ICT based	Green board, PPTs
	9	3b. <u>Define</u> Linear dependence of set of vectors 3c. <u>Explain</u> basis of vector space	Convention al, ICT based	Green board, PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit-IV Vector Differential Calculus	10	4a. <u>Interpret</u> the concepts of vectors 4b. <u>Evaluate</u> the gradient and directional derivatives of problems	Convention al, ICT based	Green board, PPTs
	11	4c. <u>Find</u> the divergence and curl of vector fields 4d. <u>Justify</u> the terms irrotational and solenoidal 4e. <u>Evaluate</u> the scalar potential function of vectors	Convention al, ICT based	Green board, PPTs
	12	5a. <u>Find</u> the Z-transform of <u>the given</u> sequences 5b. <u>Explain</u> various properties of the Z-transform	Convention al, ICT based	Green board, PPTs
Unit – V Z-transforms	13	5c. <u>Compute</u> the inverse Z-transform of <u>some</u> functions	Convention al, ICT based	Green board, PPTs
Unit-VI Fourier Transforms	14	6a. <u>Compute</u> the Fourier integral of <u>the given</u> functions 6b. <u>Find</u> the Fourier transform of <u>the given</u> functions	Convention al, ICT based	Green board, PPTs
	15	6c. <u>Explain</u> various properties of the Fourier transform 6d. <u>Determine</u> the Fourier sine and cosine transform of <u>some</u> functions	Convention al, ICT based	Green board, PPTs

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Advanced Engineering Mathematics	Erwin Kreyszig	8th Ed., Jhon Wiley & Sons, India, 1999
2	Integral transforms and their applications	Debnath, Lokenath, and Dambaru Bhatta	CRC press, 2014
3	Advanced Engineering Mathematics	K. A. Stroud and D. J. Booth	Palgrave Macmillan, 2006
4	Elementary linear algebra	Howard Anton and Chris Rorres	John Wiley & Sons, 2010
5	Advanced engineering mathematics	H. K. Dass	S. Chand, 2008
6	Advanced Engineering Mathematics	M. D. Greenberg	2nd ed., Pearson, 1998
7	Higher engineering mathematics	B. S. Grewal	43 rd Ed., Khanna Publishers, New Delhi, 2014
8	Thomas' Calculus	M. D. Weir et. al	11 th Ed., Pearson Education, 2008
9	Advanced Engineering Mathematics	C. Ray Wylie and L. C. Barrett	McGraw Hill pub., 1982

XV SOFTWARE/LEARNING WEBSITES

1. <http://nptel.ac.in/courses/111107098/>
2. <http://www1.maths.leeds.ac.uk/~frank/math2420/notes.pdf>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs														
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Ethics Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1	PSO2	PSO3
MA 252	Semester III	Protection and Switchgear Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO														
252	Competency Apply mathematical techniques for the solution of various problems of engineering	3	2	2	1	1	-	1	-	-	1	-	1	2	2	-
252.1	Solve the complex problems through the concepts of Laplace transforms	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
252.2	Compute the problems related to the Fourier Series	3	2	1	1	1	-	1	-	-	1	-	1	2	2	-
252.3	Explain the vector space, linear span, linear dependence, basis	3	2	-	1	1	-	-	-	-	1	-	1	2	2	-
252.4	Explain the concepts of vectors, directional derivative, gradient, vector fields, divergence and curl	3	2	-	1	1	-	-	-	-	1	-	1	2	2	-
252.5	Solve the problems through the concepts of Z-transforms	3	2	2	1	1	-	1	-	-	1	-	1	2	2	-
252.6	Find the Fourier Integral and Fourier Transform of various functions	2	2	1	-	-	-	-	-	-	1	-	1	1	1	-

XVII NAMES OF RESOURCE PERSONS

- a) Pragnesh Makwana
- b) Rajendra Chauhan

UG Engineering Programme	Semester
Electrical Engineering	Third

I RATIONALE

The course has been designed to introduce fundamental principles of analog and digital electronics. The students completing this course will understand basic analog and digital electronics, including semiconductor properties, operational amplifiers, combinational and sequential logic and analog-to-digital digital-to-analog conversion techniques. Finally, students will gain experience in with the design of analog amplifiers, power supplies and logic devices.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the performance of various analog and digital circuits used in Industrial applications.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 5) Design & Enhance their skills in basics of analog and digital circuits.
- 6) Evaluate the performance and classification of various components and ICs.
- 7) State linear and non-linear applications of operational amplifiers.
- 8) Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).
- 9) Analyze characteristics of popular MOS and bipolar logic families, with emphasis on CMOS and TTL technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				Total Marks
				Theory Marks		Practical Marks		
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	3	(A) Apply two input voltages and find output difference voltage and Gain of an op-amp. (B) Get Summing, Scaling and Averaging Amplifier of given signal using Inverting configuration.	04
2	3	Apply sine wave, square wave and check output waveforms of Integrator and Differentiator circuit.	02
3	3	(A) Compare sine wave input and DC input using op-amp. (B) Identify Zero Crossing of sine wave using op-amp.	02
4	4	(A) Get the frequency response curve of first order low pass filter. (B) plot the frequency response curve of all-Pass filter.	04
5	3,4	(A) find the frequency for building the Square Wave using op-amp. (B) find the quasi stable states for astable Multivibrator using 555 timer	04
6	2	(A) Observe the waveform using single transistor. (B) Observe the waveform using multi transistor.	04
7	4	Get the Truth table for logic gates.	02
8	4	Build the half-adder and half subtractor circuit.	02
9	4	Build the Full Adder circuit diagram using two- half adders circuit diagram.	02
10	4	Verify truth tables for J-K Flip-Flop and D Flip-Flop.	02
11	4	Apply input switching conditions for universal Bi-directional Shift Register.	
		Total	30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Analog Trainer kit, Digital Multimeter, Digital Oscilloscope	1-6
2	Digital trainer kit	7-11

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Practice building of analog & digital circuits.
- 2) Design and analyze combinational & sequential circuit.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample *Cognitive Domain* learning outcomes denoted by unit outcomes (UOs) related to *Revised Bloom's Taxonomy* (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I BJT and FET	CO 2. Evaluate the performance and classification of various components and ICs. PrO 6. (A)Plot the waveform using single transistor. (B) Observe the waveform using multi transistor 1a. construction and working principle of transistor, VI characteristics of transistor in cutoff, saturation and active region. 1b. Transistor as a switch, commonly used transistor connection, Transistor as an amplifier in CE arrangement, Analysis of collector current. 1c. Transistor DC load line and operating point, Examples of DC load line and operating point. 1d. Semiconductor devices & numbering system, Power rating of transistor, Transistor testing, Transistor as an differential amplifier, faithful amplification. 1e. Transistor biasing methods:Base resistor method, feedback resistor and voltage divider. 1f. Transistor Stabilization, Examples of Transistor biasing methods. 1g. Design of biasing circuits, Transistor stability and stability factor, Testing and transistor data sheet. 1h. Types of FET, construction and working principle of JFET, Difference between BJT and JFET, Voltampere characteristics. 1i. Pinch off voltage, Shorted gate drain current, gate source cut off voltage, Parameters of JFET, Examples. 1j. Biasing of JFET, Working of MOSFET and VI characteristics, Difference between JFET and MOSFET.	1.1 Transistor DC load line and Operating point, V-I Characteristics 1.2 Transistor as a switch and an amplifier, Faithful Amplification 1.3 Transistor biasing and its Methods, Transistor Stability and Stability Factor, Testing and Transistor data-sheet 1.4 Types of FET, Construction and working principle of JFET, Difference between BJT and JFET, V-I characteristics, Pinch-off Voltage, Shorted-gate Drain Current, Parameters of JFET, Biasing of JFET 1.5 Working of MOSFET and V-I Characteristics, Difference between JFET and MOSFET.
Unit– II Operational Amplifiers	CO 2. Evaluate the performance and classification of various components and ICs. 2a. Integrated Circuits, IC classifications, packages and symbols. 2b. Pin diagram of 741 Opamp, Internal block diagram of a Typical opamp, Ideal differential amplifier, Examples of finding voltage gain & CMRR. 2c. Ideal opamp and its equivalent circuit, Specification and Datasheet parameters of opamp. 2d. Differential Input Resistance(Ri), Output Resistance(Ro), Common Mode Rejection Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio(PSRR), Input offset, Voltage, input offset Current, input bias Current, Output offset Voltage, Bandwidth. 2e. Opamp in open loop and closed loop, DC and AC Amplifier, Voltage follower.	2.1 Introduction to Integrated Circuits, IC classifications, packages and symbols 2.2 Pin diagram of 741 Op-amp, Internal block diagram of a Typical op-amp, Ideal op-amp and its equivalent circuit 2.3 Specification and Datasheet parameters of op-amp (Differential Input Resistance (Ri), Output Resistance(Ro), Common Mode Rejection Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio (PSRR), Input offset Voltage, Input offset Current, Input bias Current, Output offset Voltage) 2.4 Feedback Configurations, An op-amp with and without Feedback, Closed-loop Voltage Gain.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit– III Applications of op-amp	CO 3. State linear and non-linear applications of operational amplifiers. PrO 1._(A) Apply two input voltages and find output difference voltage and Gain of an op-amp. (B) Get Summing, Scaling and Averaging Amplifier of given signal using Inverting configuration. PrO 2. Apply sine wave, square wave and check output waveforms of Integrator and Differentiator circuit. PrO 3. (A) Compare sine wave input and DC input using op-amp. (B) Identify Zero Crossing of sine wave using op-amp. PrO 5. (A) find the frequency for building the Square Wave using op-amp. 3a. Differential Amplifiers and Instrumentation Amplifier, Peaking, Summing, Scaling, Averaging Amplifier. 3b. Integrator, Differentiator Using op-amp. 3c. V to I converter, I to V converter. 3d. Basic Comparator, Zero crossing Detector, Schmitt trigger. 3e. Basic Positive and Negative Clippers and Clampers.	3.1. Linear applications: DC and AC Amplifier, Volt age follower, Peaking, Summing, Scaling, Averaging Amplifier, Differential Amplifiers and Instrumentation Amplifier, V to I converter, I to V converter, Integrator, Differentiator 3.2. Active filters & Oscillators: First order Low-Pass and High-Pass Filters, second order Low-Pass and High-Pass Filters, Band-Pass and Band-Reject filters, All-Pass filters, Phase Shift, Wien Bridge Oscillators; Square, Triangular, Saw-tooth Wave Generators. 3.3. Comparators & Converters: Basic Comparator, zero crossing Detector, Schmitt Trigger, Basic Positive and Negative Clippers and Clampers, Absolute-Value Output Circuit, Peak Detector, ADC and DAC using op-amp.
Unit-IV Specialized ICs	CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates, timer and counters. PrO 4. (Get the frequency response curve of first order low pass filter. (B) plot the frequency response curve of all-Pass filter. PrO 5._(B) find the quasi stable states for astable Multivibrator using 555 timer 4a. 555 Timer IC pin diagram & block diagram. 4b. 555 timer IC as Monostable, Bistable Multivibrator, 555 timer IC as an Astable Multivibrator. 4c. Voltage regulator, Positive & Negative Fixed voltage regulators using three terminals, adjustable Positive & Negative voltage regulators.	4.1 555 Timer IC pin diagram, Block diagram 4.2 555 timer IC as Monostable, bistable and astable Multivibrator. 4.3 Three terminal regulator ICs (Fixed Voltage Regulator ICs, Adjustable output voltage regulator ICs, their use and basic design considerations for designing regulated power supplies) 4.4 Analog Multiplier MPY634.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – V Introduction to Digital Electronics and Numbering Systems	CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits). 5a. Digital circuits, Digital signals, advantages, applications. 5b. Introduction, Binary number system, conversion to other systems, Decimal, Hexa decimal number system, Octal number system, Conversion to other number systems 5c. Binary arithmetic: addition, subtraction, division, multiplication, Signed & unsigned binary number system, advantages & Disadvantages. 5d. Complements: 1's and 2's complement, Binary subtraction using 1's and 2's complement. 5e. Advantages, applications, and classifications, BCD code conversion, Gray code conversion, applications of gray code, Alphanumeric codes, ASCII codes, BCD to seven segment display.	5.1 Digital circuits, Digital signals. 5.2 Binary number system, octal, hexadecimal and decimal Number systems and their inter conversion, Binary arithmetic 5.3 Signed & unsigned binary number system 5.4 1's complement and 2's complement 5.5 BCD numbers, ASCII and EBCDIC codes, Grey codes.
Unit – VI Digital Logic families and Design	CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits). PrO 7. Get the Truth table for logic gates. 6a. Boolean laws, Boolean expression simplification, logic gates building circuit. 6b. Universal gates 6c. Karnaugh Map simplification, Kmap structure, SOP form, POS form, simplification of Boolean expressions using Kmap, Way of grouping, examples 6d. Examples of Minimization of SOP expressions. 6e. Don't care condition, Examples of Minimization of POS expressions. 6f. classification based on circuit complexity: SSI, MSI, LSI, VLSI, classification based on devices used, voltage & current parameters, Fanin and fanout, Noise margin, Propagation delay, Current sourcing and sinking. 6g. Resistor transistor logic, Diode transistor logic, Integrated injection logic circuit. 6h. Two input TTL NAND gate logic, TTL NOR gate logic circuit, CMOS inverter, PMOS inverter, CMOS NAND gate logic, CMOS NOR gate logic circuit.	6.1 Basic Logic gates and their truth tables, Universal Gates 6.2 Laws of Boolean algebra, De-Morgan's theorem 6.3 Min term, Max term, POS, SOP, Karnaugh Maps, Minimization of logical function, Don't care condition 6.4 Introduction, Resistor transistor logic, Direct coupled transistor logic, Integrated injection logic, DTL logic, TTL logic, CMOS Logic
Unit – VII Combinational Logic	CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits). PrO 8. Build the half-adder and half subtractor circuit. PrO 9. Build the Full Adder circuit diagram using two-half adders circuit diagram. 7a. Analysis of a combinational circuit, Types of binary adders, Half adder logic circuit, Full adder logic	7.1 Half adder, Full adder, Binary subtraction, Arithmetic logic unit 7.2 Digital comparators, Priority encoder, decoders, Multiplexer & demultiplexer.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	circuit, Binary subtraction: Half & full subtractor 7b. Multiplexer (data selector), Necessity of Multiplexers, advantages of Multiplexers, Types of Multiplexer, Examples of implementation of combinational circuit using Multiplexer, demultiplexer. 7c. Priority encoder, Decoders: BCD to seven segment deoders.	
Unit – VIII Flip-flops, Timers and Counters	<i>CO 4. Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).</i> PrO 10. Verify truth tables for J-K Flip-Flop and D Flip-Flop. PrO 11. Apply input switching conditions for universal Bi-directional Shift Register. 8a. Priority encoder, Decoders: BCD to seven segment deoders. 8b. NAND latch (SR latch using NAND gates), triggering methods, concept of level triggering, types of level triggered flipflops, concept of Edge triggering. 8c. Level triggered SR flipflop, Level triggered D flipflop, 8d. Level triggered JK flipflop, Race around condition in JK latch, difference between latch & FlipFlop, MasterSlave JK FlipFlop. 8e. Registers, Data formats, Classification of registers, Shift register. 8f. Types of counters, Classification of counters, comparison of synchronous and asynchronous counters, Applications of counters, comparison of counters & Registers.	8.1 S-R flip-flop, D- flip-flop, T- flip-flop, J-K flip-flop, Clocked flip-flop design 8.2 Registers 8.3 Asynchronous counter, Synchronous counter

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	BJT and FET	04	2	3	0	0	2	0	07
II	Operational Amplifiers	04	2	2	0	0	0	0	7
III	Applications of op-amp	09	0	4	0	3	7	0	14
IV	Specialized ICs	05	0	2	4	4	6	0	7
V	Introduction to Digital Electronics and Numbering Systems	04	2	1	0	0	0	3	6
VI	Digital Logic families and Design	07	0	2	2	4	3	0	11
VII	Combinational Logic	06	1	2	1	2	3	0	9
VIII	Flip-flops, Timers and Counters	06	0	2	2	2	3	0	9
Total		45	0	10	14	20	16	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Transistor:** choose the circuit using BJT, FET, MOSFET FOR Electronics applications,
- 2) **555 Timer:** build the circuit using 555 timers and operate in different modes and electronics applications.
- 3) **Op-amp:** make the circuit and perform the various applications.
- 4) **Different types of sensor:** use any electronics component along with sensor and apply for electronics applications.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- e) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- f) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- g) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- h) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- e) Library survey regarding different analog and digital devices and circuits are used in industries.
- f) Prepare 15 min. power point presentation on any analog and digital devices application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I BJT and FET	1	PrO 6. (A) Observe the waveform using single transistor. (B) Observe the waveform using multi transistor. 1a. construction and working principle of transistor, V-I characteristics of transistor in cut-off, saturation and active region. 1b. Transistor as a switch, commonly used transistor connection, Transistor as an amplifier in CE arrangement, Analysis of collector current. 1c. Transistor DC load line and operating point, Examples of DC load line and operating point.	Lecture with media	PPTs, black board
	2	PrO 1. (A) Apply two input voltages and find output difference voltage and Gain of an op-amp. 1d. Semiconductor devices & numbering system, Power rating of transistor, Transistor testing, Transistor as a differential amplifier, faithful amplification. 1e. Transistor biasing Methods-Base resistor method, feedback resistor and voltage divider. 1f. Transistor Stabilization, Examples of Transistor <u>biasing methods</u>	Lecture With media	PPTs, black board
	3	PrO 1. (B) Get Summing, Scaling and Averaging Amplifier of given signal using Inverting configuration. 1g. Design of biasing circuits, Transistor stability and stability factor, Testing and transistor data sheet. 1h. Types of FET, construction and working principle of JFET, Difference between BJT and JFET, Volt-ampere characteristics. 1i. Pinch off voltage, Shorted gate drain current, gate-source cut off voltage, Parameters of JFET, Examples.	Lecture with media	PPTs, black board
Unit– II Operational Amplifiers	4	PrO 2 Apply sine wave, square wave and check output waveforms of Integrator and Differentiator circuit. 2a. Biasing of JFET, Working of MOSFET and V-I characteristics, Difference between JFET and MOSFET. 2b. Integrated Circuits, IC classifications, packages and symbols. 2c. Pin diagram of 741 Op-amp, Internal block diagram of a Typical op-amp, Ideal differential amplifier, Examples of finding voltage gain & CMRR.	Lecture with media	PPTs, black board
	5	PrO 3. (A) Compare sine wave input and DC input using op-amp. (B) Identify Zero Crossing of sine wave using op-amp.	Lecture	black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		2d. Ideal op-amp and its equivalent circuit, Specification and Datasheet parameters of op-amp. 2e. Differential Input Resistance(R_i), Output Resistance(R_o), Common Mode Rejection, Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio(PSRR), Input offset, Voltage, input offset Current, input bias Current, Output offset Voltage, Bandwidth. 2f. Op-amp in open-loop and closed loop, DC and AC Amplifier, Voltage follower.		
Unit– III Applications of op-amp	6	PrO 4. (A) Get the frequency response curve of first order low pass filter. (B) plot the frequency response curve of all-Pass filter. 3a. Differential Amplifiers and Instrumentation Amplifier, Peaking, Summing, Scaling, Averaging Amplifier. 3b. Integrator, Differentiator Using op-amp. 3c. V to I converter, I to V converter with floating load & grounded load.	Lecture with media	PPTs, black board
	7	3d. Basic Comparator, Zero crossing Detector, Schmitt trigger 3e. Basic Positive and Negative Clippers and Clampers.	Lecture with media	PPTs, black board
Unit– IV Specialized ICs	8	PrO 5. (A) find the frequency for building the Square Wave using op-amp. (B) find the quasi stable states for astable Multivibrator using 555 timer 4a. 555 Timer IC pin diagram & block diagram. 4b. 555 timer IC as Monostable, Bistable Multivibrator. 4c. 555 timer IC as an Astable Multivibrator. 4d. Voltage regulator, Block diagram, Positive & Negative Fixed voltage regulators using three terminal, adjustable Positive & Negative voltage regulators	Lecture with media	PPTs, black board
Unit– V Introduction to Digital Electronics and Numbering Systems	9	PrO 7. Get the Truth table for logic gates. 5a. Digital circuits, Digital signals, advantages, applications. 5b. Binary number system, conversion to other systems. 5c. Decimal, Hexa decimal number system, Octal number system, Conversion to other number systems	Lecture	PPTs, black board
	10	PrO 8. Build the half-adder and half subtractor circuit. 5d. Binary arithmetic: addition, subtraction, division, multiplication. 5e. Signed & unsigned binary number system, advantages & Disadvantages. 5f. Complements: 1's and 2's complement, Binary subtraction using 1's and 2's complement 5g. Binary codes: Advantages, applications, and	Lecture	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		classifications, BCD code conversion, Gray code conversion, applications of gray code, Alphanumeric codes, ASCII codes, BCD to seven segment display.		
Unit– VI Digital Logic families and Design	11	PrO 9. Build the Full Adder circuit diagram using two- half adders circuit diagram. 6a. classification based on circuit complexity: SSI, MSI, LSI, VLSI, classification based on devices used, voltage & current parameters, Fan-in and fan-out, Noise margin, Propagation delay, Current sourcing and sinking. 6b. Resistor transistor logic, Diode transistor logic, Integrated injection logic circuit. 6c. Two input TTL NAND gate logic, TTL NOR gate logic circuit. 6d. CMOS inverter, PMOS inverter, CMOS NAND gate logic, CMOS NOR gate logic circuit.	Lecture with media	PPTs, black board
Unit– VII Combinational Logic	12	PrO 10. Verify truth tables for J-K Flip-Flop and D Flip-Flop. 7a. Boolean laws, Boolean expression simplification, logic gates building circuit. 7b. Karnaugh Map simplification, K-map structure, SOP form, POS form, simplification of Boolean expressions using K-map, Way of grouping, 7c. Sop form examples	Lecture	black board
	13	7d. Examples of Minimization of SOP expressions, Don't care condition, 7e. Examples of Minimization of POS expressions. 7f. Analysis of a combinational circuit, Types of binary adders, Half adder logic circuit, Full adder logic circuit, Binary subtraction: Half & full subtractor.	Lecture	black board
	14	7g. Binary subtraction: Half & full subtractor. 7h. Multiplexer (data selector), Necessity of Multiplexers, advantages of Multiplexers, Types of Multiplexer, Examples of implementation of combinational circuit using Multiplexer, demultiplexer.	Lecture	black board
Unit– VIII Flip-flops, Timers and Counters	15	Pro 10. Apply input combinations and switching conditions for universal Bi-directional Shift Register. 8a. Difference between combination circuit & sequential circuit, clock signal, clock skew, 1-Bit memory cell, Latch. 8b. NAND latch (S-R latch using NAND gates), triggering methods, concept of level triggering, types of level triggered flip-flops, concept of Edge triggering.	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		Level triggered S-R flip-flop, Level triggered D-flip-flop 8c. Level triggered J-K flip-flop, Race around condition in JK latch, difference between latch & Flip-Flop, Master-Slave JK Flip-Flop. 8e. Registers, Data formats, Classification of registers, Shift register, Counters: Types of counters, Classification of counters, comparison of synchronous and asynchronous counters, Applications of counters, comparison of counters & Registers.		

XVIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Principles of Electronics	V. K. Mehta	S. Chand & Company Ltd. ISBN:81-219-2450-2
2	Op-Amp and Linear Integrated Circuit technology	Ramakant A Gayakwad	PHI Publication ISBN: 0132808684, 9780132808682
3	Analog and Digital Electronics	J. S. Katre	J. S. Katre ISBN: 978-81-8492-140-3
4	Fundamental of digital circuits	A. ANANDKUMAR	PHI Publication ISBN: 8120352688, 9788120352681
5	Operational Amplifier and Linear Integrated Circuits	James M. fiore	Jaico Publishing House ISBN:8172247788, 9788172247782
6	Electronics Devices and Circuits an Introduction	A. Mottershed	PHI Publication ISBN:8120301242, 9788120301245
7	Digital Fundamentals	Morris and Mano	PHI Publication ISBN-13: 978-0-13-277420-8 ISBN-10: 0-13-277420-8

XIX SOFTWARE/LEARNING WEBSITES

- <http://nptel.ac.in/courses/108106068/>
(Videos and Notes by Prof. K. Radhakrishna Rao, IIT Madras)
- <http://nptel.ac.in/courses/108106069/>
(Videos and Notes by Dr. Amitava Dasgupta, IIT Madras)

XX PO-COMPETENCY-CO MAPPING

				POs and PSOs
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Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Project management and finance	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	PSO3
EE251	Semester III				Analog and Digital Electronics Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO											
251	Competency Evaluate the performance of various analog and digital circuits used in Industrial applications.	3	3	3	2	2	0	0	0	0	0	-	-	3	1	-
251.1	Design & Enhance their skills in basics of analog and digital circuits.	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
251.2	Understand the performance and classification of various components and ICs.	2	3	-	-	-	-	-	-	-	-	-	-	2	-	-
251.3	State linear and non-linear applications of operational amplifiers.	1	3	-	-	-	-	-	-	-	-	-	-	2	-	-
251.4	Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates (combinational circuits), timer and counters (sequential circuits).	1	2	2	3	3	-	-	-	-	-	-	-	3	1	-
251.5	Analyze characteristics of popular MOS and bipolar logic families, with emphasis on CMOS and TTL technologies.	1	1	3	3	-	-	-	-	-	-	-	-	2	1	-

XXI NAMES OF RESOURCE PERSONS

a) **Ankur H Patel**

UG Engineering Programme	Semester
Circuit Theory	Third

I RATIONALE

Electrical engineers have to work on different electrical circuits consisting of electrical elements. This course is developed such that students should be able to analyze, evaluate, design, develop, implement various circuits efficiently for circuit analysis and synthesis. Student should be able to develop, design the circuit parameter according to the system need.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Analyze the performance of various circuit elements used in electrical circuits for electrical systems.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Use the different methods to find the solutions of circuit parameters.

Evaluate the circuit performance with the help of network topology.

Evaluate the performance of the circuits for Two Port parameters.

Evaluate the performance of circuits with Time and Laplace domain for circuits having initial conditions.

Analyze the 1st, 2nd order RL, RC and RLC series and parallel circuits.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	2	0	4	70	30	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Determine</u> the Nodal voltage for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
2	1	<u>Determine</u> the Mesh current for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
3	1	<u>Determine</u> the current and voltage for DC resistive circuits with the help of superposition theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
4	1	<u>Determine</u> the load current and voltage for DC resistive circuits by designing the Thevenin's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
5	1	<u>Determine</u> the load current and voltage for DC resistive circuits by designing the Norton's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
6	1	<u>Determine</u> the load resistance value for DC resistive circuits by using Maximum Power Transfer Theorem and check the effect of change in the load resistor for the power variation.	02
7	1	<u>Determine</u> the load current and voltage for DC resistive circuits by using Reciprocity theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.	02
8	3	<u>Examine</u> the performance of the circuit by using Z parameters concept for Two Port network with the help of graphical representation using MATLAB software.	02
9	3	<u>Examine</u> the performance of the circuit by using Y parameters concept for Two Port network with the help of graphical representation using MATLAB software.	02
10	3	<u>Examine</u> the performance of the circuit by using ABCD and A'B'C'D' parameters concept for Two Port network with the help of graphical representation using MATLAB software.	02
11	4, 5	<u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software.	04
Total			24

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Bread board, DC power supply,	1 to 10
2	MATLAB Simulation software (version 2021a)	1 to 11

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.

- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I DC and AC Circuit Simplification Techniques.	CO 1. <u>Use</u> the different methods to find the solutions of circuit parameters. PrO 1. <u>Determine</u> the Nodal voltage for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink . PrO 2. <u>Determine</u> the Mesh current for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 3. <u>Determine</u> the current and voltage for DC resistive circuits with the help of superposition theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink.. PrO 4. <u>Determine</u> the load current and voltage for DC resistive circuits by designing the Thevenin's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 5. <u>Determine</u> the load current and voltage for DC resistive circuits by designing the Norton's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 6. <u>Determine</u> the load resistance value	1.1 Circuit Elements And Their Characteristics, Energy Sources, Source Transformation 1.2 Kirchhoff's laws, Current division, Voltage division in series circuits, Nodal and mesh analysis of electric circuits in DC and AC excitations. 1.3 Superposition theorem, Thevenin theorem, Norton Theorem, Reciprocity and Maximum power transfer theorem in Electrical Circuits with DC and AC excitations.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	for DC resistive circuits by using Maximum Power Transfer Theorem and check the effect of change in the load resistor for the power variation. PrO 7. <u>Determine</u> the load current and voltage for DC resistive circuits by using Reciprocity theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1a. <u>Describe</u> the Circuit Elements And Their Characteristics, Energy Sources. 1b. <u>Apply</u> Source Transformation technique for circuit simplification. 1c. <u>Illustrate</u> KCL, KVL, current division rule and voltage division rule in electrical circuits. 1d. <u>Determine</u> nodal voltage and mesh current with nodal and mesh analysis technique for DC and AC circuits. 1e. <u>Examine</u> the circuit performance with the help of theorems. 1f. <u>Design</u> the simple circuits with the help of theorems from the complex circuits. 1g. <u>Choose</u> the optimized circuit parameters for the maximum power transfer with the help of MPT theorem.	
Unit– II Network Topology	CO 2. <u>Evaluate</u> the circuit performance with the help of network topology. 2a. <u>Identify</u> the terminology used in network graph. 2b. <u>Prepare</u> Incidence, Reduced incidence matrix, Tie-Set, Fundamental Tie-Set matrix, Fundamental Cut-Set, Cut-Set matrix . 2c. <u>Solve</u> the circuits with Dot Convention method for coupled circuit. 2d. <u>Apply</u> Principle Of Duality for creating dual networks.	2.1 Concept Of Network Graph, Terminology Used In Network Graph 2.2 Incidence, Reduced incidence matrix ,Tie-Set Matrix, Fundamental Tie-Set Matrix, Fundamental Cut-Set, Cut-Set Matrix, 2.3 The Dot Convention For Coupled Circuit, Principle Of Duality
Unit– III Two Port Parameters	CO 3. <u>Evaluate</u> the performance of the circuits for Two Port parameters. PrO 8. <u>Examine</u> the performance of the circuit by using Z parameters concept for Two Port network with the help of graphical representation using MATLAB software. PrO 9. <u>Examine</u> the performance of the circuit by using Y parameters concept for Two Port network with the help of	3.1 Network Elements, Classification of Network, Network Configuration, Recurrent Network. 3.2 Z-parameters, Y-parameters, ABCD Parameters, Inverse ABCD Parameters. 3.3 Interrelationship between parameters.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	graphical representation using MATLAB software. <i>PrO 10. Examine</i> the performance of the circuit by using ABCD and A'B'C'D' parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3a. <u>Identify</u> the Network Elements and its configuration for two port networks. 3b. <u>Evaluate</u> Open circuit (Z) parameter, Short circuit (Y) parameter for two port network and review the conditions for reciprocity and symmetricity. 3c. <u>Evaluate</u> Transmission (ABCD) parameter, Inverse Transmission (A'B'C'D') parameter for two port network and review the conditions for reciprocity and symmetricity. 3d. <u>Express</u> Z, Y, T and T' parameters by each parameters.	3.4 Concept of Reciprocity & Symmetricity for network parameters
Unit-IV Laplace transformation and Initial Conditions	CO 4. <u>Evaluate</u> the performance of circuits with Time and Laplace domain for circuits having initial conditions. <i>PrO 11. Analyze</i> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4a. <u>List</u> basic types of special signals and Laplace transformation formulas used in circuit analysis. 4b. <u>Apply</u> Laplace transformation technique for representing the transform impedance and transform admittance circuits. 4c. <u>Illustrate</u> the factors of initial conditions in the circuits and its procedure to evaluate the derivatives. 4d. <u>Analyze</u> the circuit performance under initial condition.	4.1 Basic Types Of Special Signals, Laplace Transformation Of Special Signal Waveforms. 4.2 Transform impedance and transform admittance circuits. 4.3 Initial conditions in elements, Geometrical Interpretation of Derivatives. A Procedure of Evaluating Initial conditions. 4.4 R-L, R-C & R-L-C Circuits analysis using Initial Conditions.
Unit – V Steady state and Transient Response Analysis of circuit	CO 5. <u>Analyze</u> the 1st, 2nd order RL, RC and RLC series and parallel circuits. <i>PrO 11. Analyze</i> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 5a. <u>Identify</u> the 1st and 2nd order circuits and	5.1 Step Response 1st order and 2nd order circuits. 5.2 Step response of Series & Parallel Circuit 5.3 Transient Response of Series R-L, R-C, R-L-C With DC Excitation. 5.4 Transient Response of Series R-L, R-C, R-L-C With

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	its behaviors for step input. 5b. <u>Examine</u> the performance of the series and parallel circuits under step response. 5c. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with DC excitation. 5d. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with AC excitation.	AC Excitation.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	DC and AC Circuit Simplification Techniques	12	02	04	04	04	04	02	20
II	Network Topology	05	02	02	02	00	00	00	06
III	Two Port Parameters	12	04	04	04	02	04	00	18
IV	Laplace transformation and Initial Conditions	10	02	02	02	02	06	02	16
V	Steady state and Transient Response Analysis of circuits	06	00	02	02	04	02	02	10
Total		45	10	12	12	10	20	06	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Resistors, Inductors & Capacitors:** Collect the specifications of different types of electromagnetic relays used in different protection situations in different types of industries.

- 2) **Voltage and Current Sources:** Collect the specifications of different types of static relays used in different protection situations in different types of industries.
- 3) **Initial Condition based performance of RLC circuits:** Prepare case studies of protections failures in different types of industries.
- 4) **Two port network in power systems:** Prepare case studies of circuit breakers used in the protection schemes in different types of industries.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L**' in item No. 4 does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different circuit elements used in electrical circuits.
- b) Prepare 15 min. power point presentation on the various parameter configurations used modern power system analysis.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I DC and AC Circuit Simplifica tion Techniqu es.	1	CO 1. <u>Use</u> the different methods to find the solutions of circuit parameters. PrO 1. <u>Determine</u> the Nodal voltage for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1a. <u>Describe</u> the Circuit Elements And Their Characteristics, Energy Sources. 1b. <u>Apply</u> Source Transformation technique for circuit simplification. 1c. <u>Illustrate</u> KCL, KVL, current division rule and voltage division rule in electrical circuits.	Lecture with media	PPTs, black board
	2	CO 1. <u>Use</u> the different methods to find the solutions of circuit parameters. PrO 2. <u>Determine</u> the Mesh current for DC resistive	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 3. <u>Determine</u> the current and voltage for DC resistive circuits with the help of superposition theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1d. <u>Determine</u> nodal voltage and mesh current with nodal and mesh analysis technique for DC and AC circuits. 1e. <u>Examine</u> the circuit performance with the help of theorems.		
	3	PrO 4. <u>Determine</u> the load current and voltage for DC resistive circuits by designing the Thevenin's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. PrO 5. <u>Determine</u> the load current and voltage for DC resistive circuits by designing the Norton's Equivalent circuit and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1e. <u>Examine</u> the circuit performance with the help of theorems. 1f. <u>Design</u> the simple circuits with the help of theorems from the complex circuits	Lecture with media Tutorial and Field visit	PPTs, black board
	4	PrO 6. <u>Determine</u> the load resistance value for DC resistive circuits by using Maximum Power Transfer Theorem and check the effect of change in the load resistor for the power variation. PrO 7. <u>Determine</u> the load current and voltage for DC resistive circuits by using Reciprocity theorem and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink 1f. <u>Design</u> the simple circuits with the help of theorems from the complex circuits. 1g. <u>Choose</u> the optimized circuit parameters for the maximum power transfer with the help of MPT theorem.		
Unit– II Network Topology	5	CO 2. <u>Evaluate</u> the circuit performance with the help of network topology.	Lecture with media, and Demonstra	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– III Two Port Parameters		2a. <u>Identify</u> the terminology used in network graph. 2b. <u>Prepare</u> Incidence, Reduced incidence matrix, Tie-Set, Fundamental Tie-Set matrix, Fundamental Cut-Set, Cut-Set matrix.	tion	
	6	CO 2. <u>Evaluate</u> the circuit performance with the help of network topology. 2c. <u>Solve</u> the circuits with Dot Convention method for coupled circuit. 2d. <u>Apply</u> Principle Of Duality for creating dual networks.	Lecture with media, Tutorial	PPTs
	7	CO 3. <u>Evaluate</u> the performance of the circuits for Two Port parameters. <i>PrO 8.</i> <u>Examine</u> the performance of the circuit by using Z parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3a. <u>Identify</u> the Network Elements and its configuration for two port networks. 3b. <u>Evaluate</u> Open circuit (Z) parameter, Short circuit (Y) parameter for two port network and review the conditions for reciprocity and symmetry.	Lecture with media, Field visit	PPTs, Video
	8	<i>PrO 9.</i> <u>Examine</u> the performance of the circuit by using Y parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3b. <u>Evaluate</u> Open circuit (Z) parameter, Short circuit (Y) parameter for two port network and review the conditions for reciprocity and symmetry.	Lecture with media, and Tutorial	PPTs
	9	<i>PrO 8.</i> <u>Examine</u> the performance of the circuit by using ABCD and A'B'C'D' parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3c. <u>Evaluate</u> Transmission (ABCD) parameter, Inverse Transmission (A'B'C'D') parameter for two port network and review the conditions for reciprocity and symmetry.	Lecture with media, and Tutorial	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	10	<i>PrO 8. Examine</i> the performance of the circuit by using ABCD and A'B'C'D' parameters concept for Two Port network with the help of graphical representation using MATLAB software. 3d. <i>Evaluate</i> Transmission (ABCD) parameter, Inverse Transmission (A'B'C'D') parameter for two port network and review the conditions for reciprocity and symmetry. 3e. <i>Express</i> Z, Y, T and T' parameters by each parameters.	Lecture with media, and Tutorial	PPTs
Unit– IV Laplace transformation and Initial Conditions	11	CO 4. Evaluate the performance of circuits with Time and Laplace domain for circuits having initial conditions. <i>PrO 11. Analyze</i> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4a. <i>List</i> basic types of special Signals and Laplace transformation formulas used in circuit analysis. 4b. <i>Apply</i> Laplace transformation technique for representing the transform impedance and transform admittance circuits.	Lecture with media, and Tutorial	PPTs
	12	CO 4. Evaluate the performance of circuits with Time and Laplace domain for circuits having initial conditions. <i>PrO 11. Analyze</i> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4c. <i>Illustrate</i> the factors of initial conditions in the circuits and its procedure to evaluate the derivatives.	Lecture with media, and Tutorial	PPTs
	13	CO 4. Evaluate the performance of circuits with Time and Laplace domain for circuits having initial conditions. <i>PrO 11. Analyze</i> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 4d. <i>Analyze</i> the circuit performance under initial condition.	Lecture with media, and Tutorial	PPTs
Unit– V Steady state and Transient Response	13	CO 4. Analyze the 1st, 2nd order RL, RC and RLC series and parallel circuits. <i>PrO 12. Analyze</i> the transient and steady state response of R-L, R-C, R-L-C series and parallel	Lecture with media, and Tutorial	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Analysis of circuits		circuits using MATLAB software. 5a. <u>Identify</u> the 1 st and 2 nd order circuits and its behaviors for step input.		
	14	CO 5. <u>Analyze</u> the 1st,2nd order RL, RC and RLC series and parallel circuits. PrO 13. <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 5b. <u>Examine</u> the performance of the series and parallel circuits under step response. 5c. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with DC excitation.	Lecture with media, and Tutorial	PPTs
	15	CO 5. <u>Analyze</u> the 1st,2nd order RL, RC and RLC series and parallel circuits. PrO 14. <u>Analyze</u> the transient and steady state response of R-L, R-C, R-L-C series and parallel circuits using MATLAB software. 5c. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with DC excitation. 5d. <u>Analyze</u> the R-L, R-C and R-L-C circuit performance with AC excitation.	Lecture with media, and Tutorial	PPTs

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Circuit Theory Analysis & Synthesis	Abhijit Chakrabarti	Dhanpat Rai Publishing Company (P) Limited, 2008, 8177000004, 9788177000009
2	Network analysis	Mac Elwyn Van Valkenburg	Pearson Publications. 9789332550131, 9332550131
	Electrical Circuit Theory and Technology	John Bird	Taylor & Francis 9781136347115, 1136347119
3	Network Analysis and Synthesis	by Ravish R Singh,	McGraw Hill Education Pvt. Ltd 9781259062957, 1259062953
4	Network analysis	G. K. Mithal,	Khanna publications 9788174090706, 8174090703
5	Electric Circuits and Networks	K. S. Suresh Kumar	Pearson Education 9788131713907, 8131713903
6	U. A. Patel, Network analysis and Synthesis	Late Ajay V. Bakshi, Uday A. Bakshi	Mahajan Publisher 9788189050443

XIV SOFTWARE/LEARNING WEBSITES

1. <http://nptel.ac.in/courses/108102042/9>
2. <http://freevideolectures.com/Course/2350/Networks-Signals-and-Systems>

XV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design /development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE252.01	Semester III			Protection and Switchgear Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO											
252.01	Competency Analyze the performance of various circuit elements used in electrical circuits for electrical systems.	3	3	3	2	2	-	-	-	-	-	-	2	3	1
CO1	Use the different methods to find the solutions of circuit parameters.	3	3	2	-	1	-	-	-	-	-	-	2	3	-
CO2	Evaluate the circuit performance with the help of network topology.	3	2	2	-	-	-	-	-	-	-	-	-	1	1
CO3	Evaluate the performance of the circuits for Two Port parameters.	3	2	2	2	-	-	-	-	-	-	-	2	3	1
CO4	Evaluate the performance of circuits with Time and Laplace domain for circuits having initial conditions.	3	2	3	2	1	-	-	-	-	-	-	-	2	1
CO5	Analyze the 1st,2nd order RL, RC and RLC series and parallel circuits.	3	3	3	2	3	-	-	-	-	-	-	-	2	1

XVI NAMES OF RESOURCE PERSONS

- a) Mahammadsoaib Saiyad
- b)

UG Engineering Programme	Semester
Electrical Engineering	Third

I RATIONALE

The process of measuring and controlling various quantities in industries by utilizing various industrial instruments is called as industrial instrumentation. Sensors and instrumentation devices are often used in industrial applications for condition monitoring, flow monitoring, stability monitoring, pressure and temperature data collection for process control. In view of this, study of principles of electrical measurement and industrial instruments become mandatory for all electrical engineers. This course deals with the principles of measurement, sensors and transducers.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Memorize different types of electrical instrumentation systems and transducers.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Memorize** the different sensors, transducers and various measuring instruments.
- 2) **Memorize** basic principle, working, characteristics and applications of the various measuring instruments and transducers.
- 3) **Interpret** the measured results and cause of any possible error.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Memorize different sensors, transducers and various measuring instruments used in industries.	02
2	2	Perform the stress and strain analysis of industrial equipment.	02
3	2	Select a temperature sensor based on its operating characteristic.	06
4	2	Explain the operating principle of flow restriction sensors.	02
5	3	Interpret the condition of rotating equipment based on torque analysis.	02
6	2	Explain the operating principle of level sensors.	02
7	2	Explain the operating principle of humidity sensors.	02
8	1	Memorize single phase energy meter.	02
9	3	Use DC ammeter and voltmeter for different ranges.	04
10	3	Calibrate ammeter, voltmeter and wattmeter as per IS.	02
11	3	Determine the different ratios and errors for potential transformer.	02
12	3	Determine the different ratios and errors for current transformer.	02
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Strain Gauge trainer kit	2
2	Temperature trainer kit	3
3	Orifice meter with flow arrangement	4
4	Air Purge System	5
5	Resistive Hygrometer trainer kit	6
6	Strain Gauge Torque transducer with single phase induction motor arrangement	7
7	DMMS	9
8	Potential Transformer	10
9	Current Transformer	11
10	Calibration Tester	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Function as a team member.
- 3) Function as a team leader.
- 4) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organizing Level' in 3rd year
- d) 'Characterizing Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction of Instrumentations and Measurement	CO 1. <u>Memorize</u> the different sensors, transducers and various measuring instruments. <i>PrO 1. <u>Memorize different sensors, transducers and various measuring instruments used in industries.</u></i> 1a. <u>Justify</u> the requirement of sensors/transducers for the <u>given</u> electrical equipment. 1b. <u>Memorize</u> the different sensors, transducers and various measuring instruments. 1c. <u>Select</u> the sensor/transducer based on its performance characteristics for <u>the given</u> application.	1.1 History of Instrumentation and Measurements 1.2 Application of Instrument systems 1.3 Classification of transducers 1.4 Performance characteristics of instruments
Unit– II Temperature and Strain Measurement	CO 2. <u>Explain</u> basic principle, working, characteristics and applications of the various measuring instruments and transducers. <i>PrO 2. <u>Perform the stress and strain analysis of industrial equipment.</u></i> <i>PrO 3. <u>Select a temperature sensor based on its operating characteristic.</u></i> 2a. Explain the basic principle, working, characteristics and applications of strain gauge transducers. 2b. Explain the basic principle, working, characteristics and application of temperature sensors. 2c. Select a temperature sensor based on its operating characteristic.	2.1 Temperature Scales 2.2 Non-electrical methods for temperature measurement 2.3 Electrical methods for temperature measurement 2.4 Radiation Pyrometers 2.5 Strain Measurement using Piezoresistive strain gauge 2.6 Applications of Strain Gauges 2.7 Load Cells
Unit– III Flow and Pressure Measurement	CO 2. <u>Explain</u> basic principle, working, characteristics and applications of the various measuring instruments and transducers. <i>PrO 2. <u>Perform the stress and strain analysis of industrial equipment.</u></i> <i>PrO 4. <u>Explain the operating principle of flow restriction sensors.</u></i> 3a. Explain the basic principle, working, characteristics and application of flow sensors. 3b. Explain the basic principle, working, characteristics and application of pressure sensors.	3.1 Flow Restriction Sensors – Orifice meter, Venturi meter, Flow nozzle, Rotameter, Pitot Tube 3.2 Electrical Flow Sensors – Turbine Flow meter, Electromagnetic Flow meter, Hot wire anemometer, Ultrasonic flow meter 3.3 Elastic Pressure Sensors – Diaphragm, Bourdon Tube, Bellows 3.4 Electrical Pressure Sensors – Variable inductance and capacitance transducers, Piezoelectric Sensor
Unit –IV Measurement of Liquid Level & Humidity	CO 2. <u>Explain</u> basic principle, working, characteristics and applications of the various measuring instruments and transducers. <i>PrO 5. <u>Explain the operating principle of level sensors.</u></i> <i>PrO 6. <u>Explain the operating principle of humidity sensors.</u></i> 4a Explain the basic principle, working, characteristics and application of level sensors.	4.1 Level Measurement – Direct Methods (Dip stick, sight glass, and float), Air purge method, inductive and capacitive transducers, ultrasonic level gauge, and radiation method. 4.2 Humidity measurement – Dew point meter, Hair hygrometer, Psychrometer, Resistive and

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
	4b Explain the basic principle, working, characteristics and application of humidity sensors.	capacitive hygrometer
Unit – V Measurement of Electrical Parameters	CO 2. <u>Explain</u> basic principle, working, characteristics and applications of the various measuring instruments and transducers. CO 3. <u>Interpret</u> the measured results and cause of any possible error. PrO 7. <u>Memorize single phase energy meter.</u> PrO 8. <u>Use DC ammeter and voltmeter for different ranges.</u> PrO 9. <u>Calibrate ammeter, voltmeter and wattmeter as per IS.</u> 5a Explain the basic principle, working, characteristics and application of analog instruments. 5b Explain the basic principle, working, characteristics and application of different bridge circuits. 5c Calibrate ammeter, voltmeter and wattmeter.	5.1 Instruments for measuring voltage, current, power, energy, frequency, power factor 5.2 Extension of Instrument Ranges and Multipliers, problems 5.3 Calibration of ammeter, voltmeter, wattmeter and energy meter. 5.4 Measurement of low, medium and high resistances - Wheatstone Bridge, Making balanced Wheatstone Bridge measurement, Kelvin Double Bridge, problems 5.5 Measurement of inductance & capacitance with the help of AC Bridges (Hays Bridge, Schering Bridge, Maxwell bridge, Anderson Bridge), problems
Unit – VI Instrument Transformer	CO 2. <u>Explain</u> basic principle, working, characteristics and applications of the various measuring instruments and transducers. CO 3. <u>Interpret</u> the measured results and cause of any possible error. PrO 10. <u>Determine the different ratios and errors for potential transformer.</u> PrO 11. <u>Determine the different ratios and errors for current transformer.</u> 6a Explain the basic principle, working, characteristics and application of potential transformers. 6b Explain the causes of errors and its solution in potential transformers. 6c Explain the basic principle, working, characteristics and application of current transformers. 6d Explain the causes of errors and its solution in current transformers.	6.1 Current Transformer – Types, Characteristics, Errors, Vector diagram, problems 6.2 Potential Transformer - Types, Characteristics, Errors, Vector diagram, problems

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Instrumentations and Measurement	04	2	2	2	0	0	0	06
II	Temperature and Strain Measurement	12	2	2	4	0	8	0	16
III	Flow and Pressure Measurement	08	2	2	4	0	0	0	08

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
IV	Measurement of Liquid Level & Humidity	04	2	2	0	0	0	0	04
V	Measurement of Electrical Parameters	16	2	2	4	4	6	0	18
VI	Instrument Transformer	16	2	2	4	2	8	0	18
Total		60	12	12	16	08	20	02	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- 'L' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XI SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Library survey regarding different sensors used in industries.
- Prepare 15 min. power point presentation on how to choose sensor for the particular application.

XII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction of Instrumentation s and Measurement	1	Pro 1. <u>Memorize different sensors, transducers and various measuring instruments used in industries.</u> 1a. <u>Justify the requirement of sensors/transducers for the given electrical equipment.</u> 1b. <u>Memorize the different sensors, transducers and various measuring instruments.</u>	Lecture with media	PPTs, black board
	2	Pro 2. <u>Perform the stress and strain analysis of industrial equipment.</u> 2a. Explain the basic principle, working, characteristics and applications of strain gauge transducers.	Lecture With media	PPTs, black board
Unit– II Temperature and Strain Measurement	3 & 4	Pro 3. <u>Select a temperature sensor based on its operating characteristic.</u> 2b. Explain the basic principle, working, characteristics and application of temperature sensors. 2c. Select a temperature sensor based on its	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		operating characteristic.		
Unit– III Flow and Pressure Measurement	5	Pro 2. <u>Perform the stress and strain analysis of industrial equipment.</u> 3a. Explain the basic principle, working, characteristics and application of flow sensors.	Lecture With media	PPTs, black board
	6	Pro 4. <u>Explain the operating principle of flow restriction sensors.</u> 3b. Explain the basic principle, working, characteristics and application of pressure sensors.	Lecture With media	PPTs, black board
Unit – IV Measurement of Liquid Level & Humidity	7	Pro 5. <u>Explain the operating principle of level sensors.</u> 5a. Explain the basic principle, working, characteristics and application of level sensors.	Lecture With media	PPTs, black board
		Pro 6. <u>Explain the operating principle of humidity sensors.</u> 5b. Explain the basic principle, working, characteristics and application of humidity sensors.	Lecture With media	PPTs, black board
Unit – V Measurement of Electrical Parameters	8 & 9	Pro 7. <u>Memorize single phase energy meter.</u> Pro 8. <u>Use DC ammeter and voltmeter for different ranges.</u> 7a. Explain the basic principle, working, characteristics and application of analog instruments.	Lecture With media	PPTs, black board
	10 & 11	Pro 9. <u>Calibrate ammeter, voltmeter and wattmeter as per IS.</u> 7b. Explain the basic principle, working, characteristics and application of different bridge circuits. 7c. Calibrate ammeter, voltmeter and wattmeter.	Lecture With media	PPTs, black board
Unit – VI Instrument Transformer	12 & 13	Pro 10. <u>Determine the different ratios and errors for potential transformer.</u> 8a. Explain the basic principle, working, characteristics and application of potential transformers. 8b. Explain the causes of errors and its solution in potential transformers.	Lecture With media	PPTs, black board
	14 & 15	Pro 11. <u>Determine the different ratios and errors for current transformer.</u> 8c. Explain the basic principle, working, characteristics and application of current transformers. 8d. Explain the causes of errors and its solution in current transformers.	Lecture With media	PPTs, black board

XIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electrical and Electronics Measurement	Rajput, R. K.	S. Chand, , ISBN 978-9380386348

S. No.	Title of Book	Author	Publication including ISBN
	and Instrumentation		
2	Electrical and Electronics Measurement and Instrumentation	Shawhney, A.K.	Dhanpat Rai Publisher, New Delhi

XIV SOFTWARE/LEARNING WEBSITES

- www.isa.org
- <http://nptel.iitm.ac.in/video.php?courseId=1062>
- http://www.mywbut.com/syllabus.php?mode=SM&paper_id=57&dept_id=6

XV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances Artificial intelligence, machine optimization, design, simulation and analysis, electrical provide solutions in real world
EE243 .01	Semester III	Electrical Measurement and Industrial Instrumentation Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
243.0 1	Competency Maintain different types of electrical instrumentation systems and transducers.	3	2	-	-	-	1	-	-	2	1	-	1	2	
243.0 1.1	Memorize the different sensors, transducers and various measuring instruments.	3	-	1	1	-	2	-	-	2	1	-	2	2	
243.0 1.2	Explain basic principle, working, characteristics and applications of the various measuring instruments and transducers.	3	-	2	-	-	2	-	-	2	2	-	2	2	
243.0 1.3	Interpret the measured results and cause of any possible error.	3	2	-	2	-	2	-	-	2	2	-	2	2	

XVI NAMES OF RESOURCE PERSONS

- Prof. Mihir A. Bhatt

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS121.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION

I. Credits and Schemes:

Se m	Course Code	Course Name	Credit s	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
III	HS121.02 A	Creativity, Problem Solving and Innovation	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

III. Detailed Syllabus:

1.	Introduction to Creativity, Problem Solving and Innovation	06 Hours	20%
	Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills		
2.	Questioning, Learning and Visualization	6 Hours	20%
	Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3.	Creative Thinking and Problem Solving	6 Hours	20%
	Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		

4.	Logic, Language and Reasoning	6 Hours	20%
	Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5.	Contemporary Issues and Practices in Creativity and Problem Solving	6 Hours	20%
	Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

IV. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

V. Evaluation:

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them. There may also be a Computer Based Test at the end.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Attendance	100 %	--	20
2	Individual Activity Participation	As stipulated by the Resource Person(s) in the Training		20
3	Group Activity Participation			20
4	Presentation			30
5	Feedback on Improvement			10
Total				100

VI. Course Outcomes (COs)

At the end of the course, learners will be able to:

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
--	-----	-----	-----	-----	-----	-----	-----	-----	-----	------	------	------	------	------

CO1	-	-	-	-	-	-	-	-	-	2	-	2	-	-
CO2	-	-	-	2	-	-	-	-	-	2	-	3	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	2	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO6	-	-	-	3	-	-	-	-	-	2	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VII. Books / Reference Books / Reading

Text Books

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference Books

3. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
4. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
5. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
6. Tony Buzan, How to Mind Map, Thorsons Publication 2002
7. Scott Berkum, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
8. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
9. Ira Flatow, The all Laughed, Harper Publication 1992
10. Paul Sloane, Des Mac Hale & M.A. Di Spezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

11. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
12. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
13. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
14. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
15. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
16. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
17. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
18. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
19. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
20. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
21. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
22. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
23. Howard Gardner, Creating minds, Basic Books Publication 1993

24. MihalyCsikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention,Harper Publication 1996
25. Martin Gerdner,W. H.,Ahal Insight, Freeman Publication 1978
26. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication1994
27. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

- Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – IV)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

MATLAB is considered as one of the most important tools and modern computer language, and this course enable the students to learn many of MATLAB commands and how to use them in programming to solve many problems in different mathematical subjects, specially in numerical analysis and other subjects which connected to computer oriented mathematics.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Creates valuable transferable skills, sets individuals apart in the job market, and can help accelerate professional growth.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. **Validate** program for real world problems.
2. **Understand** the use of MATLAB software for numerical methods.
3. **Implement** solution for mathematical problems.
4. **Debug** and **Troubleshoot** MATLAB code.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
0	0	2	2	0	0	70	30*	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Practical PA**, out of 30 marks, **10 marks** are for **Presentation** to facilitate integration of COs and the remaining 15 marks is converted from lab manual to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Introduction</u> To MATLAB basic Mathematical, trigonometric, Array and Matrix operations	02
2	1	<u>Implement</u> the various logical and Iterative loops in MATLAB programming.	02
3	2	<u>Solve</u> Linear equations with Gauss elimination & Gauss-Jordan elimination method using MATLAB programming.	02
4	2	<u>Solve</u> Linear equations with RLSE & Gauss Seidel method using MATLAB programming.	02
5	3	<u>Computation</u> of Matrices and Eigen values with Decomposition methods using MATLAB programming.	02
6	3	<u>Computation</u> of Matrices and Eigen values with Power methods using MATLAB programming.	02
7	1	<u>Implementation</u> of basic Curve Plotting Commands.	02
8	4	<u>Computation</u> of Lagrange Polynomial method and Newtons Polynomial method for interpolation of curve.	02
9	3	<u>Solve</u> nonlinear equation with Bisection method using MATLAB functions.	02
10	3	<u>Solve</u> nonlinear equation with Newton-Raphson method using MATLAB functions.	04
11	3	<u>Solve</u> ordinary differential equations with Euler's method using MATLAB programming.	02
12	3	<u>Solve</u> ordinary differential equations with Runge-Kutta method using MATLAB programming.	02
		Total	26

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,6,7,8,9,10,11,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Function as a team member.
- 3) Function as a team leader.
- 4) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

- i) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- j) '**P**' in **item No. 4** does not mean only the traditional practical method, but different types of teaching methods and media that are to be employed to develop the outcomes.

- k) About **15-20% of the simulation** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

IX SUGGESTED STUDENT ACTIVITIES

Other than the laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Library survey regarding different software used in industries.
- Prepare 15 min. power point presentation on how to choose software for the particular application.

X SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Applied Numerical Methods Using MATLAB	Won Young Yang, Wenwu Cao Tae-Sang Chung, John Morris,	John Willey, 2005 ISBN: 9780471698333
2	Computer Oriented Numerical Methods	V. Rajaraman	PHI, 2006 ISBN: 9388028317

XI SOFTWARE/LEARNING WEBSITES

- <http://www.nptel.ac.in/courses/122104018/node109.html>
- http://www.geos.ed.ac.uk/~yliu23/docs/lect_spline.pdf
- https://mat.iitm.ac.in/home/sryedida/public_html/caimna/interpolation/lagrange.html

XII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE287	Semester IV														
287	Competency	3	2	2	2	1	-	-	-	-	-	-	1	3	-

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	<i>Creates valuable transferable skills, sets individuals apart in the job market, and can help accelerate professional growth</i>														
287.1	Validate program for real world problems	3	3	2	-	2	-	-	-	-	-	-	-	3	2
287.2	Understand the use of MATLAB software for numerical methods	3	3	-	2	2	-	-	-	-	-	-	-	3	2
287.3	Implement solution for mathematical problems	3	2	2	2	-	-	-	-	1	-	-	-	3	3
287.4	Debug and Troubleshoot MATLAB code	-	3	3	2	2	-	-	-	-	-	-	-	2	-

XIII NAMES OF RESOURCE PERSONS

a) Dr. Jigar Sarda

Course Code: EE288 (University Elective)

Course Title: Maintenance of Household Apparatus

UG Programme	Semester
All UG Program of CHARUSAT	Forth

I RATIONALE

All the houses needed major or minor electrical maintenances over period of time. Also, small maintenance needed for household apparatus. The major component of repairing cost is the labour/skill and minor component is cost of item to be replaces (Eg. Carbon brush in mixture machine, switch, capacitor in fan). If the students get the repairing/replacement skill, he/she can save money and time, most important is; he/she doesn't have to wait for repairing time taken by external person.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Repair the household apparatus with minimum cost.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Safely Operate** the electrical/electronic devices and safeguard their selves from electrical shock.
- 2) **Repair** the domestic wiring.
- 3) **Repair/maintain/troubleshoot** the household electrical appliances.
- 4) **Develop** small electronic project.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
0	0	2	2	0	0*	70	30	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **10 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is based on lab activities by assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Operate</u> the multi-meter to measure <u>the said</u> electrical quantity.	02
2	1,2	<u>Wiring</u> the tube light, fan, Extension board (Power Socket), Staircase wiring (Two-way wiring).	02
3	1,2	<u>Choose</u> the correct rating of Fuse, MCB, ELCB. <u>Connect</u> the earthing pin wiring to the earth pit.	01 01
4	2	<u>Prepare</u> the drawing for wiring a newly built room, without any electrical wiring along with a bill of materials and specifications.	02
5	3	<u>Solder and de-solder</u> electronic components on PCB.	02
6	4	<u>Develop</u> simple mobile charger and DC voltage controller circuit.	02
7	3	<u>Maintenance and Troubleshooting</u> of Ceiling FAN.	02
8	3	<u>Maintenance and Troubleshooting</u> of Tube light & LED light.	02
9	3	<u>Maintenance and Troubleshooting</u> of Pump (Household Motor).	02
10	3	<u>Maintenance and Troubleshooting</u> of Water Purifier (RO).	02
11	3	<u>Maintenance and Troubleshooting</u> of Air-Conditioner.	02
12		<u>Interpret</u> the home electricity bill and can calculate the energy calculation of any equipment.	02
13	4	<u>Choose</u> the correct battery for <u>the given</u> application.	02
14			
15			
16			
		Total	26

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Multi-meter	1
2	Multi-meter, Tube light set, extension board, wires.	2, 8
3	Fuse, MCB, ELCB	3
4	PCS, Soldering Iron, Soldering Wire.	5
5	230/9 V Transformer, Diodes, Capacitor	6
6	Ceiling fan, Multi-meter, capacitor	7
7	Electric Motor (Single Phase), Capacitor	9
8	RO System	10
9	Air Conditioner, Capacitor	11
10	Energy meter	12
11	Various types of batteries	13

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- 'Responding Level' in 1st year
- 'Valuing Level' in 2nd year
- 'Organising Level' in 3rd year
- 'Characterising Level' in 4th year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- 'P' in **item No. 4** does not mean only the traditional practical method, but different types of hands-on practices to be employed to develop the outcomes.
- About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

IX SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Library survey regarding the evolution of arc quenching mechanism for different type of circuit breakers.
- Prepare 15 min. power point presentation on how to choose the specific circuit breaker for the particular application.

X SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electrical Design Estimation and Costing	Raina K.B. & Bhattacharya S.K.	New Age International Private Limited, Second Edition, 2018-19, ISBN : 978-8122443585
2	India Electrical Rules 1956 Hand book	Government of India and Vishal Singh	Kindle Edition
3	A Textbook of Electrical Technology in S.I. Units Volume III	B L Thareja	S Chand, 2007 ISBN : 978-8121924900 .
4	National Building code of India Group 1 and Group 4, Bureau of Indian standard,		Government of India
5	Troubleshooting and Repairing Major Appliances	Eric Kleinert	McGraw Hill, 2012 ISBN : 978-0071770187

XI SOFTWARE/LEARNING WEBSITES

- <https://www.instructables.com/id/How-to-solder/>
- <https://www.udemy.com/course/electronic-electronics-maintenance-electronic-devices-maintenance/>

XII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE288	Semester IV	Fault Analysis and Switchgear													
		Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO													
288	Competency Repair the household apparatus with minimum cost.	2	2	1	1	-	1	-	-	-	-	-	-	1	-
288.1	Safely Operate the electrical/electronic devices and safeguard their selves from electrical shock.	3	2	-	1	-	3	-	-	-	-	-	-	2	-
288.2	Repair the domestic wiring.	2	3	1	1	-	1	-	-	1	-	-	-	2	-
288.3	Repair/maintain/troubleshoot the household electrical appliances.	2	3	1	1	-	1	-	-	-	-	-	-	1	-
288.4	Develop small electronic project.	1	1	3	1	1	-	-	-	2	-	1	-	1	-

XIII NAMES OF RESOURCE PERSONS

- Dr. Nilaykumar A. Patel
-

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

As electrical power generation and economy is essential gradient for the industrial and all around development of any country This curriculum course structure is developed such that the student is able to evaluate the fundamentals of electromechanical energy conversion, concepts of electrical power generation, basic layout and function of conventional as well as non-conventional power generation technology, constructional & operational aspects of various power plant, basic knowledge of transmission line components, transmission line, various types of insulated cables and its selection criteria, economic aspects of power system operation, and required reactive power factor for most economical operation.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the functioning and performance of various power generation plants and basic power system components.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) Use the knowledge and describe the working of various power plants and various equipment of power generation plants.
- 2) Develop skill and Identify the key point to improve the performance of the power plants.
- 3) Aware with the new era of energy generation technology like wind, solar and fuel cell etc.
- 4) Analyze and select the proper cable for particular application.
- 5) Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
3	0	0	3	70	30	00	00	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

V AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.

- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VI UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction of renewable and non-renewable energy sources	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. 1a. Power and Energy, sources of renewable & non-renewable energy such as coal, water, nuclear, wind, solar, tidal	1.1 Power and Energy sources of energy such as coal, water, nuclear, wind, solar, tidal etc.
Unit– II Steam Power Plants	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. CO 2. Develop skill and Identify the key point to improve the performance of the power plants. 2a. Classification of steam power plants, layout of steam power plant, Component of steam power plants, Essential requirement for steam power plant, selection of site. 2b. Choice of steam condition, fuel handling, Combustion equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers. 2c. steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment, advantages and disadvantages of steam power plant	2.1 Classification of steam power plants 2.2 layout of steam power plant 2.3 Component of steam power plants 2.4 Essential requirement for steam power plant 2.5 selection of site 2.6 Choice of steam condition, fuel handling, Combustion equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers, steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment 2.7 advantages and disadvantages of steam power plant
Unit– III Hydro Electric Power Plant	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. CO 2. Develop skill and Identify the key point to improve the performance of the power plants. 3a. Layout and elements of hydro power plant, Classification of hydro power plant, Hydraulic turbines	3.1 Classification of hydro power plant 3.2 layout and elements of hydro power plant 3.3 Site selection for hydro power plant

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
	3b. Application, advantages and disadvantages of hydro power plant, Site selection for hydro power plant.	3.4 advantages and disadvantages of hydro power plant 3.5 Application Hydraulic turbines
Unit-IV Nuclear Power Plant	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. CO 2. Develop skill and Identify the key point to improve the performance of the power plants. 4a. General aspect of nuclear engineering, layout of nuclear power plant 4b. Nuclear reactors, Main components of nuclear power plants, advantages and disadvantages of nuclear power plant	4.1 General aspect of nuclear engineering 4.2 layout of nuclear power plant 4.3 Main components of nuclear power plants 4.4 advantages and disadvantages of nuclear power plant
Unit – V Combined cycle power plant	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. CO 2. Develop skill and Identify the key point to improve the performance of the power plants. 5a. Various combined cycle power plant, working of open cycle, close cycle & combine cycle 5b. advantages and disadvantages, comparison of various power plant	5.1 Various combined cycle power plant 5.2 working of combined cycle power plant 5.3 advantages and disadvantages 5.4 comparison of various power plant
Unit – VI Wind Power Generation	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. CO 2. Develop skill and Identify the key point to improve the performance of the power plants. CO 3. Aware with the new era of energy generation technology like wind, solar and fuel cell etc. 6a. Nature and origin of wind, variables in wind energy conversion system, wind velocities and height from ground and site selection 6b. Types of wind energy system, wind turbine generator unit with battery storage facilities, solar wind hybrid 6c. Wind farm siting, Applications, Merits and demerits of wind energy	6.1 nature and origin of wind 6.2 variables in wind energy conversion system 6.3 Types of wind energy system 6.4 wind velocities and height from ground and site selection 6.5 wind turbine generator unit with battery storage facilities 6.6 solar wind hybrid, Wind farm siting
Unit – VII Solar Power Generation	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. CO 2. Develop skill and Identify the key point to improve the performance of the power plants. CO 3. Aware with the new era of energy generation technology like wind, solar and fuel cell etc. 7a. Solar energy routes and prospects, Types of solar thermal collectors 7b. comparison between conventional and solar thermal power plant, ratings of solar power plant heat transfer fluids	7.1 Solar energy routes and prospects 7.2 Types of solar thermal collectors 7.3 solar photovoltaic cell conversion 7.4 comparison between conventional and solar thermal power plant 7.5 ratings of solar power plant heat transfer fluids 7.6 solar pond and binary cycle

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	7c. solar parameters calculation 7d. Solar photovoltaic energy conversion 7e. solar pond and binary cycle solar thermal power plant, merits and limitations of solar energy conversion and utilization	solar thermal power plant
Unit – VIII Other Renewable Energy Sources	CO 1. Use the knowledge and describe the working of various power plants and various equipment of power generation plants. CO 2. Develop skill and Identify the key point to improve the performance of the power plants. CO 3. Aware with the new era of energy generation technology like wind, solar and fuel cell etc. 8a. Layout of Fuel cell based power plant types, Electrochemical Reactions, Hydrogen, Oxygen Fuel cells 8b. Principles of MHD Power Generation, MHD Systems, Advantages of MHD Systems, Electrical conditions. 8c. Origin of geothermal resources, Geothermal energy conversion, utilization of geothermal energy, liquid dominated geothermal electric power plant, Applications. 8d. Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC, Tidal energy conversion	8.1 Fuel cell based power plant: Introduction, concept, types 8.2 Electrochemical Reactions, Hydrogen, Oxygen Fuel cells. 8.3 MHD Power Generation: Principles, MHD Systems, Advantages of MHD Systems, Electrical conditions 8.4 Geothermal energy: Applications, utilization of geothermal energy, origin of geothermal resources, liquid dominated geothermal electric power plant. 8.5 Ocean and tidal energy: Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC. Tidal energy conversion : High and low tides, tidal energy conversion
Unit – IX Insulated Cables	CO 4. Analyze and select the proper cable for particular application. 9a. Requirement of insulated cables, properties of conductor and insulating material for cables, construction of cables, types of cables, types of insulating material for cable. 9b. Insulation resistance of a single core cable 9c. capacitance of single and three core cable 9d. Electrostatic stresses in a single core cable, most economical conductor size in cable 9e. Capacitance grading and inter sheath grading 9f. Calculation of losses in cable and factors affecting it, current rating of a cable	9.1 Requirement of insulated cables 9.2 properties of conductor and insulating material for cables 9.3 construction of cables, types of cables, types of insulating material for cable 9.4 Insulation resistance of a single core cable 9.5 capacitance of single and three core cable 9.6 Electrostatic stresses in a single core cable 9.7 most economical conductor size in cable 9.8 Capacitance grading and inter sheath grading 9.9 Calculation of losses in cable and factors affecting it, current rating of a cable

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – X Components of Overhead Lines	CO 5. Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor. 10.1 Basic power system structure, generation, transmission, substation, sub transmission, distribution, Complex power. 10.2 single phase transmission, three phase transmission. Conductors and line supports, Types of Insulators 10.3 potential distribution over a string of suspension insulators, mathematical expression for String efficiency 10.4 Methods of improving String efficiency, by using longer cross-arms, by insulator grading, by using guard rings.	10.1 Basic power system structure, generation, transmission, substation, sub-transmission, distribution, 10.2 Complex power 10.3 single phase transmission, three phase transmission 10.4 Conductors and line supports 10.5 Types of Insulators 10.6 potential distribution over a string of suspension insulators, String efficiency and methods to improve it, Examples on string efficiency
Unit – XI Mechanical Design of Transmission Lines (Sag-Tension Relationship)	CO 5. Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor. 11.1 Sag-tension relation, calculation of sag at equal & unequal level support. 11.2 sag parameters calculation	11.1 Sag-tension relation and its example with different configurations
Unit – XII Economics of Power System	CO 5. Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor. 12.1 Types of Loads in power system, effects of variable load on power system, load curves and load duration curves. 12.2 Important terms and factors related to power system economics of power system, importance of those factors. 12.3 calculation of load curve parameters 12.4 Load curves and selection of generating units, Cost of electrical energy and its expression 12.5 Define depreciation, methods of determining the depreciation 12.3 depreciation methods calculation 12.4 Define Tariff, Types of Tariffs 12.5 Calculation of tariff parameters	12.1 Loads in power system, effects of variable load on power system, load curves and load duration curves 12.2 Important terms and factors related to power system economics of power system, importance of those factors 12.3 Load curves and selection of generating units 12.4 Cost of electrical energy and its expression 12.5 methods of determining the depreciation, examples 12.6 Types of Tariffs, Examples
Unit – XIII Power Factor Improvement in Power System	CO 5. Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor. 13.1 Power factor basics, Causes of low power factor, disadvantages of low power factor 13.2 power factor improvement, Power factor improvement equipments 13.3 Power factor improvement equipment's, calculations of power factor correction, most	13.1 Causes of low power factor, disadvantages of low power factor 13.2 power factor improvement, Power factor improvement equipments 13.3 calculations of power factor correction 13.4 most economical power

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	economical power factor 13.4 power factor improvement calculation	factor

VII TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	01	1	0	0	0	0	0	01
II	Steam Power Plants	06	1	3	2	2	2	0	10
III	Hydro Electric Power Plant	03	0	1	2	0	2	0	5
IV	Nuclear Power Plant	03	0	1	2	0	2	0	5
V	Combined cycle power plant	02	0	1	0	1	1	0	3
VI	Wind Power Generation	03	0	1	1	1	1	0	4
VII	Solar Power Generation	02	0	1	1	0	1	0	3
VIII	Other Renewable Energy Sources	03	0	2	1	1	1	0	5
IX	Insulated Cables	04	1	1	0	2	2	0	6
X	Components of Overhead Lines	08	1	2	2	3	4	0	12
XI	Mechanical Design of Transmission Lines (Sag-Tension Relationship)	02	0	2	1	0	0	0	3
XII	Economics of Power System	05	0	1	1	2	4	0	8
XIII	Power Factor Improvement in Power System	03	0	1	1	1	2	0	5
Total		45	04	17	14	13	22	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

VIII MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects *for a minimum Learning effort of 15 hours per semester* are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Make the small scale module of different power generation plant.**
- 2) **Make the small scale module of integrated power plant.**
- 3) **Design the underground cable and select the parameters accordingly.**
- 4) **Make the power factor correction unit.**
- 5) **Design the smart meter for electricity billing.**

IX SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- l) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- m) '**L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- n) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- o) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

X SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- g) Do the survey of power generation state wise, national and internationally.
- h) Prepare the poster of generation, transmission and distribution data.

XI COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introducti on of renewabl e and non- renewabl e energy sources Unit– II Steam Power Plants	1	1a. Power and Energy, sources of renewable & non-renewable energy such as coal, water, nuclear, wind, solar, tidal 2a. Classification of steam power plants, layout of steam power plant, Component of steam power plants, Essential requirement for steam power plant, selection of site. 2b. Choice of steam condition, fuel handling, Combustion equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers.	Lecture with media	PPTs, Video
Unit– II Steam Power Plants Unit– III Hydro Electric Power	2	2c. steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment, advantages and disadvantages of steam power plant. 3a. Layout and elements of hydro power plant, Classification of hydro power plant, Hydraulic turbines 3b. Application, advantages and disadvantages of	Lecture With media	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Plant		hydro power plant, Site selection for hydro power plant.		
Unit-IV Nuclear Power Plant	3	4a.General aspect of nuclear engineering, layout of nuclear power plant 4b.Nuclear reactors, Main components of nuclear power plants, advantages and disadvantages of nuclear power plant	Lecture with media	PPTs, black board, Video
Unit – V Combine d cycle power plant	4	5a. Various combined cycle power plant, working of open cycle, close cycle & combine cycle 5b. advantages and disadvantages, comparison of various power plant 6a. Nature and origin of wind, variables in wind energy conversion system, wind velocities and height from ground and site selection	Lecture with media,	PPTs, Video
Unit – VI Wind Power Generation	5	6b. Types of wind energy system, wind turbine generator unit with battery storage facilities, solar wind hybrid 6c. Wind farm siting, Applications, Merits and demerits of wind energy	Lecture with media,	PPTs, Video
Unit – VII Solar Power Generation	6	7a. Solar energy routes and prospects, Types of solar thermal collectors 7b. comparison between conventional and solar thermal power plant, ratings of solar power plant heat transfer fluids 7c. solar parameters calculation	Lecture with media	PPTs
	7	7d. Solar photovoltaic energy conversion 7e. solar pond and binary cycle solar thermal power plant, merits and limitations of solar energy conversion and utilization	Lecture with media, Field visit	PPTs, Video
Unit – VIII Other Renewabl e Energy Sources	8	8a. Layout of Fuel cell based power plant types, Electrochemical Reactions, Hydrogen, Oxygen Fuel cells 8b. Principles of MHD Power Generation, MHD Systems, Advantages of MHD Systems, Electrical conditions	Lecture with media	PPTs
Unit – VIII Other Renewabl e Energy Sources	9	8c. Origin of geothermal resources, Geothermal energy conversion, utilization of geothermal energy, liquid dominated geothermal electric power plant, Applications. 8d. Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC, Tidal energy conversion	Lecture with media	PPTs
Unit – IX Insulated Cables	10	9a. Requirement of insulated cables, properties of conductor and insulating material for cables, construction of cables, types of cables, types of insulating material for cable. 9b. Insulation resistance of a single core cable 9c. capacitance of single and three core cable	Lecture with media	PPTs, black board
Unit – IX Insulated Cables	11	9d. Electrostatic stresses in a single core cable, most economical conductor size in cable 9e. Capacitance grading and inter sheath grading 9f. Calculation of losses in cable and factors affecting it, current rating of a cable	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – XI Mechanical Design of Transmission Lines (Sag-Tension Relationship)	12	10.1 Basic power system structure, generation, transmission, substation, sub transmission, distribution, Complex power. 10.2 single phase transmission, three phase transmission. Conductors and line supports, Types of Insulators 10.3 potential distribution over a string of suspension insulators, mathematical expression for String efficiency 10.4 Methods of improving String efficiency, by using longer cross-arms, by insulator gradings, by using guard rings.	Lecture	black board
Unit– V Circuit Breakers	13	11.1 Sag-tension relation, calculation of sag at equal & unequal level support. 11.2 sag parameters calculation	Lecture	black board
Unit – XII Economics of Power System Unit – XIII Power Factor Improvement in Power System	14	Types of Loads in power system, effects of variable load on power system, load curves and load duration curves. 12.2 Important terms and factors related to power system economics of power system, importance of those factors. 12.3 calculation of load curve parameters 12.4 Load curves and selection of generating units, Cost of electrical energy and its expression 12.5 Define depreciation, methods of determining the depreciation 12.3 depreciation methods calculation 12.4 Define Tariff, Types of Tariffs 12.5 Calculation of tariff parameters	Lecture	black board
Unit – XIII Power Factor Improvement in Power System	15	13.1 Power factor basics, Causes of low power factor, disadvantages of low power factor 13.2 power factor improvement, Power factor improvement equipments 13.3 Power factor improvement equipment's, calculations of power factor correction, most economical power factor 13.4 power factor improvement calculation	Lecture	black board

XII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Principles of Power system	V.K. Mehta	S. Chand & Company Ltd. ISBN: 978-81-219-2496-2
2	Energy Technology	S. Rao & Dr. B.B.Parulekar	Khanna Publication, New Delhi. ISBN: 978-81-7409-040-9
3	Renewable energy sources and conversion technology	N.K. Bansal	Tata McGraw Hill, New Delhi. ISBN: 0074600230, 9780074600238
4	A Course in electrical power	Soni & Bhatnagar	S. K. Kataria & Sons publications ISBN: 8188458538
5	Energy sources	G.D. Rai	Khanna Publication, New Delhi, 2019, ISBN:

S. No.	Title of Book	Author	Publication including ISBN
			978-8174090737

XIII SOFTWARE/LEARNING WEBSITES

1. Power System Generation, Transmission and Distribution: <https://nptel.ac.in/courses/108102047>
2. Power System Engineering: [EE254 EPGE Syllabus EE 4th sem.docx](#)

XIV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs														
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Project management and finance	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	PSO3
EE254	Semester IV	ELECTRICAL POWER GENERATION AND ITS ECONOMICS Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO														
254	Competency Evaluate the performance of various protection systems used in Industry	3	3	3	2	1	1	3	-	1	-	-	-	3	-	-
254.1	Use the knowledge and describe the working of various power plants and various equipment of power generation plants.	3	2	2	1	1	1	2	-	-	1			3	1	-
254.2	Develop skill and Identify the key point to improve the performance of the power plants.	3	2	2	1	1	1	3	-	1	-			3	2	-
254.3	Aware with the new era of energy generation technology like wind, solar and	2	2	1	1	1	1	3	-	1	-			3	2	-

Course Code	Competency and COs	POs and PSOs														
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Project management and finance	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	PSO3
	fuel cell etc.															
254.4	Analyze and select the proper cable for particular application.	2	2	1	2	1	1	-	-	-	-			2	1	-
254.5	Use the knowledge of basic power system structure and determine tariff means to reduce the electricity charges by improving the power factor.	3	2	1	1	1	1	-	-	1	1			2	-	-

XV NAMES OF RESOURCE PERSONS

a) Ankur Patel

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

With advancement in electrical and electronics industry all the electrical machine nowadays can be controlled with electronics devices. This type of electronics system requires controller or processor to think and act quickly. This curriculum course is developed such that the student is able to design a system to controller electrical devices as per the application's need.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Create an embedded system to solve real life problem**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Understand** the basic building blocks & architecture of a microcontroller/ microprocessor
- 2) **Develop** a program using embedded C
- 3) **Demonstrate** use of Input, Output, Timers, Interrupts and serial communication.
- 4) **Analyse, Evaluate** and **Integrate** peripheral devices
- 5) **Design** and **create** hardware and software solution to real life problems.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of Cos and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Understand</u> 8085 and <u>Create</u> Assembly programs for the given task.	4
2	3	<u>Demonstrate</u> Digital input and Digital Output example with simulation and hardware.	2
3	2	<u>Understand</u> C programming concept for embedded C in Keil Microvision IDE.	2
4	3	<u>Demonstrate</u> the use of Timer in different modes of operation and their <u>use cases</u> with simulation and hardware.	2
5	3	<u>Demonstrate</u> use of Counter in different modes of operation and their <u>use cases</u> with simulation and hardware.	2
6	3	<u>Demonstration</u> of External interrupt and it's <u>use cases</u> with simulation and hardware.	2
7	3	<u>Demonstrate</u> use of serial port of 8051 for serial communication.	2
8	4	<u>Demonstrate</u> interfacing a keypad.	2
9	4	<u>Develop</u> a system with 8051 and liquid crystal & 7-Segment LED display to print <u>given</u> message on screen.	4
10	4	<u>Develop</u> system to measure temperature/pressure/moisture using ADC.	2
11	4	<u>Demonstrate</u> use of DAC with <u>use case</u> .	2
12	5	<u>Create</u> system to monitor and control temperature of room.	2
Total			28

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Dynalog 8085 kit	1
2	Keil uvision for C51	2,3,4,5,6,7,8,9,10,11,12
3	8051 Controller board	2,3,4,5,6,7,8,9,10,11,12
4	LEDs, Toggle Switch, Push Buttons, Connecting wires	3,4,5,6,12
5	Digital Signal Oscilloscope	3,4,5,6,10
6	4x4 Matrix Keypad	8
7	16x2 LCD, 7-Segment LED	9,12
8	ADC 0804 or 0808, DAC 0808	10,11,12
9	Temperature Sensor	10,12
10	Relay Card, Contactor	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Basic of 8085	CO 1. <u>Understand</u> the basic building blocks & architecture of a microcontroller/ microprocessor Pro 2. <u>Understand</u> 8085 Dynalog Kit and <u>Create</u> Assembly programs for <u>the given</u> task. 1a. <u>Justify</u> the need of Microprocessor for <u>the given</u> application 1b. <u>Evaluate</u> the working of <u>the given</u> microprocessor 1c. <u>Create</u> the program for <u>the given</u> application 1d. <u>Choose</u> the memory according to <u>the given</u> application	1.1 Introduction to Microprocessor, Block Diagram of 8085 and it's various registers. 1.2 Bus organization, Pin Diagram and its functions. 1.3 Assembly level programming. 1.4 Types of memory and memory interfacing
Unit– II Introduction to Microcontroller	CO 1. <u>Understand</u> the basic building blocks & architecture of a microcontroller/ microprocessor Pro 3. <u>Demonstrate</u> Digital input and Digital Output example with simulation and hardware. 2a. <u>Determine</u> the possibility of application for the <u>given</u> microcontroller 2b. <u>Understand</u> the working of <u>the given</u> microcontroller 2c. <u>Understand</u> the function of different pins for the hardware interfacing 2d. <u>Understand</u> the requirement of different type of register and their function 2e. <u>Understand</u> the operation of on chip memory 2f. <u>Evaluate</u> speed of execution for <u>the given</u> crystal 2g. <u>Understand</u> the different operation available through port	2.1 8051 Microcontroller family and its salient feature 2.2 Architectural block Diagram of INTEL 8051 2.3 Pin diagram and Pin Functions 2.4 General Purpose and Special Function Registers 2.5 Memory organization 2.6 Crystal Frequency & Crystal Connection and Speed of Execution 2.7 -PORT Structure
Unit– III 8051 Programming	CO 2. <u>Develop</u> a program using embedded C Pro 4. <u>Understand</u> C programming concept for embedded C in Keil Microvision IDE. 3a. <u>Differentiate</u> operation of IDE and use specific <u>module for the given</u> situation 3b. <u>Create</u> the hardware to load microcontroller 3c. <u>Create</u> the program for <u>the given</u> application 3d. <u>Differentiate</u> and <u>understand</u> the need of assembly and C language for creating software as	3.1 Concept of Assembler, Editor, Linker, Loader, Debugger in IDE 3.2 Hardware Loader Circuit for ISP featured 8051. 3.3 Assembly Instruction Sets:

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
	per the given application 3e. <u>Understanding</u> format used to write program in C language 3f. <u>Justify</u> the need of special keyword for assembly level language	Data Transfer & Addressing modes, Logical, Branching, Arithmetic & Bit wise Operation, 3.4 Comparison of Assembly and C Programming 3.5 Data type , Keywords, Various Operator , Control loops, Functions and Arrays in C 3.6 Special Keywords & Interrupt function in Keil Microvision IDE
Unit-IV On chip Hardware Modules/Peripherals in 8051	CO 4. <u>Demonstrate</u> use of Input, Output, Timers, Interrupts and serial communication. PrO 5. <u>Demonstrate</u> the use of Timer in different modes of operation and their <u>use cases</u> with simulation and hardware. PrO 6. <u>Demonstrate</u> use of Counter in different modes of operation and their <u>use cases</u> with simulation and hardware. PrO 7. <u>Demonstration</u> of External interrupt and it's <u>use cases</u> with simulation and hardware. PrO 8. <u>Demonstrate</u> use of serial port of 8051 for serial communication. 4a. <u>Evaluate</u> the operation of Timer/Counter and use it to create different type of application 4b. <u>Evaluate</u> the operation of Interrupts and use it to create different type of application 4c. <u>Evaluate</u> the operation of UART and use it to create different type of application	4.1 Internal Architecture of Timer/Counter and Related C Programs with all modes 4.2 Basics of Interrupt: Mask-able/ Non mask-able Interrupt, Enabling the mask-able Interrupt, Interrupt vector table, interrupt latency time and interrupt priority, External Interrupt and Related C Programs 4.3 Basics of Serial communication Protocol, Difference Between Synchronous and Asynchronous Communication, Internal Architecture of Serial Communication module in 8051 (UART) and related C programs with all modes
Unit – V Interfacing Microcontroller & Applications	CO 4. <u>Analyse, Evaluate and Integrate</u> peripheral devices PrO 9. <u>Demonstrate</u> interfacing a keypad. PrO 10. <u>Develop</u> a system with 8051 and liquid crystal & 7-Segment LED display to print <u>given</u> message on screen. PrO 11. <u>Develop</u> system to measure temperature/pressure/moisture using ADC.	

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
	PrO 12. <u>Demonstrate</u> use of DAC with <u>use case</u>. CO 5. <u>Design and create</u> hardware and software solution to real life problems. PrO 13. <u>Create</u> system to monitor and control temperature of room. 5a. <u>Understand</u> operation of relay and use it to control the given device 5b. <u>Understanding</u> memory interfacing to fulfill memory demand of <u>the given</u> application 5c. <u>Preparing</u> hardware and software to use keyboard/switch for <u>the given</u> application 5d. <u>Preparing</u> hardware and software to use LED/LCD display for <u>the given</u> application 5e. <u>Preparing</u> hardware and software to use ACD/DAC for <u>the given</u> application 5f. <u>Evaluate</u> operation of ADC using LM35 to measure temperature. 5g. <u>Develop</u> hardware and software to count the cars 5h. <u>Develop</u> hardware and software to control the DC motor	5.1 Interfacing: Hardware interfacing diagram for driving a relay by ULN 2803/Transistor ,solid state relay(162341), Controlling a Lamp using Opto-isolator(ILD74) 5.2 External Memory Interfacing (RAM and ROM) 5.3 Keyboard Interfacing (Key Debouncing Concept, Simple Key Interfacing Hardware & Software program, Matrix key board) 5.4 Displays Interfacing (LED, 7-Segment display and LCD display) 5.5 ADC & DAC interfacing (Serial and parallel output ADC/DAC). 5.6 Temperature measurement(LM35) with LM35 5.7 Counting Cars passing through passage/bottles on belt 5.8 Unidirectional and Bidirectional DC Motor Control with PWM using L293 and IL74, Stepper Motor Drive with ULN2003.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Basic of 8085	10	6		2			4	12
II	Introduction to Microcontroller	10	8	3					11
III	8051 Programming	08		3		3	3		9
IV	On chip Hardware Modules/Peripherals in 8051	16	4	2			3	10	19
V	Interfacing Microcontroller & Applications	16	2	2	3	2	4	6	19
Total		60	20	10	5	5	10	20	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Protection of Motor:** Create hardware and software to monitor different parameter of motor and to control motor under any abnormal condition.
- 2) **Green house:** Create hardware and software to create green house, in which soil moisture, humidity, temperature are continuously monitored and based on this parameters sprinklers, irrigation and cooling system can be controlled.
- 3) **Fire safety system:** Create a hardware and software to detect fire and controlling fire by turning on water sprinklers to control fire.
- 4) **Autonomous car:** Prepare hardware and software for autonomus vehicle which can follow line.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- i) Visiting project expo organized by electrical or electronics department to understand application of microcontrollers used solve real life problems.
- j) Prepare small application based on 8051, demonstrate the model with presentation/poster in front of class or at project expo.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Basic of 8085	1	PrO 1. <u>Understand</u> 8085 and <u>Create</u> Assembly programs for <u>the given</u> task. 1d. <u>Justify</u> the need of Microprocessor for <u>the given</u> application 1e. <u>Evaluate</u> the working of the <u>given</u> microprocessor	Lecture with media	Black board, PPT
	2	PrO 1. <u>Understand</u> 8085 and <u>Create</u> Assembly programs for <u>the given</u> task. 1f. <u>Create</u> the program for <u>the given</u> application	Lecture	Black board
	3	1g. <u>Choose</u> the memory according to <u>the given</u> application	Lecture	Black board
Unit– II Introduct ion to Microcon troller	3	2a. <u>Determine</u> the possibility of application for <u>the given</u> microcontroller 2b. <u>Understand</u> the working of <u>the given</u> microcontroller	Lecture	Black board
	4	2c. <u>Understand</u> the function of different pins for the hardware interfacing 2d. <u>Understand</u> the requirement of different type of register and their function 2e. <u>Understand</u> the operation of on chip memory	Lecture with media	Black board, PPT
	5	2f. <u>Evaluate</u> speed of execution for <u>the given</u> crystal PrO 2. <u>Demonstrate</u> Digital input and Digital Output example with simulation and hardware. 2g. <u>Understand</u> the different operation available through port	Lecture with media, Demonstration	Black board, Software: Keil μ vision
Unit– III 8051 Program ming	6	3a. <u>Differentiate</u> operation of IDE and use specific module for the given problem 3b. <u>Create</u> the hardware to load microcontroller 3c. <u>Create</u> the program for <u>the given</u> application	Lecture with media, Demonstration	Black board, Software: Keil μ vision
	7	PrO 3. <u>Understand</u> C programming concept for embedded C in Keil Microvision IDE. 3d. <u>Differentiate</u> and understand the need of assembly and C language for creating software as per <u>the given</u> application 3e. <u>Understanding</u> format used to write program in C language 3f. <u>Justify</u> the need of special keyword for assembly level language	Lecture with media, Demonstration	Black board, Software: Keil μ vision
Unit– IV On chip Hardware Modules/ Peripherals in 8051	8	PrO 4. <u>Demonstrate</u> the use of Timer in different modes of operation and their <u>use cases</u> with simulation and hardware. PrO 5. <u>Demonstrate</u> use of Counter in different modes of operation and their <u>use cases</u> with simulation and hardware.	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil μ vision
	9	4a. <u>Evaluate</u> the operation of Timer/Counter and use it to <u>create</u> different type of application		
	10	PrO 6. <u>Demonstration</u> of External interrupt and it's <u>use cases</u> with simulation and hardware. 4b. <u>Evaluate</u> the operation of Interrupts and use it to <u>create</u> different type of application	Lecture with media, Hardware and	Black board, Software: Keil

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
			Software Demonstration	µvision
	11	<i>PrO 7. Demonstrate use of serial port of 8051 for serial communication.</i> 4c. Evaluate the operation of UART and use it to create different type of application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
Unit– V Interfacing Microcontroller & Applications	12	5a. Understand operation of relay and use it to control the given device 5b. Understanding memory interfacing to fulfill memory demand of the given application <i>PrO 8. Demonstrate interfacing a keypad.</i> 5c. Preparing hardware and software to use keyboard/switch for the given application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	13	<i>PrO 9. Develop a system with 8051 and liquid crystal & 7-Segment LED display to print given message on screen.</i> 5d. Preparing hardware and software to use LED/LCD display for the given application	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	14	<i>PrO 10. Develop system to measure temperature /pressure /moisture using ADC.</i> <i>PrO 11. Demonstrate use of DAC with use case.</i> 5e. Preparing hardware and software to use ACD/DAC for the given application 5f. Evaluate operation of ADC using LM35 to measure temperature.	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision
	15	<i>PrO 12. Create system to monitor and control temperature of room.</i> 5g. Develop hardware and software to count the cars 5h. Develop hardware and software to control the DC motor	Lecture with media, Hardware and Software Demonstration	Black board, Software: Keil µvision

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Microprocessor Architecture, Programming, and Applications with the 8085	Ramesh Gaonkar	Penram International Publishing (India) Ltd.
2	Exploring C for Microcontrollers	Jivan Parab, Vinod Shelake, Rajanish Kamat, Gourish Naik	Springer
3	The 8051 Microcontroller Architecture	Kenneth Ayala	West Publishing Company

S. No.	Title of Book	Author	Publication including ISBN
4	The 8051 Microcontroller and Embedded Systems Using Assembly And C,	M A Mazidi, Janice Mazidi, Rolin Kinlay	Pearson Publication
5	Microcontrollers and PLCs	Sanjay Attri	Dhanpatrai& Co
6	MCS@51 Microcontroller Family User's Manual		INTEL

XV SOFTWARE/LEARNING WEBSITES

1. http://old.zntu.edu.ua/base/lection/rpf/lib/zhzh03/8051_tutorial.pdf
2. <https://nptel.ac.in/courses/108105102>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE246	Semester VI	Microprocessor & Microcontroller													
		Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
246	Competency Create an embedded system to solve real life problem	1	2	3	1	3	1			1			1	3	1
246.1	Understand the basic building blocks & architecture of a microcontroller/microprocessor	3	1										1		
246.2	Develop a program using embedded C	1	2	3	1	3				1			1		
246.3	Demonstrate use of Input, Output, Timers, Interrupts and serial communication.	1	2	1	1	3				1					
246.4	Analyse, Evaluate and Integrate peripheral devices	3	2	2	1	3				1				1	
246.5	Design and create hardware and software solution to real life problems	1	3	3	1	3	1			1			1	3	1

XVII NAMES OF RESOURCE PERSONS

- a) Jignesh S Patel
- b)

UG Engineering Programme	Semester
Electrical Engineering	Fourth

I RATIONALE

The Transformers and Induction Machines are used in power Transmission and Distribution. The course is developed such that student would get the knowledge from basics to control schemes of these machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the performance of various control methods used in field.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Use** in depth knowledge about the construction details, operating principle, performance characteristics of Transformers and induction machines.
- 2) **Evaluate** the performance of the Transformers and Induction machines.
- 3) **Interpret** the suitable control technique for induction motors.
- 4) **Analyze** with the recent developments in transformers and induction machines.
- 5) **Choose** the type of testing method for Transformers and Induction machines.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	200
4	0	4	6	70	30*	50	50	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	2	Open circuit and short circuit tests on single phase transformer	05

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
2	1	Determination of equivalent circuit parameters of transformer.	05
3	4	Load test on single-phase transformer and three phase transformers.	05
4	4	Parallel operation of single phase transformer and Three phase Transformers	05
5	5	Scott Connection of transformers.	05
6	5	No load and Blocked rotor test of single phase induction motor	05
7	1	Determination of equivalent circuit parameters and performance of single phase induction motor	05
8	2	No load and Blocked rotor test of ploy phase induction motor.	05
9	2	Load test of ploy phase induction motor	05
10	3	Speed control of wound rotor induction motor	05
11	3	Speed control of squirrel cage induction motor	05
12	3	Computation of circle diagram for induction motor	05
		Total	60

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Single Phase Transformer, Three Phase Transformers, Single and Three Phase load, Voltage regulators (Variac), Meters,	1,2,3,4,5,6
2	Induction machine with prime-mover/load	7,8,9,10,11,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the 'Apply' and above level sample *Cognitive Domain* learning outcomes denoted by unit outcomes (UOs) related to *Revised Bloom's Taxonomy* (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – I Fundamentals,	CO1. Use in depth knowledge about the construction details, operating principle,	1.1 Concept of an ideal transformer, assumptions;

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Testing and performance analysis of single phase transformers	performance characteristics of Transformers and induction machines CO2. <u>Evaluate</u> the performance of the Transformers and Induction machines CO4. <u>Analyze</u> with the recent developments in transformers and induction machines. CO5. <u>Choose</u> the type of testing method for Transformers and Induction machines. PrO1 . Open circuit and short circuit tests on single phase transformer PrO.2 Determination of equivalent circuit parameters of transformer. PrO3. Load test on single-phase transformer and three phase transformers. PrO 4. Parallel operation of single phase transformer and Three phase Transformers 1a. <u>Justify</u> the concept of static machines and fundamental component roles. 1b. <u>Evaluate</u> the performance of Single Phase Transformer 1c. <u>Justify</u> the output characteristics of Single phase Transformers 1d. <u>Setup</u> the circuit to determine output characteristics of single phase transformers. 1e. <u>Obtain</u> open circuit and short circuit characteristics of three phase transformer.	ideal transformer at no load and on load 1.2 Phasor diagram, voltage current and power relations; Transformer on DC. 1.3 EMF equation 1.4 Voltage and current transformation ratio, Effects of voltage and frequency variations. 1.5 Constructional details of a practical single phase transformer, core, winding, insulation and cooling methods 1.6 Circuit model of a practical transformer incorporating an ideal transformer; 1.7 Analysis of a practical single phase transformer, exact and approximate equivalent circuits. 1.8 Phasor diagram, relation between referred variables and parameters, per unit system of representation 1.9 Open circuit and short circuit tests 1.10 Experimental determination of single phase transformer equivalent circuit parameters 1.11 Percentage regulation of a single phase transformer 1.12 Efficiency of a single phase transformer, maximum efficiency 1.13 All day efficiency 1.14 Sumpner's test for experimental determination of a single phase transformer efficiency 1.15 Auto transformer
Unit– II Fundamentals, Operation and performance of three phase transformers	CO1. <u>Use</u> in depth knowledge about the construction details, operating principle, performance characteristics of induction machines CO2. <u>Evaluate</u> the performance of the Transformers and Induction machines CO4. <u>Analyze</u> with the recent developments in	a. Bank of three single phase transformers b. voltage, current and kVA rating with different connections c. open Δ connection

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	transformers and induction machines. PrO.4 Parallel operation of single phase transformer and Three phase Transformers. PrO.5 Scott Connection of transformers. 2a. <u>Determine</u> the output from three phase transformer. 2b. <u>Evaluate</u> the performance of Three Phase Transformer	d. Three phase transformer as a single unit e. constructional features, three limb, five limb, core type and shell type constructions f. comparison with bank of three single phase transformers g. Three phase transformer connections h. polarity and terminal convention i. vector groups; Y/Y and Δ/Δ connections, connection and phasor diagram, per phase equivalent circuit based analysis j. Vector groups Y/Δ and Δ/Y connections. Connection and phasor diagrams, per phase equivalent circuit based analysis k. Y/Z and Δ/Z connections l. Connection and phasor diagrams m. voltage, current relations n. Phase conversion: 3 to 6 phase and 3 to 2 phase conversions (Scott Connections) o. Tap changing transformer and voltage control p. Parallel operation of single and three phase transformers. q. per phase equivalent circuit based analysis with equal and unequal no load voltages r. load sharing; Harmonics and switching transients in transformers effect of transformer connections, inrush current

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit-III Construction, operation, Testing and analysis of three phase induction machines.	CO1.<u>Use</u> in depth knowledge about the construction details, operating principle, performance characteristics of Transformers and induction machines CO2.<u>Evaluate</u> the performance of the Transformers and Induction machines CO3.<u>Interpret</u> the suitable control technique for induction motors. CO4.<u>Analyze</u> with the recent developments in transformers and induction machines. CO5.<u>Choose</u> the type of testing method for Transformers and Induction machines. PrO.8 No load and Blocked rotor test of ploy phase induction motor. PrO.9 Load test of ploy phase induction motor PrO.12 Computation of circle diagram for induction motor 3a.<u>Choose</u> the relevant operating parameters for load operation of three phase induction motor. 3b. <u>Evaluate</u> the performance of three phase induction motor for the specified condition.	- Introduction, Classification of three phase induction motor, - Production of three phase rotating magnetic field - Locking of stator and rotor field - Construction, Principle of operation, slip, Frequency of rotor current, Speed of rotor field - Rotor EMF, Rotor current, Rotor reactance under starting and running condition - Relation between torque and rotor power factor - Torque equation, Torque under running condition, Torque-Speed curves and effect of change in rotor resistance - Operating region, Starting torque, Full load torque, Breakdown torque, Condition for maximum torque - Effect of change in supply voltage and frequency on torque and slip, Braking of induction motor - Losses in induction machines - Power flow diagram in a three phase induction machine - air gap power, slip power, mechanical power; Torque, Synchronous watt, Efficiency - Induction motor as generalized transformer - Phasor diagram of induction motor - Exact and approximate per phase equivalent circuit - Induction motor no load

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
		test and blocked rotor test q. determining equivalent circuit parameters; losses and efficiency r. Concept of circle diagram, Series circuit and current locus, Construction of circle diagram s. Induction machine performance computation from circle diagram t. Effect of space harmonic fields, harmonics induction torques, Harmonic synchronous torques u. Crawling and Cogging, Deep bar and double cage induction motor (High torque motors)
Unit-IV Starting and speed control of three phase induction machines	CO3.<u>Interpret</u> the suitable control technique for induction motors. CO2.<u>Evaluate</u> the performance of the Transformers and Induction machines PrO.10 Speed control of wound rotor induction motor PrO.11 Speed control of squirrel cage induction motor 4.1 <u>Interpret</u> the machine data with respect to the <u>given</u> power system. 4.2 <u>Select</u> the relevant control method for the <u>specified</u> application.	4.1 Starting of squirrel cage and slip ring induction motor 4.2 Direct on line starting 4.3 stator resistance and reactance based starting 4.4 rotor resistance based starting 4.5 Y/ Δ and auto transformer based starting; examples 4.6 Speed control of induction motors by stator voltage variation and pole changing techniques 4.7 Variable frequency speed control of induction machines, v/f control method, slip power control method
Unit-V Single phase induction motor	CO1.<u>Use</u> in depth knowledge about the construction details, operating principle, performance characteristics of induction machines CO2.<u>Evaluate</u> the performance of the Transformers and Induction machines CO4.<u>Analyze</u> with the recent developments in transformers and induction machines. PrO.6 No load and Blocked rotor test of single phase induction motor PrO.7 Determination of equivalent circuit parameters and performance of single phase	5.1 General description of the motor and constructional features 5.2 Rotating and pulsating field, development of equivalent circuit based on double revolving field theory 5.3 starting of single phase induction motor, 5.4 Determination of parameters by test 5.5 performance analysis

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	induction motor 5a. <u>Choose</u> the relevant operating parameters for load operation of single phase induction motor. 5b. <u>Evaluate</u> the performance of single phase induction motor for the specified condition.	5.6 Types of induction motor: split phase or resistance start, Capacitor-start, Capacitor start capacitorrun, Permanent capacitor, Shaded pole motor: Construction, Working, Starting and running performance 5.7 Characteristics and Applications; selection of capacitor value for starting and running conditions 5.8 Selection capacitance values for maximum torque

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Fundamentals, Testing and performance analysis of single phase transformers	15	2	6	2	6	0	0	16
II	Fundamentals, Operation and performance of three phase transformers	15	2	2	4	4	2	0	14
III	Construction, operation, Testing and analysis of three phase induction machines.	15	0	6	4	4	4	0	18
IV	Starting and speed control of three phase induction machines	9	0	4	4	2	4	0	14
V	Single phase induction motor	6	0	2	2	2	2	0	8
Total		60	4	20	16	18	12	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the

projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **OC/SC test on transformers to draw OCC and SCC.**
- 2) **No-load and Blocked rotor test of Induction motors.**
- 3) **Speed control of induction motor in variable torque condition.**
- 4) **Direct torque control of induction motor.**

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different methods of voltage regulation.
- b) Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Fundamentals, Testing and performance analysis of single phase transformers	1	PrO1. Open circuit and short circuit tests on single phase transformer 1a. <u>Justify</u> the concept of static machines and fundamental component roles. 1b. <u>Evaluate</u> the performance of Single Phase Transformer 1c. <u>Setup</u> the circuit to determine output characteristics of single phase transformers. 1d. <u>Obtain</u> open circuit and short circuit characteristics of three phase transformer.	Lecture with media	PPTs, black board
	2	PrO.2 Determination of equivalent circuit parameters of transformer.	Lecture With	PPTs, black

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		2a. <u>Evaluate</u> the performance of Single Phase Transformer 2b. <u>Justify</u> the output characteristics of Single phase Transformers	media	board
	3	PrO3. Load test on single-phase transformer and three phase transformers. 1a. <u>Evaluate</u> the performance of Single Phase Transformer 1b. <u>Justify</u> the output characteristics of Single phase Transformers 1c. <u>Setup</u> the circuit to determine output characteristics of single phase transformers.	Lecture With media	PPTs, black board
	4.1	PrO 4. Parallel operation of single phase transformer. 1a. <u>Justify</u> the concept of static machines and fundamental component roles. 1b. <u>Evaluate</u> the performance of Single Phase Transformer 1c. <u>Justify</u> the output characteristics of Single phase Transformers 1d. <u>Setup</u> the circuit to determine output characteristics of single phase transformers.	Lecture With media	PPTs, black board
Unit-2 Fundamentals, Operation and performance of three phase transformers	4.2	PrO.4 Parallel operation of Three phase Transformers. 2a. <u>Determine</u> the output from three phase transformer.	Lecture With media	PPTs, black board
	5	PrO.5 Scott Connection of transformers. 2a. <u>Evaluate</u> the performance of Three Phase Transformer	Lecture With media	PPTs, black board
Unit-3 Construction, operation, Testing and analysis of three phase induction machines.	6	PrO.8 No load and Blocked rotor test of ploy phase induction motor. 3a. <u>Choose</u> the relevant operating parameters for load operation of three phase induction motor. 3c. <u>Evaluate</u> the performance of three phase induction motor for the specified condition.	Lecture With media	PPTs, black board
	7	PrO.9 Load test of ploy phase induction motor 3a. <u>Choose</u> the relevant operating parameters for load operation of three phase induction motor. 3c. <u>Evaluate</u> the performance of three phase	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		induction motor for the specified condition.		
	8	PrO.12 Computation of circle diagram for induction motor 3a. <u>Choose</u> the relevant operating parameters for load operation of three phase induction motor. 3c. <u>Evaluate</u> the performance of three phase induction motor for the specified condition.	Lecture With media	PPTs, black board
Unit-4 Starting and speed control of three phase induction machines	9	PrO.10 Speed control of wound rotor induction motor 4.3 <u>Interpret</u> the machine data with respect to <u>the given</u> power system. 4.4 <u>Select</u> the relevant control method for <u>the specified</u> application.	Lecture With media	PPTs, black board
	10	PrO.11 Speed control of squirrel cage induction motor 4.5 <u>Interpret</u> the machine data with respect to <u>the given</u> power system. 4.6 <u>Select</u> the relevant control method for <u>the specified</u> application.	Lecture With media	PPTs, black board
Unit-5 Single phase induction motor	11	PrO.6 No load and Blocked rotor test of single phase induction motor 3a. <u>Choose</u> the relevant operating parameters for load operation of single phase induction motor. 3c. <u>Evaluate</u> the performance of single phase induction motor for the specified condition.	Lecture With media	PPTs, black board
	12	PrO.7 Determination of equivalent circuit parameters and performance of single phase induction motor 3a. <u>Choose</u> the relevant operating parameters for load operation of single phase induction motor. 3c. <u>Evaluate</u> the performance of single phase induction motor for the specified condition.	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Theory and performance of	J.B. Gupta	S.K. Kataria & Sons, ISBN: 978-93-5014-277-6

S. No.	Title of Book	Author	Publication including ISBN
	electrical machines		
2	A textbook of electrical technology VOL II	B.L. Theraja	S. Chand, ISBN: 9788121924375
3	Electric Machinery Fundamentals	Stephen. J. Chapman	McGraw Hill, ISBN: 10. 978007352954

XV SOFTWARE/LEARNING WEBSITES

- <http://nptel.ac.in/courses/108106072/>
(video lectures Prof:D. Kastha, IIT Kharagpur)

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work;	PO 10 Communication	PO11 Project management	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
CO 1	<u>Use</u> in depth knowledge about the construction details, operating principle, performance characteristics of Transformers and induction machines.	3	1	1	3	1				2		2	3	3	1
CO 2	<u>Evaluate</u> the performance of the Transformers and Induction machines.	3	2	1	1	1			1	2		2	3	3	1
CO 3	<u>Interpret</u> the suitable control technique for induction motors.	3	2	3	2	3			1	1		2	2	1	
CO 4	<u>Analyze</u> with the recent developments in transformers and induction machines.	2		2	2	3				2				2	3
CO	<u>Choose</u> the type of	2		2	1	2							1	2	

Course Code	Competency and COs	PO 1 Used of mathematical method	PO 2 Analysis mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work:	PO 10 Communication	PO11 Project management	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
5	testing method for Transformers and Induction machines.														

XVII NAMES OF RESOURCE PERSONS
Prof. Pratik B Panchal

UG Engineering Programme	Semester
Control Systems	Fourth

I RATIONALE

Electrical engineers have to work on various electrical equipment with their dedicated control system. This course is developed such that student should be able to design, develop, implement and operate efficiently various controllers, compensators and closed loop control systems for different electrical applications for efficient performance.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the performance of various control systems used for linear and no linear systems.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Use the different methods to develop mathematical models for systems.

Evaluate the performance of the system by means of stability.

Evaluate the performance of the system by means of state space analysis.

Evaluate the performance of linear and nonlinear control system with time and frequency response analysis under *steady state and dynamic condition*.

Choose the appropriate control scheme, controller, compensator for different applications.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	2	0	5	70	30	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Set</u> the value of controller gain for the effective performance of the feedback control system.	01
2	1	<u>Determine</u> the transfer function for the given SISO and MIMO system using block diagram reduction technique by using MATLAB software.	02
3	1	<u>Determine</u> the transfer function for the given SISO and MIMO system using Mason's gain Formula by using MATLAB software.	02
4	2	<u>Determine</u> the stability of the systems by using RH Criteria and Root locus with the help of standard procedure and compare the same with MATALB software.	02
5	3	<u>Examine</u> the controllability, observability for the systems with the help of MATLAB software.	02
6	3	<u>Determine</u> the Eigen values and Eigen vectors for the system using conventional and with the help of MATLAB software.	01
7	4	<u>Determine</u> the value of various transient & steady state response specifications for different type and order control systems with the help of MATALB software.	02
8	4	<u>Determine</u> the value of ζ for various 1 st and 2 nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software.	02
9	4	<u>Determine</u> the most efficient values of P, D & I constants in P, PD, PI and PID controllers for 2 nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software.	02
10	5	<u>Examine</u> the stability of the systems using BODE, POLAR and NYQUIST plots with the help of MATLAB software.	02
11	5	<u>Setup</u> the relevant compensator for the given specific control system requirement using MATLAB software.	02
Total			20

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5
2	Test kit for feedback control system.	1

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year

d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the 'Apply' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
Unit – I Introduction to control system.	CO 1. <u>Use</u> the different methods to develop mathematical models for systems. Pro 1. <u>Set the value of controller gain for the effective performance of the feedback control system.</u> Pro 2. <u>Determine the transfer function for the given SISO and MIMO system using block diagram reduction technique by using MATLAB software.</u> Pro 3. <u>Determine the transfer function for the given SISO and MIMO system using Mason's gain Formula by using MATLAB software. Justify the need of protection scheme for the given situation.</u> 1a. <u>Evaluate</u> the various control system various parameters. 1b. <u>Justify</u> the requirements of good control system. 1c. <u>Determine</u> the transfer function of the system using Block diagram reduction. 1d. <u>Determine</u> the transfer function of the system using Mason's Gain Formula. 1e. <u>Evaluate</u> the transfer function of electrical, mechanical and electromechanical system.	1.1 Control system parameters 1.2 Types of control systems 1.3 Methods of finding transfer function of the system. 1.4 Analysis of Electrical, Mechanical and Electromechanical systems. 1.5 Need of servomechanism and servomotors
Unit– II Stability of Control Systems	CO 2. <u>Evaluate</u> the performance of the system by means of stability. Pro 4. <u>Determine</u> the stability of the systems by using RH Criteria and Root locus with the help of standard procedure and compare the same with MATALB software. 2a. <u>Evaluate</u> the various factors which are relevant for the stability. 2b. <u>Evaluate</u> the various methods for determining the stability of the system. 2c. <u>Determine</u> the stability of the higher order system with the help pf RH criteria. 2d. <u>Choose</u> the appropriate value of DC gain in order to achieve the stability condition. 2e. <u>Evaluate</u> the stability condition using root locus diagram method for stability condition.	2.1 Basic stability conditions 2.2 R-H Criteria for higher order control system. 2.3 Root locus construction and stability analysis. 2.4 Effect of adding poles and zeroes in the system
Unit– III State Space	CO 3. <u>Evaluate</u> the performance of the system by means of state space analysis.	

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
Analysis of Control Systems	PrO 5. <u>Examine</u> the controllability, observability for the systems with the help of MATLAB software. PrO 6. <u>Determine</u> the Eigen values and Eigen vectors for the system using conventional and with the help of MATLAB software. 3a. <u>Evaluate the various performance indices related to state space analysis.</u> 3b. <u>Determine</u> the controllability & observability by means of Kalman's Test. 3c. <u>Determine</u> the Eigen values and Eigen vectors for the system.	3.1 Concept of state, state variable and state model 3.2 Concept of controllability and Observability, State – Transition Matrix 3.3 Computation of State Transition Matrix, Eigen Values and Eigen Vectors.
Unit-IV Time response analysis	CO 4. <u>Evaluate</u> the performance of linear and nonlinear control system with time and frequency response analysis under <i>steady state and dynamic condition</i>. PrO 7. <u>Determine</u> the value of various transient & steady state response specifications for different type and order control systems with the help of MATLAB software. PrO 8. <u>Determine</u> the value of ζ for various 1st and 2nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software. PrO 9. <u>Determine</u> the most efficient values of P, D & I constants in P, PD, PI and PID controllers for 2nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software. PrO 10. <u>Examine</u> the stability of the systems using BODE, POLAR and NYQUIST plots with the help of MATLAB software 4a. <u>Examine the various factors associated with steady state and transient state performance of the system.</u> 4b. <u>Choose</u> the relevant values of ζ for the efficient performance of the system. 4c. <u>Choose</u> the relevant values of controller parameters for the efficient performance of the system. 4d. <u>Examine</u> the various factors associated with Time response analysis. 4e. <u>Examine</u> the various factors associated with frequency response analysis. 4f. <u>Determine</u> the stability of the system under frequency response approach.	4.1 Time response of first order and second order system 4.2 Damping ratio, natural undamped frequency, Transient response specifications 4.3 Steady state error and error constant. 4.4 Effect of integral and derivative control action on system. 4.5 Bode plot, Procedure for drawing bode plot and determination of gain margin, phase margin and stability 4.6 Polar plot Procedure for drawing bode plot and determination of gain margin, phase margin and stability

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – V Frequency response analysis	CO 5. <u>Choose</u> the appropriate control scheme, controller, compensator for different applications. PrO 11. <u>Setup</u> the relevant compensator for the given specific control system requirement using MATLAB software 5a. <u>Examine</u> the the various factors associated with compensation. 5b. <u>Examine</u> the various compensators for efficient performance of the system. 5c. <u>Determine</u> the compensators for efficient performance of the system breaker. 5d. <u>Prepare</u> the specification of a compensator required for the given situation. 5e. <u>Select</u> the relevant compensator for the efficient performance of control system.	5.1 Phase lead compensation, Phase lag compensation, Phase lead lag compensation. 5.2 Selection of Compensator, Design of Phase Lag, Phase Lead and Phase Lag-Lead Compensator.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction to control Systems	10	02	04	04	04	02	0	16
II	Stability of Control Systems	06	02	02	02	00	04	0	10
III	State Space Analysis of Control Systems	06	02	02	02	00	04	0	10
IV	Time response analysis	08	02	02	02	02	06	02	16
V	Frequency response analysis	15	02	02	02	02	08	02	18
Total		45	10	12	12	08	24	04	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

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The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary

consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Mathematical modelling:** Collect the specifications of different types of electromagnetic relays used in different protection situations in different types of industries.
- 2) **Steady state and Dynamic Response :** Collect the specifications of different types of static relays used in different protection situations in different types of industries.
- 3) **Stability Analysis:** Prepare case studies of protections failures in different types of industries.
- 4) **Controllers and Compensators:** Prepare case studies of circuit breakers used in the protection schemes in different types of industries.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different control system used in industries.
- b) Prepare 15 min. power point presentation on the implementation of closed loop control system.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduct ion to control system.	1	<i>PrO 1. <u>Set</u> the value of controller gain for the effective performance of the feedback control system.</i> <i>PrO 2. <u>Determine</u> the transfer function for the given SISO and MIMO system using block diagram reduction technique by using MATLAB software.</i> 1a. <u>Evaluate</u> the various control system various parameters. 1b. <u>Justify</u> the requirements of good control system. 1c. <u>Determine</u> the transfer function of the system using Mason's Gain Formula.	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	2	PrO 3. <u>Determine</u> the transfer function for the <u>given</u> SISO and MIMO system using Mason's gain Formula by using MATLAB software. <u>Justify</u> the need of protection scheme for the <u>given</u> situation. 1a. <u>Determine</u> the transfer function of the system using Block diagram reduction.	Lecture With media	PPTs, black board
	3	PrO 3. <u>Determine</u> the transfer function for the <u>given</u> SISO and MIMO system using Mason's gain Formula by using MATLAB software. <u>Justify</u> the need of protection scheme for the <u>given</u> situation. 1a. <u>Determine</u> the transfer function of the system using Mason's Gain Formula. 1b. <u>Evaluate</u> the transfer function of electrical, mechanical and electromechanical system.	Lecture with media Tutorial and Field visit	PPTs, black board
Unit– II Stability of Control Systems	4	CO 2. <u>Evaluate</u> the performance of the system by means of stability. PrO 4. <u>Determine</u> the stability of the systems by using RH Criteria and Root locus with the help of standard procedure and compare the same with MATALB software. 2a. <u>Evaluate</u> the various factors which are relevant for the stability. 2b. <u>Evaluate</u> the various methods for determining the stability of the system.	Lecture with media, and Demonstration	Real samples of relays, PPTs, Videos
	5	2c. <u>Evaluate</u> the various methods for determining the stability of the system. 2d. <u>Determine</u> the stability of the higher order system with the help pf RH criteria. 2e. <u>Choose</u> the appropriate value of DC gain in order to achieve the stability condition.	Lecture with media, Tutorial	PPTs
	6	2e. <u>Choose</u> the appropriate value of DC gain in order to achieve the stability condition.	Lecture with media	PPTs
Unit– III State Space Analysis of Control Systems	7	CO 3. <u>Evaluate</u> the performance of the system by means of state space analysis. PrO 7. <u>Examine</u> the controllability, observability for the systems with the help of MATLAB software. 3a. <u>Evaluate</u> the various performance indices related to state space analysis. 3b. <u>Determine</u> the controllability & observability by means of Kalman's Test.	Lecture with media, Field visit	PPTs, Video
	8	PrO 8. <u>Determine</u> the Eigen values and Eigen vectors for the system using conventional and with the help of MATLAB software.	Lecture with media, and	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		3c. <u>Determine</u> the transfer function of the system. 3d. <u>Determine</u> the Eigen values and Eigen vectors for the system.	Tutorial	
Unit– IV Time response analysis	9	CO 4. <u>Evaluate</u> the performance of linear and nonlinear control system with time and frequency response analysis under <i>steady state and dynamic condition</i>. <i>PrO 13.</i> <u>Determine</u> the value of various transient & steady state response specifications for different type and order control systems with the help of MATALB software. <i>PrO 14.</i> <u>Determine</u> the value of ζ for various 1st and 2nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software. <i>PrO 15.</i> <u>Determine</u> the most efficient values of P , D & I constants in P, PD ,PI and PID controllers for 2nd order control system and compare the performance of the system with the help of graphical representation using MATLAB software. <i>PrO 16.</i> <u>Examine</u> the stability of the systems using BODE, POLAR and NYQUIST plots with the help of MATLAB software 4g. <u>Examine</u> the various factors associated with steady state and transient state performance of the system. 4h. <u>Choose</u> the relevant values of ζ for the efficient performance of the system. 4i. <u>Choose</u> the relevant values of controller parameters for the efficient performance of the system. 4j. <u>Examine</u> the various factors associated with Time response analysis. 4k. <u>Examine</u> the various factors associated with frequency response analysis. <u>Determine</u> the stability of the system under frequency response approach.	Lecture with media, Tutorial	PPT, Video
	11		Lecture with media	PPTs, black board
	12		Lecture with media, case study	PPT, Pictures , black board
Unit– V Frequency	13	CO 5. <u>Choose</u> the appropriate control scheme, controller, compensator for different applications.	Lecture with media,	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
response analysis		<i>PrO 17. Setup</i> the relevant compensator for the given specific control system requirement using MATLAB software 5f. <u>Examine</u> the the various factors associated with compensation. 5g. <u>Examine</u> the various compensators for efficient performance of the system. 5h. <u>Determine</u> the compensators for efficient performance of the system breaker. 5i. <u>Prepare</u> the specification of a compensator required for the given situation. <u>Select</u> the relevant compensator for the efficient performance of control system.	Tutorial	
	14		Lecture with media, Tutorial	PPTs, Video
	15		Lecture with media, Tutorial	PPTs, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Automatic control system	B.C. Kuo	Khanna Publishers, New Delhi, 2008, ISBN: 9788174092323
2	Modern control engineering.	K. Ogata	
3	Control System Engineering	J Nagrath & M Gopal	New Age International Publishers
4	Control Systems	A. Anand Kumar	PHI Learning Pvt. Ltd.
5	Automatic Control Systems	S. Hasan Saeed	Katson Books
6	Control Systems	Ashfaq Hussain, Haroon Ashfaq	Dhanpat Rai & Co.

XV SOFTWARE/LEARNING WEBSITES

1. http://nptel.iitg.ernet.in/courses/Elec_Engg/IIT%20Delhi/Control%20Systems.htm
2. <http://nptel.ac.in/courses/108101037/>
3. <http://nptel.ac.in/courses/108102043/>
4. <http://nptel.ac.in/courses/108103007/>

XVI PO-COMPETENCY-CO MAPPING

Cours e Code	Competency and COs	PO 1 Used of mathe matica l metho d	PO 2 Analys is mecha nical system	PO 3 Used engine ering soft.	PO 4 Solve proble m in machi ne	PO 5 The engine er and society	PO 6 Enviro nment and sustain ability	PO 7 Ethics	PO 8 Individua l and team work:	PO 9 Comm unicati on	PO 10 Life- long learnin g	PSO1 Evaluate the performance of Electrical Machines and Power systems Power Systems	PSO2 Evaluate the performance of Electrical and Electronic Instrumentatio n Systems	PSO3 Provide socially and ethically acceptable I technical solutions to complex electrical engineering problems
EE255 .01	Semester VI	Protection and Switchgear Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO												
255.0 1	Competency Evaluate the performance of various protection systems used in Industry	3	2	2	2	1	1	1	-	2	1	3	-	1
255.0 1.1	Use the protection schemes for apparatus protection	3	2	1	1	1	1	1	-	1	1	1	-	1
255.0 1.2	Evaluate the performance of the electromagnetic relays	3	2	2	1	1	1	1	-	1	-	2	-	1
255.0 1.3	Interpret the suitable protection system for the apparatus protection	3	2	1	1	1	1	1	-	1	-	2	-	1
255.0 1.4	Evaluate the performance of the static and Numerical relays	3	1	1	2	1	1	1	-	1	-	1	-	-
255.0 1.5	Choose the type of circuit breaker for different applications	3	2	1	1	1	1	1	-	1	1	1	-	-

XVII NAMES OF RESOURCE PERSONS

- Prof. Soaib Saiyad
-

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester IV
HS111.03: Human Values and Ethics

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	2
Marks	--	100	--	100	

Pre-requisite:

- Basic awareness of Values and Ethics

Objectives of the Course:

To facilitate learners to:

- Develop a familiarity with the Fundamental Human Values and Ethics
- Understanding Values and Ethics as per Ancient Indian Tradition
- Apply Values and Ethics in their professional, social and personal spectrum
- Understand and apply Constitutional Values and Fundamental Duties
- Explore global issues and propose solutions with reference to values and Ethics

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Ethics and Values in Ancient Indian Tradition	08
3.	Types of Values and Applied Ethics	08
4.	Constitutional Values and Fundamental Duties	05
5.	Global Issues of Values and Ethics in Contemporary Society	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Values and Ethics	5 Hours	17%
	Need, Relevance and Significance of Values, Concept and Meaning of Values and Ethics		
2.	Ethics and Values in Ancient Indian Tradition	8 Hours	26%
	Lessons from Jatakkatha Lessons from Panchtantra Tales Ethics to be followed from Ancient Hinduism, Jainism and Buddhism Selected Anecdotes from ancient mythology and Moral Learnings		
3.	Types of Values and Applied Ethics	8 Hours	26%
	Personal Values, Social, Civic & Democratic Value, Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics		
4.	Constitutional Values and Fundamental Duties	5 Hours	17%
	Basics of Indian Constitution Constitutional Values: Justice, Liberty, Equality and Fraternity Fundamental Rights Fundamental Duties		
5.	Global Issues of Values and Ethics in Contemporary Society	4 Hours	14%
	Case Studies, Presentations, Projects		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the significance of values and ethics
CO2	Understand concept of values and ethics as per Ancient Indian Tradition
CO3	Understand and apply various types of Values and Applied Ethics
CO4	Develop awareness and initiate implementation of Constitutional Values and Fundamental Duties
CO5	Get aware of the global issues and scope and application of values and ethics in problem solving

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

❖ Reference book:

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
3. Values and Ethics for 21st Century, BBVA.
(https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

❖ Web material:

- www.ethics.org

Note: The subtopics mentioned in the modules 2 and 3 are comprehensive. The faculty member can choose any 2 topics from module and 5 topics from module 3.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester-IV

HS113 A: Indian Knowledge System

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite:

- --

Objectives of the Course:

- to develop a foundational understanding of the Indian Knowledge System, its origins, and its significance in shaping Indian culture and society.
- to analyze the Vedic texts, their content, and their cultural and philosophical implications.
- to investigate major philosophical schools in India, including their core tenets and historical context.
- to understand the continuity and evolution of knowledge and wisdom in India across different time periods.
- to gain insight into the role of linguistics and languages in preserving and transmitting knowledge within the Indian context.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An overview of Indian Knowledge System (IKS)	04
2.	The Vedic Corpus	06
3.	Indian Philosophical Systems	04
4.	Wisdom through the Ages	04
5.	Linguistics	06
6.	The knowledge triangle	06
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1	An overview of Indian Knowledge System (IKS):	04
	▪ Importance of Ancient Knowledge ▪ Definition of IKS ▪ Classification framework of IKS ▪ Unique aspects of IKS	
2	The Vedic Corpus:	06
	▪ Vedas and Vedangas ▪ Distinctive features of Vedic Life	
3	Indian Philosophical Systems:	04
	▪ Introduction to Indian Philosophies ▪ Different schools of philosophy	
4	Wisdom through the Ages:	04
	▪ Puranas ▪ Ithihasas ▪ Niti shastras ▪ Subhasitas	
5	Linguistics:	06
	▪ Components of a language ▪ Paṇini's work on Sanskrit grammar ▪ Phonetics in Sanskrit ▪ The role of Sanskrit in natural language processing	
6	The knowledge triangle:	06
	▪ Prameya ▪ Pramāṇa ▪ Saṃsaya ▪ Framework for establishing Valid Knowledge ▪ Potential fallacies in the reasoning process	

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	describe the fundamental concepts and components of the Indian Knowledge System, including its historical development.
CO2	analyze select Vedic texts, extracting their philosophical and cultural significance.
CO3	differentiate between major Indian philosophical systems, such as Vedanta, Nyaya, and Sankhya, and comprehend their key principles.
CO4	trace the evolution of wisdom and knowledge in India from ancient times to the present day, recognizing its continuity.
CO5	examine the role of language and linguistics in preserving and transmitting knowledge and cultural heritage.
CO6	discuss the interconnectedness of philosophy, science, and spirituality in IKS, highlighting their contributions to holistic understanding.

Evaluation:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

❖ Text book:

--

❖ Reference book:

1. "Indian Knowledge Systems" by Kapil Kapoor
2. "Indian Knowledge Tradition" by K. Ramasubramanian
3. "The Advaita Worldview: God, World, and Humanity" by Anantanand Rambachan
4. "The Vedas: An Introduction to the Sacred Texts of India" by Roshen Dalal
5. "The Rigveda: 3-Volume Set" translated by Stephanie W. Jamison and Joel P. Brereton
6. "The Yajur Veda: A Rendering of the Jewel of the Vedic Hymns" translated by Devi Chand
7. "A History of Indian Philosophy" by Surendranath Dasgupta (Multi-volume work)
8. "Indian Philosophy and Modern Culture" by Satyendra Nath Dasgupta
9. "The Wisdom of the Upanishads" by A. Parthasarathy
10. "The Dhammapada: The Sayings of the Buddha" translated by Eknath Easwaran
11. "Prameya Mala" by Swami Tattvavidananda Saraswati
12. "Pramana: Vaisheshika Sutra of Kanada" translated by Satish Chandra Vidyabhusana
13. "Saṃsāya in Indian Philosophy: Text and Commentaries" by Bibhuti Baruah
14. "The Art of Reasoning" by David Kelley

❖ Web material:

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CE291: PYTHON PROGRAMMING SKILLS

B.TECH 4TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – I

(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	2	4	6	4
Marks	100	100	200	

A. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Basics of Python	01
2.	Variables and Data Types	04
3.	Loops and Conditional Control	06
4.	Advanced Data Structures	04
5.	Functions	02
6.	Object Oriented Programming in python	08
7.	Reading and Writing Files	03
8.	Packages and Modular Programming	02

Total Hours (Theory):

30

Total Hours (Lab): 60

Total Hours: 90

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1. | Basics of Python | 01 Hour | 03% |
| 1.1 | Introduction to Python | | |
| 1.2 | Installation and Environment Setup | | |
| 1.3 | Simple Python Program Practice | | |
| 2. | Variables and Data Types | 04 Hours | 13% |
| 2.1 | Variables, Operators, Precedence and Associativity | | |
| 2.2 | Data Types , Indentation, Comments, Reading Input, Print Output | | |
| 2.3 | Memory Optimization | | |

2.4	Type Conversion		
3.	Loops and Conditional Control	06 Hours	20%
3.1	The if... Decision Control Flow Statement, The if...else Decision Control Flow Statement, The if...elif...else Decision Control Statement		
3.2	Nested if Statement, The while Loop, The for Loop,		
4.	Advanced Data Structures	04 Hours	13%
4.1	Lists- Basic List Operators, Replacing, Inserting, Removing an Element, Searching and Sorting Lists, Tuples.		
4.2	Dictionaries- Dictionary Literals, Adding and Removing Keys, Accessing and Replacing Values; Traversing Dictionaries.		
4.3	Sets, Strings, Files and their Libraries		
5.	Functions	02 Hour	07%
5.1	Defining a Function, Calling a Function		
5.2	Pass by Reference vs Value, Function Arguments, Required and Default Arguments, Keyword Arguments and Variable-Length Arguments		
5.3	The Anonymous Functions, The Return Statement		
5.4	Scope of Variables		
6.	Object Oriented Programming in python	08 Hours	27%
6.1	Object-Orientation Introduction, Class Definition, Class Objects, Instance Objects, Method Objects		
6.2	Class and Instance Variables, Inheritance, Private Variables, The Polymorphism		
7.	Reading and Writing Files	03 Hour	10%
7.1	Reading Files, Parsing and Processing Files		
7.2	File storage, File operations (delete, copy)		
8.	Packages and Modular Programming	02 Hour	07%
8.1	Executing modules as Scripts, Import and Use of Built-in Modules, Accessing Third Party Packages and Modules		
8.2	Creating Custom Modules and Packages, Importing and Using Custom Modules		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Practicals/Tutorials related to course content will be carried out in the laboratory.

D. Course Outcomes (COs):

At the end of the course the students will be able to:

CO1	Interpret the fundamental python syntax, semantics and fluent in the use of python control flow statements. (Understand)
CO2	Determine the methods to create and manipulate python programs by utilizing the data structures like lists, dictionaries, tuples and sets. (Apply)
CO3	Express proficiency in the handling of strings and functions. (Apply)
CO4	Articulate the Object-Oriented Programming concepts such as inheritance and polymorphism as used in Python. (Create)
CO5	Identify and use the commonly used operations involving file systems, and regular expressions (Create)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	-	-	-	1	-	1	2	-	-	2	-	-
CO2	-	1	-	2	2	3	2	-	-	-	2	-	2	-
CO3	1	-	2	1	3	2	3	2	2	2	-	2	-	3
CO4	-	1	2	2	3	3	2	3	-	3	3	-	-	-
CO5	1	2	-	2	3	3	2	2	2	-	3	3	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

E. Recommended Study Material:

❖ **Text Books:**

1. Nagar, Sandeep, “Introduction to Python for Engineers and Scientists - Open Source Solutions for Numerical Computation”, Apress, 2018.
2. Kenneth Lambert, “Fundamentals of Python: First Programs”, ISBN-13: 978-1337560092, Cengage learning publishers, first edition, 2012.
3. Magnus Lie Hetland, "Beginning Python from Novice to Professional", Third Edition, Apress, 2017
4. Reema Thareja, “Python Programming using Problem Solving Approach”, ISBN-13:978-0-19- 948017-3, Oxford University Press, 2017.

❖ **Reference Books:**

1. David Beazley, Brian K. Jones, "Python Cookbook", 3rd edition, OREILLY, 2016
2. Brett Slatkin, "Effective Python: 59 Specific Ways to Write Better Python", Novatec, 2016
3. Allen Downey, "Think Python: How to Think Like a Computer Scientist", Green Tea Press, 2015
4. Mark Lutz "Learning Python", 4th Edition, O'REILLY, 2016
5. Vamsi kurama, “Python programming: A modern approach”, ISBN-978-93-325-8752- 6, Pearson, 2018.
6. W. Chun, “Core python programming”, ISBN-13: 978-0132269933, Pearson, 2nd edition, 2016.

❖ **Web Materials:**

1. <https://www.python.org/>
2. <http://www.diveintopython3.net/>
3. <https://developer.mozilla.org/en-US/docs/Learn/Server-side/Django>
4. <https://www.fullstackpython.com/django.html>
5. <https://codelabs.developers.google.com/>
6. <https://www.tutorialspoint.com/python/index.htm>

❖ **Software:**

1. Python IDLE
2. Anaconda Python
3. PyCharm

ME29I: BASICS OF ELECTRICAL VEHICLES
B.TECH 4TH SEMESTER (Electrical)
INSTITUTE ELECTIVE – I
 (For Minor Specialization in “Electric Vehicle Systems”)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Electric Vehicles	14
2.	Automotive Design	14
3.	Design of Electric and Hybrid Electric	15
4.	Regulation and Standardization of Vehicles	02

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

- | | | |
|--------------|--|------------------------------|
| 1 | Introduction to Electric Vehicles | 14Hours 31.11% |
| 1.1 | I.C Engine powered Vehicles Vs Electrical Vehicles (EVs), How EV works, Types of EVs, Hybrid Electric Drive-train | |
| 1.2 | Tractive effort in normal driving, Energy consumption Concept of Hybrid Electric Drive Trains, Architecture of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains, Parallel hybrid electric drive trains, Electric Propulsion unit. | |
|
2 |
Automotive Design |
14Hours 31.11% |
| 2.1 | Major parts of automobiles, Chassis, Frame, Body of automobile, Transmission system, Steering and Breaking systems, Suspension, Accessories. | |
|
3 |
Design of Electric and Hybrid Electric Vehicles |
15Hours 33.34% |

- 3.1 Series Hybrid Electric Drive Train Design: Operating patterns, control strategies, Sizing of major components, power rating of traction motor, power rating of engine/generator, design of PPS
- 3.2 Parallel Hybrid Electric Drive Train Design: Control strategies of parallel hybrid drive train, design of engine power capacity, design of electric motor drive capacity, transmission design, energy storage design
- 4 Regulation and Standardization of Vehicles 02Hours 4.44%
- 4.1 Motor vehicle act, registration of motor vehicles, driving license, control of traffic, insurance against third party, claims for compensation, traffic signs, central motor vehicle rules, vehicle safety standards and regulations, classification and definition of vehicles, enforcement of emission norms, duties of surveyor.

C. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand needs and working of various systems in automobiles.
CO2:	Practically identify different automotive systems and subsystems.
CO3:	Practically identify different automotive components.
CO4:	Understand market and businesses of automobile industry.
CO5:	This course will give the students some awareness to recent trends and research areas in Automobiles.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2:	3	1	1	-	-	-	-	-	1	-	-	1	-	-
CO3:	3	1	1	-	-	-	-	-	1	-	-	2	-	-
CO4:	-	-	-	-	-	-	-	-	-	-	2	1	-	-

CO5:	1	1	1	1	-	-	-	-	1	-	-	1	-	-
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1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial Visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of assignment.

E. Recommended Study Material:

Text Books:

1. Singh Kirpal, “Automobile Engineering” Volume I & II, Standard Pub.& Dist.
2. Gupta K. M. “Automobile Engineering” Volume I & II, Umesh Pub.
3. Gupta R. B. “Automobile Engineering”, Satya Prakashan.
4. Giri N. K., “Automobile Technology”, Khanna Pub.

Reference Books:

1. Crouse W., “Automotive Mechanics”, Tata Mc Graw Hill.
2. Narang G. B. S. “Automobile Engineering”, Khanna Pub.
3. John Lowry and James Larminie , “Electric Vehicle Technology Explained”, Wiley

Web Material:

1. <http://nptel.iitm.ac.in>
2. <http://www.sae.org/pubs/automotive/>
3. <https://www.springer.com/journal/12239/>

Other Material:

1. International Journal of Automotive Technology (www.ijat.net/)
2. International Journal of Automotive Technology

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Th.	Pr.	Total		Int.	Ext.	Int.	Ext.	
TY Sem- V	EE351	Electrical Power Transmission and Distribution	3		3	3	30	70			100
	EE342	Synchronous and DC Machines	4	4	8	6	30	70	50	50	200
	EE353	Power Electronics and Drives – I	3	2	5	4	30	70	25	25	150
	EE344	Minor Project I		4	4	2			50	50	100
	EE345	Electrical Product Survey		2	2	2			50	50	100
	EE350	Summer Internship I				3			75	75	150
	HSI31.02A, HSI42A, HSI43A	Communication and Soft Skill, French, German		2	2	2			30	70	100
	EE371-EE375, OCEE3002	Advanced Microcontrollers/ VLSI Technology and Design/Optimization Techniques/Distributed Generation and Microgrid/Energy Conservation, Audit and Management/Computer Vision and Image Processing	4	2	6	5	30	70	25	25	150
	TOTAL:		14	16	30	27	120	280	305	345	1050
TY Sem - VI	EE346	Digital Signal Processing	3	2	5	4	30	70	25	25	150
	EE347	Programmable Logic Controllers and Industrial Automation	2	4	6	4	30	70	50	50	200
	EE348	Power Electronics and Drives II	3	2	5	4	30	70	25	25	150
	EE349	Fault Analysis and Switchgear	4	2	6	5	30	70	25	25	150
	EE360	Minor Project II		4	4	2			50	50	100
	HSI32.02A	Contributor Personality Development	2		2	2			30	70	100
	EE376-EE381, OCEE3001, OCEE3003	Special Electrical Machines and Applications/ Embedded Systems/Power Electronics Applications in Power System/High Voltage Engineering/Internet of Things/Hybrid and Electric Vehicles/Microprocessors and Interfacing, State Space Approach To Control System Analysis and Design	3	2	5	4	30	70	25	25	150
	TOTAL:		15	18	33	25	150	350	230	270	1000

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – V)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Code: EE351

Course Title: ELECTRICAL POWER TRANSMISSION AND DISTRIBUTION

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

In today's time The electricity is generated in bulk at remote places using different type of generating station and transmitted to long distances, then distributed at various places as per consumer requirements. The transmission and distribution of electric power is a complex men made system which requires knowledge of different types of transmission lines and equipment's. Technicians are required to operate and maintain the power transmission and distribution system so that electrical energy is continuously available to the end users economically. So engineer or technician should able to work independently in the various area of transmission and distribution system. They can able to operate various control equipment's independently in normal and abnormal conditions. So at design and execution level skill development in student incorporate via this course.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Design, Operate and Troubleshoot various transmission and distribution system.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Identify** and Calculate the different parameters of transmission line.
- 2) **Analyze** & calculate the performance of various types of transmission & distribution systems.
- 3) **Chose** the size and location of generating station, sub-station. with bus-bar arrangement.
- 4) **Design** and Connection schemes of A.C. & D.C. distribution system.
- 5) **Acquire** the knowledge of Power system planning for Generation, Transmission & Distribution System

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2022)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
3	0	0	3	70	30*	0	0	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, 30 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain	Topics and Sub-topics
Unit – I Transmission Line Parameters	CO 1. Identify and Calculate the different parameters of transmission line. 1a. learn details of different parameters 1b. calculate the different parameters of transmission line with different configuration of transmission line. 1c. learn about importance of bundled conductor and its calculation.	1.1 Basics of Electric field. 1.2 Resistance of line, basic concepts of inductance and capacitance. 1.3 Flux linkages of an isolated current carrying conductor. 1.4 Inductance of a single phase two wire line, examples. 1.5 Flux linkages of one conductor in a group and inductance of composite conductor lines, examples. 1.6 Inductance of three phase line with unsymmetrical spacing and symmetrical spacing, 1.7 importance of transposition of tower, examples. Inductance of double circuit line, examples. 1.8 Inductance of bundled conductors and importance of bundled conductors, examples. 1.9 Electric field of a long straight conductor, potential difference between two conductors of a group of parallel conductors. 1.10 Capacitance of two wire line. Capacitance of three phase line with equal and unequal spacing, examples.
Unit– II Characteristic and Performance of Transmission Line	CO 2. Analyze & calculate the performance of various types of transmission & distribution systems. 2a. analyze the general relation of a transmission line and its ABCD parameter. 2b. analyze the performance of long transmission line and power transferred through transmission line. 2c. acquire the knowledge regarding the	2.1 Voltage regulation of short transmission line, example. 2.2 T and π representation of medium transmission line and their Phasor diagram, example. 2.3 Performance analysis of long transmission line and its ABCD parameter representation. 2.4 Surge impedance and surge impedance loading, 2.5 Equivalent T and π representation of long line, 2.6 Interpretation of long line equation,

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain	Topics and Sub-topics
	surge impedance and surge impedance loading of transmission line.	2.7 Ferranti effect. Examples.
Unit– III Design of Transmission system	CO 2. Analyze & calculate the performance of various types of transmission & distribution systems. 2a. State the need for EHV transmission 2b. design Electrical side of EHV transmission line with tower. 2c. gain the knowledge of RI and TVI. 2d. Analyze transmission towers design	3.1 Requirement of Transmission lines, 3.2 selection of voltage for high-voltage transmission lines. 3.3 choice of conductors, spacing of conductor, 3.4 corona. 3.5 Surge impedance loading of transmission lines. 3.6 Electrical design of transmission lines, 3.7 Earth Wires and towers. 3.8 Transmission of Electrical Power at extra-high voltage. Design consideration of EHV lines. 3.9 Selection & spacing of conductors for EHV line. 3.10 Radio & television interference.
Unit-IV Design of Power System and Substation	CO 3. Chose the size and location of generating station, sub-station. with bus-bar arrangement. 3a. State the need for electrical substations. 3b. Sketch the elevation layout of a typical 11/33/66/110 kV electrical substation with various switchgear and typical spacing between them and the ground level as well. 3c. State the selection of the bus bar and their arrangement. 3d. With sketches describe the various types of Earthing adapted for substations.	4.1 Introduction, Selection of size and location of generation stations. 4.2 Selection of specification of transmission lines. 4.3 Size and locations of substations. 4.4 Role of vector group in designing of sub-station. 4.5 Sub-station Grounding, Earthing, 4.6 step and touch potential. 4.7 Bus-bar arrangements in sub-station.
Unit – V Design of Distribution system	CO 4. Design and Connection schemes of A.C. & D.C. distribution system. 4a. State the need for distribution system 4b. With sketches describe the various connection schemes of the distribution system 4c. Describe the Impact of wind power on the distribution system	5.1 Distribution system, requirements of distribution system, 5.2 design consideration of distribution system, 5.3 classification of distribution system. 5.4 Connection schemes of Distribution system. 5.5 D.C. distribution system, 5.6 Types of distribution system. 5.7 A.C. distribution system, A.C. distribution system calculations.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain	Topics and Sub-topics
	4d. Describe the measures to be adapted to take of the distributed generation in the distribution system 4e. Solve simple numerical problems	
Unit-VI Power System Planning	CO 5. Acquire the knowledge of Power system planning for Generation, Transmission & Distribution System. 5a. prepare flow chart of generation, transmission and distribution planning. 5b. Acquire the knowledge about the reliable power system.	6.1 Introduction, Methods of Power system planning. 6.2 Forecasting load and energy requirements. 6.3 Generation Planning 6.4 Transmission System planning. 6.5 Distribution System Planning. 6.6 Reliability of electrical power system, 6.7 Method of measuring power system reliability. 6.8 Trends in power system Planning in India.

VI TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Transmission Line Parameters	10	2	2	2	2	2	0	10
II	Characteristic and Performance of Transmission Line	10	2	2	2	2	2	0	10
III	Design of Transmission system	8	2	2	3	3	2	0	12
IV	Design of Power System and Substation	6	2	3	3	3	2	2	15
V	Design of Distribution system	8	2	3	3	3	2	2	15
VI	Power System Planning	3	2	2	2	0	0	2	08
Total		45	12	14	15	13	10	06	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

VII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- '**L**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.

- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

VIII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Library survey regarding power system planning.
- Prepare 15 min. power point presentation on different topics of syllabus.

IX COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Transmissi on Line Parameters	1-2-3	1a. understanding Basics of Electric field. 1b. study and calculation of different parameter. 1c. calculation of inductance and capacitances in various cases.	Lecture with media, Tutorial	PPTs, black board
Unit– II Characteri stic and Performan ce of Transmissi on Line	4-5-6	2a. Learning performance of short, medium and long transmission line. 2b. Calculate voltage regulation and efficiency of system	Lecture With media Demonstra tion in software Tutorial	PPTs, black board
Unit– III Design of Transmissi on system	7-8	3a. Understand Requirement of Transmission lines. 3b. electrical design of EHV transmission line.	Lecture with media, Tutorial	PPTs, black board
Unit-IV Design of Power System and Substation	9-10	4a. Selection of size, location and different specification generating station, sub-station and transmission line. 4b. Basic of Earthing & grounding with details of step and touch potential.	Lecture with media, Case study	PPTs, black board
Unit – V Design of Distributio n system	11-12	5a. Connection scheme of different distribution system 5b. AC & DC distribution study and calculation	Lecture with media, + PPT + Tutorial	PPTs, black board
Unit – VI Power System Planning	13	6a. Methods of generation, transmission and distribution system planning.	Lecture with media, + PPT	PPTs, black board

X SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
4	Electrical Power System Design	M. V. Deshpande	Tata Mcgraw Hill
5	Principles of Power system	V.K.Mehta	S.Chand & Company Ltd.
6	A Course in electrical power	Soni & Bhattnagar	Dhanpat Rai & Sons.
7	Sub-station Design and Equipment	P.V. Gupta and P.S.Satnam,	Dhanpat Rai & Sons.
8	Handbook of Electrical Power Distribution	Ramamurthy University	Press (I) Pvt. Ltd
9	Power System Analysis and Design	B.R. Gupta	S.Chand Publications
10	Power system analysis	Hadi Saadat,	Tata McGraw Hill Publishing Company, New Delh
11	Modern Power System Analysis	D.P. Kothari & I. J. Nagrath,	Tata Mcgraw Hill
12	Power System Analysis	Grainger & Stevenson,	Tata Mcgraw Hill

XI SOFTWARE/LEARNING WEBSITES

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPUR/powersystem/ui/TOC.htm>
2. http://en.wikipedia.org/wiki/Electric_power_transmission

XII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronic s, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE351	Semester VI														

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE351	Competency Design, Operate and Troubleshoot various transmission and distribution system.	3	3	3	2	3	2	2	-	2	2	3	1	3	1
EE351.1	<u>Identify</u> and Calculate the different parameters of transmission line.	3	3	-	-	-	-	-	-	-	-	-	-	2	1
EE351.2	<u>Analyze</u> & calculate the performance of various types of transmission & distribution systems.	2	2	3	-	3	-	-	-	3	3	-	-	3	3
EE351.3	<u>Chose</u> the size and location of generating station, sub-station. with bus-bar arrangement.	-	-	3	3	3	3	-	2	3	2	-	-	3	3
EE351.4	<u>Design</u> and Connection schemes of A.C. & D.C. distribution system.	-	3	3	3	-	-	-	1	-	-	-	-	3	3
EE351.5	<u>Acquire</u> the knowledge of Power system planning for Generation, Transmission & Distribution System.	-	-	-	2	3	3	2	-	3	3	3	3	3	3

XIII NAMES OF RESOURCE PERSONS

c) Vibha Parmar

d)

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

The DC machines & synchronous machines are used power generation plants. The course is developed such that student would get the knowledge from basics to control schemes of these machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the performance of various control methods used in field.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 10) Use the control technique for electrical machines.
- 11) Evaluate the performance of the synchronous and DC machines
- 12) Interpret the suitable control technique for synchronous generator.
- 13) Evaluate the performance of the DC and synchronous generator.
- 14) Choose the type of turbine for synchronous generator.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	200
4	0	4	6	70	30*	50	50	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
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S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	To Study operating principle, working and cut model of DC machine.	04
2	2	To perform the magnetization (open circuit) characteristics and load test on a DC shunt generator to obtain the internal and external load characteristics.	04
3	3	To Perform the load test on a D.C. series generator to obtain the internal and external load characteristics.	04
4	3	To perform the speed control of DC series and DC shunt motor.	06
5	3	To find out the voltage regulation of three phase alternator by direct load test.	04
6	4	To find out voltage regulation of three phase alternator by Synchronous Impedance Method (EMF Method).	04
7	4	To find out voltage regulation of three phase alternator by Ampere Turn Method (MMF Method).	04
8	4	To find out voltage regulation of three phase alternator by ZPFC Method	06
9	4	To Perform slip test on salient pole alternator for determining direct axis synchronous reactance- X_d and quadrature axis synchronous reactance - X_q .	04
10	5	To perform parallel operation of two alternators and study method of synchronization of two alternators. .	06
11	5	To Obtain V Curve & Inverted V Curve of a Synchronous Motor.	04
12	5	To obtain constant excitation circle of synchronous motor.	04
13	1	TO study DC and AC machine windings.	06
Total			60

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	DC series/shunt/compound machine with prime-mover/load	1,2,3,4
2	Synchronous machine with prime-mover/load	5,6,7,8,9,10,11,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Fundamentals of DC Machines	CO 1. Able to recognize the electric and magnetic circuits of AC and DC machines. CO 2. Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines. PrO 14. To Study operating principle, working and cut model of DC machine. PrO 13. To study DC and AC machine windings 1k. <u>Justify</u> the concept of rotating machines and fundamental component roles.	1.6 Classification of rotating electrical machines 1.7 Construction of DC machines, Comparison of AC and DC Generators 1.8 Principle of DC generator and motor, Comparison of motor and generator action, Elementary DC generator 1.9 essential parts, Types of DC machines 1.10 EMF equation, Voltage build up process, Causes of its failure 1.11 Critical resistance and speed, Armature reaction and its effect 1.12 Commutation in DC machines, Reactance voltage 1.13 methods of improving commutation, compensating winding
Unit– II Operation and performance Characteristics of DC Machine	CO 4. Aware with the performance characteristics and applications of Synchronous and DC machines CO 6. Understand the role and importance of DC and AC machine in power generation plants PrO 2. To perform the magnetization (open circuit) characteristics and load test on a DC shunt generator to obtain the internal and external load characteristics. PrO3. To Perform the load test on a D.C. series generator to obtain the internal and external load characteristics. PrO 4. To perform the speed control of DC series and DC shunt motor. 2f. <u>Evaluate</u> the performance of DC generator 2g. <u>Evaluate</u> the performance of DC motor 2h. <u>Justify</u> the output characteristics of DC machines 2i. <u>Setup</u> the circuit to determine output characteristics of DC machines.	2.5 Characteristics of separately excited DC machines (motor and generator), 2.6 Characteristics of shunt, series and compound machines (generators & motors) 2.7 Power flow diagram of DC machines 2.8 Losses in DC machines, Efficiency , condition for maximum efficiency 2.9 Voltage regulation in DC generator 2.10 Speed regulation in DC motor 2.11 Parallel operation of DC machines, Examples 2.12 Necessity of starter, types of starter 2.13 Speed control of DC shunt and series motors PV-Grid

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
		connected system design 2.14 Braking of DC motors, Examples 2.15 Applications of all types of DC generators and DC motors.
Unit– III Construction and Operation of Synchronous Generator	CO 2. Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines. 3f. <u>Determine</u> the output from synchronous generator. 3g. <u>Obtain</u> open circuit and short circuit characteristics of synchronous generator.	3.5 Introduction, classification of synchronous machine 3.6 details of construction, advantages of rotating field type of construction 3.7 difference between salient and cylindrical alternator 3.8 damper winding, operating principle, Speed and Frequency, Synchronous speed 3.9 pitch factor, distribution factor and emf equation, emf waveform for concentrated and distributed winding, Examples 3.10 armature leakage reactance, synchronous reactance and impedance 3.11 armature reaction 3.12 Equivalent circuit, phasor diagram 3.13 Synchronous generator operating alone, operation of alternator on no load and on load 3.14 effect of variation in load on synchronous generator, Laboratory methods of determination of synchronous reactance 3.15 oc/sc test, determination of effective resistance of armature, Short circuit ratio 3.16 Losses and efficiency, power flow diagram, Examples,
Unit-IV Voltage Regulation, Power Equations and Synchronization of Alternators	CO 4. Aware with the performance characteristics and applications of Synchronous and DC machines. CO 5. Create and test the synchronization of alternators to grid <i>PrO 5.</i> To find out the voltage regulation of three phase alternator by direct load test. <i>PrO 6.</i> To find out voltage regulation of three phase	4.5 Voltage Regulation, methods of calculating voltage regulation, Direct load test 4.6 EMF method 4.7 MMF method 4.8 ZPFC method

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	alternator by Synchronous Impedance Method (EMF Method). PrO 7. To find out voltage regulation of three phase alternator by Ampere Turn Method (MMF Method). PrO 8. To find out voltage regulation of three phase alternator by ZPFC Method PrO 9. To Perform slip test on salient pole alternator for determining direct axis synchronous reactance-X_d and quadrature axis synchronous reactance – X_q PrO 10. To perform parallel operation of two alternators and study method of synchronization of two alternators 4d. <u>Choose</u> the relevant operating parameters for load operation of alternator. 4e. <u>Select</u> the relevant control method for <u>the specified</u> application. 4f. <u>Evaluate</u> the performance of <u>alternator</u> for <u>the specified</u> condition.	4.9 Comparison of methods, examples 4.10 Active, Reactive and Complex Input and Output Power Equations for cylindrical synchronous generator 4.11 Maximum input and output power and torque developed, Power Angle Characteristics 4.12 Two reactance theory for salient pole synchronous machine, Phasor Diagram, Power Equations and Torque Angle Characteristics 4.13 Determination of X_d and X_q by slip test 4.14 Capability Curves for Synchronous Generators, Prime mover characteristics 4.15 Frequency Power and Voltage Reactive Power relations (F-P and Q-V), Power shared by two alternators 4.16 Necessity of parallel operation, condition required for parallel operation, the general procedure for synchronization of alternators 4.17 Infinite Bus, Synchronizing current, power, torque, Time period of Oscillations, effect of increasing the driving torque of one the alternator 4.18 effect of change in excitation of one of the alternator, load sharing between two alternators 4.19 operation of synchronous generator with infinite bus bar, operation of synchronous generator with other generator 4.20 various types of alternators based on prime movers, Factors affecting the size and efficiency of alternators
Unit – V Synchronous Motor	CO 3. Acquire the practical exposure of rotating electrical machines. PrO 11. To Obtain V Curve & Inverted V Curve of a Synchronous Motor. PrO 12. To obtain constant excitation circle of synchronous motor.	5.6 Introduction, construction, principle of operation, Equivalent circuit and phasor diagram for cylindrical and salient rotor 5.7 synchronous motor with

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	5f. <u>Interpret</u> the machine data with respect to the <u>given</u> power system. 5g. <u>Determine</u> machine charging control strategy. 5h. <u>Select</u> the method for machine control.	different excitation, starting methods of synchronous motor 5.8 power flow equations for synchronous motor, examples 5.9 steady state synchronous motor operation - effect of load change on synchronous motor 5.10 effect of field current change on synchronous motor, power flow diagram, examples 5.11 Different torques of synchronous motor, stability and maximum load angle 5.12 construction of V curves and inverted V curves, O curves 5.13 Synchronous condenser, synchronous phase modifier 5.14 Hunting 5.15 speed control of synchronous motor, merits, demerits and application 5.16 Comparison of three phase synchronous and induction motors

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Fundamentals of DC Machines	12	2	4	2	4	0	0	12
II	Operation and performance Characteristics of DC Machine	12	2	2	4	4	2	0	14
III	Construction and Operation of Synchronous Generator	12	0	2	4	4	2	0	12
IV	Voltage Regulation, Power Equations and Synchronization of Alternators	14	0	4	4	4	6	0	18
V	Synchronous Motor	10	2	2	4	4	2	0	14
Total		60	6	14	18	20	12	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **OC/SC test on alternator to draw OCC and SCC.**
- 2) **Calculation of synchronous impedance of synchronous machine.**
- 3) **Create setup of regenerative braking of DC motor.**
- 4) **Create a setup for load condition of synchronous motor for various power factor.**

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different methods of voltage regulation.
- b) Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
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Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Fundamentals of DC Machines	1	PrO1. To Study operating principle, working and cut model of DC machine. <u>1a. Justify</u> the concept of rotating machines and fundamental component roles.	Lecture with media	PPTs, black board
	2	PrO 13. To study DC and AC machine windings	Lecture With media	PPTs, black board
Unit II- Operation and performance Characteristics of DC Machine	3	<i>PrO 2. To perform the magnetization (open circuit) characteristics and load test on a DC shunt generator to obtain the internal and external load characteristics.</i> <u>2a. Evaluate</u> the performance of DC generator <u>2c. Justify</u> the output characteristics of DC machines <u>2d. Setup</u> the circuit to determine output characteristics of DC machines.	Lecture With media	PPTs, black board
	4	<i>PrO3. To Perform the load test on a D.C. series generator to obtain the internal and external load characteristics.</i> <u>2b. Evaluate</u> the performance of DC motor <u>2c. Justify</u> the output characteristics of DC machines <u>2d. Setup</u> the circuit to determine output characteristics of DC machines.	Lecture With media	PPTs, black board
	5	<i>PrO 4. To perform the speed control of DC series and DC shunt motor.</i> <u>2c. Justify</u> the output characteristics of DC machines <u>2d. Setup</u> the circuit to determine output characteristics of DC machines.	Lecture With media	PPTs, black board
Unit– IV - Voltage Regulation, Power Equations and Synchronization of Alternators	6	<i>PrO 5. To find out the voltage regulation of three phase alternator by direct load test.</i> <u>4a. Choose</u> the relevant operating parameters for load operation of alternator.	Lecture With media	PPTs, black board
	7	<i>PrO 6. To find out voltage regulation of three phase alternator by Synchronous Impedance Method (EMF Method)</i> <u>4c. Evaluate</u> the performance of <u>alternator</u> for the <u>specified</u> condition.	Lecture With media	PPTs, black board
	8	<i>PrO 7. To find out voltage regulation of three phase alternator by Ampere Turn Method (MMF Method).</i> <u>4c. Evaluate</u> the performance of <u>alternator</u> for the <u>specified</u> condition.	Lecture With media	PPTs, black board
	9	<i>PrO 8. To find out voltage regulation of three phase alternator by ZPFC Method</i> <u>4c. Evaluate</u> the performance of <u>alternator</u> for the <u>specified</u> condition.	Lecture With media	PPTs, black board
	11	<i>PrO 9. To Perform slip test on salient pole alternator for determining direct axis synchronous reactance-</i>	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit V- Synchronous Motor		Xd and quadrature axis synchronous reactance – Xq 4b. <u>Select</u> the relevant control method for <u>the specified</u> application.		
	12	PrO 10. To perform parallel operation of two alternators and study method of synchronization of two alternators 4b. <u>Select</u> the relevant control method for <u>the specified</u> application.	Lecture With media	PPTs, black board
	13	PrO 11. To Obtain V Curve & Inverted V Curve of a Synchronous Motor. 5a. <u>Select</u> the method for machine control.	Lecture With media	PPTs, black board
	14	PrO 12. To obtain constant excitation circle of synchronous motor. 5b. <u>Interpret</u> the machine data with respect to <u>the given</u> power system.	Lecture With media	PPTs, black board
	15	5c. <u>Determine</u> machine charging control strategy. <u>Select (self)</u>	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Theory and performance of electrical machines	J.B. Gupta	S.K. Kataria & Sons, ISBN: 978-93-5014-277-6
2	A textbook of electrical technology VOL II	B.L. Theraja	S. Chand, ISBN: 9788121924375
3	Electric Machinery Fundamentals	Stephen. J. Chapman	McGraw Hill, ISBN: 10. 978007352954

XV SOFTWARE/LEARNING WEBSITES

- <http://nptel.ac.in/courses/108106072/>
(video lectures Prof:D. Kastha, IIT Kharagpur)

XVI PO-COMPETENCY-CO MAPPING

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Cour se Code	Competency and COs	PO 1 Used of math emati cal meth od	PO 2 Analy sis mech anical syste m	PO 3 Used engin eering soft.	PO 4 Solve probl em in machi ne	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Enviro nmen t and sustai nabilit y	PO 8 Ethics	PO 9 Individual and team work:	PO 10 Communi cation	PO11 Project managem ent	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE34 2	Semester V														
CO1	Able to recognize the electric and magnetic circuits of AC and DC machines.	3				1									
CO2	Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines.	2	2	1	1	1			1						
CO3	Acquire the practical exposure of rotating electrical machines.	3	2	3	2	1						2	2	1	
CO4	Aware with the performance characteristics and applications of Synchronous and DC machines.		3	2	2					1					
CO5	Create and test the synchronization of alternators to grid.	2	3	2	3	2							1		
CO6	Understand the role and importance of DC and AC machine in power generation plants.														1

XVII NAMES OF RESOURCE PERSONS

- Dr. Pratik Mochi
-

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

Electrical engineers have to deal with various types of *semiconductor devices*, power electronics circuits and converters are used for different dc drives. This curriculum course structure is developed such that the student is able to evaluate the performance of the various *semiconductor devices*, types of power electronic converters and drives for different industry drive applications.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the performance of various **SEMICONDUCTOR DEVICES, TYPES OF Controlled rectifier, Chopper and DC DRIVES**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. Evaluate the performance of the various types of SEMICONDUCTOR DEVICES and its use for different converters.
2. Evaluate the performance of the various types the Triggering, Commutation and Driver Circuit
3. Evaluate the performance of the various types of Chopper for different application
4. Evaluate the performance of the various types of Controlled Rectifier for different application
5. Evaluate the performance of the various types of DC Drives for different application

1 TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

2 SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of**

Dave's Psychomotor Domain taxonomy by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1,2	(A)To study the DC Gate control characteristics of a SCR	02
	1,2	(B)To study and plot the Anode Current Characteristics of a SCR.	02
2	3	A)To measure the holding current of a SCR.	02
	1,2	B)To Study the phase firing characteristics of a SCR.	02
3	3	To Study the V-I characteristics of a DIAC.	02
4	1,2	To Study application of DIAC as Sawtooth Wave Generator and Pulse Train Generator.	02
5	1,2	To Study the operation of UJT as Relaxation Oscillator.	02
6	1,2	To Study the single-phase Fully Controlled Bridge Rectifier.	02
7	1,2	A)To study the gate characteristics of a TRIAC.	02
	3,4	B)To study the Terminal characteristics of a TRIAC.	02
	1,2	C)To measure the Holding current of a Triac.	02
8	3	A)To study the application of a Triac as a Static Switch with DC control.	02
	1,2	B)To study the application of a Triac in control of AC with AC signal.	02
9	3	To Study the step down voltage commutated thyristorised chopper.	02
10	3	To Study and simulate the DC Drive in MATLAB	02
Total			30

3 MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	6
2	SCR characteristic kit with DC and AC gate control	1 (A), (B) and 2 (A), (B)
3	DIAC VI characteristic kit	3
4	DIAC as Sawtooth Wave Generator and Pulse Train Generator KIT	4
5	UJT as Relaxation Oscillator Circuit kit	5
6	single-phase Fully Controlled Bridge Rectifier kit with variac	6
7	TRIAC VI characteristic kit	7 (A), (B) , (C) and 8 (A), (B)
8	Step down voltage commutated thyristorised chopper KIT	9
9	DSO with isolation and its probes	4,5,6
10	Multimeter	1,2,3,7,8

4 AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.

5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- 'Responding Level' in 1st year
- 'Valuing Level' in 2nd year
- 'Organising Level' in 3rd year
- 'Characterising Level' in 4th year.

5 UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction	CO 1. Evaluate the performance of the various types of SEMICONDUCTOR DEVICES and its use for different converters. 1l. <u>Justify</u> the need of SEMICONDUCTOR DEVICES 1m. <u>Justify</u> the need of different converters	1.1 Introduction, 1.2 Scope and applications, Classification of power electronic converters
Unit– II Power Semiconductor Devices	CO 1. Evaluate the performance of the various types of SEMICONDUCTOR DEVICES and its use for different converters. <i>PRO 1. (A)To study the DC Gate control characteristics of a SCR (B)To study and plot the Anode Current Characteristics of a SCR.</i> <i>PRO 2. A)To measure the holding current of a SCR. B)To Study the phase firing characteristics of a SCR.</i> <i>PRO 3. To Study the V-I characteristics of a DIAC.</i> <i>PRO 7. A) To study the gate characteristics of a TRIAC. B) To study the Terminal characteristics of a TRIAC.</i> <i>C)To measure the Holding current of a Triac.</i> 2a. Evaluate the VI characteristics of SCR 2b. Evaluate the VI characteristics of DIAC 2c. Evaluate the VI characteristics of TRIAC	2.16 Diode: Classification and its application 2.17 SCR: VI characteristics, Switching characteristics, Gate characteristics, Ratings, Effect of dv/dt and di/dt, Snubber circuit and its design 2.18 Thyristor Family: Triac, DIAC, LASCR, SUS, SCS 2.19 Gate commutated devices: GTO, MOSFET and IGBT

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit– III Triggering, Commutation and Driver Circuit	CO 2. Evaluate the performance of the various types the Triggering, Commutation and Driver Circuit PrO 4. To Study application of DIAC as Sawtooth Wave Generator and Pulse Train Generator. PrO 5. To Study the operation of UJT as Relaxation Oscillator. 2a. Evaluate the application of DIAC as Sawtooth Wave Generator and Pulse Train Generator. 2b. Evaluate the operation of UJT as Relaxation Oscillator.	2.20 Triggering circuits: Resistance triggering circuit, Resistance capacitance trigger circuit, DIAC trigger circuit, UJT based trigger circuit; Micro-controller based Trigger circuit 2.21 Commutation circuits: Line commutation, Load commutation, Forced commutation, External pulse commutation 2.22 Driver Circuit: Optocoupler and pulse transformer based SCR driver circuit, Gate drive circuit for power MOSFET, Driver circuit for IGBT and BJT
Unit– IV DC-DC Converters (Choppers)	CO 3 Evaluate the performance of the various types of Chopper for different application <i>PrO 9. To Study the step down voltage commutated thyristorised chopper.</i> <i>3a. Evaluate the performance of step down voltage commutated thyristorised chopper</i>	4.1 Introduction, Principle of chopper operation, 4.2 Control strategies, 4.3 Step down (buck) converter, 4.4 Step up(boost) converter 4.5 Step up down converter, 4.6 Cuk DCDC converter, Chopper configuration
Unit-V AC-DC Converters (Controlled Rectifier)	CO 4. Evaluate the performance of the various types of Controlled Rectifier for different application <i>PrO 6. To Study the single-phase Fully Controlled Bridge Rectifier</i> <i>4a. Evaluate the performance of single-phase Fully Controlled Bridge Rectifier</i>	4.21 Introduction, Principle of phase control, 4.22 Single phase half wave converter with different loads, 4.23 Single phase full wave converter with different loads, 4.24 Single phase semi converter, 4.25 Three phase half wave converter with different loads, 4.26 Three phase full converter, 4.27 Three phase semi converter with different loads, 4.28 Dual converter, 4.29 Multi-pulse Rectifier
Unit – VI DC Motor Drives	CO 5. Evaluate the performance of the various types of DC Drives for different application <i>PrO 10. To Study and simulate the DC Drive in MATLAB</i> 5i. Understand the operation of DC Drive in different mode condition. 5j. Justify the need of <i>DC Drives for different application</i>	5.17 Concept of Electric Drive, 5.18 Basic DC Machine Equations and its characteristics, 5.19 Control of DC separately excited motor from single phase and three phase controlled rectifier, 5.20 Rectifier control of DC

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
		series motor, 5.21 Chopper control of separately excited DC motor, 5.22 Chopper control of DC series motor, 5.23 Multi quadrant operation of, 5.24 DC separately excited motor

1 TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	1.5	0	1	1	2	1	0	05
II	Power Semiconductor Devices	13	0	1	1	6	4	0	12
III	Triggering, Commutation and Driver Circuit	7.5	0	1	4	4	1	0	10
IV	DC-DC Converters (Choppers)	7.5	0	2	2	4	6	0	14
V	AC-DC Converters (Controlled Rectifier)	7.5	1	2	2	4	6	0	15
VI	DC Motor Drives	08	0	2	2	6	4	0	14
Total		45	1	9	12	26	22	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

2 MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Rectifier:** Simulate the rectifier for particular one application.
- 2) **Chopper:** Design and Develop the hardware of Chopper.
- 3) **Triggering circuit:** Design and simulate the SCT triggering circuit
- 4) **DC drive:** Design the simulation of DC drives in MATLAB.

3 SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

4 SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different Semiconductor switches used in converters.
- b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

5 COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introducti on	1	<i>1a. Justify the need of SEMICONDUCTOR DEVICES</i> <i>1b. Justify the need of different converters</i>	Lecture with media	PPTs, black board, MATLAB Software
Unit– II Power Semicon ductor Devices	6	<i>PrO 1. (A)To study the DC Gate control characteristics of a SCR</i> <i>(B)To study and plot the Anode Current Characteristics of a SCR.</i> <i>PrO 2. A)To measure the holding current of a SCR.</i> <i>B)To Study the phase firing characteristics of a SCR.</i> <i>PrO 3. To Study the V-I characteristics of a DIAC.</i> <i>PrO 7. A) To study the gate characteristics of a TRIAC.</i> <i>B) To study the Terminal characteristics of a TRIAC.</i> <i>C)To measure the Holding current of a Triac.</i> <i>2a. Evaluate the VI characteristics of SCR</i> <i>2b. Evaluate the VI characteristics of DIAC</i>	Lecture with media, and Demonstra tion	PPTs, black board, MATLAB Software, Practical demonstra tion

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		2c. Evaluate the VI characteristics of TRIAC		
Unit– III Triggering, Commutation and Driver Circuit	8	<u>PrO 4. To Study application of DIAC as Sawtooth Wave Generator and Pulse Train Generator.</u> <u>PrO 5. To Study the operation of UJT as Relaxation Oscillator.</u> 2a. Evaluate the application of DIAC as Sawtooth Wave Generator and Pulse Train Generator. 2b. Evaluate the operation of UJT as Relaxation Oscillator.	Lecture with media, Field visit	PPTs, black board, MATLAB Software, Practical demonstration
Unit– IV DC-DC Converters (Choppers)	10	<u>PrO 9. To Study the step down voltage commutated thyristorised chopper.</u> 4e. 3a. Evaluate the performance of step down voltage commutated thyristorised chopper	Lecture with media, Tutorial	PPTs, black board, Practical demonstration
Unit-V AC-DC Converters (Controlled Rectifier)	15	<u>PrO 6. To Study the single-phase Fully Controlled Bridge Rectifier</u> 4a. Evaluate the performance of single-phase Fully Controlled Bridge Rectifier	Lecture with media, Tutorial	PPTs, black board, MATLAB Software, Practical demonstration
Unit – VI DC Motor Drives		<u>PrO 10. To Study and simulate the DC Drive in MATLAB</u> 5a. Understand the operation of DC Drive in different mode condition. 5b. Justify the need of DC Drives for different application	Lecture with media, Tutorial	PPTs, black board, MATLAB Software

XXII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
4	Power Electronics	Dr.P.S.Bimbhra	Khanna publishers
5	Power electronics	M D Singh and K B K Khanchandani,	Tata Mcgraw-Hill Publication
6	Power electronics handbook	by M.H. Rashid	Academic press
7	High power converters and AC drives	IEEE press	IEEE press
8	Power Electronics	Mohan, Undeland,	Wiley publication

S. No.	Title of Book	Author	Publication including ISBN
	Converters, Applications and design	Robbins	

XXIII SOFTWARE/LEARNING WEBSITES

1. <http://www.nptel.ac.in/courses/108105066/>
(Notes from IIT Kharagpur)
2. <http://www.nptel.ac.in/courses/108101038/1>
(Videos and Notes by Prof. B.G. Fernandes, IIT Bombay)
3. <http://www.nptel.ac.in/courses/108108035/1>
4. <http://www.nptel.ac.in/courses/108108036/>
(Videos and Notes by Prof. L. Umanand, IIT Bangalore)
5. <http://www.nptel.ac.in/courses/108102046/>
6. <http://www.nptel.ac.in/courses/108108077/>
(Videos and Notes by Prof. K. Gopakumar, IISC, Bangalore)

XXIV PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE353	Semester VII														
353	Competency • Evaluate the performance of various SEMICONDUCTOR DEVICES, TYPES OF Controlled rectifier, Chopper and DC DRIVES	3	2	2	2	1	-	1	-	-	-	-	1	3	-
353.1	Evaluate the performance of the various types of SEMICONDUCTOR DEVICES and its use for different converters.	3	2	1	-	2	-	-	-	-	1	-	-	-	-

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
353.2	Evaluate the performance of the various types the Triggering, Commutation and Driver Circuit	3	3	2	1	2	-	-	-	-	-	-	-	2	-
353.3	Evaluate the performance of the various types of Chopper for different application	3	3	3	1	2	-	-	-	-	-	-	-	2	-
353.4	Evaluate the performance of the various types of Controlled Rectifier for different application	3	1	3	1	2	-	-	-	-	-	-	-	2	-
353.5	Evaluate the performance of the various types of DC Drives for different application	3	1	-	-	-	-	-	-	-	-	-	-	-	-

XXV NAMES OF RESOURCE PERSONS

a) Dr. Dharmesh Dabhi

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

This course enables the students to apply some of the knowledge and/or skills developed during the programme to new problem for which there are number of engineering solutions. This course include planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, , use, monitor and control resources optimally and economically. By studying this course abilities like creativity, imitativeness and performance qualities are also developed in students. Leadership development and supervision skills are also integrated objectives of learning this course.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Find the solution of a real-life problem.**

III COURSE OUTCOMES (COs)

Depending upon the nature of the projects undertaken, the following could be some of the minor course outcomes that could be attained, although, in case of some projects few of the following course outcomes may not be applicable.

Apply knowledge and skills learned from project to real-world problems.

Solve engineering problems by the team work and work with other discipline.

Use experience related to professional and ethical issues in the work environment.

Apply impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.

Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
0	0	04	02	00	00	50	00	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

V COURSE DETAILS

- Minor Project showcases that, a student during the fifth semester will able to apply the engineering knowledge to a practical problem.
- Students can carry out their project work in-house at institute/ industry/ research organization/govt. organization.
- Project should be related to the Product/prototype design and development or Experimentation, Simulations and evaluation of any system etc.

- iv. A student is required to carry out the project work related to Electrical Engineering or some interdisciplinary work under the guidance of a faculty member and the supervisor of the concerned industry/institute/organization.
- v. The minor project work provides student an opportunity to develop something on their own or group of maximum 3 students. Final decision related to the number of students in one group is based on internal supervisor recommendation.
- vi. Project will be evaluated at least twice during the semester by internal faculty as a part of continuous evaluation and final evaluation will be carried out by external examiner at the end of semester.
- vii. Student should submit the project report at the end of the semester duly signed by project supervisor and head of department. Report comprising literature review, objective, methodology, scope of the project work undertaken, outcome of project and conclusions.
- viii. The students are required to identify their problem and they are required to follow all the rules and instruction issued by department/industry.

VI PROJECT REPORT

At the end of the semester student should prepare project report. Guideline of the Project report is as follow

- a) Certificate
- b) Acknowledgements
- c) Abstract (in one paragraph not more than 150 words)
- d) Content Page
- e) Chapter-1 Introduction and background of the Problem/Task
- f) Chapter-2 Literature Survey for Problem/Task Identification and Definition,
- g) Chapter-3 Proposed Detailed Methodology of solving the problem/completing the task
- h) Chapter-4 Methodology actually followed to solve the problem/complete the task
- i) Chapter-5 The solution, its development and its implementation/execution (with details including drawings, software, used for or developed for the solution etc. in detail). In case of survey/experiment type of project this chapter will include data analysis, information presentation and analysis.
- j) Chapter-6 Final conclusions and recommendations if any for future projects.

Note:

- a) The report should contain as many diagrams and figures, charts etc. as relevant for the project.
- b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VII ASSESSMENT OF PROJECT WORK

Assessment of Project work also has two components, first is Progressive Assessment, while another is End of the Term Assessment.

i. Progressive Assessment Criteria

- a) Project guide or the group of internal faculty are supposed to carry out this assessment. It is a continuous process, during which for developing desired qualities in the students, faculty should orally give informal feedback to students about their performance and interpersonal behaviour while guiding them on their project work every week.

S. No.	Criteria	Marks
First Progressive Assessment at the end of 4th week		
1	Project Proposal for work to be completed at the end of semester: Including problem definition and methodology adapted, literature review adopted (in case of working	10

S. No.	Criteria	Marks
	models prepared then quality of workman ship, appropriateness of raw materials/components/ technology being used, functioning of the prototype etc. has to be also assessed)	
	General Behaviour (initiative, resourcefulness, reasoning ability, imagination/creativity, self-reliance)	Only written feed-back/suggestions
Second Progressive Assessment at the end of 12th week		
2	Execution of Plan during semester (marks to be also given based on ability to collect relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also assessed	5
3	Log book (for work done during semester- detailed and regular entry would be basis of marks)	5
4	General Behaviour (persistence, interest, confidence, problem solving ability, decision making ability, initiative to act, team spirit, sharing of material etc., participation in discussions, completion of individual responsibilities, leadership)	Only written feed-back./suggestions
Third Progressive Assessment at the end of 14th week		
5	Portfolio for Self-learning and reflection (for work done in semester- marks based on amount of reflection and completion of the portfolio)	10
6	Project Report including documentation - for work done in semester (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	5
7	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation for work done during semester)	5
8	Defence (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce for work done during semester)	10
Total		50

Note: Oral feedback on general behaviour may also be given whenever relevant/ required during day to day guidance and supervision.

ii. END SEMESTER EXAMINATION

b) Criteria of Marks for ESE for Capstone Project Execution.

S. No.	Description	Marks
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S. No.	Description	Marks
1	Problem Identification/Project Title (Innovation /Utility of the Project for industry/ User/Academia) marks to be also given based on (i) Accuracy or specificity of the scope and (ii) Appropriateness of the work with reference to desired course outcomes.	5
2	Industrial Survey and Literature Review (marks to be given based on extent/volume and quality of the survey of Industry / Society / Institutes/Literature/Internet for Problem Identification and possible solutions)	5
3	Project Proposal (In case of working models, detailed design and planning of fabrication/assembly of the prototype has to be also assessed). Marks to be given also based on appropriateness, flexibility, detail and clarity in methods/planning.	10
4	Final Report including documentation. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	10
5	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	10
6	Defence (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	10
Total		50

VIII Guidelines for Portfolio Preparation and assessment

The main purpose of portfolio preparation is learning based on self-assessment and portfolio is not to be used for assessment in traditional sense.

- Each student has to prepare his/her portfolio separately. However, he/she can discuss with the group members about certain issues on which he/she wants to write in the portfolio.
- Whatever is written inside the portfolio is never to be used for assessment, because if teachers start giving marks based on whatever is written in the portfolio, then students would hesitate in true self-assessment and would not openly describe their own mistakes or shortcomings.
- Some marks are allocated for portfolio, these marks are to be given based on how sincerely portfolio has been prepared and not based on what strengths and weaknesses of the students are mentioned in the portfolio.
- Portfolio has to be returned back to the students after assessing it (assessment is only to see that whether portfolio is completed properly or not) by teachers. Because student is the real owner of the portfolio.
- Students mainly learn during portfolio preparation, but they can further learn if they read it after a gap. And hence they are supposed to keep the portfolios with them even after completion of the diploma because it is record of their own experiences (it is like diary some people write about their personal experiences), because they can read it again after some time and can revise their learning (about their own qualities)

IX PO-COMPETENCY-CO MAPPING

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Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE344	Semester V														
344	Competency Find the solution of a real-life problem.	3	2	2	2	2	2	1	1	2	2	2	1	2	2
344.1	Apply knowledge and skills learned from project to real-world problems.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
344.2	Solve engineering problems by the team work and work with other discipline.	3	3	2	2	1	2	-	-	3	2	1	2	2	-
344.3	Use experience related to professional and ethical issues in the work environment.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
344.4	Apply impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.	3	-	-	3	3	3	-	-	2	1	3	2	3	3
344.5	Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.	3	1	3	2	3	3	2	-	-	2	2	3	3	2

X NAMES OF RESOURCE PERSONS

a) Prof. Mahammadsoaib Saiyad

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

Objective of this course is to understand the electrical and electronics products and its market statistics. Students get aware about the product, Manufacturing method, consumer requirement and perception, product satisfaction, consumer experience (good and bad), and intent to purchase behavior.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Specification, market survey, identification of different products like wire, cable, and switchgear can be done.
- To understand various components and its behavior.
- To make them industry ready.
- To get familiar with modern tools and technologies

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. *To apply knowledge and skills learned from project to real-world engineering problems.*
2. *To function in a team work with teammates from other disciplines.*
3. *To use experience related to professional and ethical issues in the work environment.*
4. *To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context.*
5. *To demonstrates knowledge of contemporary issues related with engineering in general with the use new tools and technologies.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
0	0	2	2	00	00	50	50	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The practical outcomes (PrOs) , which are created for particular electrical product, predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of PrOs to the ***relevant level of Dave's Psychomotor Domain taxonomy*** by the students through the ***practicum*** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Electrical Product selected by student group	-

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to ***Krathwohl's 'Affective Domain Taxonomy'*** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '***Apply***' and above level sample ***Cognitive Domain*** learning outcomes denoted by unit outcomes (UOs) related to ***Revised Bloom's Taxonomy*** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Note: The course evaluation is based on the presentation of students. No theory exam is involved. However, following instructions are given to students.

- Electrical product survey has 2 credit subject and 2 hours per week.
- Students can find his/her electrical product.
- The student may take the guidance from department faculty members.
- Product must be from Electrical bag round.
- The student shall submit product presentations to the Coordinator for the evaluation of the product survey.

- After the completion of product survey, a soft copy of “presentation” must be submitted to the Coordinator by the end of the semester.
- The students must make a group of 3 or 4 and submit their proposal to departmental coordinator.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision. If the revised presentation is still unsatisfactory the student shall be requested to repeat the project

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- ‘L’ in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student’s) portfolio which will be useful for their placement interviews:

- Library survey regarding different methods of voltage regulation.
- Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

There is no theory part for which course plan is needed.

XIV SUGGESTED LEARNING RESOURCES

The user manual and product booklet of various industries.

XV SOFTWARE/LEARNING WEBSITES

- NPTEL online courses on particular electrical product might be helpful.

XVI PO-COMPETENCY-CO MAPPING

Cour se Code	Competency and COs	PO 1 Used of math ematical meth od	PO 2 Analy sis mech anical syste m	PO 3 Used engin eering soft.	PO 4 Solve probl em in machi ne	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Enviro nmen t and sustai nabilit y	PO 8 Ethics	PO 9 Individual and team work:	PO 10 Communi cation	PO11 Project managem ent	PO12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE34 5	Semester V														
CO1	To apply knowledge and skills learned from project to real-world engineering problems.	3	2											2	
CO2	To function in a team work with teammates from other disciplines	1		1	1	2				3					1
CO3	To use experience related to professional and ethical issues in the work environment	1					1		2						1
CO4	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context	1						2		1		2		1	1
CO5	To demonstrates knowledge of contemporary issues related with engineering in general with the use new tools and technologies.	3	1		1								1		2

XVII NAMES OF RESOURCE PERSONS

- a) Pratik Mochi
- b)

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

This course offers the opportunity for the young students to acquire on job the skills, knowledge, attitudes, and perceptions along with the experience needed to constitute a professional identity. This course includes planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, use, monitor and control resources optimally and economically. By studying this course ability like creativity, imitateness and performance qualities are also developed in students. Also give an insight into the working of the real organizations. Leadership development and supervision skills are also integrated objectives of learning this course. To appreciate the linkages among different functions and departments. This course helps the students in exploring career opportunities in their areas of interest. By attending this course student get deeper understanding in specific functional areas.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Develop professional habit, attitude and ethics related to the industrial working environment.**

III COURSE OUTCOMES (COs)

On successful completion of this course, the student will be able to:

CO1: Integrate class room theory with real-world engineering problems of industry/organization.

CO2: Develop work habits, attitudes and ethics for professional environment.

CO3: Develop communication, interpersonal and other critical skills.

CO4: Develop technical report writing and presentation skill.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)			
				Theory Marks		Practical Marks	Total Marks
L	T	P	C	ESE	PA	ESE	150
0	0	0	3	00	00	150	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

V Course Details: Instructional Method and Pedagogy

Format of Summer Internship Report:

At the start of the semester student should prepare project report. Guideline of the Project report is as follow

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- Chapter-1: Overview of company: Detail of the company including Location and spread of the company, Number of employees, engineers, technicians, administrators in the company, Divisions of the company, Administrative tree (if available), Customer profile and market share, details of process or services, various department and role of each department, detail of your department.
- Chapter-2: Overview of Internship Experience: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- Chapter 3: Observations: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization. Present your observations.
- Chapter 4: Learning Outcomes: In the last section, summarize the summer internship activities, learning outcomes, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- References: List any source you have used in the document including books, articles and web sites in a consistent format.
- Appendices: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

Note:

- The report should contain as many diagrams and figures, charts etc. as relevant for the internship.
- Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VI ASSESSMENT OF Summer Internship

Assessment of summer internship is done at the beginning of term by the internal examiner and external examiner after successful completion of summer internship

A) Assessment of summer internship by Internal Examiner.

S. No.	Description	Marks
1	Final Report including documentation. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	25
2	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	25
3	Viva (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	25
Total		75

B) Assessment of summer internship by External Examiner.

S. No.	Description	Marks
1	Final Report including documentation. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	25
2	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	25
3	Viva (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	25
Total		75

VII ASSESSMENT OF Summer Internship

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Cours e Code	Competency and COs	PO 1 Engine ering knowl edge	PO 2 Proble m analysi s	PO 3 Design /devel opme nt of solutio ns.	PO 4 Condu ct investi gation s of compl ex proble ms	PO 5 Moder n tool usage	PO 6 The engine er and society	PO 7 Enviro nment and sustai nabilit y	PO 8 Ethics	PO 9 Individ ual and teamw ork	PO 10 Comm unicati on	PO 11 Projec t manag ement and financ e	PO 12 Life- long learnin g	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE350	Semester V														
350	Competency The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:	3	2	2	2	2	2	1	1	2	2	2	1	2	2
350.1	Solve engineering problems by working in a team with across other disciplines.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
350.2	Use experience related to professional and ethical issues in the work environment	3	3	2	2	1	2	-	-	3	2	1	2	2	-
350.3	Use experience related to professional and ethical issues in the work environment.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
350.4	Examine the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context.	3	-	-	3	3	3	-	-	2	1	3	2	3	3
350.5	Evaluate contemporary issues related with engineering in general by adopting new tools and technologies	3	1	3	2	3	3	2	-	-	2	2	3	3	2

VIII NAMES OF RESOURCE PERSONS

A) Mahammadsoaib Saiyad

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS131.02A: COMMUNICATION AND SOFT SKILLS

I. Credits and Schemes

Sem	Course Code	Credits	Teaching Scheme	Evaluation Scheme				
			Contact Hours/Week	Theory		Practical		Total
				Internal	External	Internal	External	
V	HS131.02 A	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

III. Detailed Syllabus:

1.	An Introduction to Communication	06 Hours	20%
	Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		
2.	Cross-cultural Communication and Globalization	03 Hours	10%
	Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication		
3.	Communication for Career Building	10 Hours	33%
	Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose		

4.	Group Dynamics and Soft Skills	05 Hours	17%
	An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset		
5.	Effective Presentation Strategies	04 Hours	14%
	Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion		
6.	Contemporary Issues in Communication and Soft Skills	02 Hours	06%
	Trends and Practices in Communication, Case Studies		

IV. Pedagogy

Teaching will be facilitated by reading material, discussions, task-based learning, projects, assignments and interpersonal activities like group work, independent and collaborative study projects and presentations, etc.

V. Evaluation

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment / Tutorial	01	25	
2	Project	01	25	
3	Quiz	02	25	
4	Case Study and Presentation	01	25	100*
5	Attendance and Class Participation			05
Total				30

* Marks will be converted into 25

External Evaluation

The University Examination will be for 70 marks and will be based on practical (computer based) tests and a viva-voce. There may also be a Computer Based Test at the end.

Details are:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical / Viva	01	70	70
Total				70

VI. Course Outcomes (COs)

At the end of the course, the students will be able to

CO1	Gain thorough understanding and proficiency in various Professional Communication Skills.
CO2	Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
CO3	Develop business writing and presentation skills to succeed in career.
CO4	Develop soft skills to stand out and take their career to the next level.
CO5	Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	3	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

VII Preference

It will be preferred that students take globally recognized tests by Cambridge, British Council or any other equivalent agency. Equivalent marks shall be considered in the University Exam.

VIII Recommended Study Material:

❖ Text book:

1. Koneru, A. Professional Communication, Tata McGraw Hill Education Private Limited
2. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education
3. Raman, M & Singh, P. Business Communication, Oxford University Press

❖ Reference book:

1. Disanza, J.R. & Legge, N. Business and Professional Communication, Pearson Education

2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGraw Hill Education Private Limited

❖ **Web material:**

1. <https://www.coursera.org/learn/careerdevelopment>
2. <https://www.futurelearn.com/courses/writing-applications>
3. <https://www.futurelearn.com/courses/workplace-englis>

Charotar University of Science and Technology (CHARUSAT)

Faculty of Humanities (FoH)

Semester V

HS142A: French

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

Objectives of the Course:

To facilitate the learners:

- to develop proficiency of a new language - French
- to learn basic grammar and sentence construction of French language
- to learn listening, speaking, reading and writing in French language
- to acquire basic conversation skills for routine activities

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to French Language	08
2.	French for Basic Social Conversations	06
3.	Formal Conversations - I	06
4.	Formal Conversations - II	06
5.	Basic Language Functions	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to French Language	08 Hours	26.66%
	• Facts and figures of French Language • Basic French Linguistics • * Alphabets * Accents * Liaison * Nasalization • French Culture, Differ between French and English • Grammar: ➤ Subject Pronoun ➤ Verbs: (être, avoir, habiter, regarder, manger ... “er” verb) ➤ Form of address ➤ Greetings ➤ Numbers (1 to 20) ➤ Nouns and plurals of nouns ➤ The expression: C’est, Il y a • Presentation: ➤ Self-Introduction ➤ Question and answering • Dialogue		
2.	French for Basic Social Conversations	06 Hours	20%
	• Grammar: ➤ Definite articles ➤ Indefinite articles ➤ Present tense ➤ Positive Forms ➤ Negative Forms ➤ Numbers (21 to 100, 100-1000) ➤ Days and Months • Social Links: ➤ My family & relations ➤ Gathering information from someone • Dialogue		
3.	Formal Conversations - I	06 Hours	20%
	• Grammar: ➤ Common Adjectives ➤ Interrogative Forms ➤ Directions, Countries, Nationalities, Seasons ➤ Professions ➤ Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “verb) • Work, Study and Travel ➤ Job/ Profession		
	➤ Ticket Reservation (At Bus/At Railway/At Airport) • Dialogue		
4.	Formal Conversations - II	06 Hours	20%

	• Grammar: ➤ Common Prepositions ➤ Common Conjunctions ➤ Future with verb “Aller” ➤ Colors, Shapes, Animals, Vegetables, Fruits • Food & Shopping ➤ Buy a vegetables and fruits ➤ Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market) • Dialogue		
5.	Basic Functions of Language	04 Hours	13.33%
	• Grammar: ➤ Comparative Adjectives ➤ Indiquer le lieu ➤ The expression: “On” ➤ Irregular Verbs • Situational Dialogue and Conversations ➤ Expressing Likes and Dislikes ➤ Asking for permissions		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	acquire understanding of French language and culture
CO2	use French for everyday social conversations
CO3	use basic French to talk in professional settings
CO4	talk in formal situations with basic French
CO5	use French for basic language functions

Evaluation:

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks

Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

***Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.**

External / Semester End Evaluation (SEE) – 25 Marks

Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

***Note: The total External / SEE score out of 100 marks will be converted to 25 marks.**

Recommended Study Material:

- ❖ **Text book:** -
- ❖ **Reference Reading:** The reference/ reading material will be provided in consultation with the expert.

Online Course:

1. Getting started with French 1 Platform: The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-french-1/contentsection-overview?active-tab=description-tab>

2. Getting started with French 2 Platform: The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-french-2/contentsection-overview?active-tab=description-tab>

3. Getting started with French 3
Platform: The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-french-3/contentsection-overview?active-tab=description-tab>

Charotar University of Science and Technology (CHARUSAT)

Faculty of Humanities (FoH)

Semester V

HS143A: German

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite courses:

Objectives of the Course:

To facilitate the learners:

- to develop proficiency of a new language - German
- to learn basic grammar and sentence construction of German language
- to learn listening, speaking, reading and writing in German language to acquire basic conversation skills for routine activities

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to German Language	08
2.	German for Basic Social Conversations	06
3.	Formal Conversations - I	06
4.	Formal Conversations - II	06
5.	Basic Language Functions	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to German Language	08 Hours	26.66%
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	• Facts and figures of German Language • Basic German Linguistics ➤ Alphabets ➤ Accents ➤ Liaison ➤ Nasalization • German Culture, Difference between German and English • Grammar: ➤ Basic Sentence Formation ➤ Personal Pronoun ➤ Basic questions & Answers ➤ Numbers (0 - 20) ➤ Verbs: (Sein, Haben, Werden) ➤ Greetings • Presentation: ➤ Self-Introduction ➤ Question and answering		
2.	German for Basic Social Conversations	06 Hours	20%
	• Grammar: ➤ Definite articles ➤ Indefinite articles ➤ Present tense (Positive Forms and Negative Forms) ➤ Numbers (21-10000) ➤ Days, Months, Family Relation ➤ Jahreszeiten ➤ Cloths and Body Parts ➤ Weather • Social Links: Gathering information from someone • Dialogue		
3.	Formal Conversations - I	06 Hours	20%
	• Grammar: ➤ Verbs: (Arbeiten, Wohnen, Kommen, Machen , Sprechen , Gehen) ➤ Possessive Adjectives ➤ Prepositions ➤ Conjunctions: und, aber, oder, denn, dann ➤ Profession ➤ W – Frage		

	➤ Im – Am -Um • Work, Study and Travel ➤ Ticket Reservation (At Bus/At Railway/At Airport) • Dialogue		
4.	Formal Conversations - II	06 Hours	20%
	• Grammar: ➤ Modal Verbs ➤ Nomminative ➤ Akkusative ➤ Negation: nicht, kein, ➤ Irregular Verbs ➤ Common Conjunctions ➤ Colors • Food & Shopping ➤ Buy vegetables and fruits ➤ Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market • Dialogue		
5.	Basic Functions of Language	04 Hours	13.33%
	• Grammar: ➤ Dative ➤ Grocery Items ➤ Mode of Transport ➤ Extra Vocabulary • Situational Dialogue and Conversations ➤ Expressing Likes and Dislikes ➤ Asking for permissions		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	acquire understanding of German language and culture
CO2	use German for everyday social conversations
CO3	use basic German to talk in professional settings
CO4	talk in formal situations in basic German language
CO5	use German for basic language functions

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks

Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

***Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.**

External / Semester End Evaluation (SEE) – 25 Marks

Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

***Note: The total External / SEE score out of 100 marks will be converted to 25 marks.**

Recommended Study Material:

- ❖ **Text book:** -
- ❖ **Reference Reading:** The reference/ reading material will be provided in consultation with the expert.

Online Course:

1. Getting started with German 1
Platform: The Open University
Link <https://www.open.edu/openlearn/languages/getting-started-german-1/contentsection-overview?active-tab=description-tab>
2. Getting started with German 2
Platform: The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-german-2/content-section-overview?active-tab=description-tab>

3. Getting started with German 3

Platform: The Open University

Link: <https://www.open.edu/openlearn/languages/getting-started-german-3/contentsection-overview?active-tab=description-tab>

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

Controllers are an essential part of modern electronic systems, providing the processing power and intelligence required to control and monitor various functions. They are widely used in a variety of industries, from automotive and aerospace to medical and consumer electronics, to control everything from engines and motors to temperature and lighting. Controllers offer a compact and cost-effective solution for a wide range of applications, with built-in peripherals, such as timers, ADCs, and DACs, that make it easy to interface with sensors and actuators. With the ever-increasing demand for automation and connectivity, the importance of controllers in modern systems is only expected to grow in the coming years.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Ability to design, program, and integrate microcontrollers into electronic systems for the purpose of controlling and monitoring various functions.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) *Develop software for microcontrollers.*
- 2) *Understand the architecture of advanced microcontrollers*
- 3) *Design and implement conventional timer-based systems and PCA based systems.*
- 4) *Design and implement ADC and DAC-based systems*
- 5) *Analyse and Understand communication protocol.*

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, 15 marks are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1,2	Understand operation of CrossBar, SFR paging using C-programming and debugging on keil microvision platform.	4
2	1,2	Configure the internal oscillator and PLL for various SYSCLK.	4
3	1,3	Develop C program for timer T0 and T1 in various modes of operations.	2
4	1,3	Develop C program for timer T2, T3 and T4 in various modes of operation.	4
5	1,4	Develop C program for ADC0 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).	4
6	1,4	Develop C program for ADC2 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).	2
7	1,4	Develop C program for using window comparator mode of ADC0.	2
8	1,4	Develop C program for on chip analog comparator.	2
9	1,4	Develop C program for DAC and generating signals.	2
10	1,3	Develop C program for various PCA mode of operations.	4
Total			

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	C8051f120 development board	1,2,3,4,5, 6,7,8,9,10
2	Software Tool: Keil uvision v4.0 and above,	1,2,3,4,5, 6,7,8,9,10
3	Silicon lab Flash Programming Utilities	1,2,3,4,5, 6,7,8,9,10
4	Digital Signal Oscilloscope , Digital Multi Meter	1,2,3,4,5, 6,7,8,9,10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- 'Responding Level' in 1st year
- 'Valuing Level' in 2nd year
- 'Organising Level' in 3rd year
- 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Embedded C Language Programming for Microcontroller	CO 1. Develop software for microcontrollers 1a. <u>Choose</u> appropriate programming language for <u>the given</u> scenario. 1b. <u>Evaluate</u> operation of super loop. 1c. <u>Analyse</u> the round robin concept and <u>implement</u> concept in <u>the given</u> application. 1d. <u>Understand</u> working of IDE platform and use them for <u>developing</u> and <u>simulating</u> code for <u>the given</u> application.	1.1 Comparison of Assembly & High Level Programming 1.2 Concept of Super loop/Infinite loop used in Embedded C Programs 1.3 Concept of Round Robin Execution in Real Time Operating System based Programming Environment. 1.4 Introduction to Keil Microvision IDE & Related debugging Techniques, Basic Working of various Control Loops, Variables: Global & Local, Various types of Variables & it's Memory Scope, Mathematical & Logical Operators, Array, Structure & Union, Pointers and its use in programming. Various C library and its use in programming.
Unit– II Introduction to c8051f120 Microcontroller	CO 1. Develop software for microcontrollers CO 2. Understand the architecture of advanced microcontrollers <i>PrO.1 Understand operation of CrossBar, SFR paging using C-programming and debugging on keil microvision platform.</i> <i>PrO.2 Configure the internal oscillator and PLL for various SYSCLK.</i> 2a. <u>Justify</u> the need of different microcontrollers and select the appropriate microcontroller for <u>the given</u> embedded system. 2b. <u>Evaluate</u> the importance of WATCHDOG timer and use them in the <u>hardware debugging</u> . 2c. <u>Choose</u> the appropriate speed of operation for <u>the given</u> application.	2.1 Comparison of 8051 & CIP-51 core Microcontrollers, Peripherals available in c8051f120 2.2 Microcontroller Watchdog Timer & it's Reset codes

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	2d. <u>Understand</u> Memory management. 2e. <u>Justify</u> the need of multifunctional pin and <u>select</u> appropriate function for <u>the given</u> application.	2.3 System Clock configuration with & without Internal PLL 2.4 SFR & SFR Paging 2.5 Concept of Crossbar & Crossbar Decoder, Internal Structure & Configuration of Port in Open drain & Push pull mode with & without Weak pull up disabled.
Unit– III Timers in c8051f120 Microcontroller	CO 1. Develop software for microcontrollers CO 3. Design and implement conventional timer-based systems and PCA based systems. <i>PrO.3 Develop C program for timer T0 and T1 in various modes of operations.</i> <i>PrO.4 Develop C program for timer T2, T3 and T4 in various modes of operation.</i> 3a. <u>Understand</u> and <u>Evaluate</u> operation of timer and use them in <u>appropriate</u> mode in <u>the given</u> application. 3b. <u>Justify</u> need of multiple timer and use them in appropriate mode in <u>the given</u> application.	3.1 Block diagram & Operation of Timer0 & 1 with Associated SFR, Calculation of input clock frequency for various SYSCLKOUT, Various Modes of Timer & Associated Programs With & Without Interrupt enabled. 3.2 Block diagram & Operation of Timer2,3 & 4 with Associated SFR. Various modes of Operations & Associated Programs.
Unit-IV Analog Peripherals of c8051f120 Microcontroller	CO 1. Develop software for microcontrollers CO 4. Design and implement ADC and DAC-based systems. <i>PrO.5 Develop C program for ADC0 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).</i> <i>PrO.6 Develop C program for ADC2 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).</i> <i>PrO.7 Develop C program for using window comparator mode of ADC0.</i> <i>PrO.8 Develop C program for on chip Analog comparator.</i> <i>PrO.9 Develop C program for DAC and generating signals.</i> 4a. <u>Understand</u> and <u>Program</u> on chip ADC for <u>the given</u> application. 4b. <u>Understand</u> and <u>Program</u> on chip DAC for <u>the given</u> application. 4c. <u>Understand</u> and <u>Program</u> on chip Comparator for <u>the given</u> application.	4.1 Block diagram & Working of 12 bit ADC0 & 8 bit ADC2 A/D Converters, Start of Conversion using Various modes, ADC in Window Comparator Mode, Temperature sensing using Onchip Temperature Sensor, Programs Related to ADC Operation 4.2 Digital to Analog Converter in c8051f120 with various mode of conversion, Generation of waveforms using DAC 4.3 On-chip Comparator.
Unit – V PCA and Application	CO 1. Develop software for microcontrollers CO 3. Design and implement conventional timer-based systems and PCA based systems. <i>PrO.10 Develop C program for various PCA mode of operations.</i> 5a. <u>Justify</u> need of PCA and use them in <u>the given</u>	5.1 Block diagram of PCA module, PCA Timer in Capture Compare, High Speed output, Frequency Output & 8/16 Bit Pulse Width Modulation Mode.: Associated Register & Working, Associated

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	application.	Programs.
Unit – VI Communication Protocol	CO 1. <i>Develop software for microcontrollers</i> CO 5. <i>Analyse and Understand communication protocol.</i> 6a. <u>Justify</u> need of different communication protocol and <u>select</u> appropriate protocol for <u>the given</u> situation.	6.1 SPI, SMBUS and I2C, UART Communication

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Embedded C Language Programming for Microcontroller	12	1	2	2	3	4	2	14
II	Introduction to c8051f120 Microcontroller	08	1	1	2	3	1	1	9
III	Timers in c8051f120 Microcontroller	12	1	2	3	3	3	2	14
IV	Analog Peripherals of c8051f120 Microcontroller	12	1	1	3	3	3	3	14
V	PCA and Application	10	1	2	2	3	2	2	12
VI	Communication Protocol	06	1	2	2	2	0	0	7
Total			6	10	14	17	13	10	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Inverter:** Use knowledge of power electronics and microcontrollers to implement hase inverter.
- 2) **Buck-Boost converter:** Implement buck boost converter to control DC voltage of system.
- 3) **3-ph Inverter:** Use knowledge of power electronics and microcontrollers to implement three phase inverter.
- 4) **Over current protection:** Implement current monitoring system using current sensor and ADC to prevent faulty operation of equipment under abnormal conditions.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different protection schemes used in industries.
- b) Prepare small application based on 8051, demonstrate the model with presentation/poster in front of class or at project expo.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Embedde d C Language Program ming for Microcon troller	1	1a. Choose appropriate programming language for the given scenario. 1b. Evaluate operation of super loop.	Lecture with media	PPTs, black board, Computer Demonstration
	2	1c. Analyse the round robin concept and implement concept in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	3	1d. Understand working of IDE platform and use them for developing and simulating code for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit– II Introduct	4	PrO.1 Understand operation of CrossBar, SFR paging using C-programming and debugging on	Lecture with media	PPTs, black board,

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
ion to c8051f12 0 Microcon troller		keil microvision platform. 2a. Justify the need of different microcontrollers and select the appropriate microcontroller for the given embedded system. 2b. Evaluate the importance of WATCHDOG timer and use them in the hardware debugging.		Computer Demonstration
	5	PrO.2 Configure the internal oscillator and PLL for various SYSCLK. 2c. Choose the appropriate speed of operation for the given application. 2d. Understand Memory management. 2e. Justify the need of multifunctional pin and select appropriate function for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit– III Timers in c8051f12 0 Microcon troller	6	PrO.3 Develop C program for timer T0 and T1 in various modes of operations. 3a. Understand and Evaluate operation of timer and use them in appropriate mode in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	7	PrO.3 Develop C program for timer T0 and T1 in various modes of operations. PrO.4 Develop C program for timer T2, T3 and T4 in various modes of operation. 3a. Understand and Evaluate operation of timer and use them in appropriate mode in the given application. 3b. Justify need of multiple timer and use them in appropriate mode in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	8	PrO.4 Develop C program for timer T2, T3 and T4 in various modes of operation. 3b. Justify need of multiple timer and use them in appropriate mode in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit-IV Analog Periphera ls of c8051f12 0 Microcon troller	9	PrO.5 Develop C program for ADC0 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method). PrO.6 Develop C program for ADC2 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method). 4a. Understand and Program on chip ADC for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	10	PrO.7 Develop C program for using window comparator mode of ADC0. PrO.8 Develop C program for on chip Analog comparator. 4a. Understand and Program on chip ADC for the given application. 4b. Understand and Program on chip DAC for the given application.	Lecture with media	PPTs, black board, Computer Demonstration

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	11	PrO.9 Develop C program for DAC and generating signals. 4b. Understand and Program on chip Comparator for the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit – V PCA and Application	12	PrO.10 Develop C program for various PCA mode of operations. 5a. Justify need of PCA and use them in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	13	PrO.10 Develop C program for various PCA mode of operations. 5a. Justify need of PCA and use them in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
	14	PrO.10 Develop C program for various PCA mode of operations. 5a. Justify need of PCA and use them in the given application.	Lecture with media	PPTs, black board, Computer Demonstration
Unit – VI Communication Protocol	14	6a. Justify need of different communication protocol and select appropriate protocol for the given situation.	Lecture with media	PPTs, black board
	15	6a. Justify need of different communication protocol and select appropriate protocol for the given situation.	Lecture with media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Embedded System Design using C8051	Han-Way Huang	Cengage Engineering 9788131512241
2	Introduction to Embedded Systems	Shibu K V	Tata McGraw Hill 9780070678798
3	Microprocessors & Interfacing	Douglas Hall	Tata McGraw Hill 9781283188982
4	The 8051 Microcontroller and Embedded Systems using Assembly and C	Muhammad Ali Mazidi	Pearson 9780199681273
5	Exploring C for Microcontrollers: A Hands on Approach	Jivan S. Parab, Vinod G. Shelake	Springer 9781402060663

XV SOFTWARE/LEARNING WEBSITES

- http://www.keil.com/dd/docs/datashts/cast/cast_c8051.pdf (Data sheet of C8051)
- <https://www.silabs.com/documents/public/data-sheets/C8051F12x-13x.pdf>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs														
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply real world like Artificial intelligence, machine learning, and optimization for design, simulation, analysis of systems to provide solutions to real world problems.	
EE371	Semester V	Advanced Microcontrollers														
		Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO														
371	Competency: Ability to design, program, and integrate microcontrollers into electronic systems for the purpose of controlling and monitoring various functions	1	2	3	1	3	1			1			1	3		1
371.1	Develop software for microcontrollers.	1	2	3	3	3	1			1		1	1	1		1
371.2	Understand the architecture of advanced microcontrollers	2	1	1	1					1			1	1		
371.3	Design and implement conventional timer-based systems and PCA based systems.	1	2	3	2	2				1			1	3		
371.4	Design and implement ADC and DAC-based systems	1	2	3	2	2				1			1	3		
371.5	Analyse and Understand communication protocol.	1	2	3	2	2				1			1	3		

XVII NAMES OF RESOURCE PERSONS

- Jignesh S Patel
-

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

This syllabus has been designed to make the students know about the fundamental IC fabrication technology, NMOS and CMOS design of digital circuits. This subject aims to give the basic design idea of digital circuits using MOS technology and different parameter associated with CMOS technologies. The course curriculum is supported by all of the VLSI/Electronics industries such as Cadence, Arrow, Einfochips, Sankalp semiconductors, Nvidia, ST Microelectronics etc.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- To learn IC fabrication process and different parameter associated with it.
- To learn the detail operation of MOS transitions and inverters
- Design the digital circuits using different MOS technologies.
- To learn advance technology in IC fabrication

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- CO-1) Identify the various IC Fabrication methods. To Understand the VLSI design flow and get aware about the trends in VLSI technology.
- CO-2) Analyses the operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology.
- CO-3) Design and Analyze combination circuits and sequential circuits using CMOS and NMOS Technology that realize specified digital functions.
- CO-4) To understand and design the dynamic logic using the CMOS technologies. To understand and evaluate the Domino logic circuits.
- CO-5) Explain and understand the advance technologies in MOS fabrication and compare it with the current CMOS technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Introduction to Xilinx ISE suite Tools.	02
2	2,3	A. Introduction to switch level modeling. Write a Verilog module using switch-level modeling to implement the following: CMOS Inverter, CMOS NAND Gate, CMOS NOR gate, CMOS AND, CMOS OR	04
3	3	A. Implement CMOS 2X1 MUX using switch level modeling B. Implement CMOS EXOR gate and also implement $X=AB+C$ using CMOS	04
4	3	A. Implement CMOS 4X1 MUX using "Tran" switches B. Implement Pseudo NMOS NOR gate..	04
5	3	Implement Full adder using transistor level modeling.	02
6	1	A. Introduction to LTSpice Tool. B. LTSpice Simulation of AND, OR, NOT gate.	04
7	3	LTSpice Simulation of NAND and NOR gate using CMOS.	02
8	2	Obtain V-I characteristics of NMOS/PMOS MOSFET using LT Spice (DC Analysis).	02
9	2	LTSpice Simulation of NOT gate using NAND & NOR with CMOS Technology.	02
10	3	LTSpice Simulation of Flip-flops using NAND gates.	04
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Xilinx 14.9	1,2,3,4,5
2	LT Spice tool	6,7,8,9,10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- 'Responding Level' in 1st year
- 'Valuing Level' in 2nd year
- 'Organising Level' in 3rd year
- 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
Introduction and fabrication of MOSFET	CO-1) Identify the various IC Fabrication methods. To Understand the VLSI design flow and get aware about the trends in VLSI technology. Pro1: Introduction to Xilinx ISE suite Tool. Pro6: A. Introduction to LTSpice Tool. 1a. Introduction to switch level modeling in Xilinx and basic simulation in LT Spice	1. Introduction and fabrication of MOSFE 1.1 VLSI Design Flow 1.2 Design Hierarchy 1.3 Design Methodology 1.4 NMOS, PMOS and CMOS fabrication process.
MOS Transistor	CO-2) Analyses the operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology. PRO2: Write a Verilog module using switch-level modeling to implement the following: CMOS Inverter, CMOS NAND Gate, CMOS NOR gate, CMOS AND, CMOS OR PRO8: Obtain V-I characteristics of NMOS/PMOS MOSFET using LT Spice (DC Analysis). 1b. Design CMOS basic gates in Xilinx with its test bench. 1c. Obtain V-I characteristics of MOSFET and obtain net list in LT spice	2. MOS Transistor 2.1 Metal Oxide Semiconductor (MOS) structure. 2.2 The MOS system under external bias. 2.3 Structure and operation of MOS transistor 2.4 MOSFET current-voltage characteristics. 2.5 MOSFET scaling & Small-geometry effects, 2.6 MOSFET capacitance
MOS Inverters: static characteristic and switching characteristic	CO-2) Analyses the operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology. PRO9: LTSpice Simulation of NOT gate using NAND & NOR with CMOS Technology. 1d. Explain the design of NOT gate using NAND and NOR gates	3. MOS Inverters: static characteristic and switching characteristic 3.1 Introduction and Resistive load inverter 3.2 Inverter with n-type MOSFET load (Enhancement & Depletion type MOSFET load) 3.3 CMOS inverter 3.4 Definition and calculation of delay times 3.5 Inverter design with delay constraints

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
		3.6 Estimation of interconnect parasitic and calculation of interconnect delay 3.7 Switching power dissipation of CMOS inverters
MOS combinational circuits	CO-3) Design and Analyze combination circuits and sequential circuits using CMOS and NMOS Technology that realize specified digital functions. PRO10 LTSpice Simulation of Flip-flops using NAND gates 1f. Design various flip-flop using CMOS technology.	4. MOS combinational circuits 4.1 Introduction 4.2 MOS logic circuit with Pseudo NMOS loads 4.3 CMOS logic circuit 4.4 Complex logic circuits 4.5 CMOS Transmission gate
MOS sequential logic circuits	CO-3) Design and Analyze combination circuits and sequential circuits using CMOS and NMOS Technology that realize specified digital functions. PRO10 LTSpice Simulation of Flip-flops using NAND gates 1f Design various flip-flop using CMOS technology.	5. MOS sequential logic circuits 5.1 Introduction 5.2 Behavior of Bi-stable elements 5.3 The S-R latch circuit 5.4 Clocked latch and Flip-flop circuits 5.5 CMOS D-latch and edge triggered Flip-flop
MOS dynamic logic circuits	CO-4) To understand and design the dynamic logic using the CMOS technologies. To understand and evaluate the Domino logic implementation.	6. MOS dynamic logic circuits 6.1 Principles of pass transistor circuits 6.2 Synchronous Dynamic Circuit Techniques 6.3 CMOS Dynamic Circuit Techniques

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction and fabrication of MOSFET	05	0	2	4	4	0	0	8
II	MOS Transistor	08	0	2	4	4	2	0	12
III	MOS Inverters: static characteristic and switching characteristic	16	0	4	4	4	4	0	20
IV	MOS combinational Circuits	06	0	0	0	2	2	6	10
V	MOS sequential logic Circuits	04	0	0	4	0	2	4	10
VI	MOS dynamic logic Circuits	04	2	0	0	4	2	2	10

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
Total		45	4	10	14	16	12	12	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) Implement the combinational circuit using Xilinx software and Spartan 3E board.
- 2) Implement the sequential circuit using Xilinx software and Spartan 3E board.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Prepare specifications of some electronic devices.
- Undertake micro-projects in groups (Max. Limit-4)

- Give seminars on any relevant topic

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introducti on and fabricati on of MOSFE T	1	Pro1: Introduction to Xilinx ISE suite Tool. 1a. Introduction to switch level modeling in Xilinx and basic simulation in LT Spice	Lecture with media	PPTs, black board
	9,10	Pro6: A. Introduction to LTSpice Tool. 1a. Introduction to switch level modeling in Xilinx and basic simulation in LT Spice	Lecture With media	PPTs, black board
Unit– II MOS Transisto r	2,3	PRO2: Write a Verilog module using switch- level modeling to implement the following: CMOS Inverter, CMOS NAND Gate, CMOS NOR gate, CMOS AND, CMOS OR	Lecture with media, and Demonstra tion	Real sample s of relays, PPTs, Videos
	12	PRO8: Obtain V-I characteristics of NMOS/PMOS MOSFET using LT Spice (DC Analysis). 1b. Design CMOS basic gates in Xilinx with its test bench. 1c. Obtain V_I characteristics of MOSFET and obtain net list in LT spice.	Lecture with media, and Demonstra tion	Real sample s of relays, PPTs, Videos
Unit– III MOS Inverte rs: static character istic and switching character istic	13	PRO9: LTSpice Simulation of NOT gate using NAND & NOR with CMOS Technology. 1d. Explain the design of NOT gate using NAND and NOR gates	Lecture with media, Field visit	PPTs, Video
Unit– IV MOS combinat ional circuits	4,5	PRO3 A. Implement CMOS 2X1 MUX using switch level modeling B. Implement CMOS EXOR gate and also implement $X=AB+C$ using CMOS Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media, Tutorial	PPT, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	6,7	PRO4 A. Implement CMOS 4X1 MUX using "Tran" switches B. Implement Pseudo NMOS NOR gate. 1e. Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media	PPT, black board
	8	PRO5 Implement Full adder using transistor level modeling. 1e. Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media, case study	PPT, Pictures, black board
	11	PRO7 LTSpice Simulation of NAND and NOR gate using CMOS. 1e. Design various combinational circuit using CMOS and Pseudo NMOS technology.	Lecture with media, case study	PPT, Pictures, black board
Unit– V MOS sequential logic circuits	15	PRO10 LTSpice Simulation of Flip-flops using NAND gates 1f. Design various flip-flop using CMOS technology.	Lecture with media, case study	PPT, Pictures, black board
MOS dynamic logic circuits	14	Principles of pass transistor circuits Explain Voltage Bootstrapping, Synchronous Dynamic Circuit Techniques Explain CMOS Dynamic Circuit Techniques , High performance Dynamic CMOS circuits	Lecture with media,	PPT, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	CMOS digital integrated circuits: Analysis and Design	Sung-Mo-Kang, Yusuf Leblebici, Chulwoo Kim	Tata McGrawhill ISBN-13: 978-93-5316-509-3 ISBN-10: 93-5316-509-1
2	Basic VLSI Design	Douglas Pucknell	Pearson Publication ISBN-13: 978-0724801053 ISBN-10: 0724801057
3	Modern VLSI Design	Wayne Wolf	Person Education ISBN-13: 978-0134186047 ISBN-10:0134186

XV SOFTWARE/LEARNING WEBSITES

- www.edaplayground.com
- www.chipverify.com
- www.nptel.iitm.ac.in

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and Cos	POs and PSOs														
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Project management and finance	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	PSO3 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE372	Semester V															
372	Competency Evaluate the performance of various analog and digital circuits used in Industrial applications.	3	1	3	2	2	0	0	0	0	0	-	-	3	1	-
372.1	Identify the various IC Fabrication methods. To Understand the VLSI design flow and get aware about the trends in VLSI technology	3	1	1	-	3	-	-	-	-	3	3	-	1	1	-
372.2	Analyses operation of MOS transistor. Design and Evaluate the MOS inverter with different load technology	2	3	3	3	2	1	-	-	1	2	3	-	3	1	-
373.3	Design and combination circuit and sequence circuits using CMOS Technology realize specific digital functions	3	3	3	3	3	1	-	1	2	1	3	1	3	1	-
374.4	To understand and design the logic using the CMOS technologies. understand evaluate the	3	1	1	1	-	-	-	-	2	2	-	2	2	2	-

	Domino logic circuits.															
375.5	Explain understand advance technologies in MOS fabrication and compare it with the current CMOS technologies	3	1	1	1	-	-	-	-	1	1	-	3	2	2	-

XVII NAMES OF RESOURCE PERSONS
a) Ankur H Patel

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

Electrical engineers have to deal with various types of optimization problem used in many applications. This curriculum course structure is developed such that the student is able to understand and apply the basic classical optimization methods to solve the basic power system problems.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

Understand and analyze the classical optimization techniques to solve the power system problem.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. Develop a fundamental understanding of concepts of various classical methods
2. Understand and formulate the linear programming problem to solve the power system linear problems.
3. Understand and formulate the unconstrained programming problem to solve the power system unconstrained problems.
4. Understand and formulate the Nonlinear programming problem to solve the power system Nonlinear problems.
5. The students will understand different methods of unconstrained and least squares-based optimization for solving real world problem.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	3,4	Formulation of optimization problem	05
2	3,4	Solution of Linear Programming Problem by graphical method	05
3	3,4	Kuhn-Tucker method	05
4	1,2	Nonlinear programming method	05
5	1,2	Use of Penalty function method	05
6	1,2	Particle swarm optimization to solve constrained and unconstrained optimization problems.	05
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB optimization solver	1 to 6
2	GAMS Software	1 to 6

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Fundamentals of Optimization	CO 1. Develop a fundamental understanding of concepts of various classical methods PrO 1 Formulation of optimization problem 1a. <i>Understand the Convexity</i> 1b. <i>Understand the Rates of convergence.</i> 1c. <i>Understand the Feasibility and optimality,</i>	a. Introduction, Feasibility and optimality, b. Convexity, c. constraints, Rates of convergence.
Unit– II Linear Programming	CO 2. Understand and formulate the linear programming problem to solve the power system linear problems. PrO 1 Solution of Linear Programming Problem by graphical method PrO 5 Use of Penalty function method 2a <i>Understand and apply the Formulation of Linear programming problem</i> 2b. <i>Understand and analyze the Graphical method</i> 2c. <i>Understand and apply the Simplex method</i> 2d. <i>Understand and apply the Big M method</i>	2.1 Introduction, Formulation of Linear programming problem, 2.2 Graphical method, Simplex method, Basic solution, 2.3 Basic feasible solution, Simplex algorithm, Big M method (Penalty method), 2.4 Two phase method, Special cases in simplex method application. Economic load dispatch using Linear programming method.
Unit– III Unconstrained optimization	CO 3. Understand and formulate the unconstrained programming problem to solve the power system unconstrained problems. CO 5. The students will understand different methods of unconstrained and least squares-based optimization for solving real world problem. PrO 3 Kuhn-Tucker method PrO 6 Particle swarm optimization to solve constrained and unconstrained optimization problems. 3a. <i>Understand and analyze the Optimality conditions,</i> 3b. <i>Understand and analyze the Newton's method for minimization,</i> 3c. <i>Understand and analyze the Quasi-Newton method</i> 3d <i>Understand and analyze the Line search methods, Steepest-Descent method,</i>	3.1 Introduction, Optimality conditions, Newton's method for minimization, 3.2 Line search methods, Steepest-Descent method, 3.3 Quasi-Newton method.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit– IV Nonlinear Programming	CO 4. Understand and formulate the Nonlinear programming problem to solve the power system Nonlinear problems. PrO 4 Nonlinear programming method PrO 5 Use of Penalty function method <i>4a. Understand and apply the Optimality conditions for constrained problems, Kuhn Tucker conditions,</i> <i>4b. Understand and apply the Penalty function method, Barrier method,</i> <i>4c. Understand and analyze the Computing the Lagrange multipliers, Sequential quadratic programming,</i> <i>4d Understand and analyze the Interior point method. Active and reactive power optimization using NLP.</i>	4.1 Optimality conditions for constrained problems, Kuhn Tucker conditions, 4.2 Penalty function method, Barrier method, 4.3 The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, 4.4 Computing the Lagrange multipliers, Sequential quadratic programming, 4.5 Interior point method. Active and reactive power optimization using NLP.
Unit– V Least squares based methods	CO 5. The students will understand different methods of unconstrained and least squares-based optimization for solving real world problem. <i>5a. Understand and apply the LSQ fit</i> <i>5b. Evaluate the of unknown coefficients</i> <i>5c. Understand and analyze the Approximating the shifted waveform by a power series</i> <i>5d. Understand and analyze the multi-variable series LSQ technique, Derivation of the multi variable series.</i>	1.1 Introduction, Integral LSQ fit, Basic assumptions, 1.2 Determination of unknown coefficients, Power series LSQ fit, 1.3 Assumptions, Shifted waveform, Approximating the shifted waveform by a power series, 1.4 Multi-variable series LSQ technique, Derivation of the multi variable series.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Fundamentals of Optimization	05	0	2	2	0	1	0	05
II	Linear Programming	12	0	1	1	4	5	3	14
III	Unconstrained optimization	13	2	1	2	2	5	3	15
IV	Nonlinear Programming	15	2	2	3	3	6	2	18
V	Least squares-based methods	15	1	3	3	4	5	2	18
Total		60	5	9	11	13	22	10	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) Formulate the Economic load dispatch non liner problem in MATLAB and GAMS.
- 2) Formulate the Unit commitment problem in MATLAB and GAMS.
- 3) Formulate the Mathematical nonlinear and linear constrained problem and solved using the classical optimization techniques using MATLAB and GAMS.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different Semiconductor switches used in converters.
- b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

XIII COURSE PLAN

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Fundamentals of Optimization	PrO 1 Formulation of optimization problem <i>1a. Understand the Convexity</i> <i>1b. Understand the Rates of convergence.</i> <i>1c. Understand the Feasibility and optimality,</i>	1.1 Introduction, Feasibility and optimality, 1.2 Convexity, 1.3 constraints, Rates of convergence.
Unit– II Linear Programming	PrO 1 Solution of Linear Programming Problem by graphical method PrO 5 Use of Penalty function method <i>2a Understand and apply the Formulation of Linear programming problem</i> <i>2b. Understand and analyze the Graphical method</i> <i>2c. Understand and apply the Simplex method</i> <i>2d. Understand and apply the Big M method</i>	2.1 Introduction, Formulation of Linear programming problem, 2.2 Graphical method, Simplex method, Basic solution, 2.3 Basic feasible solution, Simplex algorithm, Big M method (Penalty method), 2.4 Two phase method, Special cases in simplex method application. Economic load dispatch using Linear programming method.
Unit– III Unconstrained optimization	PrO 3 Kuhn-Tucker method PrO 6 Particle swarm optimization to solve constrained and unconstrained optimization problems. <i>3a. Understand and analyze the Optimality conditions,</i> <i>3b. Understand and analyze the Newton's method for minimization,</i> <i>3c. Understand and analyze the Quasi-Newton method</i> <i>3d Understand and analyze the Line search methods, Steepest-Descent method,</i>	3.1 Introduction, Optimality conditions, Newton's method for minimization, 3.2 Line search methods, Steepest-Descent method, 3.3 Quasi-Newton method.
Unit– IV Nonlinear Programming	PrO 4 Nonlinear programming method PrO 5 Use of Penalty function method <i>4a. Understand and apply the Optimality conditions for constrained problems, Kuhn Tucker conditions,</i> <i>4b. Understand and apply the Penalty function method, Barrier method,</i> <i>4c. Understand and analyze the Computing the Lagrange multipliers, Sequential quadratic programming,</i> <i>4d Understand and analyze the Interior point method. Active and reactive power optimization</i>	4.1 Optimality conditions for constrained problems, Kuhn Tucker conditions, 4.2 Penalty function method, Barrier method, 4.3 The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, 4.4 Computing the Lagrange multipliers, Sequential quadratic programming, 4.5 Interior point method. Active and reactive power optimization using NLP.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related POs required by industry)	Topics and Sub-topics
	using NLP.	
Unit– V Least squares based methods	<i>5a. Understand and apply the LSQ fit</i> <i>5b. Evaluate the of unknown coefficients</i> <i>5c. Understand and analyze the Approximating the shifted waveform by a power series</i> <i>5d. Understand and analyze the multi-variable series LSQ technique, Derivation of the multi variable series.</i>	5.1 Introduction, Integral LSQ fit, Basic assumptions, 5.2 Determination of unknown coefficients, Power series LSQ fit, 5.3 Assumptions, Shifted waveform, Approximating the shifted waveform by a power series, 5.4 Multi-variable series LSQ technique, Derivation of the multi variable series.

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electric power applications of Fuzzy systems	M. E. El-Hawary	IEEE press, 2006
2	Artificial intelligence and intelligent systems	N.P.Padhy	Oxford University Press, 2005
3	Soft Computing Techniques and its Applications in Electrical Engineering	Chaturvedi	Springer
4	Optimization Principles	Narayan Rao	John Wiley & Sons
5	Optimization for engineering design	K. Deb	PHI

XV SOFTWARE/LEARNING WEBSITES

- <https://nptel.ac.in/courses/111/105/111105100/>
- <https://nptel.ac.in/courses/105/108/105108127/>

XVI PO-COMPETENCY-CO MAPPING

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Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE373	Semester V														
373	Competency • Understand and analyze the classical optimization techniques to solve the power system problem.	3	2	2	2	1	-	1	-	-	-	-	1	3	-
373.1	Develop a fundamental understanding of concepts of various classical methods	3	-	-	1	-	-	-	-	-	2	-	-	-	-
373.2	Understand and formulate the linear programming problem to solve the power system linear problems.	3	2	3	-	2	-	-	-	1	-	-	1	1	3
373.3	Understand and formulate the unconstrained programming problem to solve the power system unconstrained problems.	3	3	2	-	2	-	-	-	1	-	-	1	1	3
373.4	Understand and formulate the Nonlinear programming problem to solve the power system Nonlinear problems.	3	3	2	-	2	-	-	-	1	-	-	1	1	3
373.5	The students will understand different methods of unconstrained and least squares-based optimization for solving real world problem.	3	3	2	-	2	-	-	-	1	-	-	1	1	3

XVII NAMES OF RESOURCE PERSONS

a. Dr. Dharmesh Dabhi

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

The micro grid can help measure and reduce energy consumption and costs. One of the visions for the micro grid is that consumers will be able to monitor their energy consumption in real time, or at least near real time, via the Web. This curriculum course structure is developed such that the student is able to understand the basic concepts, components, architecture, various measurement technologies of micro grid and the importance of distributed generation in power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Evaluate the performance of various distributed generator, energy storage equipment and converters which are used in industry.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Understand** the basic difference between conventional and non-conventional energy resources.
- 2) **Analyze** the concept of distributed generation and installation.
- 3) **Design** the grid integration system with conventional and non-conventional energy sources.
- 4) **Evaluate** the performance of the converters in micro grid.
- 5) **Analyze** power quality issues and control operation of micro grid.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **10 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is converted from 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Prepare</u> the review report of implementing non-conventional energy resources in Micro grid.	04
2	2	<u>Design</u> the Active Distribution System with Distributed Generation.	02
3	2	<u>Design</u> the Passive cell balancing for a Lithium-ion battery pack.	04
4	2	<u>Analyze</u> the DC-DC converter that can be used to maintain a constant load voltage when drawing power from an ultra-capacitor.	02
5	2	<u>Analyze</u> the charging and discharging behavior of supercapacitor.	04
6	3	<u>Design</u> the 250-kW Grid connected PV array.	04
7	4	<u>Design</u> of DC microgrid.	04
8	5	<u>Analyze</u> the islanded operation of a remote- microgrid model using droop controllers.	04
		Total	28

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,6,7,8

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction	CO 1. <u>Understand</u> the basic difference between conventional and non-conventional energy resources. <i>Pro 15. <u>Prepare</u> the review report of implementing non-conventional energy resources in Micro grid.</i> 1a. <u>Prepare</u> the report on energy crises 1b. <u>Prepare</u> the review report on non-conventional energy sources	1.1 Conventional power generation: advantages and disadvantages 1.2 Energy crises 1.3 Non-conventional energy (NCE) resources: review of Solar PV, Wind Energy systems, Fuel Cells, micro turbines, biomass, and tidal sources.
Unit– II Basic of Distributed Generation	CO 2. <u>Analyze</u> the concept of distributed generation and installation. <i>Pro 2. <u>Design</u> the Active Distribution System with Distributed Generation.</i> <i>Pro 3. <u>Design</u> the Passive cell balancing for a Lithium-ion battery pack.</i> <i>Pro 4. <u>Analyze</u> the DC-DC converter that can be used to maintain a constant load voltage when drawing power from an ultra-capacitor.</i> <i>Pro 5. <u>Analyze</u> the charging and discharging behavior of supercapacitor.</i> 2a. <u>Understand</u> the concept of distributed generations. 2b. <u>Choose</u> the correct standards for interconnecting the distributed generation for <u>the specified</u> application. 2c. <u>Analyze</u> the security issues in DG implementations. 2d. <u>Design</u> the energy storage elements.	2.1 Concept of distributed generations, topologies, selection of sources 2.2 Regulatory standards/framework, Standards for interconnecting Distributed resources to electric power systems: IEEE 1547. 2.3 DG installation classes, security issues in DG implementations 2.4 Energy storage elements: Batteries, ultra-capacitors, flywheels.
Unit– III Impact of Grid Integration	CO 3. <u>Design</u> the grid integration system with conventional and non-conventional energy sources. <i>Pro 6. <u>Design</u> the 250-kW Grid connected PV array.</i> 3a. <u>Determine the requirements for grid interconnection for the given</u> distributed generation. 3b. <u>Evaluate</u> the operational parameters for grid integration. 3c. <u>Evaluate</u> the impact of grid integration with NCE sources on existing power system. 3d. <u>Understand</u> the islanding issues of	3.1 Requirements for grid interconnection, 3.2 Limits on operational parameters: voltage, frequency, THD 3.3 Response to grid abnormal operating conditions 3.4 Islanding issues. 3.5 Impact of grid integration with NCE sources on existing power system: reliability, stability and power quality issues.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	grid integration.	
Unit-IV Basic of Microgrid	CO 4. <u>Evaluate</u> the performance of the converters in micro grid. <i>Pro 7. <u>Design</u> of DC microgrid.</i> 4a. <u>Design</u> the microgrid for the specified application. 4b. <u>Evaluate</u> the performance of the microgrid drivers for the specified condition. 4c. <u>Evaluate</u> the performance of the converters for the specified application.	4.1 Concept and definition of microgrid, 4.2 Microgrid drivers and benefits, 4.3 Review of sources of microgrids, 4.4 Typical structure and configuration of a microgrid, 4.5 AC and DC microgrids, 4.6 Power Electronics interfaces in DC and AC microgrids
Unit – V Control and Operation of Microgrid	CO 5. <u>Analyze</u> power quality issues and control operation of micro grid. <i>Pro 8. <u>Analyze</u> the islanded operation of a remote- microgrid model using droop controllers.</i> 5a. <u>Evaluate</u> the different controls of grid connected microgrid. 5b. <u>Evaluate</u> the different controls of islanded microgrid. 5c. <u>Design</u> the active and reactive control of microgrid. 5d. <u>Design</u> the communication infrastructure for microgrid. 5e. <u>Evaluate</u> the power quality issues in microgrid.	5.1 Modes of operation and control of microgrid: grid connected and islanded mode, 5.2 Active and reactive power control, protection issues, 5.3 Anti-islanding schemes: passive, active and communication based techniques 5.4 Microgrid communication infrastructure 5.5 Power quality issues in microgrids, regulatory standards, Microgrid economics 5.6 Introduction to smart microgrids.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	08	4	4	2	2	0	0	12
II	Basic of Distributed Generation	12	4	4	4	2	0	0	14
III	Impact of Grid Integration	16	0	4	4	2	2	4	16
IV	Basic of Microgrid	08	2	2	2	2	0	2	10
V	Control and Operation of Microgrid	16	0	4	4	2	4	4	18
Total		60	10	18	16	10	6	10	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Converters:** Collect the specifications of different types of converters manufacturing by different types of industries.
- 2) **Battery:** Collect the specifications of different types of battery used in different Electric vehicle by different types of industries.
- 3) **Renewable generator:** Prepare case studies on power quality issues in renewable energy integration and microgrid.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different protection schemes used in industries.
- b) Prepare 15 min. power point presentation on how to choose relay for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction	1	Pro 1. <u>Prepare</u> the review report of implementing non-conventional energy resources in Micro grid. 1a. <u>Prepare</u> the report on energy crises.	Lecture with media	PPTs, black board
	2	1b. <u>Prepare</u> the review report on non-conventional energy sources.	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	3	1b. <u>Prepare</u> the review report on non-conventional energy sources.	Lecture with media,	PPTs, black board
Unit– II Basic of Distributed Generation	4	Pro 2. <u>Design</u> the Active Distribution System with Distributed Generation. 2a. <u>Understand</u> the concept of distributed generations.	Lecture with media, Tutorial	PPTs, black board
	5	2b. <u>Choose</u> the correct standards for interconnecting the distributed generation for the specified application. 2c. <u>Analyze</u> the security issues in DG implementations.	Lecture with media, Tutorial	PPTs, black board
	6	Pro 3. <u>Design</u> the Passive cell balancing for a Lithium-ion battery pack. Pro 4. <u>Analyze</u> the DC-DC converter that can be used to maintain a constant load voltage when drawing power from an ultra-capacitor. Pro 5. <u>Analyze</u> the charging and discharging behavior of supercapacitor. 2d. <u>Design</u> the energy storage elements.	Lecture with media	PPTs, black board
		3a. <u>Determine</u> the requirements for grid interconnection for the given distributed generation. 3b. <u>Evaluate</u> the operational parameters for grid integration.	Lecture with media,	PPTs, black board
Unit– III Impact of Grid Integration	8	3c. <u>Evaluate</u> the impact of grid integration with NCE sources on existing power system. 3d. <u>Understand</u> the islanding issues of grid integration.	Lecture with media	PPTs, black board
	9	Pro 6. <u>Design</u> the 250-kW Grid connected PV array.	Lecture with media, + PPT	PPTs, black board
Unit-IV Basic of Microgrid	10	Pro 7. <u>Design</u> of DC microgrid. 4a. <u>Design</u> the microgrid for the specified application.	Lecture with media, Tutorial	PPT, black board
	11	4b. <u>Evaluate</u> the performance of the microgrid drivers for the specified condition.	Lecture with media	PPTs, black board
	12	4c. <u>Evaluate</u> the performance of the converters for the specified application.	Lecture with media, case study	PPT, black board
Unit – V Control and Operation of Microgrid	13	Pro 8. <u>Analyze</u> the islanded operation of a remote- microgrid model using droop controllers. 5a. <u>Evaluate</u> the different controls of grid connected microgrid. 5b. <u>Evaluate</u> the different controls of islanded microgrid.	Lecture with media	PPTs, black board
	14	5c. <u>Design</u> the active and reactive control of microgrid. 5d. <u>Design</u> the communication infrastructure for microgrid.	Lecture with media, Tutorial	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	15	5e. Evaluate the power quality issues in microgrid.	Lecture with media, Tutorial	PPTs, black board, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Microgrids: Architectures and Control	N. Hatziargyriou	Wiley-IEEE Press, 2014. ISBN: 9781118720677
2	Distributed Generation-Induction and Permanent Magnet Generators	Loi Lei Lai, Tze Fun Chan	Wiley-IEEE Press, 2007. ISBN: 978-0-470-06208-1
3	Voltage Source Converters in Power Systems: Modeling, Control and Applications	Amirnaser Yezdani, Reza Iravani	Wiley-IEEE Press, 2009 ISBN: 978-0-470-52156-4
4	Renewable Energy Sources	N. Twidell, A. D. Weir	University press, 2001, ISBN: 9781315766416
5	Fuel Cell Systems	James Larminie , Andrew Dicks	Wiley, 2000 ISBN: 9780470848579

XV SOFTWARE/LEARNING WEBSITES

1. DC Microgrid and Control System: - <https://nptel.ac.in/courses/108/107/108107143/>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE374	Semester V														
374	Competency Evaluate the performance of various distributed generator, energy storage equipment and converters which are used in industry	3	2	2	2	1	-	1	-	-	-	-	1	3	-

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
374.1	Understand the basic difference between conventional and non-conventional energy resources.	3	2	2	-	2	-	2	-	-	1	-	-	3	-
374.2	Analyze the concept of distributed generation and installation	3	2	1	-	-	-	-	-	1	-	-	-	2	-
374.3	Design the grid integration system with conventional and non-conventional energy sources	3	2	3	3	2	-	-	-		-	-	-	2	2
374.4	Evaluate the performance of the converters in micro grid	3	3	1	2	-	-	2	-	1	-	-	-	2	2
374.5	Analyze power quality issues and control operation of micro grid	3	-	2	-	2	-	-	-	1	2	2	-	-	3

XVII NAMES OF RESOURCE PERSONS

a) Dr. Jigar Sarda

UG Engineering Programme	Semester
Electrical Engineering	Fifth

I RATIONALE

The pressure of technological development in all sectors on renewable energy sources has led to growing the cost of energy around the world. Efficient and judicious use of available energy sources would lead to the easing of such pressures and decrease in the operating costs of the organizations and industries. Thus it is necessary to conserve the energy. Also essential theoretical knowledge and practical skills about the concept of energy conservation is to be provided through different approaches. The process of energy audit will help to identify the various possible avenues in which savings of energy can be effectively adopted.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Perform Electrical Energy Audit for Conservation of the Energy.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Use** the different energy management tools and techniques.
- 2) **Perform** the Energy Conservation opportunities in different area.
- 3) **Interpret** the suitable energy storage system.
- 4) **Perform** the Electrical Energy Audit of HT and LT Consumer.
- 5) **Choose** the energy efficient technology in electrical systems.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme			Total Credits (L+T+P)	Examination Scheme				
(In Hours)				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, 30 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Study energy management and conservation techniques	02
2	1	Use different energy audit tools to perform audit	02
3	2	Perform energy conservation opportunity in domestic user appliances	02
4	2	Perform energy conservation opportunity in Industrial user appliances	02
5	4	Identify star labeled electrical apparatus and compare the data for various star ratings.	02
6	4	Compare power consumption of different types of light, FTL & LED.	02
7	4	Perform Bill analysis of HT consumers and suggest suitable corrective measures for conservation and reduction of it's energy bill.	02
8	4	Perform energy audit of HT consumer	02
9	5	Determine the reduction in power consumption by replacement of lightning system in class room / laboratory.	02
10	5	Determine the reduction in power consumption by replacement of Fans and AC in class room / laboratory.	02
11	5	Prepare an energy audit report.(phase I)	02
12	5	Prepare an energy audit report.(phase II)	02
Total			22

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Ammeter MI Type: AC/DC 0-5-10 Amp	1
2	Voltmeter MI Type: AC/DC 0-150/300V,0-250/500V	2
3	Wattmeter: Three Phase double element 5/10Amp,250/500V	3
4	Wattmeter: Single Phase double element 2.5/5Amp,200/400V	4
5	Automatic Power factor controller (APFC)	5
6	Lux Meter	6
7	Clip on meter (amp, volts) digital/analog	7
8	FTL,CFL,LED of different ratings	8
9	Power analyzer	9
10	Thermography meter	10
11	Electric regulators, Electronics ballast	11

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.

- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Electrical Energy Conservation	CO 1. <u>Use</u> the different energy management tools and techniques. <i>PRO 16. Study energy management and conservation techniques</i> <i>PRO 17. Use different energy audit tools to perform audit</i> 1n. <u>Interpret</u> the given energy conservation clause 1o. <u>Explain</u> the specified BEE roles 1p. <u>Describe</u> the techniques for energy conservation in given electric machine	1.1 Introduction to energy science and energy technology, various forms of energy. Law of conservation of energy. Primary and Secondary energy, Commercial and Non-commercial energy, Renewable and Non-Renewable energy. Carbon credit. 1.2 Energy conservation act-2001 and its features, key definitions as outlined in the Act, Schemes of BEE under the energy conservation Act, 2001. 1.3 National Mission for Enhanced Energy Efficiency. 1.4 Energy scenario of India, Energy conservation and its importance. 1.5 Load Management and Maximum demand control. 1.6 Potential energy conservation opportunities in: (1) HVAC System (2) Lighting systems (3) Motors and Transformers
Unit– II Energy	CO 2. <u>Perform</u> the Energy Conservation opportunities in different area.	

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Management	CO 1. <u>Use</u> the different energy management tools and techniques. PrO 3. Perform energy conservation opportunity in domestic user appliances PrO 4. Perform energy conservation opportunity in Industrial user appliances	2.1 Definition and objective of energy management, Responsibilities and Duties of energy manager, Management tools for effective Implementation like 5S, Kaizen, Small group Activity, TPM, TQM & ISO 50001. 2.2 Electrical System Optimization: Electricity rate Tariff, Key to reduction in electrical energy consumption, methods to improve plant power factor, Load Management, Conduction Loss, Switching Loss, Magnetic Loss. 2.3 Cogeneration system 2.4 Intelligent buildings 2.5 Energy planning, Energy staffing, Energy Organization, Energy Requirement, Energy Costing, Energy Budgeting, Energy Monitoring, Energy consciousness, Energy Management Professionals, Environment pollution due to energy use.
Unit– III Energy storage system	CO 3. <u>Interpret</u> the suitable energy storage system.	3.1 Energy storage: Demand for energy storage, Classification of Energy storage systems. 3.2 Mechanical energy storage: Pumped hydro, Compressed Air, Flywheels. 3.3 Electrochemical energy storage: secondary batteries, Flow batteries. 3.4 Chemical energy storage: Hydrogen, Synthetic natural gas. 3.5 Electrical and magnetic energy storage systems.
Unit-IV Energy Audit	CO 4. <u>Perform</u> the Electrical Energy Audit of HT and LT Consumer. PrO 5. <u>Identify star labeled electrical apparatus and compare the data for various star ratings.</u> <i>I</i> PrO 6. <u>Compare power consumption of different types of light, FTL & LED.</u> <i>C</i> PrO 7. <u>Perform Bill analysis of HT consumers</u> <i>P</i>	4.1 Introduction, Definition, Need for Energy Audit, 4.2 Types and Procedures for energy audit. Preliminary energy audit, detailed energy audit, Industrial Audit approaches. 4.3 Procedure for energy audit and equipment's required. 4.4 Energy Audit Report, Benchmarking,

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	<u>and suggest suitable corrective measures for conservation and reduction of it's energy bill.</u> PrO 8. <u>Perform Bill analysis of HT consumers and suggest suitable corrective measures for conservation and reduction of it's energy bill.</u> P PrO 9. <u>Perform energy audit of HT consumer</u> P 4a. <u>Suggest relevant instrument for the specified energy audit with justification</u> 4b. <u>Develop</u> the energy flow diagram of the given apparatus 4c. <u>Develop</u> questionnaire for the energy audit of the given facility 4d. Prepare the energy audit report for the given apparatus.	4.5 Electricity Billing for LT and HT consumer.
Unit – V Energy Efficient Technologies In Electrical Systems	CO 5. <u>Choose</u> the energy efficient technology in electrical systems.	5.1 Automatic Power Factor Controllers 5.2 Energy Efficient Motors. 5.3 Variable Speed Drives and variable frequency Drives. 5.4 Soft Starter. 5.5 Energy Efficient Transformers. 5.6 Electronic Ballasts. 5.7 Energy Efficient Lighting. 5.8 Electrical Furnaces.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Electrical Energy Conservation	13	3	5	5	2	0	0	15
II	Energy Management	15	3	3	5	3	2	2	18
III	Energy storage system	13	2	2	2	2	2	0	10
IV	Energy Audit	11	0	3	4	4	2	2	15
V	Energy Efficient Technologies In Electrical Systems	08	2	3	5	0	2	0	12
Total		60	06	17	17	18	06	06	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Star label calculation:** choose the different appliances for star label calculation.
- 2) **Load Calculation:** Collect the specifications of different load in university campus.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding audit methodology of different utility used in industries.
- b) Prepare 15 min. power point presentation on different topics of energy audit, conservation and management.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
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Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Electrical Energy Conservation	1-2-3	1a. Identify different methods used for energy conservation. 1b. Use different methods to perform of energy conservation. 1c. Explain energy management concept.	Lecture with media	PPTs, black board
Unit– II Energy Management	4-5-6	2a. Calculate and find costing of energy conservation project. 2b. Do Case study of energy conservation. 2c. Improve power factor in electrical system. 2d. Optimize solution steps for conservation & management	Lecture With media Demonstration	PPTs, black board
Unit– III Energy storage system	7-8-9	3a. Choose the efficient method for energy storage. 3b. Select method as per applications.	Lecture with media, Tutorial	PPTs, black board
Unit-IV Energy Audit	10-11	4a. Describe concept of energy audit in detail. 4b. Use different tools of electrical energy audit. 4c. Perform energy audit in CHARUSAT campus.(3 month)	Lecture with media, Case study	PPTs, black board
Unit – V Energy Efficient Technologies In Electrical Systems	12-13	5c. Study different technology available in market. 5d. Select energy efficient electrical equipment's in industries.	Lecture with media, + PPT	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Guide Book no 1 to 4 for national Certification Examination for Energy Manager & Auditor	Bureau of energy efficiency (BEE)	Bureau of energy efficiency
2	Art of reading Electricity bill	Talware Yogendra	Danyatavyaorakashan
3	Energy Engineering	Chakrabarti ,Amlan	PHI Learning Private Limited

S. No.	Title of Book	Author	Publication including ISBN
	and Management		
4	Energy Audit in Electrical Utilities	Rajiv Shankar	Viva books Pvt.
5	Non – conventional energy sources	G. D. Rai	G. D. Rai
6	Industrial Energy Conservation Techniques	K. Nagabhusan Raju	Atlantic Publishers & Distributors (P) Ltd.

XV SOFTWARE/LEARNING WEBSITES

1. www.energymanagertraining.com
2. www.bee-india.gov.in

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE375	Semester V														
EE375	Competency Perform Electrical Energy Audit for Conservation of the Energy.	3	3	3	2	3	2	2	-	2	2	3	1	3	1
EE375.1	<u>Use</u> the different energy management tools and techniques	2	3	3	2	3	-	2	-	3	3	2	2	3	2
EE375.2	<u>Perform</u> the Energy Conservation opportunities in different area.	2	3	-	-	-	-	-	-	-	-	-	3	3	-
EE375.3	<u>Interpret</u> the suitable energy	1	-	1	-	-	2	-	-	-	-	-	-	1	-

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	storage system.														
EE375.4	<u>Perform</u> the Electrical Energy Audit of HT and LT Consumer.	3	-	3	-	3	-	-	3	3	3	3	3	3	3
EE375.5	<u>Choose</u> the energy efficient technology in electrical systems.	2	-	-	3	-	2	-	-	-	3	-	3	3	3

XVII NAMES OF RESOURCE PERSONS

- Vibha Parmar
-

OCEE3002: Computer Vision and Image Processing - Fundamentals and Applications

Description:

This course OCEE3002 – Computer Vision and Image Processing - Fundamentals and Applications is offered from SWAYAM

URL: https://onlinecourses.nptel.ac.in/noc23_ee39/preview

Credit and Week:

Teaching Scheme	Week	Theory	Practical	Total
Marks	12	100	-	100
Credit	-	4	1	5

About this course:

The intent of this course is to familiarize the students to explain the fundamental concepts/issues of Computer Vision and Image Processing, and major approaches that address them. This course provides an introduction to computer vision including image acquisition and image formation models, radiometric models of image formation, image formation in the camera, image processing concepts, concept of feature extraction and selection for pattern classification/recognition, and advanced concepts like motion estimation and tracking, image classification, scene understanding, object classification and tracking, image fusion, and image registration, etc.

This course will cover the fundamentals of Computer Vision. It is suited for mainly students who are interested in doing research in the area of Computer Vision. After completing the course, the students may expect to have the knowledge needed to read and understand more advanced topics and current research literature, and the ability to start working in industry or in academic research in the field of Computer Vision and Image Processing. They can also apply all these concepts for solving the real-world problems.

Industry Support:

The software industries that develop computer visions apps would be benefitted from this course.

Course Layout:

Week 1: Introduction to Computer Vision and Basic Concepts of Image Formation: Introduction and Goals of Computer Vision and Image Processing, Image Formation Concepts.

Week 2: Fundamental Concepts of Image Formation: Radiometry, Geometric Transformations, Geometric Camera Models.

Week 3: Fundamental Concepts of Image Formation: Camera Calibration, Image Formation in a Stereo Vision Setup, Image Reconstruction from a Series of Projections.

Week 4: Image Processing Concepts: Image Transforms.

Week 5: Image Processing Concepts: Image Transforms, Image Enhancement.

Week 6: Image Processing Concepts: Image Filtering, Colour Image Processing, Image Segmentation

Week 7: Image Descriptors and Features: Texture Descriptors, Colour Features, Edges/Boundaries.

Week 8: Image Descriptors and Features: Object Boundary and Shape Representations.

Week 9: Image Descriptors and Features: Interest or Corner Point Detectors, Histogram of Oriented Gradients, Scale Invariant Feature Transform, Speeded up Robust Features, Saliency

Week 10: Fundamentals of Machine Learning: Linear Regression, Basic Concepts of Decision Functions, Elementary Statistical Decision Theory, Parameter Estimation, Clustering for Knowledge Representation, Dimension Reduction, Linear Discriminant Analysis.

Week 11: Applications of Computer Vision: Artificial Neural Network for Pattern Classification, Convolutional Neural Networks, Autoencoders.

Week 12: Applications of Computer Vision: Gesture Recognition, Motion Estimation and Object Tracking, Programming Assignments.

Books and references:

1. Forsyth & Ponce, “Computer Vision-A Modern Approach”, Pearson Education.
2. M.K. Bhuyan , “ Computer Vision and Image Processing: Fundamentals and Applications”, CRC Press, USA, ISBN 9780815370840 - CAT# K338147.
3. Richard Szeliski, “Computer Vision- Algorithms & Applications”, Springer.

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course .

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Guwahati .It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

CE391: PYTHON FOR DATA ANALYTICS

B.TECH. 5TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – II

(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Objectives of the Course:

The main objectives of the course are:

- To provide comprehensive knowledge of python programming paradigms required for Data Science
- To get familiar with the main elements of Python programming
- To understand the basic concepts of Python
- To become Proficient in using the Most Common Python Data Science Packages like Numpy, Pandas & Matplotlib
- To understanding how to manipulate and analyze uncurated datasets
- To execute the basic statistical analysis methods

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	NumPy	10
2	Pandas	12
3	Matplotlib	10
4	Statistics for data analytics	13

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

C. Detailed Syllabus:

1	NumPy	10 Hours	22%
1.1	NumPy Introduction		
1.2	Create NumPy Arrays		
1.3	Data Type Object, type		
1.4	Numerical Operations on NumPy Arrays		
1.5	Changing the Dimensions of Arrays		
1.6	Python, NumPy and Probability		
1.7	Creation of Synthetic Test Data		
1.8	Boolean Masking of Arrays		
1.9	Matrix Arithmetic		
1.10	Reading and Writing and arrays		
2	Pandas	12 Hours	27%
2.1	Pandas Data Frames		
2.2	Replacing Values in Data Frames and Series		
2.3	Pandas Data Files		
2.4	Dealing with NaN		
2.5	Binning Data		
2.6	Pandas Tutorial Continuation: multi-level indexing		
2.7	Data Visualization with Pandas and Python		
2.8	Python, Pandas and Timeseries		
2.9	Estimating the number of Corona Cases with Python and Pandas		
3	Matplotlib	10Hours	22%
3.1	Pyplot API		
3.2	Simple Plot		
3.3	Figure Class		
3.4	Axes Class		
3.5	Multiplot		
3.6	Subplots () Function		
3.7	Formatting Axes		
3.8	Setting Ticks and Tick Labels		
3.9	Bar Plot		
3.10	Histogram		
3.11	Pie Chart		

- 3.12 Scatter Plot
- 3.13 Three-dimensional Plotting

4	Statistics for data analytics	13Hours	29%
4.1	Mean, Median, Mode		
4.2	Standard Deviation, Variance		
4.3	Variable Distribution		
4.4	Normalization		
4.5	Statistical Modeling using Auto Regression and Auto		
4.6	Correlation		

D. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted as per pedagogy as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand mechanisms for missing data and analytic implications (Understand)
CO2	Build and code a Graphical User Interface (GUI) to run a program
CO3	Understand and use python data science libraries as a tool for data analytics (Apply)
CO4	Implement numerical programming, data handling and visualization through NumPy, Pandas and Matplot-lib modules. (Create)

CO5	Analyze the significance of python program development environment by working on real world examples (Create)
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Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	1	-		-	1	-	-	2	-	-	-	-	-
CO2	-	1	-	2	2	3	2	-	-	-	-	-	-	-
CO3	1	-	2		2	2	3	2	2	2	2	2	-	-
CO4	-	1	2	2	2	3	3	2	-	3	3	3	2	-
CO5	1	2	-	2	2	3	3	2	2	-	3	3	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Text Books:

- 1) William McKinney, “ Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Ipython” O’Reilly Media, Inc.,
- 2) Nagar, Sandeep, “Introduction to Python for Engineers and Scientists - Open Source Solutions for Numerical Computation”, Apress, 2018.
- 3) Jesus Rogel-Salazar, “Data Science and Analytics with Python”, CRC Press, Taylor & Francis Group, LLC

Reference Books:

- 1) Magnus Lie Hetland, "Beginning Python from Novice to Professional", Third Edition, Apress, 2017
- 2) Mark Lutz "Learning Python", 4th Edition, O’REILLY, 2016

Web Materials:

- 1) <https://nptel.ac.in/courses/106/106/106106212/>
- 2) <https://matplotlib.org/gallery/index.html>
- 3) <https://realpython.com/python-matplotlib-guide/>
- 4) <https://www.tutorialspoint.com/python/index.htm>

LIST OF PRACTICALS

Practical No.	<u>Name of Practical</u>
1	Write Python programs to understand control structures.
2	Write Python programs to understand list and tuples.
3	Use conditional statements and loops in Python programs.
4	Write python programs to create functions and use functions in the program.
5	Import module and use it in Python programs.
6	Write python program to plot data using PyPlot.

EE391: CONTROL OF ELECTRIC MOTORS FOR VEHICULAR APPLICATIONS
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)
Minor Specialization Subject-2
Institute Elective

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

As electrical power generation and economy is essential gradient for the industrial and all around development of any country, the objectives of the course are:

- To understand the different electrical machines used for vehicle application.
- To design the DC motor control with sensor as well as without sensor.
- To understand the control of induction motor with respect to electric vehicle.
- To understand the control strategies for PM motors.
- To understand the brushless motor drive and magnet less motor drive

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Electrical Machines for Vehicular Applications	15
2	DC Motor Dynamics & Control	10
3	Induction Motor Dynamics & Control	11
4	Special Electric Motor Dynamics & Control	24

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

E. Detailed Syllabus:

1	Electric Machines for Vehicular Applications	15	25
		Hours	%
	Overview of conventional ac and dc motors, special electric machines such as permanent magnet machines, switched reluctance machines, linear motors, machine and engine rating, requirements, torque – speed characteristics and speed control, cooling of electric machines, testing of machines and testing standards		
2	DC Motor Dynamics & Control	10	16.
		Hours	66
			%
	Various schemes of closed loop control using sensors, Sensor less operation of drive, DC motor drive using PWM rectifiers.		
3	Induction Motor Dynamics & Control	11	18.
		Hours	34
			%
	Principles of soft starting, Loss minimization, Various schemes of closed loop control, Direct and indirect field oriented control of induction motor, Direct torque control.		
4	Special Electrical Motors Dynamics & Control	24	40
		Hours	%
	- Switched Reluctance Motor Drive: Structure of SR Machines, Principle of SR Machines, Converter configurations, Modified converter topology, Control strategies, Sensor and sensor less control of SRM, Reduction of torque pulsation. - Stator-PM Motor Drives : Doubly-Salient PM Motor Drives, Flux-Reversal PM Motor Drives, Flux-Switching PM Motor Drives, Hybrid-Excited PM Motor Drives Flux-Mnemonic PM Motor Drives, Design Criteria of Stator-PM Motor Drives for EVs, Application, Advantages and Limitations for EVs. - Brushless Motor Drive: Fundamentals of permanent magnet motors, Control strategies, Vector control of the Permanent Magnet Motor drive, Direct torque control of Permanent Magnet Motor drive, Sensor less control, Reduction of torque pulsations, Parameter sensitivity of drive. - Advanced Magnet less Motor Drives: Introduction of Advanced Magnet less technology,		

Synchronous Reluctance Motor Drives, Doubly-Salient DC Motor Drives, Flux-Switching DC Motor Drives

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	25	20	15	15	5

G. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand the different electrical machines used for vehicle application.
CO2:	Design the DC motor control with sensor as well as without sensor.
CO3:	Understand the control of induction motor with respect to electric vehicle
CO4:	Understand the control strategies for PM motors
CO5:	Understand the brushless motor drive and magnet less motor drive

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	-	-	-	-	-	-	-	-	1	-	-
CO2:	3	2	3	1	2	2	-	-	1	-	-	2	1	-
CO3:	3	2	1	1	1	-	-	-	-	-	-	1	1	-
CO4:	3	2	1	1	1	-	-	-	-	-	-	1	1	-
CO5:	3	1	-	-	1	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.

- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- In lab session, the student will be going through the various motors drive control strategies and simulations which will help to make some demo drives of the various motors used for electrical vehicles.

I. Recommended Study Material:

Text Books:	
1.	Mehrdad Ehsani, Yimin Gao and Ali Emadi, "Modern Electric, Hybrid Electric and Fuel cell vehicles: Fundamentals, Theory and Design", CRC Press, 2004.
2.	Electric Vehicle Machines and Drives Design, Analysis and Application by K. T. Chau, Wiley-IEEE Press; 1st edition
3.	Advanced Electric Drive Vehicles (Energy, Power Electronics, and Machines) by Ali Emadi, CRC Press; 1st edition (9 December 2014)
Reference Books:	
1.	Electric Vehicle Technology Explained by James Larminie & John Lowry, Wiley
Web Material:	
1.	https://nptel.ac.in/courses/108/106/108106170/
2.	https://onlinecourses.nptel.ac.in/noc21_ee112/
3.	https://nptel.ac.in/courses/108/102/108102121/
Other Material:	
1.	--

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – VI)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

Computer and Programmable device can process discrete information and discrete time systems can be implemented inside. The electrical engineer need to apply signal processing algorithm and design the discrete systems for products like, Numerical protection relays, close loop power electronic controller, digital twin, measuring and analysis instruments etc. Development of a real time system covers various steps such as, developing discrete mathematical model/ set of equation, verify its stability and remodifying the operational characteristics by inculcating several signal processing algorithm and discrete compensators. So before programming the microcontroller or similar programmable discrete device, the system designer has to develop and test the algorithm in software. At-least, the basic signal processing algorithms and digital system design can be taught, leaving the capacity of individual to think out of box, correlate the learning and known things to find and develop new innovative algorithm.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Test, Verify and implement Signal processing algorithms in MATLAB and similar software: The design-development and Research industries hire 'Domain Knowledge Software Engineer' for developing/implementing and testing the research hypothesis and algorithms.
- Design, implementation and Testing of Discrete Digital Systems.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) Understand basic Signal Processing concepts and Find source of error and problem associated in Signal Processing, Select Processor and Peripherals for signal Processing application.
- 2) Design and realize/implement Discrete system from time domain equation and S domain transfer function.
- 3) Design and realize/implement Digital systems like Digital Compensator and Digital Filters.
- 4) Understand the theory and Mathematics of various signal processing terms, algorithms and Transforms. Apply it and interpret the result, simulate and implement it.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **20 marks** are for **attendance, micro-project assessment** and other **assignment** if given any, to facilitate integration of COs and the remaining 10 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Identify</u> the presence/occurrence of Aliasing Phenomenon in MATLAB environment and <u>recommend</u> appropriate solution in terms choice of sampling frequency and specification of antialiasing filter.	01
2	4	<u>Code</u> various Mathematical equations & Operation in MATLAB, that can be useful in implementing signal processing equations/transforms/ algorithms. Also <u>determine</u> numbers of operations, time and memory requirement. (Example equations: Various statistics terms, Feature extracting terms, Integration, Differentiation, Shifting, Convolution, Correlation, DFT, Continuous Wavelet Transform etc.)	04
3	2,3	<u>Realize</u> the given electrical and Mechanical engineering systems on digital platform and <u>Verify</u> the response.	04
4	2,3,4	<u>Design</u> a digital close loop system and <u>analyze-verify</u> the response after incorporating the Digital Compensator (P, PI, PID, Lag Lead Compensator).	02
5	3	<u>Design</u> various IIR and FIR Filters with Hand calculation and Excel, Python and MATLAB.	03
6	3,4	<u>Simulate</u> and <u>Verify</u> the operation of various IIR and FIR filters in MATLAB, dSPACE and Microcontroller.	03
7	3,4	<u>Design</u> and <u>Simulate</u> Mutli-Resolution Wavelet Filter bank and Wavelet Packet Filter banks in MATLAB.	02
8	2,3,4	<u>Implement</u> Signal Processing algorithms for power system and Power Electronics applications in MATLAB, dSPACE Real time digital simulator and Microcontroller. (Special interest are Fourier and Wavelet based algorithm for protection, event detection and power quality assessment)	04
9	2,3,4	<u>Test- Verify</u> and <u>choose</u> various concepts/properties related to Signal Processing algorithms for and in given application. The algorithms to be considered are: Feature extraction criteria, statistical equations, correlation, DFT, S Transform, Wavelet, IIR/FIR Filters. Various example of failure of algorithm, in giving desired output will generate insight about selection of algorithm. Few of the selection criteria needs to be covered are time resolution, frequency resolution, correct frequency domain representation, delays introduced in filtering for removal of particular harmonic, response in presence of random signal, decaying dc and non-integer multiple harmonics)	03
10	3	<u>Simulate</u> Signal processing algorithms on image in MATLAB.	04
		Total	30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

Sr. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version higher than 2016a), Python (An Open source compiler), Microsoft Excel or Libre Calculator (An Open source software)	All (1 to 10)
2	dSPACE Real Time Digital Simulator	1, 2,4,5,7
3	Fast Microcontroller with ADC and DAC (Like Texas DSP, ARM M4)	6,8
4	Signal Generator and Digital Storage Oscilloscope, Harmonic Analyzer	6,8
5	Converter Drive /Grid connected converter	8
6	Camera interface	10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Test – verify result by simulation before believing the algorithm. Develop the boundary with in which algorithm will success.
- 2) Think out of box for new invention of algorithm. Anything learned/unlearned can be useful for new research at any point of research and development. Learn and make habit of noting down the conclusions and mathematics involved.
- 3) Follow ethics and predict about the prospective damage by a wrong algorithm. Don't hide bad side of your algorithm from yourself and others.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – I Introduction to Signal Processing	CO 1. <u>Understand</u> basic Signal Processing concepts and <u>Find</u> source of error and problem associated in Signal Processing, <u>select</u> Processor and Peripherals for signal Processing application. CO 4. <u>Understand</u> the theory and Mathematics of various signal processing terms, algorithms and Transforms. <u>Apply</u> it and interpret the result, simulate and implement it.	1.1 Basic block diagram of Digital processing system, 1.2 Sampling and Sampling Theorem for Reconstruction of signal in Digital domain, Aliasing and Solution, Re-construction Filter, 1.3 Errors and noise,

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
	PrO 18. <u>Identify</u> the presence/occurrence of Aliasing Phenomenon in MATLAB environment and <u>suggest</u> and <u>apply</u> appropriate solution in terms choice of sampling frequency and specification of antialiasing filter. PrO 19. <u>Code</u> various Mathematical equations & Operation in MATLAB, that can be useful in implementing signal processing equations/transforms/ algorithms. Also <u>determine</u> numbers of operations, time and memory requirement. PrO 9. <u>Test- Verify</u> and choose various concepts/properties related to Signal Processing algorithms for and in given application. PrO 10. <u>Simulate</u> Signal processing algorithms on image in MATLAB. 1a. <u>Justify</u> the need of proper sampling rate and <u>decide</u> it for given case. 1b. <u>Identify</u> source of error and <u>Quantify</u> various error in processing the signal on a given hardware. 1c. <u>Design</u> Antialiasing Analog Filter. 1d. <u>Code</u> basic signal processing math like signal statistics, Integration, Differentiation, Interpolation, Time Shifting Convolution, correlation etc.	1.4 Mathematical and graphical representation of various signals and basic operations over it, 1.5 Various Discrete systems and their characterization, Convolution and Correlation. 1.6 Introduction to DSP Processor.
Unit– II Z Transform and Realization of Discrete System	CO 2. <u>Design</u> and realize/implement Discrete system from time domain equation and S domain transfer function. CO 3. <u>Design</u> and <u>realize/implement</u> Digital systems like Digital Compensator and Digital Filters. PrO 3. <u>Realize</u> the given electrical and Mechanical engineering systems on digital platform and <u>Verify</u> the response. PrO 4. <u>Design</u> a digital close loop system and <u>observe-verify</u> the response after incorporating the Digital Compensator (P, PI, PID, Lag Lead Compensator). 2a. <u>Devise Discrete systems from continuous equations or system.</u> 2b. <u>Design</u> digital compensator for a close loop system. 2c. <u>Determine</u> the stability of Discrete System.	2.1 Definition and ROC for Z Transform, 2.2 S plane to Z plane mapping using various techniques like impulse invariant and Bilinear Transformation, 2.3 Inverse Z Transform, 2.4 Realization of IIR& FIR Systems (Direct, Cascade and Parallel form Realization), 2.5 Discrete Integrator and Differentiator
Unit– III Fourier Analysis	CO 4. Understand the theory and Mathematics of various signal processing terms, algorithms and Transforms. Apply it and interpret the result, simulate and implement it.	3.1 Continuous time and Discrete time Fourier Series (CFS and DFS) and their properties,

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
	PrO 8. <u>Implement</u> Signal Processing algorithms for power system and Power Electronics applications in MATLAB, dSPACE Real time digital simulator and Microcontroller. PrO 9. <u>Test- Verify</u> and choose various concepts/properties related to Signal Processing algorithms for and in given application. PrO 10. <u>Simulate</u> Signal processing algorithms on image in MATLAB. 3a. <u>Code the Discrete Fourier Transform, draw spectrogram/harmonic spectrum, of signal and calculate numbers of operation required in DFT Operation.</u> 3b. <u>Determine</u> the Sampling frequency for extraction of harmonic components. 3c. Write equation of each stage of FFT for given total N point sample set (Using DIT and DIF butterfly diagram, means without deriving the equations), and determine numbers of operation required. 3d. <u>Implement</u> the DFT, STFT and FFT on Micro-controller 3e. <u>Judge</u> the performance of Fourier Transform for power system signals.	3.2 Discrete Time Fourier Transform(DTFT), 3.3 Discrete Fourier Transform(DFT), 3.4 Computational Advantage of Fast Fourier Transform(FFT) Over DFT, Bit Reversed ordering, 3.5 Decimation in time (DIT) & Decimation in frequency (DIF) Algorithm for implementation of FFT.
Unit-IV Filter Design	CO 3. <u>Design and realize/implement</u> Digital systems like Digital Compensator and Digital Filters. PrO 5. <u>Design</u> various IIR and FIR Filters with Hand calculation and Excel, Python and MATLAB. PrO 6. <u>Simulate</u> and Verify the operation of various IIR and FIR filters in MATLAB, dSPACE and Microcontroller. PrO 7. <u>Design and Simulate</u> Multi-resolution Wavelet Filter bank and Wavelet Packet Filter banks in MATLAB. PrO 8. <u>Implement</u> Signal Processing algorithms for power system applications in MATLAB, dSPACE Real time digital simulator and Microcontroller. PrO 9. <u>Test- Verify</u> and choose various concepts/properties related to Signal Processing algorithms for and in given application. PrO 10. <u>Simulate</u> Signal processing algorithms on image in MATLAB. 4a. <u>Select the specification of filter for filtering the signal.</u> 4b. <u>Design</u> different types of IIR Filter for given specification.	4.1 Introduction to Digital Filters and their classification, Applications, 4.2 Design of IIR and FIR Filters using various methods, their comparison. 4.3 Introduction to adaptive Filtering concepts. 4.4 Multi-rate Signal Processing: Concept and Applications, decimation, Interpolation, Sampling rate conversion by rational factor, Multi-rate filter banks.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	4c. <u>Design</u> different types of FIR filter for given specification. 4d. <u>Implement</u> the IIR and FIR Filter on Microcontroller and dSPACE Real Time Simulator. 4e. <u>Decide</u> if application really requires adaptive filtering. 4f. <u>Implement</u> Multi rate Filters.	

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction to Signal Processing	10	1	3	2	2	2	0	10
II	Z Transform and Realization of Discrete System	10	2	4	10	2	2	0	20
III	Fourier Analysis	10	2	4	10	2	2	0	20
IV	Filter Design	15	10	4	12	1	1	0	30
Total		45	15	15	34	7	7	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. It should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

Just to give idea, A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here:

- 1) **Object identification:** Review various object identification methods. Write Matlab code and C code that is based on related mathematics or algorithms (Don't use ready libraries available in software) and verify the result for various images and videos.
- 2) **Posture identification:** Identify, alarm and announce who is sleeping (or degree on concentration and involvement) in class (where lectures are conducted) . Use camera and code image processing algorithms for the same. (CO 2,4)

- 3) **Harmonic analyzer:** code STFT and wavelet in microcontroller and give display on computer screen or GLCD.(co 3)
- 4) **Wavelet based Power Quality measurement:** Download research papers for Wavelet based Power quality indices and implement it in microcontroller. (CO 3,4)
- 5) **Impedance Relay:** Use Fourier transform (STFT) to find fundamental component of voltage current and implement the Impedance relay on Microcontroller. (CO 4)
- 6) **VHDL/Verilog implementation of Specialized DSP algorithm:** Select the algorithm and code on FPGA. The algorithm selected should be much more complex to be executed by microcontroller in real time and hence the use of FPGA should be justified. (CO 1,2,3,4)

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

The teacher can use following strategies, to accelerate the attainment of the various types of learning outcomes in this course:

- a) Tutorials that need hand calculation and testing in Software.
- b) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance may to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) The teacher can suggest several research paper to read and refer. This can be assign to specially to evaluate the capacity of individual, to understand topics related to signal processing but not in syllabus. The capacity can be judge by the way or quality of efforts put, the simulation developed to verify the algorithm and brain storming done and presented to evaluator in form of presentation.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Make a report on new signal processing algorithms, their application in Power Engineering and their benefit over classical Research publication. You may refer IEEE Transaction on signal processing and similar high impacting journals for the same.
- b) Identify a hypothetical non existing as well as existing phenomenon which can fail certain signal processing algorithm.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introducti on to Signal Processing	1	<i>Pro 1. Identify the presence/occurrence of Aliasing Phenomenon in MATLAB environment and give appropriate solution in terms choice of sampling frequency and specification of antialiasing filter. (1 hour)</i> 1a. <u>Get</u> overall idea of signal Processing hardware. 1b. <u>Know</u> Role of Signal Processing in various engineering domain, to explore the application of this field, which can help student to identify and plan his/her dream signal processing related job.	Lecture + Experimentation	Mixture of black board and ICT

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		1c. <u>Justify</u> the need of proper sampling rate and calculate/Decide it. 1d. <u>Design</u> Antialiasing Analog Filter and Reconstruction Filter.		
	2	PrO 2. <u>Code</u> various Mathematical equations & Operation in MATLAB, that can be useful in implementing signal processing equations/transforms/ algorithms. Also <u>determine</u> numbers of operations, time and memory requirement. (2 Hours) 1e. <u>Determine</u> source of problem and <u>quantify</u> various error involved in processing the signal on a given hardware.	Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.	Mixture of black board and ICT
	3	PrO2. Continued. For 2 Hours. The outcomes will be achieved by end of this week. 2a. <u>Devise</u> Discrete Systems from continuous equations and systems.	Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.	Mixture of black board and ICT
Unit– II Z Transform and Realization of Discrete System	4	PrO 3. <u>Realize</u> the Given electrical and Mechanical engineering systems on digital platform and Verify the response. (2 Hours) 2b. <u>Design</u> digital compensator for a close loop system.	Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.	Mixture of black board and ICT
	5	PrO3. (Continued from last lab session for 2 Hours). Outcome will be achieved by end of this week. PrO 4. <u>Design</u> a digital close loop system and observe-verify the response after incorporating the Digital Compensator (P, PI, PID, Lag Lead Compensator). 2c. <u>Determine</u> the stability of Discrete System.	Lecture + Experimentation + Solved/Unsolved Tutorials from Resource Person.	Mixture of black board and ICT
	6	PrO 8. <u>implement</u> Signal Processing algorithms for power system and Power Electronics applications in MATLAB, dSPACE Real time digital simulator and Microcontroller. (Part 2: The DFT FFT based application for Power system) 3a. <u>Code</u> the Discrete Fourier Transform, draw spectrogram and harmonic spectrum, and calculate numbers of operation required. 3b. <u>Determine</u> the Sampling frequency for extraction of harmonic components using DFT.	Lecture + Experimentation	Mixture of black board and ICT
Unit– III Fourier	7	Note: Lab hours will be utilized in this week for lecture.	Lecture only.	Mixture of black

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Analysis		3c. <u>Draw</u> out equation of each stage of FFT for given total N point sample set (Using DIT and DIF butterfly diagram, means without deriving the equations), and determine numbers of operation required. 3d. <u>Judge</u> the performance of Fourier Transform for given signal (e.g for power system signals) . 3e. Decide if FT based algorithm can be used in required application or not. 3f. <u>Find</u> out time domain information from frequency domain information.		board and ICT
	8	PrO8. Part 1: Statistical equation, correlation and other such algorithm based applications (Continued from week 6 for 1 hour) 3a. <u>Find</u> out time domain information from frequency domain information. 3b. <u>Implement</u> the DFT, STFT and FFT on Micro Controller	Lecture + Experimentation.	Mixture of black board and ICT
	9	Note: Lab hours in this week will be utilized for lecture. 4a. <u>Select</u> the specification of filter for filtering the signal. 4b. <u>Design</u> different types of IIR Filter for given specification.	Lecture only.	Mixture of black board and ICT
Unit– IV Filter Design	10	Note: Lab hours in this week will be utilized for lecture. 4b. continued.... Outcome will be achieved by end of this week.	Lecture + Experimentation	Mixture of black board and ICT
	11	Note: 1 Hour of lecture will be utilized in laboratory during this week. Pro 5. <u>Design</u> various IIR and FIR Filters with Hand calculation and Excel, Python and MATLAB. (Part 1: Design of IIR Filters: 3 Hours) 4c. <u>Design</u> different types of FIR filter for given specification.	Lecture	Mixture of black board and ICT
	12	Pro 6. <u>Simulate</u> and Verify the operation of various IIR and FIR filters in MATLAB, dSPACE and Microcontroller. 4d. <u>Implement</u> the IIR and FIR Filter on Microcontroller and dSPACE Real Time Simulator.	Lecture + Experimentation	Mixture of black board and ICT
	13	Pro 7. <u>Design</u> and Simulate Multi-resolution Wavelet Filter bank and Wavelet Packet Filter banks in MATLAB. 4e. <u>Decide</u> if application really requires adaptive filtering. 4f. <u>Implement</u> Multi rate Filters.	Lecture + Experimentation	Mixture of black board and ICT
	14	Pro8. Part 3: Wavelet based algorithm for power system application (Continued from week 8, for 1 hour). The outcome of Pro 8 will be achieved by end of this week. 4f. continued.	Lecture + Experimentation + Resource Person's	Mixture of black board and ICT

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		The outcome of 4f will be achieved by end of this week.	Material	
	15	Note: Lecture hours in this week will be utilized for lab. PrO 9. <u>Test-Verify</u> and choose various concepts/properties related to Signal Processing algorithms for and in given application. PrO 10. <u>Simulate</u> Signal processing algorithms on image in MATLAB.	Experimentation only.	Mixture of black board and ICT

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Digital Signal Processing	Anand Kumar	PHI Learning Private Ltd. ISBN-978-81-203-5071-7
2	"Real Time Digital Signal Processing: Fundamentals, Algorithms and Implementation using TMS Processor"	Udayashankara	Prentice-Hall Of India Pvt Limited, 2010 ISBN-10 8120340493 : ISBN-13 : 978-8120340497
3	Digital Signal Processing	Ramesh Babu	SCITECH Publications ISBN-10 818371630X : ISBN-13 978-8183716307 :
4	Digital Signal Processing Principles", Algorithms and Application	John Proakis, Dimitris G Manolakis	PHI, 3rd Edition (2000). ISBN-10 9788131710005 : ISBN-13 : 978-8131710005

XV SOFTWARE/LEARNING WEBSITES

1. nptel video course by prof.s.c. duttaroy, iitdelhi <http://nptel.iitm.ac.in/video.php?subjectid=117102060>
2. DSP Video Lectures by Prof. E. Ambikairajah: http://onlinevideolecture.com/?course_id=373
3. A beginner's guide to digital signal processing
http://www.analog.com/en/processorsdsp/processors/beginners_guide_to_dsp/fca.html

XVI PO-COMPETENCY-CO MAPPING

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Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE 346	Semester VI	Digital Signal Processing Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
346.1	Understand basic Signal Processing concepts, Find source of error and problem associated in Signal Processing, Select Processor and Peripherals for signal Processing application.	2	2	1	-	-	-	-	1	-	2	-	-	1	-
346.2	Design and realize/implement Discrete system from time domain equation and S domain transfer function.	2	3	2	2	3	-	-	1	-	2	-	1	3	-
346.3	Design and realize/implement Digital systems like Digital Compensator and Digital Filters.	2	3	3	2	3	-	-	1	-	2	-	1	3	-
346.4	Understand the theory and Mathematics of various signal processing terms, algorithms and Transforms. Apply it and interpret the result, simulate and implement it.	2	3	3	3	3	-	-	2	2	3	2	2	-	3

XVII NAMES OF RESOURCE PERSONS

a) Jivanadhar A. Joshi

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

In modern industry all machines are connected with each other and they operate automatically. This automated system requires PLCs, HMI and SCADA system to operate satisfactorily. This curriculum course structure is developed such that the student is able to understand basic functions of PLCs, Sensor and Actuators; and can design small scale automated systems.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Understand and Evaluate the operation of automated systems in industry

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- Enhance** skills in basics of PLC
- Know** the performance and classification of Input and Output Devices.
- Understand** different Programming language used in PLC
- Design** any automate control system for any machinery
- Program** PLC and apply their knowledge in the field of automation and control

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (Anna 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	200
2	0	4	4	70	30	50	50	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of Cos and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Understand basics of PLC and SCADA	4
2	2	Understand different programming language used to program PLC and merits and demerits of each language.	6
3	3	Gain knowledge of sensors and actuators.	4
4	4,5	Implement logic gates in Ladder language and on PLC (AND, OR, NOT, EX-OR, EX-NOR, NAND, NOR)	6
5	4,5	Design and implement motor latch and unlatch Ladder on PLC (with and without using instruction)	4
6	4,5	Design and implement Liquid level control of tank using PLC	4
7	4,5	Design and implement flashing of Lamp on PLC.	4
8	4,5	Design and implement traffic light control sequence using ladder logic on PLC.	4
9	4,5	Design and implement flashing of Lamp 10 times on PLC.	4
10	4,5	Design and implement Pattern of Lamp based on ADC (battery level indicator).	4
11	4,5	Design and implement flashing of LED with adjustable ON time	4
12	4,5	Interfacing of RTD with PLC.	4
13	4,5	Implement SCADA screens with PLC programs.	8
		Total	60

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Schneider Modicon M340 (PLC Controller) with its digital and analog peripherals(DDM16025,AMI0800, ART0414,AMO0410)	1,3,4,5,6,7,8,9,10,11,12,13
2	Unity pro XL (any version)	2, 4,5,6,7,8,9,10,11,12,13
3	Push Button, Toggle Switch	4,5,6,7,8,9,10,11,12,13
4	LEDs	4,5,6,7,8,9,10,11,12,13
5	10K POT with Digital Meter	10,11
6	Programmable Digital Voltmeter	10,11
7	RTD	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year

- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Basics of Control	CO 1. Enhance skills in basics of PLC 1a. Differentiate control scheme 1b. Evaluate the control strategy for the specified system 1c. Differentiate Relay logic and PLC logic 1d. Differentiate Analog , Automation and Sequential Logic 1e. Understand evolution in control strategy	1.1 Introduction to Sequential/ Logic Control 1.2 control strategy, control philosophy and control algorithm, 1.3 Difference between Relay logic control and PLC 1.4 Difference between Analog/ Automatic and Sequential/ Logic Control 1.5 Evolution of Control System
Unit– II Introduction to Programmable Logic Control (PLC)	CO 1. Enhance skills in basics of PLC <i>PrO 20. Understand basics of PLC and SCADA</i> 2a. Evaluate performance of PLC for given Application 2b. Understand need of PLC 2c. Evaluate operation of given PLC 2d. Justify need of PLC for given application	2.1 Definition of PLC, introduction to standard (IEC61131) used for PLC and basics of process automation 2.2 History and Evolution of PLC 2.3 Block Diagram of PLC 2.4 Advantages and Disadvantages of PLC
Unit– III Input and Output Devices	CO 2. Know the performance and classification of Input and Output Devices. <i>PrO 3. Gain knowledge of sensors and actuators</i> 3a. Evaluate operation of sensors for given application. 3b. Select the transmitter wiring under given situation. 3c. Evaluate operation of output device for given application. 3d. Evaluate operation of control valve for given application.	3.1 A brief overview of sensors and its types, Overview of Transmitter 3.2 difference between two wire Transmitter and four wire transmitter 3.3 output devices connected to PLC 3.4 Control Valves and its classification
Unit-IV PLC input and	CO 2. Know the performance and classification of Input and Output Devices.	

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Output Module	<i>PrO 1. Understand basics of PLC and SCADA</i> <i>PrO 3. Gain knowledge of sensors and actuators</i> <i>4a. <u>Select</u> the power supply for the <u>given</u> application</i> <i>4b. <u>Select</u> the appropriate input and output module for <u>given</u> application</i>	4.1 Power supply circuit/Wiring Diagram, Selection of power supply 4.2 Function of input and output modules, Classification and block diagram of Input and output Modules(AC input module, Discrete/Digital input module ,Special input module cards TTL Logic, Relay and Triac output module)and selection of cards/modules
Unit – V PLC Operation	<i>CO 5. <u>Program</u> PLC and apply their knowledge in the field of automation and control</i> <i>PrO 1. Understand basics of PLC and SCADA</i> <i>5a. <u>Understand</u> type of memory used in the <u>given</u> scenario</i> <i>5b. <u>Prepare</u> addressing table for the <u>given</u> application</i> <i>5c. <u>Determine</u> the operation of PLC in the <u>given</u> situation</i>	5.1 PLC memory types and its mapping, PLC Register basics 5.2 Addressing: Internal and External 5.3 PLC Scan Cycle and response time
Unit – VI PLC Programming	<i>CO 3. <u>Understand</u> different Programming language used in PLC</i> <i>CO 5. <u>Program</u> PLC and apply their knowledge in the field of automation and control</i> <i>PrO 4. Implement logic gates in Ladder language and on PLC (AND, OR, NOT, EX-OR, EX-NOR, NAND, NOR)</i> <i>PrO 5. Design and implement motor latch and unlatch Ladder on PLC (with and without using instruction)</i> <i>PrO 6. Design and implement Liquid level control of tank using PLC</i> <i>PrO 7. Design and implement flashing of Lamp on PLC.</i> <i>PrO 8. Design and implement traffic light control sequence using ladder logic on PLC.</i> <i>PrO 9. Design and implement flashing of Lamp 10 times on PLC.</i> <i>PrO 10. Design and implement Pattern of Lamp based on ADC (battery level indicator).</i> <i>PrO 11. Design and implement flashing of LED with adjustable ON time</i> <i>PrO 12. Interfacing of RTD with PLC.</i> <i>6a. <u>Understand</u> types of programming language available and use any of the language to</i>	6.1 Types of programming

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	<i>implement the given application</i> 6b. <u>Evaluate</u> operation of basic block of ladder logic for <u>the given program</u> 6c. <u>Design</u> ladder logic program for <u>the specified application</u>.	languages 6.2 Various PLC Instruction: NO, NC and output contacts and coils, Set, Reset, Timer (along with types), Counters (its types), PID, Logical, Arithmetic, Data handling, skip, MCR, jump, Bit operations and move instructions 6.3 Programming related to above instructions
Unit – VII Designing Automation System	CO 4. <u>Design</u> any automated control system for any machinery CO 5. <u>Program</u> PLC and <u>apply</u> their knowledge in the field of automation and control PrO 13. Implement SCADA screens with PLC programs. 7a. <u>Select</u> PLC for <u>the given</u> application/machinery. 7b. <u>Design</u> and <u>Prepare</u> detailed document for <u>the given</u> application/machinery. 7c. <u>Implement</u> automated system for the <u>given</u> application/machinery.	7.1 Selection of PLC 7.2 Documentation: System architecture, Piping and Instrumentation Diagram (P&ID) (Flow sheet symbols as per standard ISA S5.1 1984(R1992)), General Arrangement (GA) Drawing, Wiring Diagram, I/O Listing and program flowchart, Debugging (Simulation) 7.3 Commissioning, troubleshooting and maintenance of PLC system

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Basics of Control	02	3	2	0	0	0	0	5
II	Introduction to Programmable Logic Control (PLC)	02	3	2	0	0	0	0	5
III	Input and Output Devices	03	2	3	1	1	0	0	7
IV	PLC input and Output Module	04	3	2	1	3	0	0	9
V	PLC Operation	03	1	0	2	2	2	0	7
VI	PLC Programming	10	2	5	2	4	6	4	23
VII	Designing Automation System	06	0	0	1	4	6	3	14
Total		30	14	14	7	14	14	7	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Temperature Monitoring System:** Implement Temperature control of liquid in tank using PLC.
- 2) **Speed control of Induction Motor:** Control speed of induction motor using VFD. PLC will act as remote unit commanding/controlling VFD to control speed of motor.
- 3) **Automatic bottle filler:** Implement bottle filler using PLC. Where, each bottle will position automatically under dispenser and will be filled automatically without any human intervention.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different protection schemes used in industries.
- b) Prepare 15 min. power point presentation on how to implement control strategy for the machinery in particular industry.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Basics of Control	1	1a. <u>Differentiate</u> control scheme 1b. <u>Evaluate</u> the control strategy for <u>the specified</u> system 1c. <u>Differentiate</u> Relay logic and PLC logic 1d. <u>Differentiate</u> Analog , Automation and Sequential Logic 1e. <u>Understand</u> evolution in control strategy	Lecture with media	PPTs, black board
Unit– II Introduction to Programmable Logic Control (PLC)	2	<i>PrO 1. Understand basics of PLC and SCADA</i> 2a. Evaluate performance of PLC for given Application 2b. <u>Understand</u> need of PLC 2c. <u>Evaluate</u> operation of <u>given</u> PLC 2d. <u>Justify</u> need of PLC for <u>given</u> application	Lecture with media	PPTs, black board
Unit– III Input and Output Devices	3	<i>PrO 3. Gain knowledge of sensors and actuators</i> 3a. Evaluate operation of sensors for given application. 3b. <u>Select</u> the transmitter wiring under <u>given</u> situation. 3c. <u>Evaluate</u> operation of output device for <u>given</u> application.	Lecture with media	PPTs, black board
	4	<i>PrO 3. Gain knowledge of sensors and actuators</i> 3d. <u>Evaluate</u> operation of control valve for <u>given</u> application.	Lecture with media	PPTs, black board
Unit-IV PLC input and Output Module	4	<i>PrO 1. Understand basics of PLC and SCADA</i> 4a. <u>Select</u> the power supply for <u>the given</u> application	Lecture with media	PPTs, black board
	5	<i>PrO 1. Understand basics of PLC and SCADA</i> <i>PrO 3. Gain knowledge of sensors and actuators</i> 4a. <u>Select</u> the power supply for <u>the given</u> application 4b. <u>Select</u> the appropriate input and output module for <u>given</u> application	Lecture with media	PPTs, black board
	6	<i>PrO 3. Gain knowledge of sensors and actuators</i> 4b. <u>Select</u> the appropriate input and output module for <u>given</u> application	Lecture with media	PPTs, black board
Unit – V PLC Operation	6	<i>PrO 1. Understand basics of PLC and SCADA</i> 5a. <u>Understand</u> type of memory used in <u>the given</u> scenario	Lecture with media	PPTs, black board
	7	<i>PrO 1. Understand basics of PLC and SCADA</i> 5a. <u>Prepare</u> addressing table for <u>the given</u> application 5b. <u>Determine</u> the operation of PLC in the <u>given</u> situation	Lecture with media	PPTs, black board
Unit – VI PLC	8	<i>PrO 4. Implement logic gates in Ladder language and on PLC (AND, OR, NOT, EX-OR, EX-NOR, NAND,</i>	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Program ming		NOR) 6a. <u>Understand</u> types of programming language available and use any of the language to implement <u>the given</u> application 6b. <u>Evaluate</u> operation of basic block of ladder logic for <u>the given</u> program		
	9	PrO 5. <i>Design and implement motor latch and unlatch Ladder on PLC (with and without using instruction)</i> PrO 6. <i>Design and implement Liquid level control of tank using PLC</i> 6b. <u>Evaluate</u> operation of basic block of ladder logic for <u>the given</u> program 6c. <u>Design</u> ladder logic program for <u>the specified</u> application.	Lecture with media	PPTs, black board
	10	PrO 7. <i>Design and implement flashing of Lamp on PLC.</i> PrO 8. <i>Design and implement traffic light control sequence using ladder logic on PLC.</i> 6b. <u>Evaluate</u> operation of basic block of ladder logic for <u>the given</u> program 6c. <u>Design</u> ladder logic program for <u>the specified</u> application.	Lecture with media	PPTs, black board
	11	PrO 9. <i>Design and implement flashing of Lamp 10 times on PLC.</i> PrO 10. <i>Design and implement Pattern of Lamp based on ADC (battery level indicator).</i> 6b. <u>Evaluate</u> operation of basic block of ladder logic for <u>the given</u> program 6c. <u>Design</u> ladder logic program for <u>the specified</u> application.	Lecture with media	PPTs, black board
	12	PrO 11. <i>Design and implement flashing of LED with adjustable ON time</i> PrO 12. <i>Interfacing of RTD with PLC.</i> 6b. <u>Evaluate</u> operation of basic block of ladder logic for <u>the given</u> program 6c. <u>Design</u> ladder logic program for <u>the specified</u> application.	Lecture with media	PPTs, black board
Unit – VII Designing Automation System	13	PrO 13. <i>Implement SCADA screens with PLC programs.</i> 7a. <u>Select</u> PLC for <u>the given</u> application/machinery.	Lecture with media	PPTs, black board
	14	PrO 13. <i>Implement SCADA screens with PLC programs.</i> 7b. <u>Design</u> and <u>Prepare</u> detailed document for <u>the given</u> application/machinery.	Lecture with media	PPTs, black board
	15	PrO 13. <i>Implement SCADA screens with PLC programs.</i> 7c. <u>Implement</u> automated system for the <u>given</u> application/machinery.	Lecture with media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Programmable Logic Controllers	John W. Webb and Ronald A. Reis	PHI Publication 978-8120323087
2	Programmable Logic Controllers	W. Bolton	Newnes 978-0750681124
3	Programmable Logic Controllers & their Engineering Application	A.J. Crispin	McGraw Hill 978-0077093174
4	Programmable Logic Controllers	Thomas A. Hughes	ISA 978-1556178993
5	Programmable Logic Controllers	John R. Hackworth	Pearson Education 978-0130607188
6	Exploring Programmable Logic Controllers with Applications	Pradeep Kumar Srivastava	BPB Publications 978-8176569330
7	Programmable Logic Control Principles and Applications	NIIT	PHI Publication 978-8120325258

XV SOFTWARE/LEARNING WEBSITES

- <http://coep.vlab.co.in/?sub=33&brch=97>
- <http://www.plcdev.com/book/export/html/9>
- <http://www.plcmanual.com/>
- <http://nptel.ac.in/courses/108105062>
- <http://nptel.ac.in/courses/108105088>
- <http://nptel.ac.in/courses/108105063>
- www.siemensplcweb.com
- <http://www.ab.com/programmablecontrol/plc>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE347	Semester VI	Programmable Logic Controllers and Industrial Automation Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
347	Understand and Evaluate the operation of automated systems in industry	1	2	3	1	3				1			1	3	1

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
347.1	Enhance skills in basics of PLC	3													
347.2	Know the performance and classification of Input and Output Devices.	3												1	
347.3	Understand different Programming language used in PLC	2													
347.4	Design any automate control system for any machinery		3	3	3	2				2				3	
347.5	Program PLC and apply their knowledge in the field of automation and control		3	3	3	2				2				3	

XVII NAMES OF RESOURCE PERSONS

- Jignesh S Patel
-

UG Engineering Programme	Semester
Electrical Engineering	sixth

I RATIONALE

Electrical engineers have to deal with various types of *semiconductor devices*, power electronics circuits and converters are used for different dc drives. This curriculum course structure is developed such that the student is able to evaluate the performance of the various *semiconductor devices*, types of power electronic converters and drives for different industry drive applications.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the performance of various **SEMICONDUCTOR DEVICES, TYPES OF POWER ELECTRONICS CIRCUITS AND AC DRIVES**.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. Evaluate the performance of the various types of INVERTERS for different application
2. Evaluate the performance of the various types of power SUPPLIES for different application
3. Evaluate the performance of the various types of AC Voltage controllers for different application
4. Evaluate the performance of the various types of Cycloconverters for different application
5. Evaluate the performance of the various types of AC Drives for different application

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of**

Dave's Psychomotor Domain taxonomy by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1,2	Create the simulation of a single phase voltage source inverter and Current source inverter for different load condition using MATLAB software.	02
2	1,2	Create the simulation of 180°, 120° and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB software.	04
3	3	Analyze the waveform of 180°, 120° and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB	02
4	1,2	Create the simulation of Unipolar and Bi- polar Sine PWM (SPWM) technique using MATLAB software.	02
5	3	Analyze the waveform of Unipolar and Bi- polar Sine PWM (SPWM) technique in MATLAB.	02
6	1,2	Create the simulation of half bridge topology of SMPS using MATLAB software.	02
7	1,2	Create the simulation of full bridge topology of SMPS using MATLAB software.	02
8	1,2	Create the simulation of uninterruptible power supply (UPS) using MATLAB software.	02
9	1,2	Create the simulation of different types of AC voltage controller using MATLAB software.	04
10	3,4	Evaluate the Performance of different types of AC voltage controller for different load condition and firing angle delay using MATLAB software.	02
11	1,2	Create the simulation of step-up and step-down single- phase Cycloconverters using MATLAB software.	04
12	3	Analyze the Performance of the operation of variable frequency drive on hardware.	02
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,6,7,8,9,10,11
2	variable frequency drive hardware kit	12
3	DSO with isolation and its probes	12
4	Multimeter	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.

5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- 'Responding Level' in 1st year
- 'Valuing Level' in 2nd year
- 'Organising Level' in 3rd year
- 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Inverters	<u>CO 1. Evaluate the performance of the various types of INVERTERS for different application</u> <i>PrO 1. Create the simulation of a single phase voltage source inverter and Current source inverter for different load condition using MATLAB software.</i> <i>PrO 2. Create the simulation of 180°, 120° and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB software.</i> <i>PrO 3. Analyze the waveform of 180°, 120° and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB.</i> <i>PrO 4. Create the simulation of Unipolar and Bi-polar Sine PWM (SPWM) technique using MATLAB software.</i> <i>PrO 5. Analyze the waveform of Unipolar and Bi-polar Sine PWM (SPWM) technique in MATLAB.</i> 1a. <u>Justify</u> the need of three phase and single phase voltage source inverter 1b. <u>Justify</u> the need of three phase and single phase current source inverter. 1c. <u>Evaluate</u> the operation of inverters for different load condition. 1d. <u>Determine</u> switches used for inverters 1e. <u>Choose</u> Harmonic reduction techniques in inverter	1.1 Introduction and revision: Overview of switches 1.2 Introduction to inverters Voltage Source Inverter : Single-phase half bridge and full bridge inverter with different loads 1.3 Three-phase 180 and 120 degree mode inverter 1.4 Fourier analysis of Single-phase and Three-phase inverter 1.5 Output Voltage control in inverters 1.6 Harmonic reduction techniques in inverter output voltage 1.7 Current Source Inverter: Concept of CSI, CSI with different loads 1.8 Load and Self-commutated inverter: Single-phase Series Inverter and Parallel Inverter

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit– II Power Supplies	CO 2. Evaluate the performance of the various types of power SUPPLIES for different application PrO 6. Create the simulation of half bridge topology of SMPS using MATLAB software. PrO 7. Create the simulation of full bridge topology of SMPS using MATLAB software. PrO 8. Create the simulation of uninterruptible power supply (UPS) using MATLAB software. 2a. <u>Justify</u> the need of SMPS 2b. <u>Justify</u> the need of UPS 2c. <u>Evaluate</u> the operation of different SMPS	2.1 DC Power Supplies: Switched-Mode DC Power Supplies, 2.2 Half bridge converter, Full bridge converter, Fly back converter, Push-pull converter 2.3 AC Power Supplies: Uninterruptible Power Supplies (UPS) , Batteries used in UPS 2.4 Static switches and Solid State relays
Unit– III AC Voltage controllers	CO 3 <u>Evaluate the performance of the various types of AC Voltage controllers for different application</u> <u>PrO 9. Create the simulation of different types of AC voltage controller using MATLAB software</u> <u>PrO 10. Evaluate the Performance of different types of AC voltage controller for different load condition and firing angle delay using MATLAB software</u> 3a. <u>Evaluate</u> the performance of different types of AC Voltage controllers for different load condition. 3b. <u>Justify</u> the need of AC Voltage controllers	3.1 Voltage control techniques: Principle of voltage controllers, Phase control, Integral Cycle control, 3.2 Single-phase Voltage controllers with R and RL load Sequence control of AC voltage controllers: Two-stage and Multistage sequence control of voltage controllers
Unit-IV Cycloconverters	CO 4. <u>Evaluate the performance of the various types of Cycloconverters for different application</u> <u>PrO 11. Create the simulation of step-up and step-down single- phase Cycloconverters using MATLAB software</u> 4a. <u>Choose</u> the appropriate cycloconverter control strategy as per required application 4b. <u>Evaluate</u> the performance of different types of cycloconverter for different load condition. 4c. . <u>Justify</u> the need of cycloconverter	4.1 Introduction: Principle and Applications of Cycloconverters, Types of Cycloconverters, 4.2 Single-phase to single-phase Cycloconverters: Bridge-type Cycloconverters, Midpoint Cycloconverters, 4.3 Three-phase Half-wave Cycloconverters: Three-phase to Single-phase, Threephase to Three-phase, Output voltage Equation of a Cycloconverter
Unit – V AC Drives	CO 5. Evaluate the performance of the various types of AC Drives for different application PrO 12. Analyze the Performance of the operation of variable frequency drive on hardware. 5a. <u>Understand</u> the operation of VFD Drive in different mode condition. 5b. <u>Justify</u> the need of AC Drives for different application	5.1 Basic machine equations: Analysis and Performance of AC motors, 5.2 Speed Control of three-phase Squirrel-cage Induction Motors (SCIMs): Stator Voltage Control, Stator Frequency Control, Stator Voltage and Frequency Control, Stator Current Control, 5.3 Speed Control of three-phase Wound Rotor Induction Motors (WRIMs): Static Rotor Resistance Control, Slip-energy recovery

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
		Control.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Inverters	15	0	3	3	10	7	0	23
II	Power Supplies	05	0	1	1	4	2	0	8
III	AC Voltage controllers	06	0	1	4	4	1	0	10
IV	Cycloconverters	07	0	2	2	4	6	0	14
V	AC Drives	12	1	2	2	4	6	0	15
Total		45	1	9	12	26	22	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PROs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Inverter:** Simulate the inverter for particular one application.
- 2) **Power supply:** Design and Develop the hardware of full wave SMPS.
- 3) **Ac voltage controller:** Design and Develop the hardware of **Ac voltage controller**
- 4) **Cycloconverter:** Design and Develop the hardware of **Cycloconverter**.
- 5) **Ac drive:** Design the simulation of AC drives in MATLAB.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.

- b) '**1**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different Semiconductor switches used in converters.
- b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Inverters	1	PrO 1. Create the simulation of a single phase voltage source inverter and Current source inverter for different load condition using MATLAB software. PrO 2. Create the simulation of 180°, 120° and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB software. PrO 3. Analyze the waveform of 180°, 120° and 150 mode of conduction for 3-Φ Voltage source Inverter using MATLAB PrO 4. Create the simulation of Unipolar and Bi- polar Sine PWM (SPWM) technique using MATLAB software. PrO 5. Analyze the waveform of Unipolar and Bi- polar Sine PWM (SPWM) technique in MATLAB. 1h. <u>Justify the need of three phase and single phase voltage source inverter</u> 1i. <u>Justify the need of three phase and single phase current source inverter.</u> 1j. <u>Evaluate the operation of inverters for different load condition.</u> 1k. <u>Determine switches used for inverters</u> 1l. <u>Choose Harmonic reduction techniques in inverter</u>	Lecture with media	PPTs, black board, MATLAB Software
Unit– II Power	6	PrO 6. Create the simulation of half bridge topology of SMPS using MATLAB	Lecture with media,	PPTs, black board,

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Supplies		software. PrO 7. Create the simulation of full bridge topology of SMPS using MATLAB software. PrO 8. Create the simulation of uninterruptible power supply (UPS) using MATLAB software. 2a. Justify the need of SMPS 2b. Justify the need of UPS 2c. Evaluate the operation of different SMPS	and Demonstration	MATLAB Software
Unit– III AC Voltage controllers	8	PrO 9. Create the simulation of different types of AC voltage controller using MATLAB software PrO 10. Evaluate the Performance of different types of AC voltage controller for different load condition and firing angle delay using MATLAB software 3a. Evaluate the performance of different types of AC Voltage controllers for different load condition. 3b. Justify the need of AC Voltage controllers	Lecture with media, Field visit	PPTs, black board, MATLAB Software
Unit-IV Cycloconverters	10	PrO 11. Create the simulation of step-up and step-down single- phase Cycloconverters using MATLAB software 4a. Choose the appropriate cycloconverter control strategy as per required application 4b. Evaluate the performance of different types of cycloconverter for different load condition. 4c. Justify the need	Lecture with media, Tutorial	PPTs, black board, MATLAB Software
Unit – V AC Drives	15	PrO 12. Analyze the Performance of the operation of variable frequency drive on hardware. 5a. Understand the operation of VFD Drive in different mode condition. 5b. Justify the need of AC Drives for different application	Lecture with media, Tutorial	PPTs, black board, MATLAB Software, Practical demonstration

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power Electronics	Dr.P.S.Bimbhra	Khanna publishers
2	Power electronics	M D Singh and K B K Khanchandani,	Tata Mcgraw-Hill Publication

S. No.	Title of Book	Author	Publication including ISBN
3	Power electronics handbook	by M.H. Rashid	Academic press
4	High power converters and AC drives	IEEE press	IEEE press
5	Power Electronics Converters, Applications and design	Mohan, Undeland, Robbins	Wiley publication

XV SOFTWARE/LEARNING WEBSITES

- 1). <http://nptel.ac.in/courses/108101038/>
2. http://cusp.umn.edu/power_electronics.php
3. <http://ecee.colorado.edu/~ecen5797/notes.html>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE348	Semester VII														
348	Competency Evaluate the performance of various SEMICONDUCTOR DEVICES, TYPES OF POWER ELECTRONICS CIRCUITS AND AC DRIVES.	3	2	2	2	1	-	1	-	-	-	-	1	3	-
348.1	Evaluate the performance of the various types of INVERTERS for different application	3	2	1	-	2	-	-	-	-	1	-	-	-	-
348.2	Evaluate the performance of	3	3	2	1	2	-	-	-	-	-	-	-	2	-

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	the various types of power SUPPLIES for different application														
348.3	Evaluate the performance of the various types of AC Voltage controllers for different application	3	3	3	1	2	-	-	-	-	-	-	-	2	-
348.4	Evaluate the performance of the various types of Cycloconverters for different application	3	1	3	1	2	-	-	-	-	-	-	-	2	-
348.5	Evaluate the performance of the various types of AC Drives for different application	3	1	-	-	-	-	-	-	-	-	-	-	-	-

XVII NAMES OF RESOURCE PERSONS

a) Dr. Dharmesh Dabhi

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

Electrical engineers have to deal with various types of circuit breakers. Different types of circuit breakers are used for various applications for different voltage levels. This curriculum course structure is developed such that the student is able to evaluate the performance of the circuit breaker and calculate the different types of fault current at any location of power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Evaluate the performance of various circuit breakers employed for various applications in power system.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Evaluate** the symmetrical fault current and voltage at any location of power system.
- 2) **Evaluate** the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools.
- 3) **Interpret** the *phenomena of current interruption in circuit breaker*.
- 4) **Evaluate** the different parameters for arc formation and arc interruption in circuit breaker.
- 5) **Choose** the type of circuit breaker for different applications.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **10 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in

integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Formulate</u> the mathematical model for <u>the given</u> power system network.	02
2	1	<u>Find</u> the power system symmetrical fault current for <u>the given</u> power system network.	02
3	2	<u>Find</u> the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase transformer.	02
4	2	<u>Find</u> the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase Synchronous machine.	02
5	2	Verification of sequence network for <u>the given</u> connections of 3-phase transformer.	02
6	2	<u>Develop</u> the mathematical model of symmetrical components for <u>the given</u> power system network.	02
7	2	<u>Find</u> the power system unsymmetrical fault current for <u>the given</u> power system network.	02
8	2	<u>Simulate</u> the power system unsymmetrical fault current for <u>the given</u> power system network.	02
9	3	<u>Classify</u> the fault clearing process and problems associated with fault clearing process for <u>the given</u> conditions.	02
10	4	<u>Evaluate</u> the arc extinction process for <u>the given</u> circuit breaker.	02
11	5	<u>Compare</u> the performance of air break, air blast and SF6 for <u>the given</u> data.	02
12	5	<u>Evaluate</u> the applications of vacuum and direct current circuit breaker for <u>the given</u> voltage levels.	02
Total			24

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Power World Simulation software	2,7,8
2	Air break circuit breaker	11
3	Test rig of fault analysis	2
4	Test rig of 3 single-phase transformer.	3,5
5	Test rig of three phase synchronous generator.	4

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Per Unit System	CO 1. Evaluate the symmetrical fault current and voltage at any location of power system. PrO 1. FFor mulate the mathematical model for <u>the given</u> power system network. 1a. <u>Develop</u> the impedance and reactance diagram for <u>the given</u> power system network. 1b. <u>Determine</u> the per unit values of reactances of equipment for <u>the given</u> base.	1.1 Single phase solution to three phase system, one line diagram, impedance 1.2 and reactance diagram 1.3 Per unit system and examples
Unit– II Symmetrical Fault Analysis	CO 1. Evaluate the symmetrical fault current and voltage at any location of power system. PrO 2. Find the power system symmetrical fault current for <u>the given</u> power system network. 2a. <u>Determine</u> the making current of the circuit breaker for <u>the specified</u> fault current levels. 2b. <u>Determine</u> the 3-phase short circuit current for <u>the given</u> power system network. 2c. <u>Choose</u> the correct value of synchronous machine reactance for fault current calculation for <u>the given</u> condition.	2.1 sudden short circuit on R-L series circuit 2.2 3 – phase short circuit current calculation, examples. 2.3 Sub-transient, transient and steady state model of synchronous machine
Unit– III Symmetrical Components	CO 2. Evaluate the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools. PrO 3. Find the sequence impedances (Positive, Negative and Zero-sequence) of the three- phase transformer. PrO 4. Develop the mathematical model of symmetrical components for <u>the given</u> power system network. PrO 5. <u>Find</u> the sequence impedances (Positive, Negative and Zero-sequence) of the three- phase Synchronous machine.	3.1 Symmetrical Component transformation and its examples. 3.2 Sequence impedance and

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
	PrO 6. Verification of sequence network for <u>the given</u> connections of 3-phase transformer. 3a. <u>Find</u> the symmetrical components for the <u>given</u> phase values. 3b. <u>Develop</u> sequence networks from <u>the given</u> one line diagram of power system. 3c. <u>Justify</u> the phase shift to be consider for <u>the specified</u> transformer connection for unsymmetrical fault calculation.	sequence network of various equipment of power system. 3.3 Phase shift in star delta Transformers.
Unit-IV Unsymmetrical Fault Analysis	CO 2. <u>Evaluate</u> the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools. PrO 7. Find the power system unsymmetrical fault current for <u>the given</u> power system network. PrO 8. Simulate the power system unsymmetrical fault current for <u>the given</u> power system network. 4a. <u>Formulate</u> the fault current equation for the <u>given</u> type of fault for the <u>specified</u> condition. 4b. <u>Evaluate</u> the unsymmetrical fault current and voltage <u>the given</u> power system network for <u>the specified</u> condition.	4.1 Types of unsymmetrical faults and its analysis 4.2 Examples on unsymmetrical fault analysis
Unit – V Fundamentals of Fault Clearing, Switching Phenomena and Circuit Breaker (CB) Ratings.	CO 3. <u>Interpret</u> the phenomena of current interruption in circuit breaker. PrO 9. <u>Classify</u> the fault clearing process and problems associated with fault clearing process for <u>the given</u> conditions. 5a. <u>Interpret</u> the data with respect to the <u>given</u> interruption method. 5b. <u>Determine</u> RV and RRRV for <u>the specified</u> condition of <u>the given</u> type circuit breaker. 5c. <u>Prepare</u> the specification of a circuit breaker required for <u>the given</u> situation. 5d. <u>Determine</u> the rating of the circuit breaker for <u>the given</u> conditions.	5.1 Current interruption in AC circuit breaker. 5.2 Transient recovery voltage, rate of rise of restriking voltage and Examples. Different factors affecting TRV. 5.3 Problems associated with circuit breaker for different types of faults, applications and configurations. 5.4 Resistance switching, current chopping and interruption of capacitive current 5.5 Rating and selection of circuit breakers
Unit – VI Arc Extinction Process in Circuit Breaker	CO 4. <u>Evaluate</u> the different parameters for arc formation and arc interruption in circuit breaker.	

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
	PrO 10. <u>Evaluate</u> the arc extinction process for <u>the given</u> circuit breaker. 6a. <u>Interpret</u> the various modes of ionization and deionization of gases under <u>the specified</u> conditions. 6b. <u>Interpret</u> the phenomena of arc formation and arc extinction in <u>the given</u> condition.	6.1 Ionization and Deionization of gases 6.2 Formation of electric arc and methods of arc extinction. 6.3 Arc interruption theories.
Unit – VII Types of Circuit Breakers	CO 5. <u>Choose</u> the type of circuit breaker for different applications. PrO 11. <u>Compare</u> the performance of air break, air blast and SF6 for <u>the given</u> data. PrO 12. <u>Evaluate</u> the applications of vacuum and direct current circuit breaker for <u>the given</u> voltage levels. 7a. <u>Select</u> the relevant type of CB for the minimum cost for <u>the given</u> situation.	7.1 Types of circuit breakers. 7.2 Constructional details and principals of operations.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Per Unit System	04	0	0	6	0	0	0	06
II	Symmetrical Fault Analysis	08	0	7	0	7	0	0	14
III	Symmetrical Components	10	0	7	0	4	0	0	11
IV	Unsymmetrical Fault Analysis	10	0	0	0	4	7	0	11
V	Fundamentals of Fault Clearing, Switching Phenomena and Circuit Breaker (CB) Ratings.	10	0	2	7	4	0	0	13
VI	Arc Extinction Process in Circuit Breaker	05	0	0	5	0	0	0	5
VII	Types of Circuit Breakers	13	0	4	0	0	6	0	10
Total		60	0	20	18	19	13	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Per Unit System:** Collect the nameplate rating of different types of equipment (Power transformer, Distribution Transformer, alternators etc) and compare the pu value of each type of equipment.
- 2) **Symmetrical components:** Google the different protective relays and list the relays work based on symmetrical components principals.
- 3) **Fault Clearing Process:** Develop MATLAB model which shows the waveform of voltage at time of contact separation.
- 4) **Circuit Breakers:** Prepare case studies of circuit breakers used in the protection schemes in different types of industries.
- 5) Undertake the survey in CHARUSAT campus and submit a report on different type of circuit breaker used with their rating.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding the evolution of arc quenching mechanism for different type of circuit breakers.
- b) Prepare 15 min. power point presentation on how to choose the specific circuit breaker for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Per Unit System	1	1a. <u>Develop</u> the impedance and reactance diagram for <u>the given</u> power system network. 1b. <u>Determine</u> the per unit values of reactances of equipment for <u>the given</u> base.	Lecture with media, field visit	PPTs, black board
Unit– II Symmetrical Fault Analysis	2	PrO 1. <u>Formulate</u> the mathematical model for the <u>given</u> power system network. 2a. <u>Determine</u> the making current of the circuit breaker for <u>the specified</u> fault current levels.	Lecture With media	PPTs, black board
	3	PrO 2. <u>Find</u> the power system symmetrical fault current for <u>the given</u> power system network. 2b. <u>Determine</u> the 3-phase short circuit current for <u>the given</u> power system network. 2c. <u>Choose</u> the correct value of synchronous machine reactance for fault current calculation for <u>the given</u> condition.	Lecture with media Tutorial	PPTs, black board
Unit– III Symmetrical Components	4	PrO 3. <u>Find</u> the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase transformer. PrO 4. Develop the mathematical model of symmetrical components for the given power system network. PrO 5. Find the sequence impedances (Positive, Negative and Zero-sequence) of the three-phase Synchronous machine.	Lecture with media,	PPTs, Videos
	5	PrO 6. <u>Verification</u> of sequence network for <u>the given</u> connections of 3-phase transformer. 3a. <u>Develop</u> sequence networks form <u>the given</u> one line diagram of power system.	Lecture with media, Tutorial and Demonstration	PPTs
	6	3b. <u>Justify</u> the phase shift to be consider for <u>the specified</u> transformer connection for unsymmetrical fault calculation.	Lecture with media	PPTs
Unit– IV Unsymmetrical Fault Analysis	7	PrO 7. <u>Find</u> the power system unsymmetrical fault current for <u>the given</u> power system network. 4a. <u>Formulate</u> the fault current equation for <u>the given</u> type of fault for <u>the specified</u> condition.	Lecture with media, Field visit	PPTs, Video
	8	PrO 8. <u>Simulate</u> the power system unsymmetrical fault current for <u>the given</u> power system network. 4b. <u>Evaluate</u> the unsymmetrical fault current and voltage <u>the given</u> power system network for <u>the</u>	Lecture with media, and Tutorial	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		<u>specified</u> condition.		
Unit – V Fundamentals of Fault Clearing , Switching Phenomena and Circuit Breaker (CB) Ratings.	9	5a. <u>Interpret</u> the data with respect to <u>the given</u> interruption method.	Lecture with media, + PPT	PPTs, Video
	10	PrO 9. <u>Classify</u> the fault clearing process and problems associated with fault clearing process for <u>the given</u> conditions. 5b. <u>Determine</u> RV and RRRV for the specified condition of <u>the given</u> type circuit breaker. 5c. <u>Prepare</u> the specification of a circuit breaker required for <u>the given</u> situation.	Lecture with media, Tutorial	PPT, Video
	11	5d. <u>Determine</u> the rating of the circuit breaker for <u>the given</u> conditions.	Lecture with media	PPTs, black board
Unit – VI Arc Extinction Process in Circuit Breaker	12	PrO 10. <u>Evaluate</u> the arc extinction process for <u>the given</u> circuit breaker. 6a. <u>Interpret</u> the various modes of ionization and deionization of gases under <u>the specified</u> conditions. (self-learn) 6b. <u>Interpret</u> the phenomena of arc formation and arc extinction in <u>the given</u> condition.	Lecture with media, case study	PPT, Pictures , black board
Unit – VII Types of Circuit Breakers	13	PrO 11. <u>Compare</u> the performance of air break, air blast and SF6 for <u>the given</u> data. 7a. <u>Select</u> the relevant type of CB for the minimum cost for <u>the given</u> situation.	Lecture with media, Tutorial	PPTs, black board
	14	PrO 12. <u>Evaluate</u> the applications of vacuum and direct current circuit breaker for <u>the given</u> voltage levels. 7a. <u>Select</u> the relevant type of CB for the minimum cost for <u>the given</u> situation.	Lecture with media, Tutorial	PPTs, Video
	15	7a. <u>Select</u> the relevant type of CB for the minimum cost for <u>the given</u> situation.	Lecture with media, Tutorial	PPTs, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Analysis	Hadi Saadat	PSA Publishing , Third Edition, ISBN : 9780984543809, 9780984543823
2	Principles of Power system	V. K. Mehta , Rohit Mehta	S. Chand & Company Ltd, 2005 ISBN: 8121924960, 9788121924962
3	Modern Power System Analysis	D.P. Kothari & I. J. Nagrath	McGraw Hill Education; 5th edition ISBN: 9789354600968
4	Power System Analysis	Grainger & Stevenson	McGraw Hill Education, 2017 ISBN: 9780070612938

S. No.	Title of Book	Author	Publication including ISBN
5	Power System Analysis and Design	Glover and Sharma	Nelson Engineering; 4th edition ISBN: 9780534548841
6	Analysis of Faulted Power System	P.M. Anderson	IEEE Press, 1995 ISBN: 9780780311459
7	Computer Techniques In Power System Analysis	M A Pai and Chatterjee	McGraw Hill Education; 3rd edition (1 July 2017) ISBN: 978-9332901131
8	J&P Switchgear	R T Lythall	J&P Limited London, 1963 Identifier: in.ernet.dli.2015.474861 ISBN: -

XV SOFTWARE/LEARNING WEBSITES

- <https://nptel.ac.in/courses/108105067>
- <https://nptel.ac.in/courses/117105140>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE349	Semester VI	Fault Analysis and Switchgear Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO													
349	Competency Evaluate the performance of various circuit breakers employed for various applications in power system.	3	2	1	2	1	-	-	-	-	-	-	-	-	-
349.1	Evaluate the symmetrical fault current and voltage at	3	3	1	2	2	-	-	-	-	-	-	-	-	-

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	any location of power system.														
349.2	Evaluate the unsymmetrical fault current and voltage at any location of power system with mathematical model and with software tools.	3	3	2	2	2	-	-	-	-	-	-	-	-	-
349.3	Interpret the phenomena of current interruption in circuit breaker.	3	1	-	1	-	-	-	-	-	-	-	-	-	-
349.4	Evaluate the different parameters for arc formation and arc interruption in circuit breaker.	3	1	-	1	-	-	-	-	-	-	-	-	-	-
349.5	Choose the type of circuit breaker for different applications.	3	1	2	2	-	2	-	-	-	-	-	-	-	-

XVII NAMES OF RESOURCE PERSONS

- a) Dr. Nilaykumar A. Patel
- b)

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

This course enables the students to apply some of the knowledge and/or skills developed during the programme to new problem for which there are number of engineering solutions. This course include planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, , use, monitor and control resources optimally and economically. By studying this course abilities like creativity, imitativeness and performance qualities are also developed in students. Leadership development and supervision skills are also integrated objectives of learning this course.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- Find the solution of a real-life problem.

III COURSE OUTCOMES (COs)

Depending upon the nature of the projects undertaken, the following could be some of the minor course outcomes that could be attained, although, in case of some projects few of the following course outcomes may not be applicable.

Apply knowledge and skills learned from project to real-world problems.

Solve engineering problems by the team work and work with other discipline.

Use experience related to professional and ethical issues in the work environment.

Apply impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.

Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
0	0	04	02	00	00	50	00	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

V COURSE DETAILS

- Minor Project showcases that, a student during the fifth semester will able to apply the engineering knowledge to a practical problem.
- Students can carry out their project work in-house at institute/ industry/ research organization/govt. organization.
- Project should be related to the Product/prototype design and development or Experimentation, Simulations and evaluation of any system etc.

- iv. A student is required to carry out the project work related to Electrical Engineering or some interdisciplinary work under the guidance of a faculty member and the supervisor of the concerned industry/institute/organization.
- v. The minor project work provides student an opportunity to develop something on their own or group of maximum 3 students. Final decision related to the number of students in one group is based on internal supervisor recommendation.
- vi. Project will be evaluated at least twice during the semester by internal faculty as a part of continuous evaluation and final evaluation will be carried out by external examiner at the end of semester.
- vii. Student should submit the project report at the end of the semester duly signed by project supervisor and head of department. Report comprising literature review, objective, methodology, scope of the project work undertaken, outcome of project and conclusions.
- viii. The students are required to identify their problem and they are required to follow all the rules and instruction issued by department/industry.

VI PROJECT REPORT

At the end of the semester student should prepare project report. Guideline of the Project report is as follow

- a) Certificate
- b) Acknowledgements
- c) Abstract (in one paragraph not more than 150 words)
- e) Content Page
- f) Chapter-1 Introduction and background of the Problem/Task
- g) Chapter-2 Literature Survey for Problem/Task Identification and Definition,
- h) Chapter-3 Proposed Detailed Methodology of solving the problem/completing the task
- d) Chapter-4 Methodology actually followed to solve the problem/complete the task
- e) Chapter-5 The solution, its development and its implementation/execution (with details including drawings, software, used for or developed for the solution etc. in detail). In case of survey/experiment type of project this chapter will include data analysis, information presentation and analysis.
- f) Chapter-6 Final conclusions and recommendations if any for future projects.

Note:

- a) The report should contain as many diagrams and figures, charts etc. as relevant for the project.
- b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VII ASSESSMENT OF PROJECT WORK

Assessment of Project work also has two components, first is Progressive Assessment, while another is End of the Term Assessment.

i. Progressive Assessment Criteria

- a) Project guide or the group of internal faculty are supposed to carry out this assessment. It is a continuous process, during which for developing desired qualities in the students, faculty should orally give informal feedback to students about their performance and interpersonal behaviour while guiding them on their project work every week.

S. No.	Criteria	Marks
First Progressive Assessment at the end of 4th week		
1	Project Proposal for work to be completed at the end of semester: Including problem definition and methodology adapted, literature review adopted (in case of working	10

S. No.	Criteria	Marks
	models prepared then quality of workman ship, appropriateness of raw materials/components/ technology being used, functioning of the prototype etc. has to be also assessed)	
	General Behaviour (initiative, resourcefulness, reasoning ability, imagination/creativity, self-reliance)	Only written feed-back/suggestions
Second Progressive Assessment at the end of 12th week		
2	Execution of Plan during semester (marks to be also given based on ability to collect relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also assessed	5
3	Log book (for work done during semester- detailed and regular entry would be basis of marks)	5
4	General Behaviour (persistence, interest, confidence, problem solving ability, decision making ability, initiative to act, team spirit, sharing of material etc., participation in discussions, completion of individual responsibilities, leadership)	Only written feed-back./suggestions
Third Progressive Assessment at the end of 14th week		
5	Portfolio for Self-learning and reflection (for work done in semester- marks based on amount of reflection and completion of the portfolio)	10
6	Project Report including documentation - for work done in semester (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	5
7	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation for work done during semester)	5
8	Defence (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce for work done during semester)	10
Total		50

Note: Oral feedback on general behaviour may also be given whenever relevant/ required during day to day guidance and supervision.

ii. END SEMESTER EXAMINATION

b) Criteria of Marks for ESE for Capstone Project Execution.

S. No.	Description	Marks
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S. No.	Description	Marks
1	Problem Identification/Project Title (Innovation /Utility of the Project for industry/ User/Academia) marks to be also given based on (i) Accuracy or specificity of the scope and (ii) Appropriateness of the work with reference to desired course outcomes.	5
2	Industrial Survey and Literature Review (marks to be given based on extent/volume and quality of the survey of Industry / Society / Institutes/Literature/Internet for Problem Identification and possible solutions)	5
3	Project Proposal (In case of working models, detailed design and planning of fabrication/assembly of the prototype has to be also assessed). Marks to be given also based on appropriateness, flexibility, detail and clarity in methods/planning.	10
4	Final Report including documentation. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	10
5	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	10
6	Defence (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	10
Total		50

XI Guidelines for Portfolio Preparation and assessment

The main purpose of portfolio preparation is learning based on self-assessment and portfolio is not to be used for assessment in traditional sense.

- Each student has to prepare his/her portfolio separately. However, he/she can discuss with the group members about certain issues on which he/she wants to write in the portfolio.
- Whatever is written inside the portfolio is never to be used for assessment, because if teachers start giving marks based on whatever is written in the portfolio, then students would hesitate in true self-assessment and would not openly describe their own mistakes or shortcomings.
- Some marks are allocated for portfolio, these marks are to be given based on how sincerely portfolio has been prepared and not based on what strengths and weaknesses of the students are mentioned in the portfolio.
- Portfolio has to be returned back to the students after assessing it (assessment is only to see that whether portfolio is completed properly or not) by teachers. Because student is the real owner of the portfolio.
- Students mainly learn during portfolio preparation, but they can further learn if they read it after a gap. And hence they are supposed to keep the portfolios with them even after completion of the diploma because it is record of their own experiences (it is like diary some people write about their personal experiences), because they can read it again after some time and can revise their learning (about their own qualities)

XII PO-COMPETENCY-CO MAPPING

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Cours e Code	Competency and COs	PO 1 Engine ering knowl edge	PO 2 Proble m analysi s	PO 3 Design /devel opme nt of solutio ns.	PO 4 Condu ct investi gation s of compl ex proble ms	PO 5 Moder n tool usage	PO 6 The engine er and society	PO 7 Enviro nment and sustai nabilit y	PO 8 Ethics	PO 9 Individ ual and teamw ork	PO 10 Comm unicati on	PO 11 Projec t manag ement and financ e	PO 12 Life- long learnin g	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE360	Semester VI														
360	Competency Find the solution of a real-life problem.	3	2	2	2	2	2	1	1	2	2	2	1	2	2
360.1	Apply knowledge and skills learned from project to real-world problems.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
360.2	Solve engineering problems by the team work and work with other discipline.	3	3	2	2	1	2	-	-	3	2	1	2	2	-
360.3	Use experience related to professional and ethical issues in the work environment.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
360.4	Apply impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.	3	-	-	3	3	3	-	-	2	1	3	2	3	3
360.5	Implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies.	3	1	3	2	3	3	2	-	-	2	2	3	3	2

XIII NAMES OF RESOURCE PERSONS

b) Prof. Mahammadsoaib Saiyad

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS132.02 A: CONTRIBUTORY PERSONALITY DEVELOPMENT

I. Credits and Schemes:

Sem.	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				Total
					Theory		Practical		
					Internal	External	Internal	External	
VI	HS133.02 A	CONTRIBUTORY PERSONALITY DEVELOPMENT	02	02	--	--	30	70	100

II. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

III. Detailed Syllabus:

1.	Concept of Personality	06 Hours	20%
	Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling		
2.	Soft Skills and Personality Development	08 Hours	26%
	Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player		
3.	Developing Contributory Personality	06 Hours	20%
	Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities		

4.	Life skills and Personality Development	06 Hours	20%
	Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving		
5.	Contemporary Issues in CPD	04 Hours	14%
	Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations		

IV. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by personality profiling, reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies (Great Personality) , critical reading, group work, independent and collaborative research, presentations etc.

V. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The practical evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Components	Number	Marks per incidence	Total Marks
1	Assignments	2	25	
2	Project work	2	25	
3	Personality Profiling and Presentation	1	25	
4	Quiz	2	25	100*
5	Attendance and Classroom participation			05
			Total	30

*** Total Marks will be converted to 25**

External Evaluation

The University Exam will be for 70 marks and will be based on practical computer based tests and a viva-voce. It will test the contributory personality aspects and their applications. There may also be a Computer Based Test at the end.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical Exam / Viva	01	70	70
			Total	70

VI. Course Outcomes (COs)

At the end of the course, learners will be able to:

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	2	-	2	2	-	-	-	-
CO6	-	-	-	-	-	2		-	-	2	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VII. Preference

It will be preferred that at the end of the course, students take an official Personality Profiling test like MBTI through CHARUSAT. Equivalent marks shall be considered in the University Exam.

VIII. Recommended Study Material:

❖ Text book:

1. Personality Development & Soft Skills, Oxford University Press
2. Soft Skills, Bookboon
3. Personality Development, Swami Vivekananda; Advaita Ashrama

❖ Reference book:

1. Contributor Personality Program Workbook (Volume 1,2),
2. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

❖ Web material:

1. <https://www.coursera.org/learn/wharton-succcess>
2. <https://www.coursera.org/learn/personality-types-at-work>
3. <https://www.coursera.org/learn/self-awareness>

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

The special electrical machines are widely used in power industries. The course is developed such that student would get the knowledge from basics to control schemes of these machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the performance of various control methods used in field.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) Use the control technique for special electrical machines.
- 2) Evaluate the performance of the special electrical machines.
- 3) Interpret the suitable control technique for special electrical machines.
- 4) Evaluate the performance of the special electrical machines.
- 5) Choose the type of turbine for induction generator.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	2	To Perform the Direct Load Test On Auto Synchronous Motor.	04
2	2	To Perform the Constant Excitation Test On Auto Synchronous Motor.	04
3	3	To Observe a Change in Output Voltage with Change in Rotor Position of 3 Phase Induction Regulator.	04
4	3	To Perform the Speed Control and Power Factor Control on Commutator Motor.	04
5	1	Study of a Stepper Motor.	02
6	1	Study of a Switched Reluctance Motor and Synchronous Reluctance Motor.	02
7	1	Study of a Permanent Magnet Synchronous Motor.	02
8	4	Demonstration of an Induction Generator.	04
9	4	To Understand the Behavior of Induction Generators Considering the Impact of a Load.	02
10	4	To Understand the Working of a Doubly Fed Induction Generator.	02
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Commutator machine with load	4
2	Auto Synchronous machine with prime-mover/load	1,2
3	Induction machine with prime-mover/load	8,9
4	Induction voltage regulator	3

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample *Cognitive Domain* learning outcomes denoted by unit outcomes

(UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction of Special Electrical Machines	CO 1. Recognize & understand the performance, operation and application of various AC & DC special machines. PrO 1. To Perform the Direct Load Test On Auto Synchronous Motor. PrO 2. To Perform the Constant Excitation Test On Auto Synchronous Motor. PrO 3. To Observe a Change in Output Voltage with Change in Rotor Position of 3 Phase Induction Regulator. PrO 4. To Perform the Speed Control and Power Factor Control on Commutator Motor 1a. <u>Justify</u> the concept of special machines and fundamental component roles.	1.1 Rosenberg Generator 1.2 Dynamotors, Printed Circuit Board Motor 1.3 Special Induction Machines: Synchronous Induction Motor 1.4 Induction Voltage Regulator 1.5 Linear Induction Motor 1.6 Special Transformers: Induction Heating Transformer 1.7 Welding Transformers, Pulse Transformer 1.8 Schrage Motor 1.9 Repulsion Motor 1.10 Hysteresis Motor 1.11 Universal Motor, DC-AC Servomotor 1.12 Applications, Examples
Unit– II Special Synchronous Machines	CO 3. Acquire the knowledge regarding the fundamentals of permanent magnet, motors/generators, switched reluctance motor, linear motors and stepper motors. PrO 5. Study of a Stepper Motor. PrO 6. Study of a Switched Reluctance Motor and Synchronous Reluctance Motor. 2a. <u>Evaluate</u> the performance of stepper motor. 2b. <u>Evaluate</u> the performance of SRM.	2.1 Stepper Motor 2.2 Switched Reluctance Motor 2.3 Linear Synchronous Motor 2.4 Applications, Examples
Unit– III Permanent Magnet Machines	CO 2. Define various permanent magnet material used for manufacturing of special machine. PrO 7. Study of a Permanent Magnet Synchronous Motor. 3a. <u>Evaluate</u> the performance of PMSM.	3.1 Basics of permanent magnet property 3.2 Unique features of Permanent Magnet (PM) Machines 3.3 Permanent Magnet Material & Characteristics 3.4 Permanent magnet DC motor (PMDC) 3.5 Permanent magnet brushless DC motor (PMBLDC) : construction, operation, control, advantages, disadvantages, applications 3.6 permanent magnet synchronous motor(PMSM), Applications
Unit-IV Wind Turbine Generators	CO 4. Understand the types and operation of wind turbine generators and evaluate case studies based on renewable energy sources. PrO 8. Demonstration of an Induction Generator.	4.1 Introduction, Types of induction generator (line

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
	<i>Pro 9. To Understand the Behavior of Induction Generators Considering the Impact of a Load.</i> <i>Pro 10. To Understand the Working of a Doubly Fed Induction Generator.</i> 4a. <u>Choose</u> the relevant operating parameters for load operation of induction generator.	excited and self-excited) 4.2 Introduction of fixed speed induction generator 4.3 Introduction of doubly fed induction generator 4.4 Working of Induction generator: Principle of Operation, Reactive power requirement 4.5 voltage build up process, Load and Power factor control, Effect of capacitor 4.6 Characteristics of induction generator, Advantages, Applications, Examples
Unit – V Modelling & Application	CO 5. To understand the modeling of machines and reference frames for machine control. 5a. <u>Interpret</u> the machine data with respect to the <u>given power system</u>. 5b. <u>Select</u> the method for machine control.	5.1 Synchronous Reference Frames: dq Axis Theory 5.2 Conversions: abc to $\alpha\beta$, $\alpha\beta$ to dq, dq to $\alpha\beta$, $\alpha\beta$ to abc 5.3 Space Vector Pulse Width Modulation 5.4 discrete and analog implementation, Applications
Unit VI Case Study	CO 4. Understand the types and operation of wind turbine generators and evaluate case studies based on renewable energy sources. 6a. Evaluate performance of Wind generator application	6.1 Case Study: Grid Connected Renewable Energy Source

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Special Electrical Machines	13	2	4	4	4	0	0	16
II	Special Synchronous Machines	05	0	2	2	2	2	0	08
III	Permanent Magnet Machines	07	0	2	4	4	2	0	12
IV	Wind Turbine Generators	06	0	2	2	2	6	0	12
V	Modelling & Applications	09	2	2	4	4	2	2	16
VI	Case Study	05	0	0	0	2	4	0	06
Total		45	4	12	16	18	16	2	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Load test on induction generator.**
- 2) **Calculation of step angle of stepper motor for various applications.**
- 3) **Create setup of regenerative braking of auto synchronous motor.**
- 4) **Create a setup for load condition of auto synchronous motor for various power factor.**

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different methods of voltage regulation.
- b) Prepare 15 min. power point presentation on synchronizing of alternator.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
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Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introducti on of Special Electrical Machines	1	PrO 1. To Perform the Direct Load Test On Auto Synchronous Motor.	Lecture with media	PPTs, black board
	2	PrO 2. To Perform the Constant Excitation Test On Auto Synchronous Motor.	Lecture With media	PPTs, black board
	3	PrO 3. To Observe a Change in Output Voltage with Change in Rotor Position of 3 Phase Induction Regulator.	Lecture with media	PPTs, black board
	4	PrO 4. To Perform the Speed Control and Power Factor Control on Commutator Motor 1a. Justify the concept of special machines and fundamental component roles.	Lecture With media	PPTs, black board
Unit II- Special Synchron ous Machines	5	PrO 5. Study of a Stepper Motor. 2a. Evaluate the performance of stepper motor.	Lecture With media	PPTs, black board
	6	PrO 6. Study of a Switched Reluctance Motor and Synchronous Reluctance Motor. 2b. Evaluate the performance of SRM.	Lecture With media	PPTs, black board
Unit– III - Permane nt Magnet Machines	7	PrO 7. Study of a Permanent Magnet Synchronous Motor.	Lecture With media	PPTs, black board
	8	3a. <u>Evaluate</u> the performance of PMSM.	Lecture With media	PPTs, black board
Unit IV- Wind Turbine Generato rs	9	PrO 8. Demonstration of an Induction Generator	Lecture With media	PPTs, black board
	10	PrO 9. To Understand the Behavior of Induction Generators Considering the Impact of a Load.	Lecture With media	PPTs, black board
	11	Pro 10. To Understand the Working of a Doubly Fed Induction Generator.	Lecture With media	PPTs, black board
	12	4a. <u>Choose</u> the relevant operating parameters for load operation of induction generator.	Lecture With media	PPTs, black board
Unit V - Modelling & Applicatio ns	13	5a. <u>Interpret</u> the machine data with respect to <u>the given</u> power system.	Lecture With media	PPTs, black board
	14	5b. <u>Select</u> the method for machine control.	Lecture With media	PPTs, black board
Unit VI- Case study	15	6a. Evaluate performance of Wind generator application	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Theory and performance of electrical machines	J.B. Gupta	S.K. Kataria & Sons, ISBN: 978-93-5014-277-6
2	A textbook of electrical	B.L. Theraja	S. Chand, ISBN: 9788121924375

S. No.	Title of Book	Author	Publication including ISBN
	technology VOL II		
3	Special Electrical Machines	K. Venkataratnam	University press, ISBN: 13. 978-8173716317

XV SOFTWARE/LEARNING WEBSITES

- http://www.academia.edu/9885014/special_electrical_machines_nptel_notes

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12	PSO1	PSO2
		Used of mathematical method	Analysis mechanical system	Used engineering soft.	Solve problem in machine	Mode rn tool usage	The engineer and society	Environment and sustainability	Ethics	Individual and team work:	Communication	Project management	Life-long learning	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE376	Semester VI														
CO1	Recognize & understand the performance, operation and application of various AC & DC special machines.	3	-	1	2	-	-	-	-	1	2	3	1	-	-
CO2	Define various permanent magnet material used for manufacturing of special machine.	3	1	2	2	2	-	1	-	-	1	-	-	2	-
CO3	Acquire the knowledge regarding the fundamentals of permanent magnet, motors/generators, switched reluctance motor, linear motors and stepper motors.	3	3	3	1	2	2	1	-	-	3	2	1	2	-
CO4	Understand the types and operation of wind turbine generators and evaluate case studies based on renewable energy sources.	3	3	3	1	2	2	3	-	2	2	2	3	3	3
CO5	To understand the modeling of machines and reference frames for machine control.	3	-	2	3	3	-	-	2	2	3	1	2	3	3

XVII NAMES OF RESOURCE PERSONS

- a) Dr. Pratik Mochi

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

Looking to a large set of skills required by an Embedded System Development Engineer, there should be a course serving as Fast/Crash Course bridging the requirement like, introducing the various concepts of Embedded system and inculcating the most basic skills set just in a single course.

The Embedded software engineer and Embedded Hardware Engineer has to work in close collaboration in many or most of the project targeting Embedded system development for an application. While the Embedded Hardware Engineer read and understand the datasheets of various chips and components involved in design, integrate/interface them together and finally generate the professional Printed Circuit Board (PCB), the Embedded software engineer write wide varieties of firmware for an Embedded application. The course does not focus on skills required for an Embedded hardware engineer.

For most of application, computationally fast microcontrollers with rich set of on chip peripherals support and Memory are enough (ARM series Microcontroller are one of the popular choice of Embedded system designers). Due to several reasons, the Field Programmable Gate Array (FPGA) are selected as main or supporting chip. The Embedded software engineer needs to generate hardware inside such chips using hardware description language, like VHDL or Verilog. The course covers FPGA programming in VHDL language. It starts with developing some basic hardware block and it is extended for developing several communication hardware modules/protocols as well.

Embedded C coding for Microcontroller is already covered in a separate course and it is considered a pre-requisite for this course. But those course are not covering 32-bit microcontroller programming. The memory mapped registers/SFR in 32-bit microcontroller are generally accessed using pointer-structure-union mechanism. Writing and understanding ready driver is an art and needs some practice. often, The Embedded Software Engineer need to write Embedded Application firmware using market available Real Time Operating System (RTOS). The course teaches to develop small RTOS after discussing the basic concepts and to write code using RTOS.

The computer on board/chip (like raspberry Pi, Jetson Nano and other such boards) has revolutionaries the way of programming. Even one can program the system without knowing much architectural details of microcontroller. Specially these devices are gaining attention for systems involving some AI/ML algorithms and Internet connectivity. Such topics are also covered here briefly.

Such a course can be helpful to student to understand Embedded system programming, with- out spending much time for it. The industry may find this subject useful, for example if they want someone to offer a role requiring regular discussion with team of embedded system software developers and some time to get involved in their process to modify it. The course is not target to make a full stack embedded system engineer (Hardware + software both) but it helps and work out much for it.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Understand given Embedded Firmware/HDL coding and Explain/Present to a team working on it. Remodify it for new requirement.
- Write the Firmware/HDL coding for target Embedded system and Debug/Correct it.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes,

practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) Use the ready driver to configure the on chip peripherals and develop driver as a C callable function, find the bug in driver function.
- 2) Write/Develop VHDL program for required architecture. Test and Debug it.
- 3) Develop Small Real Time Operating System(RTOS) and Write code for Embedded application using RTOS.
- 4) Apply and Test Artificial Intelligence (AI) and Machine Learning (ML) algorithms for an Embedded Application implemented on Computer on Board Devices.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **20 marks** are for **attendance, micro-project assessment** and other **assignment** if given any, to facilitate integration of COs and the remaining 10 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Develop</u> memory map registers in header file using pointer, structure and union for Peripheral registers in ARM M4 microcontroller.	01
2	2	<u>Develop/Write</u> basic codes for FPGA in VHDL as in follows, and also write test bench program for the same. a. Combinational Circuits like Digital Gates and Digital Circuits based on Digital Gates, SR Non Clocked Flip Flop b. Sequential Clock based Circuits	03
3	2	<u>Learn</u> the following architecture in an Unknown Microcontroller, and <u>develop/Write</u> the VHDL code to implement the given architecture of peripheral in FPGA. a. GPIO b. SPI,I2C, UART, USART, CAN communication	05
4	1	<u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> C driver for it a. SPI, I2C, b. CAN communication	02
5	2	<u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> VHDL	05

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
		code to implement the given architecture of peripheral in FPGA. a. UP, Down, Up-Down Timer/Counter b. Watchdog Timer c. Pulse Width Modulator d. Capture Compare Unit	
6	1	<u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write C</u> driver for it. a. UP, Down, Up-Down Timer/Counter b. Watchdog Timer c. Pulse Width Modulator d. Capture Compare Unit	02
7	4	<u>Write</u> AI and ML codes in Prolog, Python and C, and <u>Test</u> the result and speed of execution on given Computer On chip/board device and <u>make</u> (<u>Develop</u> interfacing and <u>Write</u> code) it communicating it to main controller/FPGA chip in Embedded board.	08
8	3	<u>Test</u> the small Real Time Operating system (developed during theory hours or market available with support of documentation) for various functionality.	04
		Total	30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

Sr. No.	Equipment Name with Broad Specifications	PrO No.
1	Computer On chip Device (Raspberry Pi, Jetson Nano, Raspberry Pi Pico or similar)	7
2	<i>Data Set for AI and Machine Learning</i>	7
3	<i>FPGA kit.</i>	2,3,5
4	<i>ARM or similar 32 bit Fast Microcontroller Development boards</i>	1,4
5	<i>Unknown Microcontroller (The microcontroller which is not covered in any offered course)</i>	1,4,7
6	<i>Digital Storage Oscilloscope(DSO), Signal Generator</i>	3,4,8

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when he/she performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Understand the specification, requirements and device a plan of action for developing an Embedded Application/product.
- 2) Lead the team, distribute the task, and finally integrate them all to gether, for developing Embedded system firmware

The level of achievement of the ADOs according to *Krathwohl's 'Affective Domain Taxonomy'* should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction	Co1. <u>Use</u> the ready driver to configure the on chip peripherals and develop driver as a C callable function, find the bug in driver function. Pro1. <u>Develop</u> memory map registers in header file using pointer, structure and union for Peripheral registers in ARM M4 microcontroller. Pro4. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> C driver for it - SPI, I2C, - CAN communication Pro6. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> C driver for it. - UP, Down, Up-Down Timer/Counter - Watchdog Timer - Pulse Width Modulator - Capture Compare Unit 1a. <u>Understand</u> the Architecture of peripheral from datasheet and <u>write</u> down driver to configure and <u>use</u> the peripheral. 1b. <u>Understand</u> and <u>use</u> ready Device driver.	1.1 Peripherals available in Microcontroller, their working, modes and Controlling and configuration Registers generally available in (Various) Microcontroller (GPIO, Watchdog, Timer, ADC, DAC, Compare-capture, Pulse width Modulator), 1.2 Developing Driver using pointer, structure, Union in C and Using ready driver.
Unit– II Communication Protocol and Device driver.	Co1. <u>Use</u> the ready driver to configure the on chip peripherals and develop driver as a C callable function, find the bug in driver function. Pro3. <u>Learn</u> the following architecture in an Unknown Microcontroller, and <u>develop/Write</u> the VHDL code to implement the given architecture of peripheral in FPGA. - GPIO - SPI, I2C, UART, USART, CAN communication Pro4. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> C driver for it - SPI, I2C, - CAN communication	2.1 Working of SPI, I2C bus, CAN bus, 2.2 Device driver development in C

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
	2a. <u>identify</u> the problems with communication and <u>provide</u> the solution. 2b. <u>Understand</u> and use Device driver for various communication Protocols and <u>find</u> bug if any.	
Unit- III FPGA Programming in VHDL	Co2. <u>Write/Develop</u> VHDL program for required architecture. <u>Test</u> and <u>Debug</u> it. Pro3. <u>Learn</u> the following architecture in an Unknown Microcontroller, and <u>develop/Write</u> the VHDL code to implement the given architecture of peripheral in FPGA. - GPIO - SPI, I2C, UART, USART, CAN communication Pro5. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> VHDL code to implement the given architecture of peripheral in FPGA. - UP, Down, Up-Down Timer/Counter - Watchdog Timer - Pulse Width Modulator - Capture Compare Unit 3a. <u>Understand</u> the given VHDL program. 3b. <u>Develop</u> VHDL program for given architecture. 3c. <u>Develop</u> VHDL Program for interfacing external chip with VHDL.	3.1 History of VHDL, Introduction to VHDL and meaning of Hardware abstraction, General Architecture of a VHDL Program 3.2 Entity Declaration, Architecture body (Structural, Behavioural and Mix style of Modelling), 3.3 Lexical Elements (comments, identifiers, numbers, characters, strings, bit strings), 3.4 Data types (Integer datatypes, Arrays, Records), 3.5 Object declaration, 3.6 Signal Vs Variable, 3.7 Programming OR, AND and XOR Gate in VHDL 3.8 Loop statements, Wait statement, 3.9 Libraries, 3.10 Programming exercises (Like Adder, comparator, ALU, Latch, D and JK Flip flop, Register and shift Registers, Counter, Finite state Machine), 3.11 Conditional and selected signal Assignment, 3.12 Writing functions and procedures, 3.13 Structural modelling, 3.14 Test benches, Role of RTL engineer in Embedded industries, 3.15 Implementation of communication protocols on FPGA, Interfacing ADC with FPGA
Unit-IV Fundamental of Real Time Operating System	Co3. <u>Develop</u> Small Real Time Operating System (RTOS) and <u>Write</u> code for Embedded application using RTOS.	4.1 Basics of OS and RTOS: Code Execution in Operating system with simplest Round robin Scheduling, Round robin with interrupts, Multi Processing

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
	Pro8. <u>Test</u> the small Real Time Operating system (developed during theory hours or market available with support of documentation) for various functionality. 4a. <u>Select</u> the Operating system as per requirement. 4b. <u>Understand</u> basic concepts and Components of RTOS and <u>develop</u> small Real Time Operating system. 4c. <u>Write</u> and <u>Debug</u> RTOS code for required Embedded System Application.	with multi core and Muti tasking, Types of Operating System (Real Time, Non Real Time and Soft Real time), Selection of RTOS for an Embedded system, 4.2 Architectural Diagram and functioning of Operating System, Kernel space and user space, Task, Thread and Process State of Task (Ready, running, suspended, deleted), 4.3 Various Task scheduling algorithms (Various multi-tasking system), 4.4 Task Communication and related programs, Task synchronization, Message Passing, mailbox, signalling, semaphore, Mutex. 4.5 Problems (like Deadlock, live lock, starvation),
Unit-V Expert Systems implementation on computer On chip	Co4. <u>Apply</u> and <u>Test</u> Artificial Intelligence (AI) and Machine Learning (ML) algorithms for an Embedded Application implemented on Computer on Board Devices. Pro7. <u>Write</u> AI and ML codes in Prolog, Python and C, and <u>Test</u> the result and speed of execution on given Computer On chip/board device and <u>make</u> (<u>Develop</u> interfacing and <u>Write</u> code) it communicating it to main controller/FPGA chip in Embedded board. 5a. <u>Select</u> Embedded board for required AI ML algorithm and target Embedded application. 5b. <u>Write</u> python code to access pins of computer on board device. 5c. <u>Implement</u> AI/ML algorithm on computer on board device and develop interfacing scheme and code to communicate the result to main controller/FPGA chip.	5.1 Introduction to several computers on chip boards, Architecture, 5.2 Accessing peripherals using python, 5.3 Python implementation of AI and Machine learning on Computer on chip device 5.4 Communication of result

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit	Unit Title	Teaching	Distribution of Theory Marks
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No.		Hours	R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	06	1	3	5	0	0	1	10
II	Communication Protocol and Device driver.	09	2	3	8	2	0	0	15
III	FPGA Programming in VHDL	09	1	3	2	3	1	5	15
IV	Fundamental of Real Time Operating System	09	2	4	6	1	0	2	15
V	Expert Systems implementation on computer On chip	12	4	2	2	2	2	3	15
Total		45	10	15	23	8	3	11	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. It should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

Just to give idea, A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here:

- 1) **Smart Class Room:** The machine learning and AI algorithm learned in unit IV, can be used to identify an object or a gesture. The algorithm can be trained for particular application like detecting a particular student in class, observing his learning effort and evaluating its daily, weekly and semester performance. Report can be sent to student for his/her betterment. The prototype design can be re investigated and re design to increase the speed of response. More function that make classroom teaching smart and help teaching learning and evaluation process can be incorporated.
- 2) **VHDL/Verilog implementation of Specialized DSP algorithm:** Select the algorithm and code on FPGA. The algorithm selected should be much more complex to be executed by microcontroller in real time and hence the use of FPGA should be justified. (CO 1,2,3,4)
- 3) **VHDL implementation of Space Vector modulation:** The space Vector modulation is one of the popular modulation technique use to generate the Gate driving signal in converter application. The VHDL implementation can be useful to maintain the minimum precision required for duty cycle as high frequency PWM modulators can be implemented on FPGA platform.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

The teacher can use following strategies, to accelerate the attainment of the various types of learning outcomes in this course:

- Tutorials that need hand calculation and testing in Software.
- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance may to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- The teacher can suggest several research paper to read and refer. This can be assign to specially to evaluate the capacity of individual, to understand topics related to signal processing but not in syllabus. The capacity can be judge by the way or quality of efforts put, the simulation developed to verify the algorithm and brain storming done and presented to evaluator in form of presentation.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Make a report by literature review on new Embedded boards
- Identify the possible reduction in mathematical burden required in particular AI/ML algorithm, modify the code and test verify the speed for new algorithm.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduction	1	<i>Pro1. <u>Develop</u> memory map registers in header file using pointer, structure and union for Peripheral registers in ARM M4 microcontroller.</i> <i>1a. <u>Understand</u> the Architecture of peripheral from datasheet and <u>write</u> down driver to configure and <u>use</u> the peripheral.</i> <i>1b. <u>Understand</u> and <u>use</u> ready Device driver.</i>	<i>Lecture Only (Experiment 1, will be covered during class room exercise)</i>	<i>Mixture of black board and ICT</i>
Unit– II Communication Protocol and Device driver.	2	<i>2a. <u>identify</u> the problems with communication and <u>provide</u> the solution.</i>	<i>Lecture Only. The Lab hours will also be utilized for theory.</i>	<i>Mixture of black board and ICT</i>
	3	<i>Pro4. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> C driver for it</i> <i>a. SPI, I2C,</i> <i>b. CAN communication</i> <i>2b. <u>Understand</u> and <u>use</u> Device driver for various communication Protocols and <u>find</u> bug if any.</i>	<i>Lecture + Experimentation</i>	<i>Mixture of black board and ICT</i>
Unit– III FPGA Programming in	4	<i>Pro2. <u>Develop/Write</u> basic codes for FPGA in VHDL as in follows, and also write test</i>	<i>Lecture +</i>	<i>Mixture of black board and ICT</i>

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
VHDL		bench program for the same. a. Combinational Circuits like Digital Gates and Digital Circuits based on Digital Gates, SR Non Clocked Flip Flop b. Sequential Clock based Circuits 3a. <u>Develop</u> VHDL program for given architecture.	Experimentation One of the lecture of this week will be utilized for experiment along with regular two-hour lab in a week.	
	5	Pro3. <u>Learn</u> the following architecture in an Unknown Microcontroller, and <u>develop/Write</u> the VHDL code to implement the given architecture of peripheral in FPGA. a. GPIO b. SPI,I2C, UART, USART, CAN communication 3b. <u>Develop</u> VHDL program for given architecture.	Experimentation only. (The lecture hours of this week will be utilized as lab hours along with regular slot of lab)	Mixture of black board and ICT
	6	Pro5. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> VHDL code to implement the given architecture of peripheral in FPGA. a. UP, Down, Up-Down Timer/Counter b. Watchdog Timer c. Pulse Width Modulator d. Capture Compare Unit 3a. <u>Understand</u> the given VHDL program, and <u>modify</u> it as per requirement. 3b. <u>Develop</u> VHDL program for given architecture.	Experimentation only. (The lecture hours of this week will be utilized as lab hours along with regular slot of lab)	Mixture of black board and ICT
	7	Pro6. <u>Learn</u> the following architecture in an Unknown Microcontroller, <u>Develop/Write</u> C driver for it. a. UP, Down, Up-Down Timer/Counter b. Watchdog Timer c. Pulse Width Modulator d. Capture Compare Unit 3c. <u>Develop</u> VHDL Program for interfacing external chip with FPGA.	Lecture + Experimentation	Mixture of black board and ICT
	8	3d. <u>Develop</u> VHDL Program for interfacing external chip with FPGA.	Lecture Only. The Lab hours will also be utilized for theory.	Mixture of black board and ICT

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– V Expert Systems implementation on computer On chip	9	Pro7. <u>Write</u> AI and ML codes in Prolog, Python and C, and <u>Test</u> the result and speed of execution on given Computer On chip/board device and <u>make</u> (<u>Develop</u> interfacing and <u>Write</u> code) it communicating it to main controller/FPGA chip in Embedded board. 5a. <u>Select</u> Embedded board for required AI ML algorithm and target Embedded application. 5b. <u>Write</u> python code to access pins of computer on board device.	Lecture + Experimentation	Mixture of black board and ICT
	10	Pro7. <u>Write</u> AI and ML codes in Prolog, Python and C, and <u>Test</u> the result and speed of execution on given Computer On chip/board device and <u>make</u> (<u>Develop</u> interfacing and <u>Write</u> code) it communicating it to main controller/FPGA chip in Embedded board. 5c. Briefly <u>understand</u> various AI/ML Algorithm and the parameters related to it and parameters to be passed to python AI/ML function and <u>Apply</u> it in coding. 5d. <u>Implement</u> AI/ML algorithm on computer on board device and <u>develop</u> interfacing scheme and <u>write</u> code to communicate the result to main controller/FPGA chip.	Lecture + Experimentation	Mixture of black board and ICT
	11	Pro7. <u>Write</u> AI and ML codes in Prolog, Python and C, and <u>Test</u> the result and speed of execution on given Computer On chip/board device and <u>make</u> (<u>Develop</u> interfacing and <u>Write</u> code) it communicating it to main controller/FPGA chip in Embedded board. 5c. Briefly <u>understand</u> various AI/ML Algorithm and the parameters related to it and parameters to be passed to python AI/ML function and <u>Apply</u> it in coding. 5d. <u>Implement</u> AI/ML algorithm on computer on board device and <u>develop</u> interfacing scheme and <u>write</u> code to communicate the result to	Lecture + Experimentation	Mixture of black board and ICT

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		main controller/FPGA chip.		
	12	Pro7. <u>Write</u> AI and ML codes in Prolog, Python and C, and <u>Test</u> the result and speed of execution on given Computer On chip/board device and <u>make</u> (<u>Develop</u> interfacing and <u>Write</u> code) it communicating it to main controller/FPGA chip in Embedded board. 5c. Briefly <u>understand</u> various AI/ML Algorithm and the parameters related to it and parameters to be passed to python AI/ML function and <u>Apply</u> it in coding. 5d. <u>Implement</u> AI/ML algorithm on computer on board device and <u>develop</u> interfacing scheme and <u>write</u> code to communicate the result to main controller/FPGA chip.	Lecture + Experimentation	Mixture of black board and ICT
Unit– IV Fundamental of Real Time Operating System	13	4a. Select the Operating system as per requirement.	Lecture Only.	Mixture of black board and ICT
	14	4b. Develop Real Time small Operating system and use it for developing an Embedded application.	Lecture + Experimentation in class room	Mixture of black board and ICT
	15	Pro8. <u>Test</u> the small Real Time Operating system (developed during theory hours or market available with support of documentation) for various functionality. 4b. Develop Real Time small Operating system and use it for developing an Embedded application.	Lecture of 1 hour + Experimentation in class room and lab location for 4 hours	Mixture of black board and ICT

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Introduction to Embedded system	Shibu K V	MacGraw Hill Education (India) Pvt Ltd Thirteen Reprint

S. No.	Title of Book	Author	Publication including ISBN
			ISBN (13 digit): 978-0-07-014589-4 ISBN (10 digit):0-07-014589-X
2	<i>Embedded Systems Architecture, programming and design</i>	<i>Raj Kamal</i>	<i>Tata – MGHill, Second Edition</i> ISBN 13: 978-0-07-066764-8 ISBN 10: 0-07-066764-0
3	<i>Embedded System Architecture: A Comprehensive Guide for Engineers and Programmers</i>	<i>Tammy Noergaard</i>	<i>Elsevier, Newnes</i> ISBN: 0-7506-7792-9
4	<i>The Definitive Guide to ARM cortex M3 and M4 Processor,</i>	<i>Joseph Yiu</i>	<i>Elsevier, Newnes</i> <i>Third Edition</i> ISBN-13: 978-0-12-408082-9
5	<i>Real Time Operating Systems for ARM cortex-M Microcontrollers,.</i>	<i>Jonathan W. Valvano</i>	<i>Volume 3, 4th Edition</i> ISBN-13: 978-1466468863 ISBN-10: 1466468866
6	<i>ARM System-on-chip Architecture</i>	<i>Steve furber</i>	<i>Pearson</i> ISBN-13: 9780201675191
7	<i>Machine Learning using Python</i>	<i>Manaranjan Pradhan, U Dinesh Kumar</i>	<i>Wiley,</i> ISBN: 978-81-265-7990-7 ISBN: 978-265-8855-8(ebk)
8	<i>Artificial Intelligence</i>	<i>Elaine Rich, Kevin Knight, Shivashankar B Nair</i>	<i>Third Edition, TATA MACGRAW HILL</i> ISBN 13: 978-0-07-008770-5 ISBN 10: 0-07-008770-9
9	<i>VLSI Design</i>	<i>A. Albert Raj, T. Latha</i>	<i>PHI Learning</i> ISBN-978-81-203-3431-1
10	<i>Circuit Design and simulation with VHDL.</i>	<i>Volnei A. Pedroni</i>	<i>MIT Press</i> ISBN-10: 0262014335 ISBN-13: 978-0262014335

XV SOFTWARE/LEARNING WEBSITES

1. Free RTOS online Documentation: <https://www.freertos.org/>
2. NPTEL Courses by Prof. Santanu Chaudhary, IIT Delhi
<https://nptel.ac.in/courses/108102045>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE 377	Semester VI	Embedded Systems Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
377.1	<u>Use</u> the ready driver to configure the on chip peripherals and develop driver as a C callable function, <u>find</u> the bug in driver function.	1	2	-	-	-	-	-	-	-	-	-	-	-	-
377.2	<u>Write/Develop</u> VHDL program for required architecture. <u>Test and Debug</u> it.	1	3	3	2	1	-	-	-	2	2	2	1	2	-
377.3	<u>Develop</u> Small Real Time Operating System(RTOS) and <u>Write</u> code for Embedded application using RTOS.	1	3	3	2	1	-	-	-	2	2	2	-	2	-
377.4	<u>Apply and Test</u> Artificial Intelligence (AI) and Machine Learning (ML) algorithms for an Embedded Application implemented on Computer on Board Devices.	1	3	3	2	1	-	-	-	2	2	2	1	2	2

XVII NAMES OF RESOURCE PERSONS

a) Jivanadhar A. Joshi

UG Engineering Programme	Semester
Electrical Engineering	six

I RATIONALE

Electrical engineers have to deal with various types of electrical and electronics *circuit components, its application, functions and basic electrical circuit theorem* used in many applications. This curriculum course structure is developed such that the student is able to understand and apply the basic principles and laws of electrical and electronics engineering with emphasis on the analysis and application to simple practical engineering problems.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

Understand and analyze the Modeling the FACTS and HVDC system to solve the power system problem.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. To learn the operating principle of FACTS DEVICES.
2. Develop the skill to model the FACTS devices to solve the power system problems
3. To learn the operating principle of HVDC system .
4. Develop the skill to model the different types of HVDC system to solve the power system problems
5. To address the underlying concepts of power quality, its improvement and standards

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	3,4	To study and simulate the operation of Multi-Pulse Converter Using MATLAB Simulation.	03
2	3,4	To study and analyze the operation of 6 pulse HVDC system using MATLAB Simulation in Power System.	03
3	3,4	To study and analyze the operation of 6 pulse HVDC system using Hardware model.	03
4	1,2	To study and analyze the operation of Static VAr Compensator (SVC) using MATLAB Simulation in Power System.	03
5	1,2	To Study and Analyze the Operation of 3-Phase 3-Level Diode Clamped Multi-Level Inverter Using Hardware Model.	03
6	1,2	To Study and analyze the operation of Static Synchronous Compensator (STATCOM) using MATLAB Simulation in Power System.	03
7	1,2	To Study and analyze the operation of Thyristor Controlled Series Capacitor (TCSC) using MATLAB Simulation in Power System.	03
8	1,2	To Study and analyze the operation of VSC-based FACTS Controllers using Hardware model.	03
9	1,2	Comparison of SVC and STATCOM in Static Voltage Stability Margin Enhancement using PSAT Software.	03
10	1,2	To Study and Observe the effect of Voltage Variations on the Performance of DVR.	03
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	HVDC demo kit	1,2,3
2	DSO	1 to 10
3	FACTS demo kit with all FACTS Controller	3
4	Multimeter	3,4
5	Matlab simulink	1 to 10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.

5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- 'Responding Level' in 1st year
- 'Valuing Level' in 2nd year
- 'Organising Level' in 3rd year
- 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I FACTS Controllers	CO 1.To learn the operating principle of FACTS DEVICES. CO 2. Develop the skill to model the FACTS devices to solve the power system problems PrO 4 To study and analyze the operation of Static VAR Compensator (SVC) using MATLAB Simulation in Power System. PrO 5 To Study and Analyze the Operation of 3-Phase 3-Level Diode Clamped Multi-Level Inverter Using Hardware Model. PrO 6 To Study and analyze the operation of Static Synchronous Compensator (STATCOM) using MATLAB Simulation in Power System. PrO 7 To Study and analyze the operation of Thyristor Controlled Series Capacitor (TCSC) using MATLAB Simulation in Power System. PrO 8 To Study and analyze the operation of VSC-based FACTS Controllers using Hardware model. PrO 9 Comparison of SVC and STATCOM in Static Voltage Stability Margin Enhancement using PSAT Software. PrO 10 To Study and Observe the effect of Voltage Variations on the Performance of DVR. 1a. <i>Analyse the operation of SVC</i> 1b. <i>Analyse the operation of TCSC</i> 1c. <i>Analyse the operation of STATCOM</i>	1.1 Need of FACTS devices 1.2 classification of FACTS devices, operating principle of SVC 1.3 V-I characteristic of SVC 1.4 Advantage of slope in V-I characteristic, SVC applications for transient and voltage stability improvement, mitigation of SSR 1.5 Advantages of TCSC, different mode of operation of TCSC, different modeling concepts of TCSC, 1.6 Operating principle of STATCOM, SSSC and UPFC and their applications for power system performance improvement, 1.7 Power flow solution with SVC, TCSC, SSSC and UPFC

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	1d. Analyse the operation of SSSC and UPFC	
Unit– II HVDC Transmission	CO 3. To learn the operating principle of HVDC system . CO 4. Develop the skill to model the different types of HVDC system to solve the power system problems PrO 1 To study and simulate the operation of Multi-Pulse Converter Using MATLAB Simulation. PrO 2 To study and analyze the operation of 6 pulse HVDC system using MATLAB Simulation in Power System. PrO 3 To study and analyze the operation of 6 pulse HVDC system using Hardware model. <i>2a Analyse the operation of 6-pulse rectifier</i> <i>2b. Analyse the operation of 6-pulse inverter</i> <i>2c. Analyse the operation of bilateral HVDC system</i>	2.1 Introduction, equipment required for HVDC systems, 2.2 Comparison of AC and DC Transmission, Limitations of HVDC transmission lines, 2.3 Reliability of HVDC systems, comparison of HVDC link with EHVAC link, HVDC system configuration and components, 2.4 Fundamental equations in HVDC system, HVDC links, converter theory and performance equation, valve characteristic, converter circuits, 2.5 Converter transformer rating, multi bridge converters, abnormal operation of HVDC system, control of HVDC system, harmonics and filters. Influence of AC system strength on AC/DC system interaction, response to AC and DC system faults
Unit– III Power Quality problems & Standards	CO 5. To address the underlying concepts of power quality, its improvement and standards 3a. Understand the basic terms of power quality. 3b. Understand the standards of power quality	3.1 Introduction: Definition & Reasons of Occurrence of following Voltage Dip, Brief voltage increases, Brief voltage interruption, Transients, Voltage Notches, Flickers, Distortion, Un-balance. Power Quality Indices, 3.2 Limits of Harmonic Distortion according to IEEE, IEC, EN and NORSOK limits. 3.3 Brief Introduction of Power quality Standards: IEC 61000-2-5, IEC 61000-2-1, IEC 1159 (Categories of Power quality variation according to IEEE 1159 standard with their relevant Spectral content, Duration of occurrence & Magnitude)

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	FACTS Controllers	30	3	3	6	9	10	4	35

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
II	HVDC Transmission	20	2	3	5	6	6	2	24
III	Power Quality problems & Standards	10	2	1	3	2	3	0	11
Total		60	7	7	14	17	19	6	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) Design the any FACTS controller and Simulink in MATLAB.
- 2) Design the 12 Pulse, 18 pulse and 24 pulse controller and Simulink in MATLAB.
- 3) Determine the power quality of incoming power in terms of all parameters using power quality devices.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages

for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different Semiconductor switches used in converters.
- b) Prepare 15 min. power point presentation on how to choose the particular converter for the particular application.

XIII COURSE PLAN

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I FACTS Controllers	PrO 4 To study and analyze the operation of Static VAR Compensator (SVC) using MATLAB Simulation in Power System. PrO 5 To Study and Analyze the Operation of 3-Phase 3-Level Diode Clamped Multi-Level Inverter Using Hardware Model. PrO 6 To Study and analyze the operation of Static Synchronous Compensator (STATCOM) using MATLAB Simulation in Power System. PrO 7 To Study and analyze the operation of Thyristor Controlled Series Capacitor (TCSC) using MATLAB Simulation in Power System. PrO 8 To Study and analyze the operation of VSC-based FACTS Controllers using Hardware model. PrO 9 Comparison of SVC and STATCOM in Static Voltage Stability Margin Enhancement using PSAT Software. PrO 10 To Study and Observe the effect of Voltage Variations on the Performance of DVR. 1a. Analyse the operation of SVC 1b. Analyse the operation of TCSC 1c. Analyse the operation of STATCOM 1d. Analyse the operation of SSSC and UPFC	1.1 Need of FACTS devices 1.2 classification of FACTS devices, operating principle of SVC 1.3 V-I characteristic of SVC 1.4 Advantage of slope in V-I characteristic, SVC applications for transient and voltage stability improvement, mitigation of SSR 1.5 Advantages of TCSC, different mode of operation of TCSC, different modeling concepts of TCSC, 1.6 Operating principle of STATCOM, SSSC and UPFC and their applications for power system performance improvement, 1.7 Power flow solution with SVC, TCSC, SSSC and UPFC
Unit– II HVDC Transmission	PrO 1 To study and simulate the operation of Multi-Pulse Converter Using MATLAB Simulation. PrO 2 To study and analyze the operation of 6 pulse HVDC system using MATLAB Simulation in Power System. PrO 3 To study and analyze the operation of 6 pulse HVDC system using Hardware model. 2a Analyse the operation of 6-pulse	2.1 Introduction, equipment required for HVDC systems, 2.2 Comparison of AC and DC Transmission, Limitations of HVDC transmission lines, 2.3 Reliability of HVDC systems, comparison of HVDC link with EHVAC link, HVDC system configuration and components, 2.4 Fundamental equations in HVDC system, HVDC links, converter theory and performance equation, valve characteristic, converter circuits,

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	<i>rectifier</i> 2b. <i>Analyse the operation of 6-pulse inverter</i> 2c. <i>Analyse the operation of bilateral HVDC system</i>	2.5 Converter transformer rating, multi bridge converters, abnormal operation of HVDC system, control of HVDC system, harmonics and filters. Influence of AC system strength on AC/DC system interaction, response to AC and DC system faults
Unit– III Power Quality problems & Standards	3a. Understand the basic terms of power quality. 3b. Understand the standards of power quality	3.1 Introduction: Definition & Reasons of Occurrence of following Voltage Dip, Brief voltage increases, Brief voltage interruption, Transients, Voltage Notches, Flickers, Distortion, Un-balance. Power Quality Indices, 3.2 Limits of Harmonic Distortion according to IEEE, IEC, EN and NORSOK limits. 3.3 Brief Introduction of Power quality Standards: IEC 61000-2-5, IEC 61000-2-1, IEC 1159 (Categories of Power quality variation according to IEEE 1159 standard with their relevant Spectral content, Duration of occurrence & Magnitude)

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Direct Current Transmission,	E. W. Kimbark	Vol I, Wiley-Interscience, 1971
2	Power System Stability and Control	P. Kundur	McGraw Hill, 1994.
3	HVDC Transimission,	S. Kamakshiah and V. Kamaraju	McGraw Hill, 2011
4	,HVDC Power Transmission Systems: Technology and System Interactions,	K.R. Padiyar	New Age International
5	FACTS: Modeling and Simulation in Power Networks,	E. Acha	Wiley 2004
6	Reactive Power Control in Electric Systems,	T.J.E Miller	Wiley India Edition, 1982
7	Thyristor-based FACTS controllers for Electrical Transmission systems	R. Mohan Mathur and R. K. Varma	IEEE Press, 2002

XV SOFTWARE/LEARNING WEBSITES

1. <http://nptel.ac.in/courses/108108077/>
2. <http://nptel.ac.in/courses/108102046/>

3. <http://nptel.ac.in/courses/108104011/>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE378	Semester I														
378	Competency •Understand and analyze the Modeling the FACTS and HVDC system to solve the power system problem.	3	2	2	2	1	-	1	-	-	-	-	1	3	-
378.1	To learn the operating principle of FACTS DEVICES.	3	3	-	-	-	-	-	-	2	-	-	3	3	-
378.2	Develop the skill to model the FACTS devices to solve the power system problems	2	1	3	-	3	-	-	-	-	-	-	-	2	-
378.3	To learn the operating principle of HVDC system .	3	2	2	3	2	-	3	3	3	3	-	2	3	-
378.4	Develop the skill to model the different types of HVDC system to solve the power system problems	2	2	-	-	1	-	-	-	2	-	-	-	1	-
378.5	To address the underlying concepts of power quality, its improvement and standards	3	-	-	-	-	-	-	-	-	-	-	2	1	-

XVII NAMES OF RESOURCE PERSONS

a) Dr. Dharmesh Dabhi

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

The electrical stresses experienced by the insulations will reduce the life expectancy of high voltage equipment. Hence, a thorough understanding of the fundamental properties of the materials and their failure mechanisms during service is essential for appropriate and optimal design. This curriculum course structure is developed such that the student is able to use the knowledge of different insulation systems and perform the high voltage tests on electrical equipment under the impression of different heavy electrical stress.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Perform the different high voltage tests on electrical equipment to access the performance parameters.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Explain** the breakdown mechanisms involved in different dielectrics.
- 2) **Use** the various generation and measurement techniques of high voltages and current.
- 3) **Perform** the insulation coordination studies.
- 4) **Perform** the destructive tests on electrical equipment.
- 5) **Perform** the non-destructive tests on electrical equipment.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Create a detailed report for the different types of insulating materials.	02
2	2	Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory.	02
3	2	Examine the impact of certain conditions on the breakdown voltage using sphere gap assembly.	04
4	2	Examine the breakdown characteristic for the given rod-rod, rod-plane and plane-plane electrode configurations.	04
5	3	Examine the function of horn gap lightning arrestor.	02
6	4	Determine the dielectric strength of the given transformer oil.	02
7	4	Determine the dielectric strength of the given solid dielectric materials.	02
8	4	Determine the power frequency flashover voltage of the given insulator.	02
9	4	Perform the impulse voltage test on the given insulator.	04
10	5	Measure the dielectric loss factor and dc resistivity of the given transformer oil.	02
11	5	Trace the equipotential lines and electric field lines in the given electrode systems using electrolytic tank.	02
12	5	Examine the localized discharge activity for a given insulator using COMSOL Multiphysics.	02
Total			30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Test rig for sphere gap & rod gap assembly: 100kV AC/140kV DC high voltage tester 65mm, 100mm & 150mm spherical electrodes - Earth electrode, rod electrodes and plane electrodes	3,4,8,10
2	Horn gap lightning arrestor	5
3	0-100 kV motorized oil BDV test kit	6
4	30kV AC/40kV DC high voltage tester	7
5	5-stage, 1.5kJ, 150kV Marx Impulse voltage generator	9
6	Oil dissipation factor and dc resistivity meter	11
7	Electrolytic Tank	12
8	COMSOL Multiphysics Software	13

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Function as a team member.
- 3) Function as a team leader.
- 4) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- 'Responding Level' in 1st year
- 'Valuing Level' in 2nd year
- 'Organising Level' in 3rd year
- 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Breakdown mechanisms in gaseous, liquid and solid dielectrics	CO 1. <u>Explain</u> the breakdown mechanisms involved in different dielectrics. <i>PRO 21. Create a detailed report for the different types of insulating materials.</i> 1a. <u>Justify</u> the requirement of insulation for <u>the given</u> electrical equipment. 1b. <u>Memorize</u> the different types of insulating materials. 1c. <u>Explain</u> the ionization/conduction mechanisms involved in dielectric breakdown. 1d. <u>Evaluate</u> the dielectric strength of liquid dielectrics. 1e. <u>Evaluate</u> the dielectric strength of solid dielectrics. 1f. <u>Evaluate</u> effect of impurities over the performance of the insulation.	1.1 Need of insulation 1.2 Different types of insulation 1.3 Nature and causes of insulation failures 1.4 Ionization processes 1.5 Breakdown theories
Unit– II Generation of High Voltages and Currents	CO 2. <u>Use</u> the various generation and measurement techniques of high voltages and current. <i>PRO 4. Create a detailed report for the <u>generation and measurement of high voltage and current in high voltage laboratory.</u></i> 2a. Choose the relevant high voltage generation technique. 2b. Determine the circuit parameters to obtain the standard wave shape of test voltage as per the Indian or international standards. 2c. Evaluate the performance of generating methods for the specified conditions.	2.1 Different methods to generate high DC, AC and impulse voltages and currents. 2.2 Operating principles and working of various high DC voltage generating techniques. 2.3 Operating principles and working of various high AC power frequency voltage generating methods. 2.4 Operating principle and working of high frequency

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
		high voltage AC generation technique. 2.5 Impulse voltage wave shape and Marx impulse voltage generator. 2.6 Operating principle and working of high impulse current generator.
Unit– III Measurement of High Voltages and Currents	CO 2. <u>Explain</u> the various generation and measurement techniques of high voltages and current. PrO 2. <u>Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory.</u> PrO 3. <u>Examine the impact of certain conditions on the breakdown voltage using sphere gap assembly.</u> PrO 4. <u>Examine the breakdown characteristic for the given rod-rod, rod-plane and plane-plane electrode configurations.</u> 3a. Choose the relevant high voltage and current measurement technique. 3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.	3.1 Different methods to measure high DC, AC and impulse voltages and currents. 3.2 Operating principles and working of various high DC voltage measurement techniques. 3.3 Operating principles and working of various high AC voltage measurement techniques. 3.4 Operating principles and working of various impulse and high frequency voltage measurement techniques. 3.5 Factors influencing the high voltage measurement using sphere gap assembly. 3.6 Operating principles and working of various high current measurement techniques.
Unit-IV Overvoltage phenomenon and Insulation coordination in Electric Power System	CO 3. <u>Perform insulation coordination studies.</u> PrO 5. <u>Examine the function of horn gap lightning arrestor.</u> 4a. <u>Understand the causes of overvoltage phenomena in electric power system.</u> 4b. <u>Perform insulation coordination studies to determine basic impulse insulation level.</u> 4c. <u>Explain</u> the function of lightning arrestors. 4d. <u>Select</u> the relevant lightning arrestor for <u>the specified</u> application.	4.1 Natural cause of overvoltage – Lightning phenomena 4.2 Overvoltage due to switching surges 4.3 Overvoltage due to system faults and other specified conditions 4.4 Types of lightning arrestors 4.5 Insulation coordination

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
		4.6 Basic impulse insulation level
Unit – V Destructive Testing of Electrical Equipment	CO 4. <u>Perform</u> the destructive tests on electrical equipments. <i>PrO 6. <u>Determine the dielectric strength of the given transformer oil.</u></i> <i>PrO 7. <u>Determine the dielectric strength of the given solid dielectric materials.</u></i> <i>PrO 8. <u>Determine the power frequency flashover voltage of the given insulator.</u></i> <i>PrO 9. <u>Perform the impulse voltage test on the given insulator.</u></i> 5a. <u>Perform power frequency flashover voltage test on electrical equipment under the specified test conditions.</u> 5b. <u>Perform power frequency withstand voltage test on electrical equipment under the specified test conditions.</u> 5c. <u>Perform impulse voltage test on electrical equipment under the specified test conditions.</u> 5d. <u>Perform short circuit test on circuit breakers.</u> 5e. <u>Perform synthetic test on circuit breaker.</u>	5.1 Power frequency flashover voltage test 5.2 Power frequency voltage withstand test 5.3 Impulse flashover voltage test 5.4 Testing of insulators, bushings, capacitors, transformers, circuit breakers. 5.5 Short circuit test of circuit breakers. 5.6 Synthetic test of circuit breakers.
Unit – VI Non-destructive Testing of Electrical Equipment	CO 5. <u>Perform</u> the non-destructive tests on electrical equipments. <i>PrO 10. <u>Measure the dielectric loss factor and dc resistivity of the given transformer oil.</u></i> <i>PrO 11. <u>Trace the equipotential lines and electric filed lines in the given electrode systems using electrolytic tank.</u></i> <i>PrO 12. <u>Examine the localized discharge activity for a given insulator using COMSOL Multiphysics.</u></i> 6a. <u>Measure the dissipation factor and unknown capacitance of the given insulation.</u> 6b. <u>Measure the dc resistivity of the given insulating liquid.</u> 6c. Perform partial discharge test. 6d. Perform dissolved gas analysis test on transformer oil. 6e. Perform sweep frequency response analysis of electrical equipment. 6f. Predict remaining life assessment of the given high voltage equipment.	6.1 Determination of dissipation factor and unknown capacitance of insulation. 6.2 Determination of DC resistivity of insulation liquid. 6.3 Partial discharge test of electrical equipments. 6.4 Dissolved gas analysis of transformer oil. 6.5 Sweep frequency response analysis of electrical equipment

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Breakdown mechanisms in gaseous, liquid and solid dielectrics	12	2	2	2	4	2	0	12
II	Generation of High Voltages and Currents	12	2	2	4	4	4	2	18
III	Measurement of High Voltages and Currents	08	0	0	4	2	2	0	08
IV	Overvoltage phenomenon and Insulation coordination in Electric Power System	02	2	2	0	2	2	0	08
V	Destructive Testing of Electrical Equipment	09	0	2	4	4	4	0	14
VI	Non-destructive Testing of Electrical Equipment	02	0	2	2	2	2	2	10
Total		45	06	10	16	18	16	4	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) Undertake the survey and submit a report on different type of insulating materials.
- 2) Create a prototype model of voltage multiplier circuit for HVDC generation.
- 3) Create a prototype model of multistage impulse voltage generator.
- 4) Create a prototype model of tesla coil.
- 5) Create a prototype model of Van de Graff generator.
- 6) Examine the equipotential and electric field lines between plane-plane, plane-convex and convex-convex electrode configuration using finite element method magnetics (FEMM) software.

- 7) Perform electric field and electric potential distribution for the given insulator with AC, DC and impulse voltage using COMSOL Multiphysics software.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- '**L**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Library survey regarding different insulations used in industries.
- Prepare 15 min. power point presentation on how to choose insulation for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Breakdown mechanisms in gaseous, liquid and solid dielectrics	1	PrO 6. Create a detailed report for the different types of insulating materials. 1a. Justify the requirement of insulation for the given electrical equipment. 1b. <u>Memorize</u> the different types of insulating materials. (self-learn) 1c. <u>Explain</u> the ionization/conduction mechanisms involved in dielectric breakdown.	Lecture with media	PPTs, black board
	2	1d. Explain the ionization/conduction mechanisms involved in dielectric breakdown.	Lecture With media	PPTs, black board
	3	1e. Explain the ionization/conduction mechanisms involved in dielectric breakdown.	Lecture With media	PPTs, black board
	4	1f. <u>Evaluate</u> the dielectric strength of liquid dielectrics. 1g. <u>Evaluate</u> the dielectric strength of solid dielectrics. 1h. <u>Evaluate</u> effect of impurities over the	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		performance of the insulation.		
Unit– II Generation of High Voltages and Currents	5	PrO 2. <u>Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory.</u> 2a. Choose the relevant high voltage generation technique. 2b. Analyze the high voltage and current generation techniques.	Lecture With media	PPTs, black board
	6	2b. Analyze the high voltage and current generation techniques.	Lecture With media	PPTs, black board
	7	2b. Analyze the high voltage and current generation techniques. 2c. Determine the circuit parameters to obtain the standard wave shape of test voltage as per the Indian or international standards.	Lecture With media	PPTs, black board
	8	2c. Determine the circuit parameters to obtain the standard wave shape of test voltage as per the Indian or international standards. 2d. Evaluate the performance of generating methods for the specified conditions.	Lecture With media	PPTs, black board
Unit III Measurement of High Voltages and Currents	9	PrO 2. <u>Create a detailed report for the generation and measurement of high voltage and current in high voltage laboratory.</u> PrO 3. <u>Examine the impact of certain conditions on the breakdown voltage using sphere gap assembly.</u> PrO 4. <u>Examine the breakdown characteristic for the given rod-rod, rod-plane and plane-plane electrode configurations.</u> 3a. Choose the relevant high voltage and current measurement technique. 3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.	Lecture With media	PPTs, black board
	10	3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.	Lecture With media	PPTs, black board
	11	3b. Evaluate the performance of high voltage and current measurement methods for the specified conditions.	Lecture With media	PPTs, black board
Unit-IV Overvoltage phenomenon and Insulation coordination in Electric Power System	12	PrO 5. <u>Examine the function of horn gap lightning arrester.</u> 4a. <u>Understand</u> the causes of overvoltage phenomena in electric power system. 4b. Perform insulation coordination studies to determine basic impulse insulation level. 4c. <u>Explain</u> the function of lightning arrestors. 4d. <u>Select</u> the relevant lightning arrester for the <u>specified</u> application.	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – V Destructive Testing of Electrical Equipment		<i>PrO 6.</i> <u>Determine the dielectric strength of the given transformer oil.</u> <i>PrO 7.</i> <u>Determine the dielectric strength of the given solid dielectric materials.</u> <i>PrO 8.</i> <u>Determine the power frequency flashover voltage of the given insulator.</u> <i>PrO 9.</i> <u>Perform the impulse voltage test on the given insulator.</u> 5a. <u>Perform</u> power frequency flashover voltage test on electrical equipments under the specified test conditions.		
	13	5a. <u>Perform</u> power frequency flashover voltage test on electrical equipments under the specified test conditions. 5b. <u>Perform</u> power frequency withstand voltage test on electrical equipments under the specified test conditions.	Lecture with media	PPTs, black board
	14	5c. <u>Perform</u> impulse voltage test on electrical equipment under the specified test conditions. 5d. <u>Perform</u> short circuit test on circuit breakers.	Lecture with media	PPTs, black board
Unit – VI Non-destructive Testing of Electrical Equipment	15	5e. <u>Perform</u> synthetic test on circuit breaker. <i>PrO 10.</i> <u>Measure the dielectric loss factor and dc resistivity of the given transformer oil.</u> <i>PrO 11.</i> <u>Trace the equipotential lines and electric filed lines in the given electrode systems using electrolytic tank.</u> <i>PrO 12.</i> <u>Examine the localized discharge activity for a given insulator using COMSOL Multiphysics.</u> 6a. <u>Measure</u> the dissipation factor and unknown capacitance of the given insulation. 6b. <u>Measure</u> the dc resistivity of the given insulating liquid. 6c. Perform partial discharge test. 6d. Perform dissolved gas analysis test on transformer oil. 6e. Perform sweep frequency response analysis of electrical equipment. 6f. Predict remaining life assessment of the given high voltage equipment.	Lecture with media, Tutorial	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
13	High Voltage Engineering	Naidu, M.S., Kamaraju, V.	McGraw-Hill, USA, ISBN 0-07-462286-2
14	High Voltage	Wadhwa, C.L.	New Age International (P) Ltd., New Delhi,

S. No.	Title of Book	Author	Publication including ISBN
	Engineering		ISBN: 9788122430905

XV SOFTWARE/LEARNING WEBSITES

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPU/HighVoltageEngg/ui/TOC.htm>
- <http://www.mv.helsinki.fi/tpaulin/Text.hveng.htm>
- <http://ocw.mit.edu/courses/physics/8-02-electricity-and-magnetism-spring-2002/video-lectures/lecture-6-high-voltage-breakdown-and-lightning.htm>
- http://www.sayedasad.com/High_voltage/index_solids.htm

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs												
		PO 1 Used of mathematical method	PO 2 Analysis is mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 The engineer and society	PO 6 Environment and sustainability	PO 7 Ethics	PO 8 Individual and team work:	PO 9 Communication	PO 10 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	
EE379	Semester VI	High Voltage Engineering Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO												
379	Competency Perform the different high voltage tests on electrical equipments to access the performance parameters.	3	2	2	2	1	1	1	-	2	1	3	1	
379.1	Explain the breakdown mechanisms involved in different dielectrics.	3	2	1	1	1	1	1	-	1	1	1	1	
379.2	Explain the various generation and measurement techniques of high voltages and current.	3	2	2	1	1	1	1	-	1	-	2	1	
379.3	Perform the insulation coordination studies.	3	2	1	1	1	1	1	-	1	-	2	1	
379.4	Perform the destructive tests on electrical equipments.	3	1	1	2	1	1	1	-	1	-	1	-	
379.5	Perform the non-destructive tests on electrical equipments.	3	2	1	1	1	1	1	-	1	1	1	-	

XVII NAMES OF RESOURCE PERSONS

- Dr. Mihir A. Bhatt

UG Engineering Programme	Semester
Electrical Engineering	Sixth

I RATIONALE

Things or devices connected to internet, and system working with these devices, has lots of potential. More over to data storage and transfer, such Things/Device can bring more smartness to system operating with it and/or controlled by it. The IoT is gaining popularity and attention and is securing its requirement in various application. Even Process industries as a part of fourth revolution is welcoming and applying this concept, popularly known as Industrial Internet of Things (IIoT).

The Embedded System Engineer, in this era of technology, need to work for the IoT application, requiring hardware design and writing piece of code doing necessary communication. The course covers design of Energy Harvesting hardware, small PCB antennas, Interfacing communication chips, sensors and actuators. The content is helpful for Embedded Hardware Engineer. Similarly, course also covers coding various wireless communication protocols using C and Python libraries, for few popular microcontroller boards, a required skill for an Embedded Software Engineer.

The communication protocols are required to be coded not only in/for device but also for cloud layer and Fog Layer (i.e gateways). This is covered with several cloud offerings. Developing cloud facility or cloud computing or applying data analytics cannot be the objective of this course, but Details required for developing website is covered. Also, managing interaction of network nodes and their management is covered partially.

The course will make the student more adoptable for IoT Solution Industries with functional role in team doing development of IoT Product. They can be suitably adopted for role of Hardware or software engineer.

COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Develop Embedded IoT board.
- Write code for IoT application (Code various Embedded IoT boards for interfacing sensors, actuator, Code for Communication protocols used at various IoT Layers, Write code for Managing interaction of IoT nodes).

II COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) For given IoT System, Draw the overall architecture of complete IoT system, show interconnection of Fog and Edge device and specify the requirements of Various Device involved and Select from offering. Manage and Direct the Development of an IoT product.
- 2) Design the Antenna, Power supply and Communication module for Embedded IoT Microcontroller board. Design Hardware for interfacing Actuators and Sensors.

- 3) Write Python and C Code to establish communication among Embedded IoT Boards, Mobile and IoT Gateway.
- 4) Develop the website using HTML, CSS and Java. Write Python and Java code/script for various cloud offerings. Write code for IoT Communication, IoT System Management.

III TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (CHARUSAT 2021)				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

(*) : Under the **Theory PA**, out of 25 marks, **25 marks** are for **attendance, micro-project assessment** and other **assignment** if given any, to facilitate integration of COs and the remaining 10 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

IV SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1		Develop the website using HTML, CSS and Java.	06
2		Develop dashboards for IoT application.	06
3		Design IoT Embedded boards.	04
4		Write Program in C and Python to interface various Sensors and Actuators with various IoT Microcontroller board offering.	04
5		Write code for to Establish Physical layer communication using C and Python Libraries.	04
6		Write Python code with Various cloud offerings for various Network layer and Application layer communication Protocols.	06
7		Lead Product design team (Hardware and Software Engineers) for given IoT Application/Product/system to be developed. (to be taught by Tutorial)	planned to be covered during classroom teaching
		Total	30

V MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

Sr. No.	Equipment Name with Broad Specifications	PrO No.
1	Computer On chip Device (Raspberry Pi, Jetson Nano, Raspberry Pi Pico or similar), Sensors, Actuators	4,5,6

Sr. No.	Equipment Name with Broad Specifications	PrO No.
2	Node MCUs (ESP8266, ESP32 etc), Arduino kits	4,5,6
3	Communication Chips and Module: Wifi Modules, Ethernet module	3
4	Switches, Routers, Various IoT Gateways Supporting various Physical layer protocols.	4,5,6
5	Vector Network Analyzer for Antenna Testing	3

VI AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when he/she performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Understand the specification, requirements and device a plan of action for developing a solution to required IoT product.
- 2) Lead the team, distribute the task, and finally integrate them all together, for developing an IoT product solution. Complete individual assignment as well as work in team to finish a task efficiently and quickly.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit 1: Introduction, Framework and Product Design Philosophies	Co1. For given IoT System, <u>Draw</u> the overall architecture of complete IoT system, <u>show interconnection</u> of Fog and Edge device and <u>specify</u> the requirements of Various Device involved and <u>Select</u> from offering. <u>Manage</u> and <u>direct</u> the development of an IoT product. 1a. <u>Learn</u> various types of IoT architecture and <u>select/suggest</u> based on the requirement of product. 1b. <u>Get</u> clear understanding of role of various component in IoT, and <u>understand</u> their interconnection. <u>Select</u> appropriate by analyzing specifications. 1c. Apply IoT Product design philosophies and methodologies, Prepare specifications of components, prepare step by step plan for product development, Lead IoT development and team doing it.	1. 1 Internet of Things: Definition, Requirement of IoT and Technology, Desired attributes /Characteristics of IoT, Application examples from various fields. 1. 2 The Things in IoT, Unique Identity (Physical address) in IoT Things. 1. 3 Architecture of IoT: The Fog and Edge based Architecture and Cloud Based Architecture, Block Diagram and Working of Various Cloud based Architectures (IoT Level-1,2,3,4,5,6 types of architecture), Various Layers

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
		in IoT and their functional role, Layered Architecture of IoT(Three Layer, Five Layer Architecture etc). 1. 4 Several Devices/ Components in network and Role (Client, Server, Node Router, Switch, Hub, Router, Gateway etc) 1. 5 IoT Enabling Platforms: Various Microcontroller Boards and Comparison, Open Mote and other hardware resources, IoT Gateway Devices (Intel IoT Gateway, EuroTech IoT Gateway, DynaGATE 20-30), 1. 6 IoT Design Methodology: Purpose and Requirement, Process Model Specification, Domain Model Specification, Information model Specification, Service specification, IoT level Specification, Functional view specification, Operational View Specification, Device and Component Integration, Application Development, Example of Design (Note: This topic is suitable to be taught after end of all chapters.)
Unit 2: Designing , Selecting and Programming Embedded IoT Boards	Co2. <u>Design</u> the Antenna, Power supply and Communication module for Embedded IoT Microcontroller board. <u>Design</u> Hardware for interfacing Actuators and Sensors. Pro3. <u>Design</u> IoT Embedded boards. Pro4. <u>Write</u> Program in C and Python to interface various Sensors and Actuators with various IoT Microcontroller board offering. Pro5. <u>Write</u> code for to Establish Physical layer communication using C and Python Libraries. 2a. <u>Learn</u> various types of small antenna. <u>Select</u> appropriate antenna for IOT board to be developed and <u>design</u> the antenna. 2b. <u>Design</u> the power supply with and without energy harvesting feature. 2c. <u>Learn</u> basics of wired and wireless communication	2. 1 Basic Building Blocks for an IoT Device/IoT Embedded board, Power Supply and Antenna Design for IoT Boards, Energy Harvesting, Various sensor, Actuators and their interfacing 2. 2 Working with Physical Layer Protocols and their Hardware Interfacing: Wired and Wireless Networks and Protocols, their working, interfacing chips and design hardware, 2. 3 IoT Sensors, Working and Interfacing: Simple sensors, Sensors for Bio Medical devices and Wearable,

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related POs required by industry)	Topics and Sub-topics
	for Physical layer communication in IoT. <u>Design</u> the hardware circuit and interface various chips, for various types of communication. 2d. <u>Learn</u> various types of sensors, <u>Design</u> interfacing scheme for various sensors. 2e. <u>Understand</u> basics of programming the Node MCU and other IoT enabled board, and <u>write</u> program to collect the sensor data and send it to gateway, website and cloud. 2f. <u>Select</u> the sensor and Microcontroller for IoT Product development.	Sensors for Industrial Process, Sensors for EV connected with Internet, Sensors for Robotics Application and similar other sensors. 2. 4 Node MCU Programming with Micro python and their communication libraries 2. 5 C Programming of Arduino with various communication Libraries/APIs. 2. 6 Introduction to IoT RTOS capabilities and comparison 2. 7 Selecting Embedded IoT board and/or programming tool for IoT application.
Unit 3: Front End and Backend Programming	Co3. <u>Write</u> Python and C Code to establish communication among Embedded IoT Boards, Mobile and IoT Gateway. Co4. <u>Develop</u> the website using HTML, CSS and Java. Write Python and Java code/script for various cloud offerings. <u>Write</u> code for IoT Communication, IoT System Management. Pro1. <u>Develop</u> the website using HTML, CSS and Java. Pro2. <u>Develop</u> dashboards for IoT application. Pro6. Write Python code with Various cloud offerings for various Network layer and Application layer communication Protocols. 3a. <u>Develop</u> Website using HTML, Java and CSS. Develop interactive dashboards. 3b. <u>Learn</u> basic features and <u>select</u> the cloud as per product requirement 3c. <u>Learn</u> various communication protocols and related APIs offered by cloud. <u>Write</u> piece of code in cloud machine for the communication. 3d. <u>Learn</u> and <u>use</u> important APIs (other than communication API covered in 3c) offered by cloud.	3.1 Basic Concepts of Web Page/Website designing using HTML, JAVA, CSS. 3.2 Python Libraries for IoT 3.3 Various Data Format (XML, SQL, JSON etc.), difference between them and data parsing using it. 3.4 Brief about UI/UX design, writing code for Creating Dashboards, Using ready dash boards. 3.5 Cloud and its types type: Understanding IaaS, PaaS, SaaS, Example clouds for each, 3.6 Discussion on various IoT product/Application requiring IaaS, Paas and SaaS clods for their working, 3.7 Finding and Specifying cloud resources required 3.8 Famous Scalable (e.g. AWS) and Non Scalable Cloud Servers offerings: Their Architecture and comparison. 3.9 Examples and Role of Various types of Cloud APIs viz, Storage and Database APIs, Compute APIs, Networking APIs, Data Analytics APIs, Machine learning APIs, Management tool APIs, Operation APIs, Security and Identity APIs, Managed

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
		<i>Infrastructure APIs etc.</i> 3.10 IoT communication Models viz, Request Response, Publish Subscribe, Push Pull, Exclusive pair, IoT communication APIs (REST based and Web socket based communication APIs) 3.11 Internet/ Network Layer Protocol and Application Layer Protocols for data communication and Messaging, their Implementation with clouds offerings.

VIII TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction, Framework and Product design Philosophies	15	5	5	5	2	3	3	23
II	Designing , Selecting and Programming Embedded IoT Boards	18	7	5	6	--	--	10	28
III	Front End and Back End Programming	12	8	5	6	--	--	--	19
Total		45	20	15	17	2	3	13	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

IX MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. It should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

Just to give idea, A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here:

- 1) **Making clone of Open Source IoT board /Design IoT board:** The Open source hardware designs are available for IoT board. Modify it for Memory, Power consumption and speed requirement. Test it with battery as a power source. Make own IoT board similarly.
- 2) **Implementation of various types of PCB antenna and Measuring effectiveness:** The antenna design is covered in Unit 2. Realize the antenna in small foot print and test it at various frequency and with various enclosures. (CO 1,2,3,4)
- 3) **Design and Development of Energy Harvesting Power Supplies for IoT:** The Energy Harvesting and related designed is covered in Unit 2. Those designs can be realized and can be tested for various conditions.
- 4) **Making APIs for IoT:** There are various APIs supporting various roles/function. Where communication APIs are well covered the brief idea for role of other APIs are also covered in Unit 2 and 3. The student need to write function callable as API supporting the requirement.
- 5) **Various IoT applications/ Products:** IoT has various application in medical, Robotics, process industries, smart cities, electrical power system and many more. Student can reach the teacher with an objective and can discuss the importance. If found suitable make plan for executing the project, decide specification and procure. Develop and test the system. Test it for objective as well as w.r.t security.

X SPECIAL INSTRUCTIONAL STRATEGIES (if any)

The teacher can use following strategies, to accelerate the attainment of the various types of learning outcomes in this course:

- a) Tutorials that need hand calculation and testing in Software.
- b) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance may to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) The teacher can suggest several research paper to read and refer. This can be assign to specially to evaluate the capacity of individual, to understand topics related to signal processing but not in syllabus. The capacity can be judge by the way or quality of efforts put, the simulation developed to verify the algorithm and brain storming done and presented to evaluator in form of presentation.

XI SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Make a report by literature review on new Embedded boards
- b) Identify the possible reduction in mathematical burden required in particular AI/ML algorithm, modify the code and test verify the speed for new algorithm.

XII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
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Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit 1: Introduction	1	1a. <u>Learn</u> various types of IoT architecture and <u>select/suggest</u> based on the requirement of product.	Lecture only.	Mixture of Blackboard & ICT
	2	1b. <u>Get</u> clear understanding of role of various component in IoT, and <u>understand</u> their interconnection. <u>Select</u> appropriate by analyzing specifications. 1c. <u>Select</u> the sensor and Microcontroller for IoT Product development.	Lecture only.	Mixture of Blackboard & ICT
Unit 2: Designing , Selecting and Programmin g Embedded IoT Boards	3	Pro3. <u>Design</u> IoT Embedded boards. (Antenna Design Part) 2a. <u>Learn</u> various types of small antenna. <u>Select</u> appropriate antenna for IOT board to be developed and <u>design</u> the antenna.	Laboratory and Lecture both (Th:3 hours, Pr: 2 Hours)	Mixture of Blackboard & ICT
	4	Pro3. <u>Design</u> IoT Embedded boards. (Continued for part covering Power supply. The communication Ic interfacing will be covered during classroom in 5 th week) 2b. <u>Design</u> the power supply with and without energy harvesting feature.	Laboratory and Lecture both (Th:3 hours, Pr: 2 Hours)	Mixture of Blackboard & ICT
	5	Pro4. <u>Write</u> Program in C and Python to interface various Sensors and Actuators with various IoT Microcontroller board offering. 2c. <u>Learn</u> basics of wired and wireless communication for Physical layer communication in IoT. <u>Deign</u> the hardware circuit and interface various chips, for various types of communication.	Laboratory and Lecture both (Th:3 hours, Pr: 2 Hours)	Mixture of Blackboard & ICT
	6	Pro4. <u>Write</u> Program in C and Python to interface various Sensors and Actuators with various IoT Microcontroller board offering. (Continued from last week) 2d. <u>Learn</u> various types of sensors, <u>Design</u> interfacing scheme for various sensors.	Laboratory and Lecture both (Th:3 hours, Pr: 2 Hours)	Mixture of Blackboard & ICT
	7	2e. <u>Understand</u> basics of programming the Node MCU and other IoT enabled board, and <u>write</u> program to collect the sensor data and send it to gateway, website and cloud. 2f. <u>Select</u> the sensor and Microcontroller for IoT Product development.	Lecture only.	Mixture of Blackboard & ICT
	8	Pro5. <u>Write</u> code for to Establish Physical layer communication using C and Python Libraries. 2e. <u>Understand</u> basics of programming the Node MCU and other IoT enabled board, and <u>write</u> program to collect the sensor data and send it to gateway, website and cloud. (Continued from last week, requires 1 lecture in this week)	Laboratory and Lecture both (Th:1 hours, Pr: 4 Hours)	Mixture of Blackboard & ICT

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	9	Pro1. <u>Develop</u> the website using HTML, CSS and Java. 3a. <u>Develop</u> Website using HTML, Java and CSS. Develop interactive dashboards.	Laboratory only.	Mixture of Blackboard & ICT
	10	Pro1. <u>Develop</u> the website using HTML, CSS and Java. (1 Hour) Pro2. <u>Develop</u> dashboards for IoT application. (4 Hour)	Laboratory only.	Mixture of Blackboard & ICT
	11	Pro2. <u>Develop</u> dashboards for IoT application. (2 Hour) 3b. <u>Learn</u> basic features and <u>select</u> the cloud as per product requirement 3c. <u>Learn</u> various communication protocols and related APIS offered by cloud. <u>Write</u> piece of code in cloud machine for the communication. (Types of APIS and their role, and some communication API is plan to discuss)	Laboratory and Lecture both (Th:3 hours, Pr: 2 Hours)	Mixture of Blackboard & ICT
	12	3c. <u>Learn</u> various communication protocols and related APIS offered by cloud. <u>Write</u> piece of code in cloud machine for the communication. (Communication API is plan to discuss fully with communication protocols)	Lecture only.	Mixture of Blackboard & ICT
	13	Pro6. Write Python code with Various cloud offerings for various Network layer and Application layer communication Protocols. (2 Hours) 3c. <u>Learn</u> various communication protocols and related APIS offered by cloud. <u>Write</u> piece of code in cloud machine for the communication. 3d. <u>Learn</u> and <u>use</u> important APIs (other than communication API covered in 3c) offered by cloud.	Laboratory and Lecture both (Th:3 hours, Pr: 2 Hours)	Mixture of Blackboard & ICT
	14	Pro6. Write Python code with Various cloud offerings for various Network layer and Application layer communication Protocols. (4 Hours) 3d. <u>Learn</u> and <u>use</u> important APIs (other than communication API covered in 3c) offered by cloud.	Laboratory and Lecture both (Th:1 hours, Pr: 4 Hours)	Mixture of Blackboard & ICT
	15	Topic 1.6 in Unit 1. 1c. Apply IoT Product design philosophies and methodologies, Prepare specifications of components, prepare step by step plan for product development, Lead IoT development and team doing it.	Lecture only.	Mixture of Blackboard & ICT

XIII SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	<i>Internet of Things a Hands-On Approach</i>	Arshdeep Bahga, Vijay Madisetti	First Published in India by, Universities Press (India) Private Ltd 2015, ISBN 978 81 7371 954 7
2	<i>Designing Internet of Things</i>	Adrian McEwen & Hakim Cassimally	John Wiley and Sons, ISBN 9781118430651, 1118430654
3	<i>Internet of Things, Concepts and Applications</i>	Editors, Mansaf Alam, Kashish Ara Shakil, Samiya Khan	Springer International Publishing ISBN: 9783030374686, 3030374688
4	<i>Internet of Things, AN INDIAN ADAPTATION Concepts and Applications</i>	--	Wiley Publication, ISBN-10 9354247849 : ISBN-13 978-9354247842 :
5	<i>Internet of Things with Aurdino and Bolt</i>	Aswin Pajakar	BPB Publication, First Edition (1 January 2018) ISBN-10 9789387284265 : ISBN-13 978-9387284265 :
6	<i>Enabling the Internet of Things From Integrated Circuits to Integrated Systems</i>	Massimo Alioto, Editor	Springer International Publishing ISBN: 9783319514826, 3319514822
7	<i>Internet of Things in Automotive Industries and Road Safety Electronic Circuits, Program Coding and Cloud Servers</i>	Raghuveer Chimata, Rajesh Singh, Bhupendra Singh	River Publishers ISBN: 9788770220101, 8770220107
8	<i>CLOUD COMPUTING Principles and Paradigms</i>	Edited by Rajkumar Buyya, James Broberg, Andrzej Goscinski.	Wiley Publications, ISBN 978-0-470-88799-8
9	<i>User Experience Design for the Internet of Things: Why it's More Than UI and Industrial Design</i>	Claire Rowland, Martin Charlier	Shroff/O'Reilly; First edition ISBN-10 9352138414 : ISBN-13 : 978-9352138418

XIV SOFTWARE/LEARNING WEBSITES

1. Thonny, Aurduino IDE
2. NPTEL course on 'Design for internet of things', Prof. Prabhakar T V, IISc Bangalore https://onlinecourses.nptel.ac.in/noc21_ee85/preview
3. NPTEL Course on 'Introduction to Internet of Things' by Prof. Sudip Mishra, IIT Kharagpur https://onlinecourses.nptel.ac.in/noc22_cs96/preview

XV PO-COMPETENCY-CO MAPPING

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Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE 380	Semester VI	Internet of Things Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO													
380.1	For given IoT System, Draw the overall architecture of complete IoT system, show interconnection of Fog and Edge device and specify the requirements of Various Device involved and Select from offering. Manage and Direct the Development of an IoT product.	2	2	2	2	2	2	2	1	2	2	2	--	--	--
380.2	Design the Antenna, Power supply and Communication module for Embedded IoT Microcontroller board. Design Hardware for interfacing Actuators and Sensors.	2	2	2	--	2	--	--	--	--	--	--	--	--	--
380.3	Write Python and C Code to establish communication among Embedded IoT Boards, Mobile and IoT Gateway.	2	--	--	--	2	--	--	--	--	--	--	--	--	--
380.4	Develop the website using HTML, CSS and Java. Write Python and Java code/script for various cloud offerings. Write code for IoT Communication, IoT System Management.	2	--	--	--	2	--	--	--	--	--	--	--	--	--

XVI NAMES OF RESOURCE PERSONS

a) Jivanadhar A. Joshi

Course Title: **Hybrid and Electric Vehicles**

UG Engineering Programme	Semester
Hybrid and Electric Vehicles	Sixth

I RATIONALE

To lower the greenhouse gases and promote the ecofriendly vehicles to the urban infrastructure, the EVs will dominate the future mobility. Electrical engineer has to develop the EVs which ecofriendly, high performance and according to the industry standards. This course is developed such that students should be able to analyze, evaluate, design, develop, implement the EV, HEV and PHEV and coordinate the operation of EVs in smart grid.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Analyze the performance of Hybrid and Electric vehicles and its effect on power system grid.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Identify the various terminologies associated with EV, HEV and PHEV.

Analyze the performance of motors used in modern electrical vehicles.

Analyze the various chargers, charging schemes and batteries used for EV applications.

Evaluate the factors associated grid when integrating EVs in the smart grid.

Examine the various market policies and standards associated with EVs.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in

integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Examine</u> the specifications of MG ZS EV, also make a brief report on various systems in the car.	02
2	1	<u>Examine</u> the performance of differential gear drive for vehicular applications.	02
3	2	<u>Examine</u> the performance of DC motor drive model available in MATLAB Simulink.	02
4	2	<u>Examine</u> the performance of Induction motor drive model available in MATLAB Simulink.	02
5	2	<u>Examine</u> the performance of PMBLDC motor drive model available in MATLAB Simulink.	02
6	2	<u>Examine</u> the performance of PMSM motor drive model available in MATLAB Simulink.	02
7	2	<u>Prepare</u> the open loop control & closed loop control PMBLDC drive with the help of hall sensor based controller in MATLAB Simulink.	02
8	2	<u>Prepare</u> the open loop & closed loop control PMBLDC drive with the help of back emf based controller in MATLAB Simulink.	02
9	2	<u>Prepare</u> the open loop & closed loop control PMSM motor drive with bidirectional control in MATLAB Simulink.	02
10	2	<u>Examine</u> the performance of BMS model available in MATLAB Simulink.	02
11	3	<u>Examine</u> the performance of charger.	02
12	4	<u>Examine</u> the effect of EV charging/discharging and its effects on grid based on reactive power compensation model available in MATLAB Simulink.	02
13	4	<u>Examine</u> the effect of EV charging/discharging and its effects on grid based on frequency regulations model available in MATLAB Simulink.	02
Total			26

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	AC & DC power supply	2,10
2	MATLAB Simulation software (version 2021a)	3 to 9, 11,12
3	MG ZS EV	1
4	Differential gear setup	2
5	Off board DC charger	10

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.

- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction of Electric Vehicle	CO 1. <u>Identify</u> the various terminologies associated with EV, HEV and PHEV. PrO 8. <u>Examine</u> the specifications of MG ZS EV, also make a brief report on various systems in the car. PrO 9. <u>Examine</u> the performance of differential gear drive for vehicular applications. 1h. <u>Describe</u> the History of Electric Vehicles from gasoline to PHEVs. 1i. <u>State</u> the challenges in implementing the EVs in the world. 1j. <u>List</u> the Pure EV, Hybrid EV, PHEVs, Fuel cell EV. 1k. <u>Discuss</u> Four quadrant operation of an electric drive. 1l. <u>Explain</u> Various energy sources for EV applications. 1m. <u>List</u> Various batteries used for EVs and its behavior.	1.4 History of Electric Vehicles (EVs), EV and the environment 1.5 EV challenges, Pure EV, Hybrid EV, PHEVs, Fuel cell EV 1.6 EV Technologies: motor drive technology, energy source technology. 1.7 Battery charging technology.
Unit– II Electric Motor Drives for EV	CO 2. <u>Analyze</u> the performance of motors used in modern electrical vehicles. PrO 10. <u>Examine</u> the performance of DC motor drive model available in MATLAB Simulink. PrO 11. <u>Examine</u> the performance of Induction	2.4 DC Motor Drive: system configuration, structure, operation, DC-DC converter, control of DC motor, regenerative braking. 2.5 Induction Motor Drive:

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	motor drive model available in MATLAB Simulink. PrO 12. <u>Examine</u> the performance of PMBLDC motor drive model available in MATLAB Simulink. PrO 13. <u>Examine</u> the performance of PMSM motor drive model available in MATLAB Simulink. PrO 14. <u>Prepare the</u> open loop control & closed loop control PMBLDC drive with the help of hall sensor based controller in MATLAB Simulink. PrO 15. <u>Prepare the</u> open loop & closed loop control PMBLDC drive with the help of back emf based controller in MATLAB Simulink. PrO 16. <u>Prepare the</u> open loop & closed loop control PMSM motor drive with bidirectional control in MATLAB Simulink. 2a. <u>Explain</u> Block diagram of an electric drive system and explain each block. 2b. <u>Explain</u> the working of DC-DC converter. 2c. <u>Examine</u> the control of DC motor 2d. <u>Compare</u> the various methods for speed braking of DC motor and explain regenerative braking. 2e. <u>Examine</u> the control of induction motor drive. 2f. <u>List out</u> various PM materials. 2g. <u>Explain</u> PMBLDC motor construction and working principle. 2h. <u>Examine</u> PMBLDC motor and its working under open loop and closed loop configurations. 2i. <u>Explain</u> PMSM motor construction and working principle. 2j. <u>Examine</u> PMSM motor and its working under open loop and closed loop configurations.	system configuration, structure, operation, Inverters, control of induction motor 2.6 PMBLDC Motor Drive: PM Materials, system configuration, structure, operation, inverters, control of PMBLDC motor, Comparison of various motor drives for EV application. 2.7 PMSM Motor Drive: system configuration, structure, operation, inverters, control of PMSM motor, Comparison of various motor drives for EV application.
Unit– III EV Charging Strategy	CO 3. <u>Analyze</u> the various chargers, charging schemes and batteries used for EV applications. PrO 12. <u>Examine</u> the performance of BMS model available in MATLAB Simulink. PrO 13. <u>Examine</u> the performance of charger.	3.5 Categorizing EV charging 3.6 EV Charging Approaches: dumb charging, smart charging coordination, PV Based PEV Charging Facilities, 3.7 Control Strategies:

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	3e. <u>Identify</u> EV charging requirements. 3f. <u>List</u> various charging methods for EVs. 3g. <u>Examine</u> the control strategies for residential, commercial & photovoltaic charging. 3h. <u>Identify</u> the effect of charging and its effect on Power Demand on Grid, Voltage Profile and frequency regulations. 3i. <u>Examine</u> the concept of Battery Management System and its functionality.	residential photovoltaic charging, Power Demand on Grid, Voltage Profile.
Unit-IV Vehicle to Grid	CO 4. <u>Evaluate</u> the factors associated grid when integrating EVs in the smart grid. <i>PrO 14. Examine</i> the effect of EV charging/discharging and its effects on grid based on reactive power compensation model available in MATLAB Simulink. <i>PrO 15. Examine</i> the effect of EV charging/discharging and its effects on grid based on frequency regulations model available in MATLAB Simulink. 4e. <u>Discuss</u> the concept of V2G and G2V. concept. 4f. <u>Analyze</u> Impact of EV Charging On Electrical Distribution Systems. 4g. <u>List</u> various factors associated with Grid Integration, EV Supply Equipment Design and Power Levels. 4h. <u>Analyze</u> the Charging Station requirement and its physical connection to the Grid. 4i. <u>Identify</u> various V2X systems and its future.	4.5 Introduction to V2G and G2V concept 4.6 Communication and Control 4.7 Impact of EV Charging On Electrical Distribution Systems 4.8 Concept of V2G and G2V: Definition, Benefits, Risks, Grid Integration, EV Supply Equipment Design And Power Levels 4.9 Charging Station Environment, Physical Connection to the Grid, 4.10 V2X systems and its future.
Unit – V EV Business Trend & Market Policy	CO 5. <u>Examine</u> the various market policies and standards associated w0ith EVs. 5e. <u>Explain</u> the concept of centralized charging control. 5f. <u>Describe</u> the role of EVs in electricity market and EV in smart grid. 5g. Prepare the algorithm for EV charging scheduling, data structure and V2X control.	5.5 Introduction to centralized charging control. 5.6 Electricity market, EV Aggregator, Algorithm for EV charging scheduling, data structure and requirement.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Electric Vehicle	06	04	04	02	-	-	-	10
II	Electric Motor Drives for EV	15	10	05	05	08	-	02	30
III	EV Charging Strategy	12	04	02	04	-	-	04	14
IV	Vehicle to Grid	09	02	04	04	-	-	-	10
V	EV Business Trend & Market Policy	03	06	-	-	-	-	-	06
Total		45	06						70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eighth semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 5) **Electric motors used for EVs:** Collect the specifications of different types of electric motors used by different vehicle manufacturers.
- 6) **Batteries used for EVs:** Collect the specifications of different types of batteries used by different vehicle manufacturers.
- 7) **BMS used for EVs:** Collect the specifications of different types of BMS techniques used by different vehicle manufacturers.
- 8) **Chargers used for EVs:** Collect the specifications of different types of chargers manufactured by different manufacturers.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- e) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.

- f) '**1**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- g) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- h) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- c) Library survey regarding different policies for EVs in different countries.
- d) Prepare 15 min. power point presentation on the implementation of EVs in different countries.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Introduct ion of Electric Vehicle	1	CO 1. <u>Identify</u> the various terminologies associated with EV, HEV and PHEV. PrO 2. <u>Examine</u> the specifications of MG ZS EV, also make a brief report on various systems in the car. 1a. <u>Describe</u> the History of Electric Vehicles from gasoline to PHEVs. 1b. <u>State</u> the challenges in implementing the EVs in the world. 1c. <u>List</u> the Pure EV, Hybrid EV, PHEVs, Fuel cell EV.	Lecture with media	PPTs, black board
	2	CO 1. <u>Identify</u> the various terminologies associated with EV, HEV and PHEV. PrO 3. <u>Determine</u> the Mesh current for DC resistive circuits and check the effect of change in the circuit elements with the help of bread board and MATLAB Simulink. 1d. <u>Discuss</u> Four quadrant operation of an electric drive. 1e. <u>Explain</u> Various energy sources for EV applications. 1f. <u>List</u> Various batteries used for EVs and its behavior.	Lecture With media	PPTs, black board
Unit– II Electric Motor	3	CO 2. <u>Analyze</u> the performance of motors used in modern electrical vehicles.	Lecture with media, and	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Drives for EV		PrO 3. <u>Examine</u> the performance of DC motor drive model available in MATLAB Simulink. PrO 4. <u>Examine</u> the performance of Induction motor drive model available in MATLAB Simulink. 2a. <u>Explain</u> Block diagram of an electric drive system and explain each block. 2b. <u>Explain</u> the working of DC-DC converter.	Demonstration	
	4	CO 2. Analyze the performance of motors used in modern electrical vehicles. PrO 5. <u>Examine</u> the performance of PMBLDC motor drive model available in MATLAB Simulink. 2c. <u>Examine</u> the control of DC motor 2d. <u>Compare</u> the various methods for speed braking of DC motor and explain regenerative braking.	Lecture with media, and Demonstration	PPTs, black board
	5	CO 2. Analyze the performance of motors used in modern electrical vehicles. PrO 6. <u>Examine</u> the performance of PMSM motor drive model available in MATLAB Simulink. 2e. <u>Examine</u> the control of induction motor drive. 2f. <u>List</u> out various PM materials.	Lecture with media, and Demonstration	PPTs, black board
	6	CO 2. Analyze the performance of motors used in modern electrical vehicles. PrO 7. <u>Prepare</u> the open loop control & closed loop control PMBLDC drive with the help of hall sensor based controller in MATLAB Simulink. 2g. <u>Explain</u> PMBLDC motor construction and working principle. 2h. <u>Examine</u> PMBLDC motor and its working under open loop and closed loop configurations.	Lecture with media, and Demonstration	PPTs, black board
	7	CO 2. Analyze the performance of motors used in modern electrical vehicles. PrO 8. <u>Prepare</u> the open loop & closed loop control PMBLDC drive with the help of back emf	Lecture with media, and Demonstration	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		based controller in MATLAB Simulink. PrO 9. <u>Prepare</u> the open loop & closed loop control PMSM motor drive with bidirectional control in MATLAB Simulink. 2i. <u>Explain</u> PMSM motor construction and working principle. 2j. <u>Examine</u> PMSM motor and its working under open loop and closed loop configurations.		
Unit– III EV Charging Strategy	8	CO 3. <u>Analyze</u> the various chargers, charging schemes and batteries used for EV applications. PrO 10. <u>Examine</u> the performance of BMS model available in MATLAB Simulink. 3a. <u>Identify</u> EV charging requirements. 3b. <u>List</u> various charging methods for EVs.	Lecture with media, Field visit	PPTs, Video
	9	CO 3. <u>Analyze</u> the various chargers, charging schemes and batteries used for EV applications. PrO 10. <u>Examine</u> the performance of BMS model available in MATLAB Simulink. 3c. <u>Examine</u> the control strategies for residential, commercial & photovoltaic charging.	Lecture with media, and Tutorial	PPTs
	10	CO 3. <u>Analyze</u> the various chargers, charging schemes and batteries used for EV applications. PrO 11. <u>Examine</u> the performance of charger. 3d. <u>Identify</u> the effect of charging and its effect on Power Demand on Grid, Voltage Profile and frequency regulations.	Lecture with media, and Tutorial	PPTs
	11	CO 3. <u>Analyze</u> the various chargers, charging schemes and batteries used for EV applications. PrO 11. <u>Examine</u> the performance of charger. 3e. Examine the concept of Battery Management System and its functionality.	Lecture with media, and Tutorial	PPTs
Unit– IV Vehicle to Grid	12	CO 4. <u>Evaluate</u> the factors associated grid when integrating EVs in the smart grid. PrO 12. <u>Examine</u> the effect of EV charging/discharging and its effects on grid	Lecture with media, and Tutorial	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		based on reactive power compensation model available in MATLAB Simulink. 4e. <u>Discuss</u> the concept of V2G and G2V. concept. 4f. <u>Analyze</u> Impact of EV Charging On Electrical Distribution Systems.		
	13	CO 4. Evaluate the factors associated grid when integrating EVs in the smart grid. <i>PrO 12.</i> <u>Examine</u> the effect of EV charging/discharging and its effects on grid based on frequency regulations model available in MATLAB Simulink. 4g. <u>List</u> various factors associated with Grid Integration, EV Supply Equipment Design and Power Levels. 4h. <u>Analyze</u> the Charging Station requirement and its physical connection to the Grid.	Lecture with media, and Tutorial	PPTs
	14	CO 4. Evaluate the factors associated grid when integrating EVs in the smart grid. <i>PrO 12.</i> <u>Examine</u> the effect of EV charging/discharging and its effects on grid based on frequency regulations model available in MATLAB Simulink. 4d. <u>Analyze</u> the Charging Station requirement and its physical connection to the Grid. 4e. <u>Identify</u> various V2X systems and its future.	Lecture with media, and Tutorial	PPTs
Unit– V EV Business Trend & Market Policy	15	CO 5. Examine the various market policies and standards associated w0ith EVs. 5d. <u>Explain</u> the concept of centralized charging control. 5e. <u>Describe</u> the role of EVs in electricity market and EV in smart grid. 5f. Prepare the algorithm for EV charging scheduling, data structure and V2X control.	Lecture with media, and Tutorial	PPT

XIV SUGGESTED LEARNING RESOURCES

Sr. No.	Title of Book	Author	Publication including ISBN
1.	Plug in Electric Vehicles in Smart Grids: Charging	Sumedha Rajakaruna, Farhad Shahnia, Arindam	Springer

Sr. No.	Title of Book	Author	Publication including ISBN
	Strategy	Ghosh	
2.	Smart Energy Grid Engineering	Hossam A Gabbar	ELSEVIER
3.	Electric Vehicle Technology Explained	James Larminie & John Lowry, Wiley	
4.	Electric Vehicle Machines & Drives Design, Analysis & Application	K.T.Chau,	IEEE press, Wiley
5.	Influences of Electric Vehicles on Power System and Key Technologies of Vehicle to Grid	Canbing Li, Yijia Cao, Yonghong Kuang & Bin Zhou,	Springer

XV SOFTWARE/LEARNING WEBSITES

3. <https://www.edx.org/course/electric-cars-introduction> (Electric Cars: Introduction)
4. <https://www.edx.org/course/electric-cars-technology-0> (Electric Cars: Technology)
5. <https://www.edx.org/course/electric-cars-business> (Electric Cars: Business)
6. <https://www.edx.org/course/electric-cars-policy> (Electric Cars: Policy)

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE381	Semester VI														
381	Competency To lower the greenhouse gases and promote the ecofriendly vehicles to the urban infrastructure, the EVs	3	3	3	2	2	-	-	-	-	-	-	2	3	1

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design /development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	will dominate the future mobility. Electrical engineer has to develop the EVs which ecofriendly, high performance and according to the industry standards. This course is developed such that students should be able to analyze, evaluate, design, develop, implement the EV, HEV and PHEV and coordinate the operation of EVs in smart grid.														
CO1	Identify the various terminologies associated with EV, HEV and PHEV.	3	-	-	-	3	2	3	-	-	-	-	2	2	-
CO2	Analyze the performance of motors used in modern electrical vehicles.	3	1	1	2	1	-	-	-	-	-	-	2	1	1
CO3	Analyze the various chargers, charging schemes and batteries used for EV applications.	3	2	3	3	3	3	3	-	2	2	2	2	3	3
CO4	Evaluate the factors associated grid when integrating EVs in the smart grid.	3	2	2	3	2	-	-	-	-	-	2	2	2	3
CO5	Examine the various market policies and standards associated with EVs.	3	1	3	3	3	-	-	-	-	-	2	2	2	2

XVII NAMES OF RESOURCE PERSONS

- a) Mahammadsoaib Saiyad
b)

OCEE3001: Microprocessors and Interfacing

Description:

This course OCEE3001 – Microprocessors and Interfacing is offered from SWAYAM

URL: https://onlinecourses.nptel.ac.in/noc23_ee06/preview

Credit and Week:

Teaching Scheme	Week	Theory	Practical	Total
Marks	12	100	-	100
Credit	-	3	1	4

About this course:

To enable students to apply the knowledge of microprocessor interfacing. Students will be given an overview of 8086 microprocessor and comparison with 8-bit processor will be discussed. Later, the detailed architecture of 8086 will be discussed. The 8086 instructions will be covered with examples. Simple to complex programs using 8086 assembly language will be discussed. A peripheral device 8255 will be discussed in detail. Then, the interfacing of 8086 with several peripherals such as key board, display, stepper motor will be covered.

Industry Support:

INTEL

Course Layout:

Week 1: 8086 Architecture

Week 2: 8086 Pins and Signals

Week 3: 8086 Instruction Set I

Week 4: 8086 Instruction Set II

Week 5: Programming I

Week 6: Memory Interfacing and Programmable Peripheral Interface

Week 7: I/O Interfacing

Week 8: I/O Interfacing contd.

Week 9: I/O Interfacing and Timer

Week 10: Programmable Interrupt Controller (8259)

Week 11: Programmable DMA Controller (8237), Serial I/O

Week 12: Programmable Communication Interface (8251)

Books and references:

1. Douglas V Hall and SSSP Rao, MICROPROCESSORS AND INTERFACING, McGraw Hill Education, 3rd Edition, 2017

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course .

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Madras. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

OCEE3003: State Space Approach to Control System Analysis and Design

Description:

This course OCEE3003 – State Space Approach to Control System Analysis and Design is offered from SWAYAM

URL: https://onlinecourses.nptel.ac.in/noc24_ee62/preview

Credit and Week:

Teaching Scheme	Week	Theory	Practical	Total
Marks	12	100	-	100
Credit	-	3	1	4

About this course:

Concept of state variable, output variable and input variable Standard state-space representation of dynamic systems Mathematical Modeling of dynamic systems from first principles and representation into standard state-space form Solution of state-equations, Simulation Representation of Transfer Function into canonical state-space forms Controllability, Observability; Methods of Controllability and observability test, Controllability sub-space, Observability sub-space, Stabilizability Canonical decomposition Lyapunov stability State & Output Feedback, Observer Design, Introduction to State estimation using

Kalman filter

Multirate output feedback

Case studies: (1) CSTR (2) Nuclear Reactor

Outcome: One can learn and develop the following competence after undergoing this course:

- to express mathematical models of physical systems into the standard state space form,
- explore the open loop characteristics by obtaining the solution of state equations,
- assess the controllability, observability and stability properties,
- design of controllers based on output and state feedback techniques and implementation of the same using the states estimated by an observer/state estimator,
- design of controllers based on multirate output feedback techniques

Industry Support:

Useful for Avionics/Space, Nuclear, Chemical industries and R&D institutions engaged in research in such sectors

Course Layout:

Week 1: Concept of state variable, output variable(s) and input variable(s); Standard state-space representation of dynamic systems; Mathematical Modeling of dynamic systems from first principles and representation into standard state-space form with examples

Week 2: Representation of Transfer Function into canonical state-space forms

Week 3: Solution of state-equations, State-transition Matrix (STM), Methods to determine the STM

Week 4: Simulation, Numerical solution of Linear and Nonlinear State equations

Week 5: Controllability, Observability; Methods of Controllability and observability test

Week 6: Controllability sub-space, Observability sub-space, Stabilizability, Canonical decomposition

Week 7: Lyapunov stability

Week 8: Concept of State & Output Feedback, Compensator design, Pole-placement design, Linear Quadratic Regulator Design

Week 9: Multirate output feedback: Design of Periodic Output Feedback and Fast Output Sampling Controllers

Week 10: Full order and Reduced order Observer Design

Week 11: State Estimation: Introduction to Kalman filter as optimal observer and Examples of applications of Kalman filter

Week 12: Case-studies: Nuclear Reactor and CSTR (if time permits)

Books and references:

1. B. Friedland, "Control System Design: An Introduction to State-Space Methods"
2. John D. Azzo and CH Houpis, "Linear Control System Analysis and Design: Conventional and Modern"
3. Norman S Nise, "Control Systems Engineering"
4. Chi-Tsong Chen, "Linear System Theory and Design"

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course .

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Madras. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

CE392: MACHINE LEARNING FUNDAMENTALS

B.TECH 6TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – III

(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

A. Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Objective of the course:

The main objectives for offering the course Artificial Intelligence are:

- To learn about the most effective machine learning techniques, and gain practice implementing them.
- To able to effectively use the common neural network "tricks", including initialization, dropout regularization, Batch normalization, gradient checking,
- To understand industry best-practices for building deep learning applications.
- To learn how to quickly and powerfully apply these techniques to new problems.

C. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to AI and Machine Learning	12
2.	Supervised Learning	16
3.	Evaluation Metrics	06
4.	Unsupervised Learning	10
5.	Neural Network	16

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

D. Detailed Syllabus:

1.	Introduction to AI and Machine Learning	12 Hours	20%
1.1	Environment Setup. Introduction to Machine Learning Pipeline		
2.	Supervised Learning	16 Hours	27%
2.1	Regression, Simple Linear regression Least Square method, Simple Linear regression Using gradient Descent, Multiple Linear regression: Application,		
2.2	Classification, Logistic Regression, Log Loss. Sigmoid, K-Nearest Neighbour		
3.	Evaluation Metrics	06 Hours	10%
3.1	Regression (SSE, MSE, RMSE)		
3.2	Classification (Accuracy, precision, Recall, F1 Score, Confusion Matrix)		
4.	Unsupervised Learning	10 Hours	16%
4.1	Anomaly Detection		
5.	Neural Network	16 Hours	27%
5.1	Perceptron, Perceptron Algorithm, Single Layer NN, Back Propagation, Tuning, Application (Classification, Regression)		

E. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

F. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To solve difficult and complex problem of computer science using AI techniques (Apply).
CO2	To select any R&D field related to application of AI (Understand, Apply).

CO3	To understand soft computing and machine learning courses (Understand).
CO4	To develop software solution as per need of today's IT edge which requires high automation and less human intervention (Understand, Create, Apply)
CO5	To demonstrate working knowledge in Python in order to write and explore more sophisticated Python programs (Apply).
CO6	To apply knowledge representation, reasoning, and machine learning techniques to real-world problems (Apply).

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	2	-	-	2	-	-	1	2	2
CO2	3	3	2	2	2	-	-	-	-	-	-	-	-	1
CO3	2	-	-		-	-	-	-	-	-	-	-	1	1
CO4	3	3	3	2	1	1	-	-	-	-	-	-	1	1
CO5	2	2	2	2	2	1	-	-	-	-	-	2	-	1
CO6	3	3	3	3	2	2	-	-	-	-	-	1	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

G. Recommended Study Material:

❖ Text Books:

1. Machine Learning, Tom Mitchell, McGraw Hill, 1997. ISBN 0070428077
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, 2004

❖ Reference Books:

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
2. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification, Second Edition", Wiley & Sons, 2001.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of statistical learning", Springer, 2001.
4. Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", MIT Press, 1998.

❖ Web Materials:

1. <https://www.youtube.com/watch?v=foHSmB48rY&list=PLKvX2d3IUq586Ic9gIhZj6ubpWV-OJfl4>
2. <https://www.youtube.com/watch?v=CS4cs9xVecg&list=PLkDaE6sCZn6EcXTbcXluRg2u4xOEky0>
3. <https://www.youtube.com/watch?v=UzxYlbK2c7E>
4. <https://www.youtube.com/watch?v=fgtUFzxNztA>
5. <http://www-formal.stanford.edu/jmc/whatisai/whatisai.html>
6. https://www.webopedia.com/TERM/A/artificial_intelligence.html
7. https://en.wikipedia.org/wiki/Artificial_intelligence

❖ Software

1. Scikit Learn
2. PyTorch
3. TensorFlow
4. Colab
5. Keras

LIST OF PRACTICALS

Practical No.	<u>Name of Practical</u>
1	Application in Structural Engineering – Damage Detection, Damage Localizations, Damage Quantification, real time Structural Health Monitoring.
2	Application in Geotechnical Engineering – Soil Moisture Estimation, Soil Classification using Machine Learning
3	Application in Construction Management – Site Layout using Machine Learning, Machine Learning for preliminary estimation, Predicting change of rate of material, risk prediction and management systems, construction activity monitoring system, Site safety monitoring by Machine Learning, Project schedule optimization
4	Concrete mix design using Machine Learning
5	Material Recognition and design

EE392: EV BATTERIES AND CHARGING SYSTEM
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)
Minor Specialization Subject-3
Institute Elective
(For Minor Specialization in “Electric Vehicle Systems”)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

As electrical power generation and economy is essential gradient for the industrial and all around development of any country, the objectives of the course are:

1. To understand the basics of battery and physics of battery.
2. To understand and select the proper battery for electric vehicle.
3. To analyze the different charging strategies for EVs.
4. To analyze and design the converters for battery charging.
5. To understand about the fuel cell and design battery management system

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of Batteries	06
2	EV Batteries	08
3	EV Charging	10
4	Power Electronic Converter for Battery Charging	15
5	Battery Management System	15
6	Fuel cells	06

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

E. Detailed Syllabus:

1	Basics of Batteries	06 Hours	10 %
	Basics – Types, Parameters – Capacity, Discharge rate, State of charge, state of Discharge, Depth of Discharge, Technical characteristics, Battery pack Design, Properties of Batteries, Cell and battery voltages, Charge (or Amphour) capacity, Energy stored, Energy density, Specific power, Amphour (or charge) efficiency, Energy efficiency, Self-discharge rates, Battery geometry, Battery temperature, heating and cooling needs, Battery life and number of deep cycles		
2	EV Batteries	08 Hours	13. 33 %
	Lead Acid Batteries :Lead acid battery basics, Special characteristics of lead acid batteries, Battery life and maintenance, Battery charging, Nickel-based Batteries : Introduction, Nickel cadmium, Nickel metal hydride batteries, Lithium Batteries :Introduction, The lithium polymer battery, The lithium ion battery ,Recent development in EV batteries, Modeling of Batteries		
3	EV Charging	13 Hours	21. 67 %
	Charging Infrastructure: Domestic Charging Infrastructure, Public Charging Infrastructure, Normal Charging Station, Occasional Charging Station, Fast Charging Station, Battery Swapping Station, Move-and-charge zone. Battery Chargers: Charge equalisation, Conductive (Basic charger circuits, Microprocessor based charger circuit. Arrangement of an off-board conductive charger, Standard power levels of conductive chargers, Inductive (Principle of inductive charging, Soft-switching power converter for inductive charging), Battery indication methods		
4	Power Electronic Converter for Battery Charging	12 Hours	20 %
	Charging methods for battery, Termination methods, charging from grid, The Z-converter, Isolated bidirectional DC-DC converter, Design of Z-converter for battery charging, High-frequency transformer based isolated charger topology, Transformer less topology		
5	Battery Management System	15 Hours	25 %
	Introduction and BMS functionality, Battery pack topology, BMS Functionality, Voltage Sensing,		

Temperature Sensing, Current Sensing, BMS Functionality, High-voltage contactor control, Isolation sensing, Thermal control, Protection, Communication Interface, Range estimation, State-of charge estimation, Cell total energy and cell total power, Battery state of charge estimation (SOC), voltage-based methods to estimate SOC, Model-based state estimation, Battery Health Estimation, Lithium-ion aging: Negative electrode, Lithium ion aging: Positive electrode, Cell Balancing, Causes of imbalance, Circuits for balancing

6 Fuel cells 06 Hours 10 %

Thermodynamics of fuel cells; free energy change and cell potentials; effects of temperature and pressure on cell potential; energy conversion efficiency; factors affecting conversion efficiency; polarization losses; important types of fuel cells (hydrogen-oxygen, organic compounds-oxygen, carbon or carbon monoxide-air, nitrogen compounds-air); Equivalent circuits of fuel cells.

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	25	20	15	15	5

G. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand the basics of battery and physics of battery.
CO2:	Understand and practically select the proper battery for electric vehicle.
CO3:	Analyze the different charging strategies for EVs.
CO4:	Design the converters for battery charging.
CO5:	Understand about the fuel cell and design battery management system

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-1	-	-	-	-
CO2	3	1	1	-	1	-	-	-	-	-	-	1	-	-
CO3	3	1	1	1	2	-	-	-	1	-	-	1	1	-
CO4	3	3	3	1	2	-	-	-	1	-	1	1	2	-

CO5	3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realise the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

I. Recommended Study Material:

Text Books:	
1.	Mehrdad Ehsani, Yimin Gao and Ali Emadi, “Modern Electric, Hybrid Electric and Fuel cell vehicles: Fundamentals, Theory and Design”, CRC Press, 2004.
2.	Plug in Electric Vehicles in Smart Grids: Charging Strategy by Sumedha Rajakaruna, Farhad Shahnia, Arindam Ghosh, Springer.
3.	Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.
Reference Books:	
1.	Deshang Sha, Guo Xu, HFrequency Isolated Bidirectional Dual Active Bridge DC–DC Converters with Wide Voltage Gain, Springer Nature Singapore Pte Ltd. 2019.
2.	Battery management systems, Volume I & II: Battery modeling. Artech House, 2015
3.	John Lowry and James Larminie , “Electric Vehicle Technology Explained”, Wiley
Web Material:	
1.	https://nptel.ac.in/courses/108/106/108106170/

2.	https://onlinecourses.nptel.ac.in/noc21_ee112/
3.	https://nptel.ac.in/courses/108/102/108102121/
Other Material:	
1.	--

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Theory	Practical	Total		Internal	External	Internal	External	
VII	EE441	Power System Operations and Control	4	2	6	5	30	70	25	25	150
	EE442	Power System Protection	4	2	6	5	30	70	25	25	150
	EE443	Electrical Machine Design	4	2	6	5	30	70	25	25	150
	EE444	Simulation Lab		4	4	2			50	50	100
	EE450	Summer Internship II		3	3	3			75	75	150
	XXXXX	Elective - I	3	2	5	4	30	70	25	25	150
	Assignment Practice/ Student Counselling		0	0	4	0	0	0	0	0	0
			15	15	34	24	120	280	225	225	850
VIII	EE458	Major Project	0	36	36	20	0	0	400	400	800
	Student Counselling		0	0	2	0	0	0	0	0	0
			00	00	40	20	0	0	400	400	800

Code	Elective - I	Code	Elective - I
EE446.01	Commissioning and Testing of Electrical Equipment	EE481	Industry 4.0
EE447.01	Modeling and Control of Renewable Energy Sources	EE482	Smart Grid Technologies and Applications
EE472.01	Communication Engineering	OCEE4001	Computer-Aided Design of Electrical Machines
EE476.02	Advances in Power System		

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Charotar University of Science and Technology, Gujarat

Course Code: EE441

Course Title: **Power System Operations and Control**

UG Engineering Programme	Semester
Electrical Engineering	Sevan

I RATIONALE

Electrical engineers have to deal with the power system operations and controls. Calculations and simulations are performed to verify power system stability, power flow analysis, short circuit studies, and Economic operation of generators. This curriculum course structure is developed such that the student is able to evaluate the power flow analysis, power system stability studies, short circuit studies, economic dispatch, automatic generation control, and power system security for the interconnected power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Evaluate the performance of power system stability analysis and power flow analysis for the interconnected electric power system.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 15) **Evaluate** planning and designing of the new interconnected power system through power flow analysis with practical simulation and with software tools.
- 16) **Evaluate** short circuit studies of an inter connected power system and design the circuit breaker ratings.
- 17) **Evaluate** the small signal stability, transient stability, and voltage stability for the electric power system with practical simulation and with software tools.
- 18) **Interpret** the transients condition in power system.
- 19) **Choose** the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme (Anna 2021)		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, 15 marks are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
14	1	<u>Formulate</u> the bus admittance (Ybus) matrix using singularity and non-singular transformation method for <u>the given</u> power system network.	02
15	1	<u>Find</u> power flow analysis using NR, GS method for <u>the given</u> power system network.	02
16	2	<u>Formulate</u> the bus impedance(Zbus) matrix using building algorithm method.	02
17	4	<u>Find</u> the Transients in power system.	02
18	3	<u>Evaluate</u> the Small signal stability for <u>the given</u> power system network.	02
19	3	<u>Evaluate</u> the Transient stability for <u>the given</u> power system network.	02
20	5	<u>Evaluate</u> the economic operating schedule for <u>the given</u> generator units.	02
21	1	<u>Simulate</u> the power flow analysis in MATLAB and Power World Simulator for <u>the given</u> power system network.	02
22	3	<u>Simulate</u> the Small signal stability in MATLAB for <u>the given</u> power system network.	02
23	3	<u>Simulate</u> the Transient stability in MATLAB for <u>the given</u> power system network.	02
24	3	<u>Simulate</u> Voltage stability in MATLAB for <u>the given</u> power system network.	02
25	5	<u>Simulate</u> economic dispatch using Power World Simulator for <u>the given</u> generator units.	02
Total			24

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
3	MATLAB Simulation software (version 2021a)	8,9,10,11
4	Power World Simulator (Version 22)	8,12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy balances.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – I Load Flow Analysis	CO 1. Evaluate planning and designing of the new interconnected power system through power flow analysis with practical simulation and with software tools. <i>PrO 22.</i> Formulate the bus admittance (Ybus) matrix using singularity and non-singular transformation method for the given power system network. <i>PrO 23.</i> Find power flow analysis using NR, GS method for the given power system network. <i>PrO 8.</i> Simulate the power flow analysis in MATLAB and Power World Simulator for the given power	1.14 Introduction to Load Flow Analysis 1.15 Formulation of Bus Admittance Matrix and Examples using Singular transformation method 1.16 Gauss – Siedel Load Flow Method and examples, Newton – Raphson Load Flow Method 1.17 Fast Decoupled Load Flow Method 1.18 Computer aided simulation of load flow analysis, Examples

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	system network. 1q. <u>Justify</u> the need of Load Flow Analysis for <u>the given</u> power system network. 1r. <u>Evaluate</u> Bus Admittance Matrix for <u>the given</u> power system network. 1s. <u>Determine</u> power flow analysis for <u>the given</u> network using GS and NR method. 1t. <u>Determine</u> power flow analysis for <u>the given</u> network using FDLF method. 1u. <u>Justify</u> the method of power flow analysis for the given power system network.	
Unit– II Optimal Dispatch of Generation	CO 5. Choose the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system. PrO 7. <i>Evaluate the economic operating schedule of generator.</i> PrO 12. <i>Simulate economic dispatch using Power World Simulator for the given power system.</i> 2j. <u>Describe</u> the optimal dispatch generation 2k. <u>Determine</u> the equality and inequality constraints for <u>the given</u> type of generation unit 2l. <u>Evaluate</u> the Operating cost of thermal power plant. 2m. <u>Discriminate</u> the economic dispatch without losses and different generation limits for <u>the given</u> thermal power plant units. 2n. <u>Evaluate</u> the Economic dispatch including losses for <u>the given</u> thermal power plant units.	2.23 Introduction to optimal dispatch generation 2.24 equality and inequality constraints. 2.25 Operating cost of thermal power plant 2.26 economic dispatch neglecting losses and no generation limits and with consideration of generation limits 2.27 Economic dispatch including losses and examples
Unit– III Automatic Generation Control (AGC)	CO 5. Choose the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system. 3h. <u>Describe</u> the basic generation control loops and automatic generation control. 3i. <u>Formulate</u> generator model, prime mover model, load model, and governor model for <u>the given</u> system. 3j. <u>Propose</u> AGC in a single area and in	3.17 basic generation control loops and automatic generation control 3.18 generator model, prime mover model, load model, and governor model 3.19 AGC in a single area and in multi-area system 3.20 Reactive power and voltage control

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	multi-area power system network. 3k. <u>Demonstrate</u> Reactive power and voltage control for <u>the given</u> system. 3l. <u>Prepare</u> the amplifier model, exciter model, generator model, sensor model, and excitation system stabilizer for the power system network.	3.21 amplifier model, exciter model, generator model, sensor model, and excitation system stabilizer
Unit-IV Transients in Power System	CO 4. <u>Interpret the transients condition in power system.</u> <i>PrO 4. <u>Find the Transients in power system.</u></i> 4g. <u>Analyse</u> Traveling waves on transmission lines. 4h. <u>Examine</u> surge impedance and wave velocity. 4i. <u>Evaluate</u> the Reflection and refraction of wave for <u>the given</u> transmission line with the <u>specified</u> condition. 4j. <u>Recognize</u> the Reflection and refraction of <u>the given</u> transmission line with <u>the use</u> of Bewley lattice diagram. 4k. <u>Define</u> the Lightning phenomena and insulation coordination.	4.30 Traveling waves on transmission lines, wave equations, and specifications of traveling waves 4.31 surge impedance and wave velocity. 4.32 Reflection and refraction of traveling waves, and typical cases of line terminations 4.33 Successive reflections and Bewley lattice diagram 4.34 Lightning phenomena, protection of power system against lightning surges, and insulation coordination
Unit – V Short Circuit Studies using Bus Impedance Matrix	CO 2. <u>Evaluate short circuit studies of an inter connected power system and design the circuit breaker ratings.</u> <i>PrO 3. <u>Formulate the bus impedance(Z_{BUS}) matrix using building algorithm method.</u></i> 5k. <u>Compare</u> the Bus admittance and incidence matrices for <u>the given</u> power system network. 5l. <u>Review</u> Thevenin's theorem and Z_{BUS} for <u>the given</u> power system network. 5m. <u>Calculate</u> Modification of existing Z_{BUS} for <u>the given</u> power system network. 5n. Unbalanced fault <u>analysis</u> using Z_{BUS} .	5.25 Bus admittance and incidence matrices. 5.26 Thevenin's theorem and Z_{BUS} 5.27 Modification of existing Z_{BUS} after addition or removal of branches and Direct determination of Z_{BUS} 5.28 Unbalanced fault analysis using Z_{BUS} , Examples
Unit – VI Power System Stability	CO 3. <u>Evaluate the small signal stability, transient stability, and voltage stability for the electric power system with practical simulation and with software tools.</u> <i>PrO 5. <u>Evaluate the Small signal stability for the given power system network.</u></i> <i>PrO 6. <u>Evaluate the Transient stability for the given power system network.</u></i>	6.1 Derivation of swing equation, M and H constants, equivalent H constant. 6.2 Classification of power system stability. 6.3 Steady state stability, derive the expression for natural frequency of

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	<i>PrO 9. <u>Simulate</u> the Small signal stability in MATLAB for the given power system network.</i> <i>PrO 10. <u>Simulate</u> the Transient stability in MATLAB for the given power system network.</i> <i>PrO 11. <u>Simulate</u> Voltage stability in MATLAB for the given power system network.</i> 6a. <u>Illustrate</u> the swing equation for rotor dynamics. 6b. <u>Classification</u> of power system stability. 6c. <u>Evaluate</u> the Steady state stability analysis for given power system network. 6d. <u>Evaluate</u> the Transient stability analysis for given power system network. 6e. <u>Justify</u> Voltage stability analysis for power system stability. 6f. Voltage stability <u>analysis</u> for power system stability	oscillations, damped frequency of oscillations, and Derivation of synchronizing power coefficient, examples. 6.4 Transient stability and factors affecting it. Equal area criteria, Cases of sudden application of mechanical power input, application of three phase fault at the middle of the line or at the end of the line, Example based on equal area criteria. 6.5 Voltage stability, Classification of Voltage Stability, Reasons for Voltage instability may arise. Transmission System Characteristic, P-V Curve, Q-V Curve, Generator Characteristic, Load Characteristic, and Characteristic of reactive power compensating device. Voltage collapse, Prevention of voltage collapse.
Unit – VII Power System Security	CO 5. Choose the automatic generation control, power system security, and economic dispatch for ensuring the smooth and efficient operation of an electric power system. 7a. <u>Describe</u> the Operating states of power system and Factors affecting power system security. 7b. <u>Explain</u> Overview of security analysis. 7c. <u>Illustrate</u> the AC power flow methods and Contingency relaxation for given power system network.	7.1 Operating states of power system, Factors affecting power system security, and Contingency analysis. 7.2 Overview of security analysis and Linear sensitivity factors. 7.3 AC power flow methods and Contingency relaxation.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit	Unit Title	Teaching	Distribution of Theory Marks
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No.		Hours	R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Load Flow Analysis	10	0	2	2	4	4	0	12
II	Optimal Dispatch of Generation	08	0	2	2	2	2	0	08
III	Automatic Generation Control (AGC)	08	0	2	2	2	2	0	08
IV	Transients in Power System	04	0	2	0	0	2	0	04
V	Short Circuit Studies using Bus Impedance Matrix	06	0	2	2	2	2	0	08
VI	Power System Stability	20	2	8	6	4	6	0	26
VII	Power System Security	04	0	2	2	0	0	0	04
Total		60	0	10	14	20	16	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Formulation of bus admittance matrix:** Collect the Transmission network data and formulate the bus admittance matrix using singular and non-singular type method for power flow analysis.
- 2) **Perform power flow analysis:** Collect the real network data or the IEEE standard test system for the power flow analysis using different methods like NR, GS and FDLF.

- 3) **Formulation of bus impedance matrix:** Collect the Transmission network data and formulate the bus impedance matrix using building algorithm method for short circuit studies.
- 4) **Perform small signal stability analysis:** Collect the real network data or the IEEE standard test system for and perform small signal stability analysis.
- 5) **Perform transient stability analysis:** Collect the real network data or the IEEE standard test system for and perform transient stability analysis.
- 6) **Economic Dispatch of generator:** Collect the data from the different generator units and perform economic dispatch.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L**' in item No. 4 does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different types of power system stability.
- b) Prepare 15 min. power point presentation for voltage stability analysis.
- c) Prepare 15 min. power point presentation for automatic generation and control for power system.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Load Flow Analysis	1	PrO 1. Formulate the bus admittance (Ybus) matrix using singularity and non-singular transformation method for the given power system network. 1m. <u>Justify</u> the need of Load Flow Analysis for the	Lecture with media, Tutorial	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		given power system network. 1n. <u>Evaluate</u> Bus Admittance Matrix for <u>the given</u> power system network.		
	2	<i>Pro 2. Find power flow analysis using NR, GS method for the given power system network.</i> 1c. <u>Determine</u> power flow analysis for <u>the given</u> network using GS and NR method.	Lecture With media, Tutorial	PPTs, black board
	3	<i>Pro 8. Simulate the power flow analysis in MATLAB and Power World Simulator for the given power system network.</i> 1d. Determine power flow analysis for the given network using FDLF method. 1e. Justify the method of power flow analysis for the given power system network.	Lecture with media, Tutorial	PPTs, black board
Unit– II Optimal Dispatch of Generation	4	<i>Pro 7. Evaluate the economic operating schedule of generator.</i> 2a. <u>Describe</u> the optimal dispatch generation 2b. <u>Determine</u> the equality and inequality constraints for <u>the given</u> type of generation unit. 2c. <u>Evaluate</u> the Operating cost of thermal power plant.	Lecture with media, and Tutorial	PPTs, black board
	5	<i>Pro 12. Simulate economic dispatch using Power World Simulator for the given power system.</i> 2d. <u>Discriminate</u> the economic dispatch without losses and different generation limits for <u>the given</u> thermal power plant units. 2e. <u>Evaluate</u> the Economic dispatch including losses for <u>the given</u> thermal power plant units.	Lecture with media, Tutorial	PPTs, black board
Unit– III Automatic Generation Control (AGC)	6	3a. <u>Describe</u> the basic generation control loops and automatic generation control. 3b. <u>Formulate</u> generator model, prime mover model, load model, and governor model for <u>the given</u> system. 3c. <u>Propose</u> AGC in a single area and in multi-area power system network.	Lecture with media	PPTs
	7	3d. <u>Demonstrate</u> Reactive power and voltage control for <u>the given</u> system. 3e. <u>Prepare</u> the amplifier model, exciter model, generator model, sensor model, and excitation system stabilizer for the power system network.	Lecture with media	PPTs, Video
Unit-IV Transient s in	8	4a. <u>Analyse</u> Traveling waves on transmission lines. 4b. <u>Examine</u> surge impedance and wave velocity. 4c. <u>Evaluate</u> the Reflection and refraction of wave	Lecture with media, and	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Power System		for the <u>given</u> transmission line with- the <u>specified</u> condition.	Tutorial	
	9	<i>PrO 4. Find the Transients in power system.</i> 4d. <u>Recognize</u> the Reflection and refraction of the <u>given</u> transmission line with <u>the use</u> of Bewley lattice diagram. 4e. <u>Define</u> the Lightning phenomena and insulation coordination.(Self-study)	Lecture with media, Tutorial	PPTs, black board
Unit – V Short Circuit Studies using Bus Impedance Matrix	10	<i>PrO 3. Formulate the bus impedance(Z_{bus}) matrix using building algorithm method.</i> 5a. <u>Compare</u> the Bus admittance and incidence matrices for the <u>given</u> power system network. 5b. <u>Review</u> Thevenin's theorem and Z_{BUS} for the <u>given</u> power system network. 5c. <u>Calculate</u> Modification of existing Z_{BUS} for the <u>given</u> power system network. 5d. Unbalanced fault <u>analysis</u> using Z_{BUS} .	Lecture with media, Tutorial	PPTs, black board
Unit – VI Power System Stability	11	<i>PrO 5. Evaluate the Small signal stability for the given power system network.</i> <i>PrO 9. Simulate the Small signal stability in MATLAB for the given power system network.</i> 6a. <u>Illustrate</u> the swing equation for rotor dynamics. 6b. <u>Classification</u> of power system stability. 6c. <u>Evaluate</u> the Steady state stability analysis for <u>given</u> power system network.	Lecture with media, Tutorial	PPTs, black board
	12	<i>PrO 6. Evaluate the Transient stability for the given power system network.</i> <i>PrO 10. Simulate the Transient stability in MATLAB for the given power system network.</i> 6d. <u>Evaluate</u> the Transient stability analysis for <u>given</u> power system network.	Lecture with media, Tutorial	PPT, Pictures , black board
	13	<i>PrO 6. Evaluate the Transient stability for the given power system network.</i> <i>PrO 10. Simulate the Transient stability in MATLAB for the given power system network.</i> 6d. <u>Evaluate</u> the Transient stability analysis for <u>given</u> power system network.	Lecture with media, Tutorial	PPTs, black board
	14	<i>PrO 11. Simulate Voltage stability in MATLAB for the given power system network.</i> 6e. <u>Justify</u> Voltage stability <u>analysis</u> for power system stability. 6f. Voltage stability <u>analysis</u> for power system stability	Lecture with media	PPTs, Video
Unit – VII Power System Security	15	7a. <u>Describe</u> the Operating states of power system and Factors affecting power system security. 7b. <u>Explain</u> Overview of security analysis. 7c. <u>Illustrate</u> the AC power flow methods and	Lecture with media	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		Contingency relaxation for given power system network.		

XXVI SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Analysis	Hadi Saadat	Tata McGraw Hill
2	Power System Stability and Control	P. Kundur	Tata McGraw Hill
3	Power System Analysis and Design	B.R. Gupta	S. Chand
4	Power System Analysis	Grainger & Stevenson	Tata McGraw Hill
5	Power Generation, Operation and Control	J. Wood and B.F. Wollenberg	John Wiley & Sons, New York, USA
6	Power System Voltage Stability	C.W. Taylor	Tata McGraw Hill

XXVII SOFTWARE/LEARNING WEBSITES

1. <https://cusp.umn.edu/power-systems/power-generation-operation-control>
2. <http://nptel.ac.in/downloads/108101040/>
3. https://www.montefiore.uliege.be/cms/c_4570395/en/electric-power-and-energy-systems
4. <http://labs.ece.uw.edu/pstca/>
5. <https://www.ece.ucf.edu/~qu/ma-opendss/>

XXVIII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs												PSO1	PSO2
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12		
		Engineering knowledge	Problem analysis	Design/development of solutions.	Conduct investigations of complex problems.	Modern tool usage	The engineer and society	Environment and sustainability	Ethics	Individual and teamwork	Communication	Project management and finance	Life-long learning	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE441	Semester VII	Power System Operations and Control													
		Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
441	Competency Evaluate the performance of power system stability analysis and power flow analysis for the interconnected electric power system	3	3	3	2	2	1	1	-	2	1	2	-	1	1
441.1	Evaluate planning and designing of the new interconnected power system through power flow analysis with practical simulation and with software tools.	3	2	2	2	2	-	-	-	-	-	-	-	2	-
441.2	Evaluate short circuit studies of an inter connected power system and design the circuit breaker ratings	2	1	3	1	2	-	-	-	-	-	-	-	2	-
441.3	Evaluate the small signal stability, transient stability, and voltage stability for the electric power system with practical simulation and with software tools.	1	3	2	1	2	-	-	-	-	-	-	-	2	-
441.4	Interpret the transients condition in power system.	1	1	2	-	-	-	-	-	-	-	-	-	2	-
441.5	Choose the automatic generation control, power system security, and economic	1	-	1		1	-	-	-	-	-	-	-	2	-

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	dispatch for ensuring the smooth and efficient operation of an electric power system.														

XXIX NAMES OF RESOURCE PERSONS

- Jignesh Patel
-

Course Code: **EE442**
Course Title: **Power System Protection**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

Electrical engineers have to deal with various types of electrical machines and power system. Different types of protection systems are used to protect various types of machines and power system for safe operation and desired result. This curriculum course structure is developed such that the student is able to evaluate the performance of the protection schemes, relays, CTs & PTs, and suggest the setting of relay for different type of protective schemes.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Evaluate the performance of various electrical protection systems used in industry and power system**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Use** the protection schemes for apparatus and power system protection.
- 2) **Evaluate** the performance of the relays and Instrument transformers.
- 3) **Design** the suitable protection system for the apparatus and power system protection with relay setting.
- 4) **Conduct** the different types of tests on relay.
- 5) **Interpret** the numerical relaying.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **10 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is converted from 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Create the protection zone for the given power system network.	02
2	2	Set the operating time of the given electromagnetic relay for the given system.	02
3	2	Set the given numerical relay for the given system.	02
4	2	Set the given numerical relay for voltage-based operation.	02
5	3	Design the relay coordination for the given radial feeder.	02
6	3	Design the relay coordination for the given parallel feeder.	02
7	3	Design the protection system for the given transformer for different configuration.	02
8	3	Set the given numerical relay for the given generator.	02
9	3	Set the given numerical relay for the given induction motor.	02
10	3	Design the protection system for the given busbar configuration.	02
11			
12			
13			
14			
15			
Total			20

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a) (Optional)	1
2	Test rig of electromagnetic relays	2
3	Test rig of Numeric relay consisting of (Siemens relay REJ525) • Microprocessors, • ADC • DAC	3
4	Test rig of Numeric relay consisting of (ABB SPAU 130C) • Microprocessors, • ADC • DAC	4
5	Test rig of radial feeder with IDMT relays.	5
6	Test rig of parallel feeder with IDMT and directional relays.	6
7	Test rig of transformer differential protection schemes	7
8	Test rig of Generator protection schemes	8
9	Test rig of Induction motor protection schemes	9

S. No.	Equipment Name with Broad Specifications	PrO No.
10	Test rig of Busbar protection schemes	10
11		

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – I Protective Relaying System	CO 1. <u>Use</u> the protection schemes for apparatus and power system protection. <i>PrO 1.</i> <i>reate the protection zone for <u>the given</u> power system network.</i> 1a. <u>Justify</u> the need of protection scheme for <u>the given</u> situation 1b. <u>Evaluate</u> the basic tripping circuit for <u>the specified</u> condition 1c. <u>Justify</u> the requirements of protection in <u>the given</u> system 1d. <u>Choose</u> protection scheme for <u>the specified</u> situation.	1.1 Principles and need for protective schemes 1.2 Basic tripping circuit, Characteristic of protective system 1.3 Basics to design protective system.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit– II Instrument Transformer	CO 2. <u>Evaluate</u> the performance of the relays and Instrument transformers. 2a. <u>Determine</u> the CT and PT rating for the <u>specified</u> application. 2b. <u>Choose</u> the correct CT for the <u>specified</u> protective system for the <u>given</u> type of protection 2c. <u>Justify</u> the selection for the VT or CCVT. For the <u>specified</u> conditions.	2.1 Saturation characteristic of current transformer 2.2 Difference between measurement CT and Protective CT. 2.3 Potential Transformer
Unit– III Overcurrent Protection	CO 2. <u>Evaluate</u> the performance of the relays and Instrument transformers. <i>PrO 2. <u>Set</u> the operating time of the <u>given</u> electromagnetic relay for the <u>given</u> system.</i> <i>PrO 3. <u>Set</u> the <u>given</u> numerical relay for the <u>given</u> system.</i> <i>PrO 4. <u>Set</u> the <u>given</u> numerical relay for voltage-based operation.</i> 3a. <u>Choose</u> the relevant electromagnetic or numerical relay for the <u>given</u> situation 3b. <u>Determine</u> the relay operating time for the specified fault current levels for the <u>given</u> type of electromagnetic relay 3c. <u>Justify</u> the need for directional features of electromagnetic relay for the <u>specified</u> conditions. 3d. <u>Evaluate</u> the performance of the given electromagnetic relay for the <u>specified</u> application.	3.1 Operating principles of relays. 3.2 Different operating characteristics of different relays. 3.3 Over current and directional relays
Unit-IV Protection of Transmission Line.	CO 3. <u>Design</u> the suitable protection system for the apparatus and power system protection with relay setting. <i>PrO 5. <u>Design</u> the relay coordination for the <u>given</u> radial feeder.</i> <i>PrO 6. <u>Design</u> the relay coordination for the <u>given</u> parallel feeder.</i> 4a. <u>Select</u> the relevant relays for the <u>specified</u> application. 4b. <u>Evaluate</u> the performance of the <u>given</u> type of relay configurations for the <u>specified</u> condition. 4c. <u>Design</u> the radial feeder protection system for the <u>given</u> set of radial transmission lines.	4.1 Different configuration for the relays for transmission line protection. 4.2 Distance protection. 4.3 Carrier current protection.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	4d. <u>Design</u> the parallel feeder protection system for <u>the given</u> set of parallel transmission lines. 4e. Prepare the drawing and specification of a carrier current protection for the given situation.	
Unit – V Transformer Protection	CO 3. <u>Design</u> the suitable protection system for the apparatus and power system protection with relay setting. PrO 7. <u>Design</u> the protection system for <u>the given</u> transformer for different configuration. 5a. <u>Distinguish</u> between fault and abnormalities from <u>the given</u> set of data. 5b. <u>Select</u> the proper connection of control circuit for the given configuration. 5c. <u>Design</u> the transformer protection scheme for the given transformer and relay.	5.1 Faults and abnormalities in the transformer. 5.2 Percentage biased differential protection of transformer.
Unit – VI Protection of Generator	CO 3. <u>Design</u> the suitable protection system for the apparatus and power system protection with relay setting. PrO 8. <u>Set the given numerical relay for the given generator</u> . 6a. <u>Choose</u> the relevant protection scheme for <u>the given</u> generator 6b. <u>Develop</u> control circuit for the protection of the given generator. 6c. <u>Choose</u> the correct relay for <u>the given</u> protection of generator. 6d. <u>Differentiate</u> between class A, B and C protection from <u>the given</u> data.	6.1 Generator stator and rotor protection. 6.2 Negative phase sequence protection. 6.3 Reverse power protection and field failure protection.
Unit– VII Induction Motor Protection	CO 3. <u>Interpret</u> the suitable protection system for the apparatus protection. PrO 9. <u>Set the given numerical relay for the given induction motor</u> . 7a. <u>Develop</u> control circuit for the protection of the given induction motor. 7b. <u>Choose</u> the correct relay setting <u>the given</u> induction motor.	7.1 Protection of induction motor
Unit-VIII Bus bar Protection	CO 3. <u>Design</u> the suitable protection system for the apparatus and power	

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	system protection with relay setting. PrO 10. <u>Design</u> the protection system for <u>the given</u> busbar configuration. 8a. <u>Develop</u> control circuit for the protection of <u>the given</u> busbar configuration.	8.1 Protection of busbar
Unit – IX Testing, commissioning and Maintenance of Relays	CO 4. Conduct the different types of tests on relay. 9a. <u>Perform</u> the different types of tests on relay depending upon <u>the specified</u> conditions.	9.1 Different types of tests for relays.
Unit –X Numerical Relaying	CO 5. Interpret the numerical relaying. 10a. <u>Select</u> the best algorithm for <u>the given</u> application. 10b. <u>Develop</u> the block diagram of numerical relay for <u>the given</u> application. 10c. <u>Estimate</u> the correct phasor by <u>the given</u> method.	10.1 Numerical relay hardware 10.2 DSP and data acquisition system. 10.3 Fourier algorithm

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Protective Relaying System	04	0	2	2	0	0	0	04
II	Instrument Transformer	05	0	2	0	2	0	0	04
III	Overcurrent Protection	08	0	2	2	2	0	0	06
IV	Protection of Transmission Line	16	0	2	0	4	0	6	12
V	Transformer Protection	07	0	0	3	0	6	0	09
VI	Protection of Generator	05	0	0	5	5	0	0	10
VII	Induction Motor Protection	02	2	0	0	5	0	0	07
VIII	Bus bar Protection	02	2	0	5	0	0	0	07

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
IX	Testing, commissioning and Maintenance of Relays	03	0	6	0	0	0	0	06
X	Numerical Relaying	08	2	3	0	0	0	0	05
Total		60	06	17	17	18	06	06	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Protection schemes:** Collect the specifications of different types of electromagnetic relays used in different protection situations in different types of industries.
- 2) **Relays:** Collect the specifications of different types of static relays used in different protection situations in different types of industries.
- 3) **Equipment protection:** Prepare case studies of protections failures in different types of industries.
- 4) Undertake the survey in reputed industry and submit a report on different type of protection schemes used in that industry.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.

- b) '**L**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different protection schemes used in industries.
- b) Prepare 15 min. power point presentation on how to choose relay for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Protectiv e Relaying System	1	PrO 7. <i>Create the protection zone for <u>the given</u> power system network using MATLAB software.</i> 1a. <u>Justify</u> the need of protection scheme for the <u>given situation</u> 1b. <u>Evaluate</u> the basic tripping circuit for <u>the specified</u> condition 1c. <u>Justify</u> the requirements of protection in <u>the given</u> system 1d. <u>Choose</u> protection scheme for the specified situation.	Lecture with media	PPTs, black board
Unit– II Instrum ent Transfor mer	2	2c. <u>Determine</u> the CT and PT rating for the specified application. 2d. <u>Choose</u> the correct CT for <u>the specified</u> protective system for <u>the given</u> type of protection 2e. <u>Justify</u> the selection for the VT or CCVT. For <u>the specified</u> conditions. (Self-learn)	Lecture With media	PPTs, black board
Unit– III Overcur rent Protectio n	3	PrO 8. <i>Set the operating time of the given electromagnetic relay for the given system.</i> 3c. <u>Choose</u> the relevant electromagnetic or numerical relay for <u>the given</u> situation	Lecture with media,	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		3d. <u>Determine</u> the relay operating time for the specified fault current levels <u>for the given</u> type of electromagnetic relay	Tutorial and Demonstration	
	4	PrO 9. <i>Set the given numerical relay for the given system.</i> PrO 10. <i>Set the given numerical relay for voltage-based operation.</i> 3e. <u>Justify</u> the need for directional features of electromagnetic relay for <u>the specified</u> conditions. 3f. <u>Evaluate</u> the performance of the given electromagnetic relay for <u>the specified</u> application.	Lecture with media, Tutorial and Demonstration	PPTs, black board, Video
Unit-IV Protection of Transmission Line.	5	4c. <u>Select</u> the relevant relays for <u>the specified</u> application. 4d. <u>Evaluate</u> the performance of <u>the given</u> type of relay configurations for <u>the specified</u> condition.	Lecture with media, Tutorial	PPTs, black board
	6	PrO 11. <i>Design the relay coordination for the given radial feeder.</i> 4e. <u>Design</u> the radial feeder protection system for <u>the given</u> set of radial transmission lines.	Lecture with media	PPTs, black board
	7	PrO 12. <i>Design the relay coordination for the given parallel feeder.</i> 4f. <u>Design</u> the parallel feeder protection system for <u>the given</u> set of parallel transmission lines.	Lecture with media,	PPTs, black board
	8	4g. <u>Prepare</u> the drawing and specification of a carrier current protection for <u>the given</u> situation.	Lecture with media	PPTs, black board
Unit – V Transformer Protection	9	5e. <u>Distinguish</u> between fault and abnormalities from <u>the given</u> set of data. 5f. <u>Select</u> the proper connection of control circuit for the given configuration.	Lecture with media, + PPT	PPTs, black board
	10	PrO 13. <i>Design the protection system for the given transformer for different configuration.</i>	Lecture	PPT, Demonstration, black

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		5g. <u>Design</u> the transformer protection scheme for <u>the given</u> transformer and relay.	with media, Tutorial	board
Unit – VI Protection of Generator	11	6a. <u>Choose</u> the relevant protection scheme for <u>the given</u> generator 6b. <u>Develop</u> control circuit for the protection of the given generator.	Lecture with media	PPTs, black board
	12	PrO 14. <i>Set the given numerical relay for <u>the given</u> generator.</i> 6c. <u>Choose</u> the correct relay for <u>the given</u> protection of generator. 6d. <u>Differentiate</u> between class A, B and C protection from <u>the given</u> data.	Lecture with media, case study	PPT, black board
Unit– VII Induction Motor Protection Unit-VIII Bus bar Protection	13	PrO 15. <i>Set the given numerical relay for <u>the given</u> induction motor.</i> PrO 16. <i>esign the protection system for <u>the given</u> busbar configuration.</i> 7a. <u>Develop</u> control circuit for the protection of <u>the given</u> induction motor. 7b. <u>Choose</u> the correct relay setting <u>the given</u> induction motor. 8a. <u>Develop</u> control circuit for the protection of <u>the given</u> busbar configuration.	Lecture with media	PPTs, black board
Unit – IX Testing, commissioning and Maintenance of Relays Unit –X Numerical Relaying	14	9a. <u>Perform</u> the different types of tests on relay depending upon <u>the specified</u> conditions. 10a. <u>Select</u> the best algorithm for <u>the given</u> application.	Lecture with media, Tutorial	PPTs, black board
	15	10b. <u>Develop</u> the block diagram of numerical relay for <u>the given</u> application. 10c. <u>Estimate</u> the correct phasor by <u>the given</u> method.	Lecture with media, Tutorial	PPTs, black board, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Protection and Switchgear	B.A. Oza, N.C. Nair, R.P. Mehta and V.H. Makwana,	Mcgraw Hill, 2010 ISBN: 9780070671188, 0070671184
2	Protection and Switchgear	Bhaves Bhalja, R. P. Maheshwari and N. G. Chothani	Oxford University Press, 2018 ISBN:9780199470679, 0199470677
3	Fundamentals of Power System Protection	Paithankar, Y.G. and Bhide, S.R.	Prentice Hall of India Pvt. Ltd, New Delhi, 2010, ISBN: 9788120341234
4	Power System Protection and Switchgear	Ram, Badri and Vishwakarma, B. H.	Mc Graw Hill Education, New Delhi, 2019, ISBN: 9780071077743
5	Power System Protection Static Relays	T.S. Madhav Rao	Tata Mcgraw Hill, 1989 ISBN: 9780074603079, 0074603078
6	Switchgear and Protection	Rao, Sunil S.	Khanna Publishers, New Delhi, 2008, ISBN: 9788174092323
7	Power System Protection and Switchgear	Rabindranath, B. and Chander, N.	New Age International (P) Ltd., New Delhi, First Edition 2011, ISBN: 9780852267585
8	Switch Gear and Protection	Ingole, Arun	Pearson Education, Noida, 2017, ISBN:9789353063382

XV SOFTWARE/LEARNING WEBSITES

1. https://onlinecourses.nptel.ac.in/noc21_ee110/preview
2. Protective Relaying <https://www.youtube.com/watch?v=NEXWcOggZOI>
<https://www.youtube.com/watch?v=h8rpdVt9MUo>

XVI PO-COMPETENCY-CO MAPPING

Cour se Cod e	Competency and COs	PO 1 Engin eerin g know ledge	PO 2 Probl em anal ysis	PO 3 Desig n/dev elop ment of soluti ons.	PO 4 Condu ct invest igatio ns of comp lex probl ems	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Envir onme nt and sustai nabili ty	PO 8 Ethics	PO 9 Indivi dual and team work	PO 10 Com muni cation	PO 11 Proje ct mana gemen t and finan ce	PO 12 Life- long learn ing	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE442	Semester VII														
442	Competency Evaluate the performance of various electrical	3	2	2	2	1	1	-	-	-	-	-	1	3	-

Cour se Cod e	Competency and COs	PO 1 Engin eerin g know ledge	PO 2 Probl em analy sis	PO 3 Desig n/dev elop ment of soluti ons.	PO 4 Condu ct invest igatio ns of comp lex probl ems	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Envir onme nt and sustai nabili ty	PO 8 Ethics	PO 9 Indivi dual and team work	PO 10 Com muni cation	PO 11 Proje ct mana geme nt and finan ce	PO 12 Life- long learni ng	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	<i>protection systems used in industry and power system</i>														
442. 1	Use the protection schemes for apparatus and power system protection	3	-	-	1	-	-	-	-	-	-	-	-	1	-
442. 2	Evaluate the performance of the relays and instrument transformers	2	2	1	1	1	-	-	-	-	-	-	1	1	-
442. 3	Design the suitable protection system for the apparatus and power system protection with relay setting.	3	3	3	2	2	2	-	-	1	-	-	1	3	-
442. 4	Conduct the different types of tests on relay.	1	-	-	1	1	-	-	-	-	-	-	1	1	-
442. 5	Interpret the numerical relaying.	3	1	-	-	1	-	-	-	-	1	-	-	1	-

XVII NAMES OF RESOURCE PERSONS

- a) Dr. Nilaykumar A. Patel
- b)

Course Code: **EE443**
Course Title: **Electrical Machine Design**

UG Engineering Programme	Semester
Electrical Engineering	7th

I RATIONALE

Electrical engineers have to deal with various types of electrical machines. Overall design of rotational and stationary machines is essential. This curriculum course structure is developed such that the student is able to design transformers, induction machines and synchronous machines.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Design various components of the electrical machines.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **To introduce the students with the fundamentals of designing an equipment system.**
- 2) **To learn how to design different part of transformer.**
- 3) **To learn how to design different part of induction machine.**
- 4) **To learn how to design different part of synchronous machine.**

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
4	0	2	5	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the

students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	To study various parts of AC machines	04
2	3	To solve the design problems related to the main dimensions of Induction machine.	04
3	1	Basic consideration of electrical machine design based on heating time constant.	04
4	2	Various basic design concepts of transformer	02
5	3	To solve the problems of stator and rotor design of an induction machine.	04
6	3	To solve the rotor design of a slip ring induction machine.	04
7	2	To solve and calculate the dimensions of various parts of a transformer	04
8	4	To solve and calculate various components for an alternator.	04
		Total	30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Matlab for coding	7
2	Excel for calculating parameters	2,5,6

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Follow designing norms.
- 3) Practice energy conservation.
- 4) Function as a team member.
- 5) Function as a team leader.
- 6) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning

outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Basic Considerations in Electrical Machine Design	CO 1. To introduce the students with the fundamentals of designing an equipment system. <i>PrO 1. To study various parts of AC machines.</i> <i>PrO 3. Basic consideration of electrical machine design based on heating time constant.</i>	1.1 Design factors 1.2 Limitations in design 1.3 Modern trends in design of electric machines 1.4 Temperature rise
Unit– II Design of Transformer	CO 2. To learn how to design different part of transformer. <i>PrO 5. Various basic design concepts of transformer.</i> <i>PrO 7. To solve and calculate the dimensions of various parts of a transformer</i>	2.1 Output equation of a transformer. 2.2 Output equation – volt per turn 2.3 Stacking factor 2.4 Ratio of iron loss to copper loss 2.5 Relation between core area and weight of iron and copper 2.6 Optimum designs 2.7 variation of output and losses in transformer with linear dimensions 2.8 Design of core 2.9 Choice of flux density and current density 2.10 Choice of window space factor, window dimensions 2.11 Design of yoke 2.12 Overall dimensions 2.13 Design of high voltage and low voltage winding 2.14 Primary & Secondary resistance, Leakage reactance of windings, Regulation 2.15 Mechanical forces 2.16 No load current calculation 2.17 Change of parameters with change of frequency 2.18 Temperature rise of transformer 2.19 Design of tank 2.20 Design of tank with cooling tubes.
Unit– III Induction Machines Design	CO 3. To learn how to design different part of induction machine. <i>PrO 2. To solve the design problems</i>	3.1 Choice of specific electrical loadings 3.2 Choice of specific magnetic loadings 3.3 Output equation 3.4 Separation of D & L

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
	related to the main dimensions of Induction machine. <i>PrO 5. To solve the problems of stator and rotor design of an induction machine.</i> <i>PrO 6. To solve the rotor design of a slip ring induction machine.</i>	3.5 Stator Design: Turns per phase, Stator conductors, Shape of stator slots, Number of stator slots, Area of stator slots, Length of mean turn, Stator teeth, Stator core 3.6 Length of air gap, Relation for calculation of length of air gap. 3.7 Squirrel Cage Rotor Design: Number of rotor slots, Rules for selecting rotor slots, Reduction of harmonic torques, Design of rotor bars & slots, Design of end rings 3.8 Wound Rotor Design: Number of rotor slots, number of rotor turns, area of rotor conductors, Rotor windings, Rotor teeth, design of rotor core 3.9 Estimation of operating characteristics- No load current calculation, short circuit current calculation, Stator and rotor resistance and leakage reactance calculation 3.10 Dispersion coefficient – Effect on maximum power factor and overload capacity, Effect of change of air gap length, number of poles and frequency, Relation between D&L for best power factor
Unit-IV Synchronous Machines Design	CO 4. To learn how to design different parts of Synchronous machine. PrO 8. To solve the problems of stator design of the alternator	4.1 Choice of specific electrical loadings 4.2 Choice of specific magnetic loadings 4.3 Output equations 4.4 Main dimension 4.5 Short Circuit Ratio Effect of SCR on machine performance 4.6 Length of air gap and shape of pole face 4.7 Armature design: Number of armature slots, Coil span, Turns per phase, Conductor section, Slots dimension, Length of mean turn. 4.8 Stator Core design 4.9 Elimination of harmonics, Estimation of air gap length. 4.10 Design of rotor, Height of pole, Design of damper winding, Height of pole shoe, Pole profile drawing 4.11 Design of magnetic circuit, Determination of full load field MMF 4.12 Design of field winding, Determination of Direct and Quadrature axis synchronous reactance.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Basic Consideration in Electrical Machine Design	04	2	2	1	0	0	0	05
II	Design of Transformer	19	2	4	4	5	4	3	22
III	Induction Machines Design	19	2	4	4	5	4	3	22
IV	Synchronous Machines Design	18	2	3	5	5	3	3	21
Total		60	8	13	14	15	11	9	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Transformer Core Design:** Prepare a case study and design core of a transformer in excel and/or matlab.
- 2) **Induction Machine Design:** Prepare a case study and design the **stator** of a squirrel cage induction machine in matlab.
- 3) **Induction Machine Design:** Prepare a case study and design the **rotor** of a squirrel cage induction machine in matlab.
- 4) **Induction Machine Design:** Prepare a case study and design the **stator** of a wound rotor induction machine in matlab.
- 5) **Induction Machine Design:** Prepare a case study and design the **rotor** of a wound rotor induction machine in excel.
- 6) **Synchronous Machine Design:** Prepare a case study and design the synchronous machine in excel.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**L**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different design methods used in industries.
- b) Prepare 15 min. PowerPoint presentation on how steps of designing a particular electrical machine.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
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Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– I Basic Considerations in Electrical Machine Design	1	PrO 1. <i>To study various parts of AC machines.</i> Pro 3. <i>Basic consideration of electrical machine design based on heating time constant</i> 1o. Design factors 1p. Limitations in design 1q. Modern trends in the design of electric machines	Lecture with media	PPTs, black board
	2	1r. Temperature rise, Expression for temperature rise, heating & cooling time constants. 1s. Examples PrO 4. Various basic design concepts of transformer. PrO 7. To solve and calculate the dimensions of various parts of a transformer Specification, Output equation of transformer 2a. Output equation-Volt per turn, Stacking factor	Lecture With media	PPTs, black board
Unit – II Design of Transformer	3	2b. Ratio of iron loss to copper loss 2c. Relation between core area and weight of iron and copper 2d. Optimum designs	Lecture with media Tutorial and Field visit	PPTs, black board
	4	2e. variation of output and losses in transformer with linear dimensions with examples 2f. Design of core, Choice of flux density and current density 2g. Choice of window space factor, window dimensions	Lecture with media, and Demonstration	Real samples of relays, PPTs, Videos
	5	2h. Design of yoke, Overall dimensions with examples 2i. Design of high voltage and low voltage winding 2j. No load current calculation	Lecture with media, Tutorial	PPTs
	6	2k. Change of parameters with the change of frequency 2l. Temperature rise of the transformer 2m. Design of tank and examples	Lecture with media	PPTs

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – III Induction Machines Design	7	PrO 2. To solve the design problems related to the main dimensions of Induction machine. PrO 5. To solve the problems of stator and rotor design of an induction machine. PrO 6. To solve the rotor design of a slip ring induction machine. 3a. Choice of specific electrical loadings, Choice of specific magnetic loadings 3b. Output equation, Separation of D & L with examples	Lecture with media, Field visit	PPTs, Video
	8	3c. Turns per phase, Stator conductors 3d. Shape of stator slots, Number of stator slots, Area of stator slots 3e. Length of mean turn, Stator teeth, Stator core 3f. Length of air gap, Relation for calculation of length of air gap	Lecture with media, and Tutorial	PPTs
	9	3g. Number of rotor slots, Rules for selecting rotor slots, Reduction of harmonic torques 3h. Design of rotor bars & slots, Design of end rings 3i. Estimation of operating characteristics 3j. No load current calculation	Lecture with media, + PPT	PPTs, Video
	10	3k. Short circuit current calculation 3l. Stator and rotor resistance and leakage reactance calculation 3m. Dispersion Co-efficient 3n. Effect of dispersion coefficient on maximum power factor and overload capacity.	Lecture with media, Tutorial	PPT, Video
	11	3o. Effect of change of air-gap length 3p. Effect of change of number of poles and frequency 3q. Relation between D & L for best power factors and examples.	Lecture with media	PPTs, black board
Unit– IV Synchron ous Machine Design	12	PrO 8. To solve the problems of stator design of the alternator 4a. Choice of specific electrical loadings and Choice of specific magnetic	Lecture with media, case study	PPT, Pictures , black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		loadings 4b. Output equations and Main dimension 4c. Short Circuit Ratio, Effect of SCR on machine performance 4d. Length of air gap and shape of pole face		
	13	4e. Examples 4f. Number of armature slots and Coil span 4g. Turns per phase, Conductor section 4h. Slots dimension, Length of mean turn	Lecture with media, Tutorial	PPTs, black board
	14	4a. Stator Core, Elimination of harmonics 4b. Estimation of air gap length 4c. Height of pole, Design of damper winding, Height of pole shoe	Lecture with media, Tutorial	PPTs, Video
	15	4d. Design of magnetic circuit, Determination of full load field MMF 4e. Design of field winding 4f. Determination of Direct and Quadrature axis synchronous reactance 4g. Examples	Lecture with media, Tutorial	PPTs, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	A course in Electrical machine design	A.K. Sawhney & A. Chakrabarti	Dhanpat Rai & Co. ISBN-10. 8177001019.
2	The performance and design of alternating current machines	M. G. Say	CBS Publishers & Distributors, ISBN: 9788123910277
3	Design of rotating electrical machines	Juha Pyrhonen, Tapani Jokinen, Valeria	Wiley, ISBN:9781118581575
4	Design of electrical machines	K.G. Upadhyay	New Age International, 2011. ISBN. 8122422829

S. No.	Title of Book	Author	Publication including ISBN
5	Design of electrical machines	V.N. Mittal & A. Mittal	Publisher, N.C. Jain, 2002. ISBN, 8180141268
6	Electric Machinery	A.E. Fitzgerald, Charles Kingsley, Stephen D. Umans	Tata Mcgraw Hill, ISBN: 9780073660097
7	Elements of electrical design	J.G. Jamnani	Mahajan publishing house, ISBN: 9789381256046

XV SOFTWARE/LEARNING WEBSITES

1. <https://www.youtube.com/playlist?list=PLux1Y9qIGZ3MAitb5PkcZXLStSPhCwnia>
2. Electrical Machine Design <https://cusp.umn.edu/electric-machines-drives/electric-machines-design>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE443	Semester VII	Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO													
443.01	Course Outcome 1	3	1	3	2	2		2		1	1	2		2	
443.02	Course Outcome 2	3	2	2	2	1		2		2		1	1	2	
443.03	Course Outcome 3	3	2	2	2	1		2		2		1	1	2	
443.04	Course Outcome 4	3	2	2	2	1		2		2		1	1	2	

XXX NAMES OF RESOURCE PERSONS

- a) Mihir Mehta

Course Code: **EE444**
Course Title: **Simulation Lab**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

Simulations are recognized as an efficient and effective way of teaching complex and dynamic engineering systems. A simulation-based teaching environment enables students to acquire experience and consider their previous results. The use of Simulation labs has been gaining currency in the domains of engineering and technology programs

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Identify and define engineering problems, conduct experiments and investigate to analyze and interpret data to arrive at substantial conclusions**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

1. **Use** a suitable package to solve the power flow problems for interconnected power systems.
2. **Develop** a program in a suitable package to assess the performance of power system parameters.
3. **Use** simulators as a learning aid and gain a profound insight into the working of various power electronic converters and drive systems.
4. **Explore** the behavior of different types of electrical machines.
5. **Validate** the dynamic models of existing as well as unknown circuits and systems.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	100
0	0	4	2	0	0	50	50*	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Practical PA**, out of 50 marks, **25 marks** are for **Research Paper Publication** to facilitate integration of COs and the remaining 25 marks is converted from lab manual to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Analyze</u> the different types of solvers used in MATLAB software	02
2	1	<u>Analyze</u> the use of symbolic Toolbox in MATLAB software.	02
3	2	Solve Linear equations and matrix solution using MATLAB software	02
4	1	<u>Creation</u> of bus Admittance and Impedance matrix Using MATLAB software.	02
5	2	<u>Computation</u> of transmission line parameter using MATLAB software.	02
6	3	<u>Analyze</u> the single phase half wave and full wave bridge uncontrolled rectifier with R, RL & RLE load.	02
7	3	<u>Analyze</u> the single phase half wave and full wave bridge controlled rectifier with R, RL & RLE load.	02
8	4	<u>Simulate</u> speed control of separately excited DC Machines	02
9	3	<u>Simulate</u> the three phase full wave circuit using MATLAB software	02
10	4	<u>Implement</u> Open Circuit, Short Circuit & Load tests on Single Phase transformer using MATLAB	04
11	5	<u>Analyze</u> the effect of fault using Power world simulator and using MATLAB software	02
12	4	<u>Design</u> of leakage inductance of transformer using FEM software	02
13	1	<u>Simulate</u> the Load flow analysis using Power World Simulator software	04
14	2	<u>Implement</u> relay Co- ordination for radial system using Mi Power software.	04
15	2	<u>Computation</u> of Long term load forecasting using Mi Power software.	04
16	3	<u>Analyze</u> the effect of Harmonic using MATLAB software	02
17	3	<u>Introduction</u> of PSIM simulation software	02
18	3	<u>Design</u> Boost convertor with PI controller	04
19	3	<u>Design</u> Multilevel inverter using PSIM software	04
20	3	<u>Design</u> Flyback converter using PSIM software	04
Total			54

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,3,4,5,9,10,16
2	PSIM Simulation Software	6,7,8,17,18,19,20
3	Power World Simulator	11, 13
4	FEM Software	12
5	Mi Power Software	14,15

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Function as a team member.
- 3) Function as a team leader.
- 4) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII SPECIAL INSTRUCTIONAL STRATEGIES (if any)

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) '**P' in item No. 4** does not mean only the traditional practical method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the simulation** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

IX SUGGESTED STUDENT ACTIVITIES

Other than the laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different software used in industries.
- b) Prepare 15 min. power point presentation on how to choose software for the particular application.

X SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electric Machinery Fundamentals	Stephen. J. Chapman,	Mcgraw Hill, 2017 ISBN: 9780071070522
2	Power System	Hadi Saadat	Mcgraw Hill, 1998

S. No.	Title of Book	Author	Publication including ISBN
	Analysis		ISBN: 0075616343

XI SOFTWARE/LEARNING WEBSITES

- 1 <http://www.prdcinfotech.com/business/software-engineering-group/software-products/mipower/>
- 2 <https://www.mathworks.com/products/matlab.html>
- 3 <https://www.powerworld.com/>
- 4 <http://www.femm.info>

XII PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE444	Semester VII														
444	Competency Identify and define engineering problems, conduct experiments and investigate to analyze and interpret data to arrive at substantial conclusions	3	2	2	2	1	-	1	-	-	-	-	1	3	-
444.1	Use a suitable package to solve the power flow problems for interconnected power systems	3	2	2	2	2	-	-	-	-	-	-	-	3	3
444.2	Develop a program in a suitable package to assess the performance of transmission line parameters	3	3	3	2	2	-	-	-	-	-	-	-	-	3
444.3	Use simulators as a learning aid and gain a profound insight into the	3	-	-	2	3	-	-	2	-	-	1	2	-	3

Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
	working of various power electronic converters and drive systems														
444.4	Explore the behavior of different types of electrical machines	3	2	3	1	-	-	-	-	-	-	-	-	3	2
444.5	Validate the dynamic models of existing as well as unknown circuits and systems	3	3	2	2	-	-	-	-	-	-	-	-	3	-

XIII NAMES OF RESOURCE PERSONS

a) Dr. Jigar Sarda

Course Code: **EE450**
Course Title: **Summer Internship-II**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

This course offers the opportunity for the young students to acquire on job the skills, knowledge, attitudes, and perceptions along with the experience needed to constitute a professional identity. This course includes planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, use, monitor and control resources optimally and economically. By studying this course ability like creativity, imitativensness and performance qualities are also developed in students. Also give an insight into the working of the real organizations. Leadership development and supervision skills are also integrated objectives of learning this course. To appreciate the linkages among different functions and departments. This course helps the students in exploring career opportunities in their areas of interest. By attending this course student get deeper understanding in specific functional areas.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Develop professional habit, attitude and ethics related to the industrial working environment.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the ***meso-level industry-oriented sub-tasks*** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

Solve engineering problems by working in a team with across other disciplines.

Apply knowledge and skills learned in company/industry/organization to real-world problems.

Use experience related to professional and ethical issues in the work environment

Examine the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context

Evaluate contemporary issues related with engineering in general by adopting new tools and technologies

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	150
0	0	0	3	00	00	150	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

V Course Details: Instructional Method and Pedagogy

Format of Summer Internship Report:

At the start of the semester student should prepare project report. Guideline of the Project report is as follow

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- Chapter-1: Overview of company: Detail of the company including Location and spread of the company, Number of employees, engineers, technicians, administrators in the company, Divisions of the company, Administrative tree (if available), Customer profile and market share, details of process or services, various department and role of each department, detail of your department.
- Chapter-2: Overview of Internship Experience: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- Chapter 3: Observations: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what

you write should be based on what you did and observed that truly belongs to the company/industry/organization. Present your observations.

- Chapter 4: Learning Outcomes: In the last section, summarize the summer internship activities, learning outcomes, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- References: List any source you have used in the document including books, articles and web sites in a consistent format.
- Appendices: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

Note:

- k) The report should contain as many diagrams and figures, charts etc. as relevant for the internship.
- l) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VI ASSESSMENT of Summer Internship

Assessment of summer internship is done at the beginning of term by the internal examiner and external examiner after successful completion of summer internship

A) Assessment of summer internship by Internal Examiner.

S. No.	Description	Marks
7	Final Report including documentation. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	25
8	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	25
9	Viva (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	25
Total		75

B) Assessment of summer internship by External Examiner.

S. No.	Description	Marks
4	Final Report including documentation. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	25
5	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	25
6	Viva (ability to give answers the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	25
Total		75

XIV PO-COMPETENCY-CO MAPPING

Cour se Code	Competency and COs	PO 1 Engin eerin g know ledge	PO 2 Probl em anal ysis	PO 3 Desig n/dev elop ment of soluti ons.	PO 4 Cond uct invest igatio ns of comp lex probl ems	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Envir onme nt and sustai nabili ty	PO 8 Ethics	PO 9 Indivi dual and team work	PO 10 Com muni catio n	PO 11 Proje ct mana gemen t and finan ce	PO 12 Life- long learn ing	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE450	Semester VII														
450	Competency The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:	3	2	2	2	2	2	1	1	2	2	2	1	2	2
450.1	Solve engineering problems by working in a team with across other disciplines.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
450.2	Use experience related to professional and ethical issues in the work environment	3	3	2	2	1	2	-	-	3	2	1	2	2	-
450.	Use experience	3	2	1	-	-	-	-	3	1	-	2	-	-	1

Course Code	Competency and COs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2
		Engineering knowledge	Problem analysis	Design/development of solutions.	Conduct investigations of complex problems	Modern tool usage	The engineer and society	Environment and sustainability	Ethics	Individual and teamwork	Communication	Project management and finance	Life-long learning	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
3	related to professional and ethical issues in the work environment.														
450.4	Examine the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context.	3	-	-	3	3	3	-	-	2	1	3	2	3	3
450.5	Evaluate contemporary issues related with engineering in general by adopting new tools and technologies	3	1	3	2	3	3	2	-	-	2	2	3	3	2

XV NAMES OF RESOURCE PERSONS

A) Mahammadsoaib Saiyad

Course Code: **EE446.01**
Course Title: **Commissioning and Testing of Electrical Equipment**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

Power Systems and Industrial Plants consist of number of electrical drives, transformers, circuit breakers and other equipment which require installation, commissioning, testing and regular maintenance to prevent permanent breakdown. This course will enable the students to understand the concepts, principles and acquire basic skills of installation, commissioning, testing and maintenance of electrical equipment in power stations, substations and industry.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Perform the different commissioning and high voltage tests on electrical equipment.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Familiar** with specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment.
- 2) **Perform** the high voltage tests on electrical equipment.
- 3) **Perform** the installation and commissioning tests on electrical equipment.
- 4) **Prepare** commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Understand the name plate and other specifications of transformer, dc machine, alternator and induction machine.	04
2	2	Determine the dielectric strength of the given transformer oil.	02
3	2	Perform the impulse voltage test on the given insulator.	04
4	2	Measure the dielectric loss factor and dc resistivity of the given transformer oil.	02
5	3	Perform the polarity test on single phase and three phase transformers.	02
6	3	Examine the reflection of load on secondary of three phase transformer.	02
7	3	Determine the number of pole of induction motor without energizing the motor.	02
8	4	Prepare a troubleshooting and maintenance chart for the given electrical equipment.	04
9	4	Prepare a commissioning test report for the given electrical equipment.	04
10	3	Perform the polarity test, ratio test and saturation characteristic on current transformer.	04
		Total	30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Induction Motor	2
2	Current Transformer	3
3	Single and Three phase transformers	4, 8
4	0-100 kV motorized oil BDV test kit	5
5	5-stage, 1.5kJ, 150kV Marx Impulse voltage generator	6
6	Oil dissipation factor and dc resistivity meter	7

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Function as a team member.
- 3) Function as a team leader.
- 4) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Extracts from Indian Electricity Rules – 1956 and Principles and Planning of Maintenance	CO 1. <u>Familiar</u> with specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment. PrO 1. <u>Understand the name plate and other specifications of transformer, dc machine, alternator and induction machine.</u> 1a. Understand the specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment.	1.1 Indian Electricity Act 1.2 Indian Electricity rules and regulations 1.3 Routine Maintenance, Periodical Maintenance & Maintenance on Fault
Unit– II Commissioning & Testing of Transformer	CO 2. <u>Perform</u> the high voltage tests on electrical equipment. CO 3. <u>Perform</u> the installation and commissioning tests on electrical equipment. CO 4. <u>Prepare</u> commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment. PrO 2. Determine the dielectric strength of the given transformer oil. PrO 3. Perform the impulse voltage test on the given insulator. PrO 4. Measure the dielectric loss factor and dc resistivity of the given transformer oil. PrO 5. Perform the polarity test on single phase and three phase transformers. PrO 6. Examine the reflection of load on secondary of three phase transformer. PrO 8. Prepare a troubleshooting and maintenance chart for the given electrical	2.1 Measurement of winding resistance of transformer 2.2 Open circuit and short circuit test of transformer 2.3 Measurement of Insulation resistance 2.4 Power frequency voltage withstand test on transformer 2.5 Impulse voltage and partial discharge test 2.6 Parallel operation of transformers 2.7 Transformer oil purification and tests 2.8 Polarization index and dissipation factor

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	equipment. <i>PrO 9.</i> Prepare a commissioning test report for the given electrical equipment. 2a. Perform different high voltage tests on the given transformer oil and transformer. 2b. Perform commissioning and installation tests on the given transformer. 2c. Prepare the detailed report on commissioning, installation and testing of the given transformer.	measurement 2.9 Drying of transformers 2.10 Commissioning of transformers 2.11 Causes of faults, troubleshooting and maintenance of transformers
Unit– III Commissioning & Testing of Induction Machines	<i>CO 3. <u>Perform</u> the installation and commissioning tests on electrical equipment.</i> <i>CO 4. <u>Prepare</u> commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.</i> <i>PrO 7.</i> Determine the number of pole of induction motor without energizing the motor. <i>PrO 8.</i> Prepare a troubleshooting and maintenance chart for the given electrical equipment. <i>PrO 9.</i> Prepare a commissioning test report for the given electrical equipment. 3a. Perform commissioning and installation tests on the given induction machine. 3b. Prepare the detailed report on commissioning, installation and testing of the given induction machine.	3.1 Insulation test and drying out of windings, Temperature rise test, 3.2 induction motor stator and rotor interaction, balance and vibration, ventilation and cooling, 3.3 Hammer test, Testing against variation of voltage/current/frequency 3.4 Testing of auxiliaries, Degree of protection (IP Grade) 3.5 AC Motor Troubles: Insulation failure due to transient voltages, low starting torque, pull out torque and stalling, low power factor, excessive slip, crawling, single phasing, reversal of one phase winding, starting transients in squirrel cage induction motors, overheating, pull over, vibration having electrical origins, slip ring wear, shaft currents, ball and roller bearing trouble. 3.6 Commissioning steps for Induction motor 3.7 Commissioning of Induction Generator.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit-IV Commissioning & Testing of Alternator	CO 3. <u>Perform</u> the installation and commissioning tests on electrical equipment. CO 4. <u>Prepare</u> commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment. PrO 8. Prepare a troubleshooting and maintenance chart for the given electrical equipment. PrO 9. Prepare a commissioning test report for the given electrical equipment. 4a. Perform commissioning and installation tests on the given alternator. 4b. Prepare the detailed report on commissioning, installation and testing of the given alternator.	4.1 Commissioning of alternator: Preparation and drying out of alternator windings before commissioning, insulation resistance measurements, high voltage tests, measurements of temperature of windings 4.2 extra high voltage alternators, alternators protective gear tests, trip circuit supplies and tripping tests, starting time and rate of picking up load, control of auxiliaries Alternator stator and rotor interaction, effect of excitation on stator current of an alternator in parallel with a large system 4.3 Alternator troubles: Instability of excitors, complete loss of field, failure of exciter voltage to build up, motor driven excitors, alternator instability, neutral inversion with grounded voltage transformer and un-grounded power system, heating of copper conductors, shaft currents and bearing troubles, slip ring wear on alternator rotor, Unbalanced stator currents, transient torques, heating of stator core end punching and clamps
Unit – V Commissioning & Testing of	CO 3. <u>Perform</u> the installation and commissioning tests on electrical equipment. CO 4. <u>Prepare</u> commissioning test report,	5.1 Bus bar: Temperature rise test, rated short time current test, Power

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Sub Station Equipment	<i>troubleshooting chart and maintenance schedule of electrical equipment.</i> PrO 10. Perform the polarity test, ratio test and saturation characteristic on current transformer. 5a. Perform commissioning and installation tests on the given substation equipment. 5o. Prepare the detailed report on commissioning, installation and testing of the given substation equipment.	frequency voltage withstand test, Impulse / surge testing, Vibration test. 5.2 Isolator Testing: Short circuit test, charging current, Making & Breaking test, Inductive current making & breaking test. 5.3 Circuit Breaker testing: Mechanical endurance test, Temperature rise test, Impulse & surge testing, Short circuit testing of circuit breakers, synthetic testing of circuit breakers, mechanical operation and adjustment, electrical auxiliaries, Troubleshooting & maintenance of circuit breakers 5.4 Protective relay tests: relay setting 5.5 C.T. & P.T. Testing, Station Batteries for D.C. Supply 5.6 Testing & Commissioning of Lightning Arrestor, and Substation Commissioning by Thermography

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Extracts from Indian Electricity Rules – 1956 and Principles and Planning of Maintenance	03	6	0	0	0	0	0	06
II	Commissioning & Testing of Transformer	12	2	2	4	6	6	2	22
III	Commissioning & Testing of Induction	09	0	0	4	2	6	0	12

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
	Machines								
IV	Commissioning & Testing of Alternator	09	2	2	0	2	2	0	08
V	Commissioning & Testing of Sub Station Equipment	12	0	2	4	4	4	8	22
Total		45	10	06	12	14	18	10	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) Undertake the survey and submit a report on specifications and manufacturer of various electrical equipment.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given

to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding specifications and manufacturers of electrical equipment.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Breakdown mechanisms in gaseous, liquid and solid dielectrics	1	PrO 1. Understand the name plate and other specifications of transformer, dc machine, alternator and induction machine. 1a. Indian Electricity Act. 1b. Indian Electricity rules and regulations (self-learn) 1c. Routine Maintenance, Periodical Maintenance & Maintenance on Fault.	Lecture with media	PPTs, black board
Unit– II Commissioning & Testing of Transformer	2	PrO 2. Determine the dielectric strength of the given transformer oil. 2.1 Measurement of winding resistance of transformer 2.2 Open circuit and short circuit test of transformer 2.3 Measurement of Insulation resistance	Lecture With media	PPTs, black board
	3	PrO 3. Perform impulse voltage test on given insulator. 2.4 Power frequency voltage withstand test on transformer 2.5 Impulse voltage and partial discharge test	Lecture With media	PPTs, black board
	4	PrO 4. Measure the dielectric loss factor and dc resistivity of the given transformer oil. 2.6 Parallel operation of transformers 2.7 Transformer oil purification and tests 2.8 Polarization index and dissipation factor measurement	Lecture With media	PPTs, black board
	5	PrO 5. Perform the polarity test on single phase and three phase transformers.	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		PrO 6. Examine the reflection of load on secondary of three phase transformer. 2.9 Drying of transformers 2.10 Commissioning of transformers 2.11 Causes of faults, troubleshooting and maintenance of transformers		
Unit– III Commissioning & Testing of Induction Machines	6	PrO 7. Determine the number of pole of induction motor without energizing the motor. 3.1 Insulation test and drying out of windings, Temperature rise test, 3.2 induction motor stator and rotor interaction, balance and vibration, ventilation and cooling,	Lecture With media	PPTs, black board
	7	3.3 Hammer test, Testing against variation of voltage/current/frequency 3.4 Testing of auxiliaries, Degree of protection (IP Grade) 3.5 AC Motor Troubles: Insulation failure due to transient voltages, low starting torque, pull out torque and stalling, low power factor, excessive slip, crawling, single phasing, reversal of one phase winding	Lecture With media	PPTs, black board
	8	PrO 8. Prepare a troubleshooting and maintenance chart for the given electrical equipment. 3.6 AC Motor Troubles: starting transients in squirrel cage induction motors, overheating, pull over, vibration having electrical origins, slip ring wear, shaft currents, ball and roller bearing trouble. 3.7 Commissioning steps for Induction motor 3.8 Commissioning of Induction Generator.	Lecture With media	PPTs, black board
Unit-IV Commissioning & Testing of Alternator	9	4.1 Commissioning of alternator: Preparation and drying out of alternator windings before commissioning, insulation resistance measurements, high voltage tests, measurements of temperature of windings	Lecture With media	PPTs, black board
	10	PrO 9. Prepare a troubleshooting and maintenance chart for the given electrical equipment. 4.2 extra high voltage alternators, alternators protective gear tests, trip circuit supplies and tripling tests, starting time and rate of picking up load, control of auxiliaries Alternator stator and rotor interaction, effect of excitation on stator current of an alternator in parallel with a large system	Lecture With media	PPTs, black board
	11	PrO 9. Prepare a commissioning test report for the given electrical equipment. 4.3 Alternator troubles: Instability of exciters, complete loss of field, failure of exciter voltage	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		to build up, motor driven exciters, alternator instability, neutral inversion with grounded voltage transformer and un-grounded power system, heating of copper conductors, shaft currents and bearing troubles, slip ring wear on alternator rotor, Unbalanced stator currents, transient torques, heating of stator core end punching and clamps		
Unit – V Commissioning & Testing of Sub Station Equipment	12	5.1 Bus bar: Temperature rise test, rated short time current test, Power frequency voltage withstand test, Impulse / surge testing, Vibration test. 5.2 Isolator Testing: Short circuit test, charging current, Making & Breaking test, Inductive current making & breaking test.	Lecture With media	PPTs, black board
	13	PrO 9. Prepare a commissioning test report for the given electrical equipment. 5.3 Circuit Breaker testing: Mechanical endurance test, Temperature rise test, Impulse & surge testing, Short circuit testing of circuit breakers, synthetic testing of circuit breakers, mechanical operation and adjustment, electrical auxiliaries, Troubleshooting & maintenance of circuit breakers	Lecture with media	PPTs, black board
	14	PrO 10. Perform the polarity test, ratio test and saturation characteristic on current transformer. 5.4 Protective relay tests: relay setting 5.5 C.T. & P.T. Testing, Station Batteries for D.C. Supply	Lecture with media	PPTs, black board
	15	5.6 Testing & Commissioning of Lightning Arrestor, and Substation Commissioning by Thermography	Lecture with media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Testing, Commissioning and Maintenance of Electrical Equipments	S. S. Rao	Khanna Publisher, India, ISBN 978-8174091857

XV SOFTWARE/LEARNING WEBSITES

- http://nptel.iitm.ac.in/courses/IITMADRAS/Electrical_Machines_II/Testing/index.php
- http://www.ece.ualberta.ca/~knight/ee332/synchronous/s_main.html
- http://www.ece.ualberta.ca/~knight/ee332/induction/i_main.html

XVI COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs														
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and team work	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.	
EE44 6.01	Semester VI	Commissioning and Testing of Electrical Equipment														
		Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO														
446.01	Competency Perform the different commissioning and high voltage tests on electrical equipment.	3	2	-	2	-	1	-	-	2	1	-	2	3	-	
CO1	Familiar with specifications, electrical safety regulations and rules during installation and maintenance of electrical equipment.	2	-	-	-	-	-	-	-	-	-	-	3	1	-	
CO2	Perform the high voltage tests on - electrical equipment.	3	3	2	1	2	-	2	-	2	-	-	3	2	-	
CO3	Perform the installation and commissioning tests on electrical equipment.	3	3	-	-	2	-	-	-	2	3	-	3	2	-	
CO4	Prepare commissioning test report, troubleshooting chart and maintenance schedule of electrical equipment.	3	3	-	-	2	-	-	-	2	3	-	3	2	-	

XVII NAMES OF RESOURCE PERSONS

a) Dr. Mihir A. Bhatt

Course Code: **EE447.01**
Course Title: **Modeling and Control of Renewable Energy Sources**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

The renewable energy sources like solar photovoltaic, wind energy and fuel cell are emerging power generation sources. The course is developed such that student would get the knowledge from basics to control schemes of renewable energy sources.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Evaluate the performance of various control methods used in field.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Use** the control technique for photovoltaic panel.
- 2) **Evaluate** the performance of the photovoltaic and wind system
- 3) **Interpret** the suitable control technique for photovoltaic system and wind system.
- 4) **Evaluate** the performance of the mppt converter used for power maximization.
- 5) **Choose** the type of converter circuit for power control.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **15 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the

students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Modeling of PV cell to determine I-V & P-V curve in MATLAB.	04
2	2	Programing of PV cell output parameters to determine I-V & P-V curve in MATLAB.	04
3	3	Simulation of Grid Connected PV system in PVSyst.	04
4	3	Simulation of stand-alone PV system in PVSyst.	04
5	3	To observe the output parameters of PV panel.	02
6	4	Simulate of Nernst equation to determine fuel cell output.	04
7	4	To study Wind Farm DFIG detailed model.	04
8	5	To Study Solid-oxide fuel cell connected to three phase-Electrical power system.	04
		Total	30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,2,5,6,7,8
2	PVsyst Simulation software (Version 6.7)	3,4

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT)

given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction of Renewable Energy Sources	CO 1. Recognize the electric circuits connected to various Renewable Energy Sources. <i>PrO 1. Modeling of PV cell to determine I-V & P-V curve in MATLAB.</i> 1a. <u>Evaluate</u> the performance of solar cell circuits. 1b. <u>Justify</u> the characteristics of solar cell for various connections.	1.1 Global and Indian renewable energy scenario 1.2 various renewable energy sources 1.3 Energy units and conversions 1.4 Energy Storage 1.5 Introduction to IEEE 1547 standard.
Unit– II Solar Photovoltaic System	CO 3. Understand the construction, working and control of solar and wind energy system CO 5. Design & Analyze the performance of PV system. <i>PrO 2. Programing of PV cell output parameters to determine I-V & P-V curve in MATLAB.</i> <i>PrO3. Simulation of Grid Connected PV system in PVSyst.</i> <i>PrO 4. Simulation of stand-alone PV system in PVSyst.</i> <i>PrO 5. To observe the output parameters of PV panel.</i> 2a. <u>Evaluate the performance of grid connected PV system</u> 2b. <u>Evaluate</u> the performance of standalone PV system 2c. <u>Justify</u> the output characteristics of PV cell. 2d. <u>Setup</u> the circuit to determine output characteristics of PV cell.	2.1 History of Solar Energy Usage 2.2 Equivalent circuit of photovoltaic cell 2.3 From cell to module and from module to array 2.4 I-V curve under standard test condition, Fill Factor 2.5 impact of temperature and insolation on I-V curve 2.6 Shading impacts on I-V curve, I-V curve for various types of loads (resistor, DC motor, battery) 2.7 MPPT Concept, DC-DC converters (boost, buck, buck-boost) for MPPT 2.8 MPPT methods: reference cell methods, Perturb & Observation method, power slope method 2.9 standalone PV system design 2.10 Grid connected PV system principle 2.11 PV-Grid connection topologies: unfold circuit, double stage,

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
		single stage 2.12 classification of grid connected inverter: centralized inverter, string inverter & micro inverter 2.13 PV-Grid connected system architecture 2.14 PV-Grid connected system design 2.15 Economics of PV System, PV system in micro-grid 2.16 Emerging PV cell technologies.
Unit– III Wind Energy System	CO 3. Understand the construction, working and control of solar and wind energy system. <i>PrO 7. To study Wind Farm DFIG detailed model.</i> 3a. <u>Determine</u> the output from wind energy system. 3b. <u>Choose</u> the control methods for <u>maximizing output power</u> .	3.1 Introduction, Wind Characteristics 3.2 Wind turbines: fixed speed & variable speed wind turbine 3.3 components of wind energy conversion system 3.4 Types of generators: Induction generator, synchronous generator, Permanent Magnet Synchronous Generator, 3.5 Doubly Fed Induction Generator 3.6 Pitch angle control, Grid Integration of wind energy system 3.7 Economics of wind energy conversion system
Unit-IV Fuel Cell	CO 4. Aware with the performance characteristics of Renewable Energy Sources. <i>PrO 6. Simulate of Nernst equation to determine fuel cell output.</i> <i>PrO 8. To Study Solid-oxide fuel cell connected to three phase-Electrical power system.</i> 4a. <u>Choose</u> the relevant operating parameters of fuel cell for <u>the given</u> condition. 4b. <u>Select</u> the relevant control method for <u>the</u>	4.1 Construction & Operation of fuel cell 4.2 types of fuel cells, comparison of different component and characteristic of fuel cell 4.3 fuel cell equivalent circuit, 4.4 need for dynamic modeling of fuel cell,

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	specified application. 4c. <u>Evaluate</u> the performance of <u>fuel cell</u> for the <u>specified</u> condition.	4.5 Polymer Electrolyte Membrane Fuel Cell (PEMFC) model structure, 4.6 Solid Oxide Fuel Cell (SOFC) model structure
Unit – V Renewable Energy Sources in Smart Grid & Electric Vehicle	CO 5. Acquire the knowledge regarding the fundamentals of Smart Grid and Electric Vehicle. 5a. <u>Interpret</u> the smart grid data with respect to the <u>given power system</u> . 5b. <u>Determine</u> vehicle charging control strategy. 5c. <u>Prepare</u> a design of solar water pumping system. 5d. <u>Select</u> the method for vehicle to grid connection.	5.1 Introduction of Smart Grid 5.2 History of Electric Vehicle 5.3 Architecture of PV based electric vehicle charging: Conventional method, Centralized architecture, Distributed architecture, single stage architecture 5.4 Control Strategy for EV charging 5.5 Impact of EV charging on Electrical Distribution System, V2G Concept 5.6 PV powered water pumping 5.7 Smart Meters

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction of Renewable Energy Sources	05	2	2	2	4	0	0	10
II	Solar Photovoltaic System	20	0	2	4	4	2	0	12
III	Wind Energy System	15	0	2	4	4	4	0	14
IV	Fuel Cell	05	0	4	4	4	6	0	18
V	Renewable Energy Sources in Smart Grid & Electric Vehicle	15	0	2	4	4	6	0	16
Total		60	2	12	18	20	18	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PROs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Solar PV MPPT:** Make and test the advanced version of solar MPPT methods.
- 2) **Design grid connected PV system and make economic evaluation.**
- 3) **Design stand alone PV system and make economic evaluation.**
- 4) **Create a solar water pumping system.**

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different designs of grid connected renewables.
- b) Prepare 15 min. power point presentation on control schemes for renewables.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduct ion of Renewab le Energy Sources Unit– II	1	<i>PrO1. Modeling of PV cell to determine I-V & P-V curve in MATLAB.</i> <u>1a. Evaluate</u> the performance of solar cell circuits.	Lecture with media	PPTs, black board
	2	<u>1b. Justify</u> the characteristics of solar cell for various connections.	Lecture With media	PPTs, black board
Solar Photovol taic System Unit– III Wind Energy System	3	<i>PrO 2. Programing of PV cell output parameters to determine I-V & P-V curve in MATLAB.</i> <u>2c. Justify</u> the output characteristics of PV cell.	Lecture With media	PPTs, black board
	4	<u>2d. Setup</u> the circuit to determine output characteristics of PV cell. (self learn)	Lecture With media	PPTs, black board
	5	<i>PrO3. Simulation of Grid Connected PV system in</i> <u>2a. Evaluate</u> the performance of grid connected PV system	Lecture With media	PPTs, black board
Unit-IV Fuel Cell Unit – V	6	<u>2c. Justify</u> the output characteristics of PV cell.	Lecture With media	PPTs, black board
	7	<i>PrO 4. Simulation of stand-alone PV system in PVSyst.</i> <u>2b. Evaluate</u> the performance of stand alone PV system	Lecture With media	PPTs, black board
	8	<u>2c. Justify</u> the output characteristics of PV cell. (self learn)	Lecture With media	PPTs, black board
Renewab le Energy Sources in Smart Grid & Electric Vehicle Unit – I Introduct ion of Renewab le Energy Sources	9	<i>PrO 5. To observe the output parameters of PV panel.</i> <u>2d. Setup</u> the circuit to determine output characteristics of PV cell.	Lecture With media	PPTs, black board
	11	<i>PrO 7. To study Wind Farm DFIG detailed model.</i> <u>3a. Determine</u> the output from wind energy system.	Lecture With media	PPTs, black board
	12	<u>3b. Choose</u> the control methods for <u>maximizing output power</u> . (self learn)	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit– II	13	PrO 6. Simulate of Nernst equation to determine fuel cell output. 4a. <u>Choose</u> the relevant operating parameters of fuel cell for <u>the given</u> condition.	Lecture With media	PPTs, black board
	14	4b. <u>Select</u> the relevant control method for <u>the specified</u> application(self learn)	Lecture With media	PPTs, black board
	15	PrO 8. To Study Solid-oxide fuel cell connected to three phase-Electrical power system 4c. <u>Evaluate</u> the performance of <u>fuel cell</u> for <u>the specified</u> condition. (self learn)	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Renewable & Efficient Electric Power System	Gilbert M. Masters	Wiley, ISBN: 9780471280606
2	Design of Smart Power Grid Renewable Energy Systems	Ali Keyhani	Wiley, ISBN:9781119573265
3	Modeling and Control of Fuel Cells	M. Hashem Nehrir, Caisheng Wang	Wiley, ISBN: 9780470233283

XV SOFTWARE/LEARNING WEBSITES

1. https://onlinecourses.nptel.ac.in/noc18_ee35/course (Design of Photovoltaic System)

XVI PO-COMPETENCY-CO MAPPING

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Cou rse Cod e	Competency and COs	PO 1 Used of math emat ical meth od	PO 2 Anal ysis mec hanic al syste m	PO 3 Used engi neeri ng soft.	PO 4 Solve probl em in mach ine	PO 5 Mod ern tool usag e	PO 6 The engi neer and socie ty	PO 7 Envir onm ent and susta inabi lity	PO 8 Ethic s	PO 9 Individu al and team work:	PO 10 Commu nication	PO11 Project manage ment	PO12 Life- long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics , power systems, electrical machines, PLC, renewable energy integration and its conservati on	PSO2 Apply recent advances like Artificial intelligenc e, machine learning and optimizatio n for design, simulation and analysis of electrical systems to provide solutions to real world problem
EE4 47. 01	Semester VII														
CO1	Use the control technique for photovoltaic panel.	3		1		1				1	1		1	2	
CO2	Evaluate the performance of the photovoltaic and wind system	3	2	2	1	1		2		1		1		2	
CO3	Interpret the suitable control technique for photovoltaic system and wind system.	3	2	2	1	1	1	2		1		1	1	2	1
CO4	Evaluate the performance of the mppt converter used for power maximization.	3	1		1	2		2	1			2		2	1
CO5	Choose the type of converter circuit for power control.	3	1	3	3	3	1	3		1		2	3	3	2

XVII NAMES OF RESOURCE PERSONS

- Dr. Pratik Mochi
-

Course Code: **EE472.01**
Course Title: **COMMUNICATION ENGINEERING**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

The course has been designed to introduce the Various Analog Communication Fundamentals, describe Amplitude Modulation and Demodulation, understand Angle Modulation and Demodulation, Noise and Performance of Various Receivers, Probability Theory & Random Variables & Error Correction Codes, Digital Modulation Techniques along with communication Systems in Presence of Noise, Digital Multiplexing, Line Coding, and PSD of different signals.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Evaluate the analog and digital communication parameters.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) To get Knowledge of Various Amplitude Modulation and Demodulation Systems.
- 2) Describe Different Angle Modulation and Demodulation Systems.
- 3) Able to get Depth Analysis in Noise Performance of Various Receivers.
- 4) Design a digital communication system.
- 5) Analyze Channel characteristics.
- 6) To simulate on MATLAB software.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks to facilitate integration of COs and the average of 2 tests, attendance and assignments to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Illustrate Double Sideband AM Generator.	01
2	1	calculate modulation index of DSB wave by trapezoidal pattern	01
3	1	Illustrate Double Sideband AM Reception.	01
4	1	Demonstrates Diode Detector.	01
5	2	Frequency Modulation using Varactor modulator.	01
6	2	Demonstrates Operation of Detuned Resonant Circuit.	01
7	2	Demonstrates Operation of Quadrature Detector	01
8	2	Demonstrates Operation of Phase-Locked Loop Detector	01
9	3	Illustrate selectivity curve for radio receiver	01
10	4	Describes Basic Block Diagram of Digital Communication System.	01
11	4	Illustrate Pulse Amplitude Modulation and observe its waveform	01
12	4	Illustrate Pulse Position Modulation and observe its waveform.	01
13	4	Describes concept of Pulse Code Modulation and to observe the performance of PCM system.	01
14	4	Illustrates concept of Pulse Width Modulation (PWM) using MATLAB, observe its waveforms and/or implement same using available trainer kit.	01
15	5,6	Demonstrates sampling theorem and reconstruction of signal using MATLAB, observe its waveforms and/or implement same using available trainer kit.	01
16	5,6	Reviews concept of Pulse Code Modulation (PCM) using MATLAB, observe its waveforms and/or implement same using available training kit.	01
17	5,6	Performs concept of Delta Modulation (DM) technique using MATLAB, observe its waveforms and/or implement same using available trainer kit.	01
18	5,6	Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit.	01
19	5,6	Analyze Huffman code using MATLAB.	02
		Total	20

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	Sintech trainer board for communication	1 to 14
2	MATLAB	15 to 19

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – I Amplitude Modulation	CO 1. To get Knowledge of Various Amplitude Modulation and Demodulation Systems. Pro 1. Illustrate Double Sideband AM Generator. Pro 2. calculate modulation index of DSB wave by trapezoidal pattern. Pro 3. Illustrate Double Sideband AM Reception. 1a. Define Amplitude Modulation 1b. Describe Amplitude Modulation Index 1c. Derive Modulation Index for Sinusoidal AM	1.1 Amplitude Modulation 1.2 Amplitude Modulation Index, Modulation Index for Sinusoidal AM 1.3 Frequency Spectrum for Sinusoidal AM 1.4 Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM 1.5 Non sinusoidal

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	1d. Derive Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM 1e. Explain Non sinusoidal Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation	Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation
Unit– II Angle Modulation	CO 2. Describe Different Angle Modulation and Demodulation Systems. PrO 6. Frequency Modulation using Varactor modulator. Pro 7. Demonstrates Operation of Quadrature Detector 2a. Define frequency modulation. 2b. Describe Frequency Spectrum for Sinusoidal FM. 2c. Derive Average Power in Sinusoidal FM. 2d. 2d. Define Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation. 2e. 2e. Differentiate Equivalence Between PM and FM 2f. 2f. Describe Sinusoidal Phase Modulation, Digital Phase Modulation 2g. 2g. Describe FM Transmitter, Angle Modulation Detectors 2h. 2h. Explain Automatic Frequency Control 2i. 2i. Describe Amplitude Limiters 2j. 2j. Differentiate Pre-Emphasis and De-Emphasis 2k. 2k. Describe FM Broadcast Receivers.	a. Frequency Modulation b. Frequency Spectrum for Sinusoidal FM, Average Power in Sinusoidal FM c. Non-Sinusoidal Modulation: Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation d. Equivalence Between PM and FM e. Sinusoidal Phase Modulation, Digital Phase Modulation f. Angle Modulator Circuits g. FM Transmitter, Angle Modulation Detectors h. Automatic Frequency Control i. Amplitude Limiters, Pre-Emphasis and De-Emphasis, FM Broadcast Receivers.
Unit– III Receivers	CO 3. Able to get Depth Analysis in Noise Performance of Various Receivers. Pro 6. Demonstrates Operation of Detuned Resonant Circuit. Pro 9. Illustrate selectivity curve for radio receiver. 3a. Select the different types of receivers. 3b. Define Tuning range, Tracking, sensitivity and gain 3c. Explain Image rejection, Spurious	3.1 Homodyne, Heterodyne and Super Heterodyne Receivers 3.2 Tuning range, Tracking, sensitivity and gain 3.3 Image Rejection, Spurious Responses, Adjacent Channel Selectivity 3.4 AGC, Double Conversion

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
	responses, Adjacent channel selectivity 3d. Describe different types of receivers.	3.5 Electronically Tuned Receivers (ETRS), Integrated-Circuit Receivers
Unit-IV Probability Theory & Random Processes	CO-5) Analyze Channel characteristics. CO-6) To simulate on MATLAB software. PrO10: Describes basic block diagram of communication system 4a. Calculates example of probability. 4b. Calculates example of random process. 4c. Analyzes communication system response in context with CDF, PDF and Gaussian PDF. 4d. Solved problems of binary schematic channel for BER issue.	4.1 Introduction to digital communication, Probability, Conditional Probability Of Independent Events 4.2 Relation between probability and probability density 4.3 CDF, PDF, Gaussian PDF 4.4 Threshold Detection, Random Variable, Variance of a random variable 4.5 Statistical Average, Chebyshev In Equality 4.6 Raleigh Probability Density, The Central Limit Theorem 4.7 correlation, random processes, PSD of random processes(along with examples) 4.8 Multiple random processes, band pass random processes
Unit – V Information Theory	<i>CO-4) Design a digital communication system.</i> <i>PrO19: Analyze Huffman code using MATLAB.</i> PrO 18. Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit. 5a. Computes capacity of channel. 5b. Evaluates Huffman code for systems.	5.1 Concept & Measure Of Information 5.2 entropy, Error Free Communication Over A Noisy Channel 5.3 The Channel Capacity Of A Discrete Memory Less Channel 5.4 Channel Capacity Of A Continuous Channel 5.5 Practical Communication Systems In Light Of Shannon's Equation

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – VI Principle of digital data Transmission	CO-6) To simulate on MATLAB software. PrO10: Describes Basic Block Diagram of Digital Communication System. Pro 14: Illustrates concept of Pulse Width Modulation (PWM) using MATLAB, observe its waveforms and/or implement same using available trainer kit. Pro 15: Demonstrates sampling theorem and reconstruction of signal using MATLAB, observe its waveforms and/or implement same using available trainer kit. Pro 16. Reviews concept of Pulse Code Modulation (PCM) using MATLAB, observe its waveforms and/or implement same using available training kit. Pro 17. Performs concept of Delta Modulation (DM) technique using MATLAB, observe its waveforms and/or implement same using available trainer kit. 6a. Evaluates various line codes efficiency. 6b. Design simplified systems using latest and effective components.	a. Line Coding, PSD of On/Off Signal, Bipolar Signal b. Duo Binary Signal, Pulse Shaping, Nyquist First and Second Criterion for Zero ISI c. Regenerative Repeaters, Detection Error Probability d. M-ary System, Scrambling e. Digital Carrier System, Digital Multiplexing

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Amplitude Modulation	10	0	2	4	5	4	0	15
II	Angle Modulation	10	0	2	4	5	4	0	15
III	Receivers	05	0	2	4	2	1	0	09
IV	Probability Theory & Random Processes	10	0	1	4	4	6	0	15
V	Information Theory	05	0	2	2	2	2	0	08
VI	Principle of digital data Transmission	05	0	0	2	3	3	0	08
Total		45	0	09	20	21	20	0	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

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The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

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- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical

evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different communication technology.
- b) Prepare 15 min. power point presentation on how to choose line code for the particular communication system.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Amplitude Modulation	1	Pro 1. Illustrate Double Sideband AM Generator. Pro 2. calculate modulation index of DSB wave by trapezoidal pattern. 1a. Define Amplitude Modulation 1b. Describe Amplitude Modulation Index 1c. Derive Modulation Index for Sinusoidal AM	Lecture With media	PPTs, black board
	2	Pro 3. Illustrate Double Sideband AM Reception. 1d. Derive Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM 1e. Explain Non sinusoidal Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation	Lecture With media	PPTs, black board
Unit– II Angle Modulation	3	Pro 4. Frequency Modulation using Varactor modulator. 2a. Define frequency modulation. 2b. Describe Frequency Spectrum for Sinusoidal FM. 2c. Derive Average Power in Sinusoidal FM.	Lecture With media	PPTs, black board
	4	Pro 6. Demonstrates Operation of Detuned Resonant Circuit. 2d. Define Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation. 2e. Differentiate Equivalence Between PM and FM 2f. Describe Sinusoidal Phase Modulation, Digital Phase Modulation	Lecture With media	PPTs, black board
	5	Pro 9. Illustrate selectivity curve for radio receiver. 2g. Describe FM Transmitter, Angle Modulation Detectors 2h. Explain Automatic Frequency Control 2i. Describe Amplitude Limiters	Lecture With media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	6	Pro 7. Demonstrates the Operation of Quadrature Detector 2j. Differentiate Pre-Emphasis and De-Emphasis 2k. Describe FM Broadcast Receivers.	Lecture With media	PPTs, black board
Unit– III Receivers	7	PrO10. Describes basic block diagram of communication system 3a. Select the different types of receivers. 3b. Define Tuning range, Tracking, sensitivity and gain	Lecture With media	PPTs, black board
	8	PrO10. Describes basic block diagram of communication system 3c. Explain Image rejection, Spurious responses, Adjacent channel selectivity 3d. Describe different types of receivers.	Lecture With media	PPTs, black board
Unit-IV Probability Theory & Random Processes	9	Pro 14: Illustrates concept of Pulse Width Modulation (PWM) using MATLAB, observe its waveforms and/or implement same using available trainer kit. 4a. Calculates example of probability. 4b. Calculates example of random process.	Lecture With media	PPTs, black board
	10	Pro 15: Demonstrates sampling theorem and reconstruction of signal using MATLAB, observe its waveforms and/or implement same using available trainer kit. 4c. Analyzes communication system response in context with CDF, PDF and Gaussian PDF. 4d. Solved problems of binary schematic channel for BER issue.	Lecture With media	PPTs, black board
Unit – V Information Theory	11	Pro 18. Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit. 5a. Computes capacity of channel.	Lecture with media	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
	12	Pro 18. Forms a digital carrier system signals like Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Binary Phase Shift Keying (BPSK) using MATLAB, observe its waveforms and/or implement same using available training kit. 5a. Computes capacity of channel. 5b. Evaluates Huffman code for systems.	Lecture With media	PPTs, black board
Unit – VI Principle of digital data Transmission	13	Pro 16. Reviews concept of Pulse Code Modulation (PCM) using MATLAB, observe its waveforms and/or implement same using available training kit. 6a. Evaluates various line codes efficiency.	Lecture With media	PPTs, black board
	14	Pro 17. Performs concept of Delta Modulation (DM) technique using MATLAB, observe its waveforms and/or implement same using available trainer kit. 6b. Design simplified systems using latest and effective components.	Lecture With media	PPTs, black board
	15	<i>PrO19: Analyze Huffman code using MATLAB.</i> 5b. Evaluates Huffman code for systems.	Lecture With media	PPTs, black board

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Electronic Communications	Dennis Roddy & John Coolen	Prentice Hall of India Pvt. Ltd, New Delhi ISBN: 9780133120837
2	Modern Digital and Analog Communication System	Rabindranath, B. and Chander, N.	New Age International (P) Ltd., New Delhi, First Edition 2011, ISBN: 978-0-19-538493-2
3	Digital and Analog communication system	B.P.Lathi	LCBS Publication ISBN:9789353063382
4	Communication system	B. Carlson	TMH Publication ISBN: 9780071321174
5	Digital and analog communication system	Simon Haykin	Wiley Publication ISBN: 978-8126536535

S. No.	Title of Book	Author	Publication including ISBN
6	Modern Digital and Analog Communication System	B.P.Lathi	Oxford Press 978-0-19-538493-2
7	Principle of communication system	Taub & Schilling,	TMH Publication ISBN: 978-1259029851
8	Embedded Systems Architecture	Tammy Noergaard	Elsevier ISBN: 9780123821966

XV SOFTWARE/LEARNING WEBSITES

1. <http://adp.mmu.edu.my/enotes/asma/dtc5038/notes/lecture%20notes%203%2020chapter%203.pdf>
2. <http://web.cecs.pdx.edu/~ece2xx/ece223/slides/communicationsx4.pdf>
3. <http://stattrek.com/Lesson2/Binomial.aspx?Tutorial=sat>
<http://www.youtube.com/watch?v=IYdiKeQ9xEI>
4. <http://www.complextoreal.com/tutorial.htm>

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													PSO1 Evaluate the performance of Electrical Machines and Power systems	PSO2 Evaluate the performance of Electrical and Electronic Instrumentation Systems	
		PO 1 Used of mathematical method	PO 2 Analysis of mechanical system	PO 3 Used engineering soft.	PO 4 Solve problem in machine	PO 5 The engineer and society	PO 6 Environment and sustainability	PO 7 Ethics	PO 8 Individual and team work:	PO 9 Communication	PO 10 Life-long learning						
EE472.01	Semester VI	COMMUNICATION ENGINEERING Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO															
472.01	Competency Evaluate the performance communication system	3	1	3	3	1	0	0	-	-	1	-	-	1	-	-	
472.01.1	To get Knowledge of Various Amplitude Modulation and Demodulation Systems.	3	1	2	2	1	-	-	-	-	-	-	-	1	3	-	
472.01.2	Describe Different Angle Modulation and Demodulation Systems.	3	2	2	1	1	-	-	-	-	-	-	-	2	2	-	
472.01.3	Able to get Depth Analysis in Noise Performance of Various Receivers.	3	2	1	2	1	-	-	-	-	-	-	-	2	2	-	
472.01.4	Design a digital communication system.	3	1	2	2	1	1	1	-	-	-	-	-	1	2	-	

472.01.5	Analyze Channel characteristics	3	2	1	1	1	-	-	-	-	1	-	-	1	1	-
472.01.6	To simulate on MATLAB software	3	2	3	2	-	-	-	-	-	-	-	-	2	2	-

XVII NAMES OF RESOURCE PERSONS

Prof. Ankur Patel

UG Engineering Programme	Semester
Electrical Engineering	Seven

I RATIONALE

Electrical engineers have to deal with the power system operations and controls. Power systems are living in an era of major changes, pushed forward by the emergence of new technical issues and the availability of innovative technologies, enabling new functions for monitoring and control of transmission network operation and stability. This curriculum course structure is developed such that the student is able to recognize the Restructuring of Power System, State Estimation in Power System, Substation Automation, Smart Grid Technology, virtual power plant, Load Forecasting for the interconnected power system.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Advances in power system through the understanding of restructuring of power system, smart grid technology and virtual power plant.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Describe** past, present and future of Power System with Restructuring of an Indian Power System, and **analysis** of power system state estimation with methodologies used for state estimation.
- 2) **Illustrate** the fundamentals and detail knowledge regarding smart grid technologies with operations and control.
- 3) **Describe** the fundamentals and detail knowledge regarding virtual power plant with operations and control.
- 4) **Apply and validate** various algorithm to predict future energy demand for smart grid and learn data analysis.
- 5) **Express** the **formulation** for substation automation and distribution, system monitoring, control and design.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme (Anna 2021)		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, 15 marks are for **micro-project assessment** to facilitate integration of COs and the remaining 15 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	<u>Illustrate</u> National Power Portal (NPP)	02
2	1	<u>Describe</u> Restricting of Power System	02
3	1	<u>Describe</u> and <u>Evaluate</u> power system state estimation	02
4	2	<u>Describe</u> Smart Grid Technology	02
5	3	<u>Describe</u> Virtual Power Plant	02
6	4	<u>Describe</u> Load Forecasting	02
7	5	<u>Describe</u> Distribution Automation	02
8	2	<u>Simulate</u> Battery energy management system for micro grid	02
9	2	<u>Simulate</u> load flow analysis for hybrid AC/DC micro grid	02
10	4	Energy Demand Forecasting for EAI data in Python: - Using Stats Model like AR, ARMA, ARIMA and SARIMA - Using ML model like RNN and LSTM	02
11	1	<u>Simulate</u> Power System State Estimation in Torrit using WLSE method <u>the given</u> power system network.	02
12	2	<u>Design</u> and <u>simulate</u> Hybrid AC/DC Micro-grid for Residential Application Using Homer PRO.	02
Total			24

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	8,9
2	Python	10
3	Torrit	11
4	Homer PRO	12

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy balances.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Restructuring of Power System	CO 1. Describe past, present and future of Power System with Restructuring of an Indian Power System, and analysis of power system state estimation with methodologies used for state estimation. <i>PrO 1. <u>Illustrate</u> National Power Portal (NPP).</i> <i>PrO 2. <u>Describe</u> Restructuring of Power System.</i> 1a. <u>Describe</u> history of electric power and Traditional power industry. 1b. <u>Justify</u> Restructuring power system network and <u>discuss</u> Electricity market entities and model. 1c. <u>Illustrate</u> Regional Load Dispatch Centre and <u>discuss</u> the Benefits of Deregulation. 1d. <u>Explain</u> Congestion management and Available Transfer capabilities for Power market.	1.1 Brief history of electric power, Indian History of Power System, Traditional power industry 1.2 Motivation for Restructuring, Players in Restructuring Process, Electricity market entities and model 1.3 Regional Load Dispatch Centre, Benefits of Deregulation, Basic terminologies in deregulation 1.4 Congestion, Available Transfer capabilities 1.5 Ancillary Services, Growth of power sector in India

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	1e. <u>Explain</u> Ancillary Services, and <u>recognise</u> the Growth of power sector in India.	
Unit– II State Estimation in Power System	CO 1. <i>Describe past, present and future of Power System with Restructuring of an Indian Power System, and analysis of power system state estimation with methodologies used for state estimation.</i> PrO 3. <u>Describe</u> and <u>Evaluate</u> power system state estimation. PrO 11. <u>Simulate</u> Power System State Estimation in Torrit using WLSE method the given power system network. 2l. <u>Describe</u> Power system state estimation. 2m. <u>Determine</u> Weighted least square estimation and <u>evaluate</u> State estimation of AC network. 2n. <u>Detect</u> and identify bad data in measurement set. 2o. Network observability and <u>demonstrate</u> state estimation.	2.1 Power system state estimation: Past, Present and Future 2.2 Weighted least square estimation, Hessian matrix formulation, State estimation of AC network. 2.3 Detection and identification of bad data measurement, estimation of quantities not being measured, 2.4 Network observability and Pseudo measurement, Application of state estimation.
Unit– III Introduction to Smart Grid Technology	CO 2. <i>Illustrate the fundamentals and detail knowledge regarding smart grid technologies with operations and control.</i> PrO 4. <u>Describe</u> Smart Grid Technology. PrO 12. <u>Design</u> and <u>simulate</u> Hybrid AC/DC Micro-grid for Residential Application Using Homer PRO. 3a. <u>Discriminate</u> the conventional and smart grid technology, and <u>discuss</u> the promoters of smart grid. 3b. <u>Identify</u> the functions, components, communication methods, and technology used in smart grid. 3c. <u>Discuss</u> benefits, disadvantages, application, and challenges for smart grid. 3d. <u>Explain</u> Microgrid architecture, components, and <u>discuss</u> its benefits. 3e. <u>Illustrate</u> application, challenges, importance for Microgrid, also, <u>outline</u> Smart Grid Projects in India.	3.1 Structure of conventional grid, Conventional grid to Smart grid, Comparison, Smart Grid Enablers/Promoters, what is smart grid? 3.2 Functions of Smart Grid, Components of Smart Grid, Communication Methods in Smart Grids, Technologies used in Smart Grid 3.3 Benefits of Smart Grid, Disadvantages of Smart Grid, Applications of Smart Grid, Base elements and challenges 3.4 Microgrid: A Review, Microgrid Architecture, Components of Microgrid, Benefits of Microgrids 3.5 Application of Microgrid, Challenges in Microgrid,

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
		R&Ds in Microgrid, Microgrid Importance, Smart Grid Projects in India
Unit-IV Introduction to Virtual Power Plant	CO 3. <u>Describe the fundamentals and detail knowledge regarding virtual power plant with operations and control.</u> PrO 5. <u>Describe Virtual Power Plant.</u> 4a. <u>Identifying current scenarios of renewable energy resources and discuss its challenges in Indian power system.</u> 4b. <u>Describe virtual power plant with its control architectures.</u> 4c. <u>Discuss VPP benefit to the various stakeholder of power system and outline the business opportunity in the worldwide as well as in India.</u>	4.1 Introducing the current scenarios of renewable energy resources and its challenges in Indian power system 4.2 Virtual Power Plant, Classification of Virtual Power Plant (VPP), Control Architectures of VPP, VPP differences due to grid location 4.3 VPP benefit to the various stakeholder of power system, the business opportunity in the worldwide as well as in India.
Unit – V Load Forecasting	CO 4. <u>Apply and validate various algorithm to predict future energy demand for smart grid and learn data analysis.</u> PrO 6. <u>Describe Load Forecasting.</u> PrO 10. <u>Energy Demand Forecasting for EAI data in Python:</u> a) <u>Using Stats Model like AR, ARMA, ARIMA and SARIMA</u> b) <u>Using ML model like RNN and LSTM</u> 5a. <u>Explain Load Forecasting and its different methods.</u> 5b. <u>Discriminate data, information, knowledge, and wisdom. Also, explain time series for data analysis.</u> 5c. <u>Describe and formulate Statistical Models (AR, ARMA, ARIMA, SARIMA) Used for Load Forecasting.</u> 5d. <u>Describe and formulate Machine Learning Methods (RNN, LSTM) Used for Load Forecasting.</u> 5e. <u>Validation of load forecasting and compare different cases studied for load forecasting.</u>	5.1 Introduction to Load Forecasting, Method Used for Load Forecasting. 5.2 what is Data, Information and Knowledge? Time Series 5.3 Statistical Models Used for Load Forecasting, AR, ARMA, ARIMA, SARIMA 5.4 Machine Learning Methods Used for Load Forecasting, RNN, LSTM 5.5 Error in forecasting, Different Cases Studied for Load Forecasting

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
Unit – VI Distribution Automation	CO 5. Express the formulation for substation automation and distribution, system monitoring, control and design. PrO 7. Describe Substation Automation. 6a. Explain distribution automation(DA) and its Functionality. 6b. Discuss typical integrated distribution utility and key design parameters of SCADA/DMS system. 6c. Explain levels, application and benefits of DAS. Also, illustrate Advanced Distribution Management System 6d. Describe Fiber Optic Technology, power line carrier communication, and explain typical components of PLCC.	6.1 Introduction to distribution automation(DA), Automation Systems and Functionality, 6.2 Typical Integrated Distribution Utility, Key Design Parameters of SCADA/DMS system, 6.3 Levels of Automation, Applications and benefits of DAS. Advanced Distribution Management System (ADMS), 6.4 Fiber Optic Technology(FOT) in dedicated communication system, power line carrier communication(PLCC) Explain typical components of PLCC.

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Restructuring of Power System	08	2	3	3	3	0	1	12
II	State Estimation in Power System	06	2	2	2	2	2	0	10
III	Introduction to Smart Grid Technology	12	2	6	4	4	1	0	17
IV	Introduction to Virtual Power Plant	06	2	4	2	2	0	0	10
V	Load Forecasting	08	2	3	2	2	2	1	12
VI	Distribution Automation	05	2	2	2	3	0	0	09
Total		45	12	20	15	16	05	02	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PROs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Prepare case study report on Indian power system:** Collect the information and prepare the case study report on Indian power system from the national power portal and ministry of power website.
- 2) **Design Hybrid AC/DC Micro-grid for Residential Application:** Collect the real network data or the IEEE standard test system and design the micro-grid for residential application using Homer PRO simulation platform.
- 3) **Energy Demand Forecasting:** Collect the past energy consumption data and using statistical/ machine learning algorithm predict next day/week/month energy demand. Any data analytics simulation tool/platform is use.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

These are sample strategies, which the teacher can use to accelerate the attainment of the various types of learning outcomes in this course:

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b) **'L' in item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) With respect to **section No.12**, teachers need to ensure by creating opportunities and provisions for **co-curricular activities**.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group

and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Library survey regarding restructure and deregulation of power system.
- Prepare 15 min. power point presentation for smart grid technology.
- Prepare 15 min. power point presentation for Micro-grid.
- Prepare 15 min. power point presentation for virtual power plant

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Restructuring of Power System	1	<i>PrO 2. <u>Illustrate</u> National Power Portal (NPP).</i> 1d. <u>Describe</u> history of electric power and Traditional power industry. 1e. <u>Justify</u> Restructuring power system network and <u>discuss</u> Electricity market entities and model. 1f. <u>Illustrate</u> Regional Load Dispatch Centre and <u>discuss</u> the Benefits of Deregulation.	Lecture with media, Tutorial	PPTs
	2	<i>PrO 3. <u>Describe</u> Restricting of Power System.</i> 1g. <u>Explain</u> Congestion management and Available Transfer capabilities for Power market. 1h. <u>Explain</u> Ancillary Services, and <u>recognise</u> the Growth of power sector in India.	Lecture With media, Tutorial	PPTs
Unit– II State Estimation in Power System	3	<i>PrO 3. <u>Describe</u> and <u>Evaluate</u> power system state estimation.</i> 2f. <u>Describe</u> Power system state estimation. 2g. <u>Determine</u> Weighted least square estimation and <u>evaluate</u> State estimation of AC network.	Lecture with media, and Tutorial	PPTs, black board
	4	<i>PrO 11. <u>Simulate</u> Power System State Estimation in Torrit using WLSE method the given power system network.</i> 2h. <u>Detect</u> and identify bad data in measurement set. 2i. Network observability and <u>demonstrate</u> state estimation.	Lecture with media, Tutorial	PPTs
Unit– III Introduction to Smart Grid Technology	5	<i>PrO 4. <u>Describe</u> Smart Grid Technology.</i> 3f. <u>Discriminate</u> the conventional and smart grid technology, and <u>discuss</u> the promoters of smart grid.	Lecture with media	PPTs, Video
	6	3g. <u>Identify</u> the functions, components, communication methods, and technology used in smart grid. 3h. <u>Discuss</u> benefits, disadvantages, application, and challenges for smart grid.	Lecture with media	PPTs
	7	<i>PrO 12. <u>Design</u> and <u>simulate</u> Hybrid AC/DC Micro-grid for Residential Application</i>	Lecture with media,	PPTs, Video

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		<i>Using Homer PRO.</i> 3i. <u>Explain</u> Microgrid architecture, components, and <u>discuss</u> its benefits.	and Tutorial	
	8	3j. <u>Illustrate</u> application, challenges, importance for Microgrid, also, <u>outline</u> Smart Grid Projects in India.	Lecture with media, Tutorial	PPTs
Unit – IV Introducti on to Virtual Power Plant	9	PrO 5. <u>Describe</u> Virtual Power Plant. 4a. <u>Identifying</u> current scenarios of renewable energy resources and <u>discuss</u> its challenges in Indian power system.	Lecture with media, Tutorial	PPTs, Video
	10	4b. <u>Describe</u> virtual power plant with its control architectures. 4c. <u>Discuss</u> VPP benefit to the various stakeholder of power system and <u>outline</u> the business opportunity in the worldwide as well as in India.	Lecture with media, Tutorial	PPTs
Unit – V Load Forecasti ng	11	PrO 6. <u>Describe</u> Load Forecasting. 5a. <u>Explain</u> Load Forecasting and its different methods. 5b. <u>Discriminate</u> data, information, knowledge, and wisdom. Also, <u>explain</u> time series for data analysis.	Lecture with media, Tutorial	PPTs, black board
	12	5c. <u>Describe</u> and <u>formulate</u> Statistical Models (AR, ARMA, ARIMA, SARIMA) Used for Load Forecasting. 5d. <u>Describe</u> and <u>formulate</u> Machine Learning Methods (RNN, LSTM) Used for Load Forecasting.	Lecture with media, Tutorial	PPT, Pictures , black board
	13	PrO 10. <i>Energy Demand Forecasting for EAI data in Python:</i> c) <i>Using Stats Model like AR, ARMA, ARIMA and SARIMA</i> d) <i>Using ML model like RNN and LSTM</i> 5e. <u>Validation</u> of load forecasting and <u>compare</u> different cases studied for load forecasting.	Lecture with media, Tutorial	PPTs, black board
Unit – VII Distributi on Automat ion	14	PrO 7. <u>Describe</u> Substation Automation. 6a. <u>Explain</u> distribution automation(DA) and its Functionality. 6b. <u>Discuss</u> typical integrated distribution utility and key <u>design</u> parameters of SCADA/DMS system.	Lecture with media, Tutorial	PPTs, Video
	15	6c. <u>Explain</u> levels, application and benefits of DAS. Also, <u>illustrate</u> Advanced Distribution Management System. 6d. <u>Describe</u> Fiber Optic Technology, power line carrier communication, and <u>explain</u> typical components of PLCC.	Lecture with media, Tutorial	PPTs, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Power System Analysis	Hadi Saadat	Tata McGraw Hill
2	Restructured Power System	S.A. Khaparde and A.R. Abhyanka	Narosa
3	Electric Power Distribution Systems	A.S. Pabla	6th Edition, Tata McGraw Hill
4	Smart Grids and Microgrids: Technology Evolution	Prajof Prabhakaran, S. Mohan Krishna, J. L. Febin Daya, Umashankar Subramaniam, P. V. Brijesh	ISBN: 978-1-119-76059-7 Willy
5	Smart Grid Fundamentals	Radian Belu	eBook ISBN 9780429174803 Taylor & Francis Group
6	Scheduling and Operation of Virtual Power Plants	Ali Zangeneh, Moein Moeini-Aghaie	eBook ISBN: 9780323852685 Paperback ISBN: 9780323852678 ELSEVIER

XV SOFTWARE/LEARNING WEBSITES

1. <https://www.ci2lab.com/>
2. <https://sourceforge.net/directory/windows/?q=smart+grid>
3. https://onlinecourses.swayam2.ac.in/arp19_ap60/announcements?force=true
4. <https://www.statsmodels.org/stable/tsa.html>
5. <https://enode.com/blog/virtual-power-plant-software>
6. <https://www.global.toshiba/ww/products-solutions/renewable-energy/products-technical-services/vpp.html>
7. <https://transparency.entsoe.eu/dashboard/show?loggedUserIsPrivileged=false>

XVI PO-COMPETENCY-CO MAPPING

		POs and PSOs
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Course Code	Competency and COs	PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE476.02	Semester VII	Advances in Power System Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1 for the relevant correlation of each PO or PSO, Competency or CO													
476.02	Competency Advances in power system through the understanding of restructuring of power system, smart grid technology and virtual power plant.	3	3	3	2	3	1	1	-	2	1	2	-	1	1
476.02.1	Describe past, present and future of Power System with Restructuring of an Indian Power System, and analysis of power system state estimation with methodologies used for state estimation.	3	3	1	1	1	-	-	-	-	-	1	-	2	1
476.02.2	Illustrate the fundamentals and detail knowledge regarding smart grid technologies with operations and control.	3	2	3	1	2	-	1	-	-	-	-	-	3	-
476.02.3	Describe the fundamentals and detail knowledge regarding virtual power plant with operations and control.	3	2	2	1	2	-	-	-	-	-	-	-	2	1
476.02.4	Apply and validate various algorithm to predict future energy demand for smart grid and learn data analysis.	3	3	2	2	3	-	-	-	-	-	-	-	2	2
476.02.5	Express the formulation for substation automation and distribution, system monitoring, control and design.	3	2	3	1	2	-	-	-	-	-	-	-	2	1

XVII NAMES OF RESOURCE PERSONS

a) Jignesh Patel

b)

Course Code: EE481
Course Title: **Industry 4.0**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

The industries are now shifting to have more power of automation and industrial revolutions is making the industrial process efficient and easy to control without special skilled man power. The old day's industries are slowly adopting the deployment of new industry revolutions and hence engineers are required for suggesting technologies for these revolutions, deploying it, train the operators and manpower for this new edge technology and efficiently operate the industrial process through it. The syllabus focuses on backbone technologies in industry 4.0, such as dynamic simulation, digital twin and cyber physical system, augmentation of AI and ML algorithm and industrial robot programming. With these all, the student will be able to help in its procurement, installation as well as to use these technologies in an industry and train the manpower to use it as well.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- Lead the industry 4.0 deployment and train the man power to effectively use it.
- Use(/work with) the smart manufacturing technologies, 3D printers, smart sensors, IIoT solutions, Industrial robot etc. by networking them and programming them.

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 5) Find opportunity of applying industry 4.0 Solutions for various industrial process, advise deployment on basis of the cost involved in deployment and possible return on investment.
- 6) Design the 3D model of an object to be 3D printed.
- 7) Write Python and C Code for smart sensor, Industrial and IIoT communications.
- 8) Develop dynamic Simulation for a physical system and test it. Develop Digital twin and Cyber Physical System(CPS).
- 9) Write code for preventive/predictive maintenance.
- 10) Program Industrial robot for required action and motion.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme (CHARUSAT 2021)				Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: **L**-Lecture; **T** – Tutorial/Teacher Guided Theory Practice; **P** - Practical; **C** – Credit, **ESE** - End Semester Examination; **PA** - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **20 marks** are for **attendance, micro-project assessment** and other **assignment** if given any, to facilitate integration of COs and the remaining 10 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	2,4	<u>Make</u> 3 D model in various freely available software supporting 3 D printing.	02
2	4	<u>Simulate</u> Mechanical and Electrical Systems in industries using software like ANSYS and MATLAB.	03
3	3	<u>Develop</u> IoT solution for Industries. a) <u>Integrate</u> sensor and <u>Write</u> code to collect the data using M2M communication. b) <u>Write</u> code to interconnect and interact among smart devices/PLC/Robots with IOT systems.	04
4	4,5	<u>Write</u> code to apply AI ML algorithm and <u>verify</u> with various data set.	Planned to be covered during class room
5	4,5	<u>Write</u> code to use statistics for feature extraction. <u>Apply</u> to various industrial process.	Planned to be covered during class room
6	4,5	<u>Apply</u> Fourier, Wavelet and similar transform to Mechanical Vibration data set, <u>generate</u> features and <u>Write</u> code for Predictive maintenance of machines, using AI ML applications.	Planned to be covered during class room
7	6	<u>Design</u> Mechanical System for various types of industrial Robots and <u>make</u> 3 D model, simulate the response of robot.	06
8	6	<u>Program</u> Industrial Robot for various types of kinematics and movements.	06
9	4	<u>Develop</u> Digital Twin(DT) and Cyber Physical Systems (CPS) for various	06

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
		industrial task and process. a) <u>Develop</u> Digital Twin for industrial processes with ANSYS, MATLAB and Amazon AWS/PTC Thingwork/or suitable cloud offering. b) <u>Develop</u> CPS and use Library support available in Amazon AWS for implementing CPS.	
10	4	<u>Write</u> MATLAB code for parameter tuning of digital twin using Least Square Estimation (LSQ), Weighted Least Square Estimation (WLS) and Maximum Like-hood(ML) methods.	03
		Total	30

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

Sr. No.	Equipment Name with Broad Specifications	PrO No.
1	IIoT Gateways, Vibration sensor, ESP series kits	3,9
2	PLC	3
3	Industrial Robot	6
4	3 D Printer	1,7

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) "One cannot always expect to goal in first attempt". Physical processes are difficult to model. Test – verify and remodify the simulation for best responses.
- 2) "Think out of box but have empathy and follow ethics". The deployment of new industry revolution needs thinking out of box. The success got because of technology must not cut the job. It is moral duty of the industry to providing proper training to already available man power and sustain them by assigning suitable task. The prospective engineers, needs to be trained for these new technologies and make them employable. The student enrolled in this course will try to look to their internship at various industries with this approach and is expected to try their best.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year
- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit 1: Background and Framework for Industry 4.0	Co1. <u>Find</u> opportunity of applying industry 4.0 Solutions for various industrial process, <u>Advise</u> deployment on basis of the cost involved in deployment and possible return on investment. Co 2: <u>Design</u> the 3D model of an object to be 3D printed. Co4. <u>Develop</u> dynamic Simulation for a physical system and test it. <u>Develop</u> Digital twin and Cyber Physical System(CPS). Pro1. <u>Make</u> 3 D model in various freely available software. (Simple model as well as several products like Ultra capacitor, BLDC motor, Fuel Cell etc) Pro 2: <u>Simulate</u> Mechanical and Electrical Systems in industries, using software like ANSYS and MATLAB. 1a. <u>Understand</u> the origin and key benefits of deploying Industry 4.0 1b. <u>Understand</u> design principles and Pillars for Industry 4.0. 1c. <u>Transform</u> a small and medium enterprises (SMEs) into Industry 4.0. Do cost benefit and ROI analysis for deployment of industry 4.0. 1d. <u>Develop</u> mathematical model/ simulation/Digital twin for the given physical process or object. 1e. <u>Select</u> appropriate additive manufacturing process/3D printer for given application. 1f. <u>Understand</u> Environmental Management and Technology potential for Environmental Management 4.0. and hence, <u>Apply</u> design principles looking to it. 1g. <u>Understand</u> Technical and Human Resource view of Industry 4.0 and <u>incorporate</u> it in deployment and management strategies.	1. 7 Basics (Covering topics like, Industry 4.0: origin and key understandings, Industry 4.0 design principles, Industry 4.0 Pillars and Role) and related other topics helping in, technology selection, cost analysis and ROI for deployment of Industry 4.0. 1. 8 Additive Manufacturing techniques, Various types of 3D printers and Example applications. 1. 9 Example implementation of Cyber Physical Systems (CPS) and digital Twin for Industries.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit 2: Industrial IoT and Industrial Communication	Co3. <u>Write</u> Python and C Code for smart sensor, Industrial and IIoT communications. Pro 3: <u>Develop</u> IoT solution for Industries. a) <u>Integrate</u> sensor and <u>Write</u> code to collect the data using M2M communication. b) <u>Write</u> code to interconnect and interact among smart devices/PLC/Robots with IOT systems. 2a. <u>Understand</u> Various Industrial Sensor and smart sensors. 2b. <u>Understand</u> various industrial communication protocols. 2c. <u>Understand</u> difference between IoT and M2M communication. Understand basics of M2M communication. 2d. <u>Understand</u> structural of IIoT and key features. Accordingly, <u>select</u> appropriate cloud and software offering for IIoT.	2. 8 Comparison for various Industrial IoT solution offerings 2. 9 Industrial Sensors 2. 10 Various Industrial Communications. 2. 11 M2M communication. 2. 12 Interconnect Industrial Sensor, PLC, Robot etc with IIoT systems.
Unit 3: AI-ML application in Industry 4.0	Co4. <u>Develop</u> dynamic Simulation for a physical system and test it. <u>Develop</u> Digital twin and Cyber Physical System(CPS). Co5. <u>Write</u> code for preventive/predictive maintenance. Pro 4: <u>Write</u> code to apply AI ML algorithm and <u>verify</u> with various data set. Pro 5: <u>Write</u> code to use statistics for feature extraction. Apply to various industrial process. Pro 6: <u>Apply</u> Fourier, Wavelet and similar transform to Mechanical Vibration data set, <u>generate</u> features and <u>Write</u> code for Predictive maintenance of machines, using AI ML applications. 3a. <u>Understand</u> role and various application of of AI and ML in Industry 4.0. 3b. <u>Understand</u> various statistical feature extraction criteria and code it. 3c. <u>Write</u> code to Apply Fourier and Wavelet on given data set. 3d. <u>Understand</u> LSTM, Random forests, Decision Trees, and logistic regression and <u>Write</u> code for it using ready libraries. 3e. Understand classification algorithms like SVM, and code it. <u>Write</u> code using ready	3.10 Application of AI ML in to industrial process. 3.11 Various AI ML algorithm (LSTM, Random forests, Decision Trees, and logistic regression, classification algorithms like SVM etc) and Their coding 3.12 Generating features from real time data for predictive maintenance, Application of various Algorithm for Predictive Maintenance.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	libraries, for Predicting machines health using classification algorithm.	
Unit 4: Industrial Robotics and Programming	Co6. <u>Program</u> Industrial robot for required action and motion. Pro 7: <u>Design</u> Mechanical System for various types of industrial Robot and make 3 D model, simulate the response of robot. Pro 8: <u>Program</u> Industrial Robot for various types of kinematics and movements. Pro 9: <u>Develop</u> Digital Twin(DT) and Cyber Physical Systems (CPS) for various industrial task and process. a) <u>Develop</u> Digital Twin for industrial processes with ANSYS, MATLAB and Amazon AWS/PTC Thingwork/or suitable cloud offering. b) <u>Develop</u> CPS and use Library support available in Amazon AWS for implementing CPS. Pro 10. <u>Write</u> MATLAB code for parameter tuning of digital twin using Least Square Estimation (LSQ), Weighted Least Square Estimation (WLS) and Maximum Likelihood(ML) methods. 4a. <u>Understand</u> various types of Industrial Robot and their expected role and Functions in various industrial processes. 4b. <u>Understand</u> various parts of robot: Robot Manipulator and Robot Controller, Sensors and actuators in Robot, Other mechanical arrangement. 4c. <u>Understand</u> various Specification of Industrial Robot. Learn key features of Various Robot (Model) from leading Industrial Robot Manufacturer. 4d. <u>Understand</u> Mathematical back-ground behind various types of motion and develop the simulation for evaluating the response. 4e. <u>Program</u> Industrial robot for various types of motion using Rapid, ROS and PLC ladder	4.1 Types of Industrial Robot and their expected role and Functions, 4.2 Various Robot (Model) from leading Industrial Robot Manufacturer 4.3 Robot Manipulator and Robot Controller, Sensors and actuators in Robot, Other mechanical arrangement, 4.4 Specification of Industrial Robot and Procurement (Payload, Reach, Max. TCP Velocity, Max TCP Acceleration, Acceleration Time, Repeatability, Services available and Programmable), 4.5 Mathematical back-ground and Programming using Rapid, ROS and PLC ladder logic.

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PrOs required by industry)	Topics and Sub-topics
	logic.	

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Background and Framework for Industry 4.0	10	--	15	--	--	--	--	15
II	Industrial IoT and Industrial Communication	10	4	4	7	--	--	--	15
III	AI-ML application in Industry 4.0	15	10	8	7	--	--	--	25
IV	Industrial Robotics and Programming	10	4	4	4	3	--	--	15
Total		45	18	31	18	3	00	00	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. It should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

Just to give idea, A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here:

- 1) **Augmented Reality for Training the Industrial work force:** the AR can help to teach the newly recruited and old days (but not much experience to work with digital equipment) man power and can make them ready to efficiently use the industry 4.0 for industrial system.

- 2) **3D Printing of complex products:** Analyze the internal of several real products like Ultra capacitor, BLDC motor, Fuel Cell, Semiconductor switch etc. Develop the 3D model after understanding the internal and manufacture it with help of industries having specialized 3D printers for it.
- 3) **Modeling and verification of industrial processes:** The MATLAB and ANSYS can help a lot to develop the simulation for a physical system. It is not much covering the ready single blocks to simulate the industrial system presently (2023). The libraries can be built and verification can be done by comparing the simulated results with real system data logging.
- 4) **Optimization routines and its deployment in industry 4.0:** Various optimization techniques are researched till date and they can be used to optimize the industrial process. Such routines can be called as API during some layer of Industry 4.0.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

The teacher can use following strategies, to accelerate the attainment of the various types of learning outcomes in this course:

- a) Tutorials that need hand calculation and testing in Software.
- b) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance may to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.
- d) The teacher can suggest several research paper to read and refer. This can be assign to specially to evaluate the capacity of individual, to understand topics related to signal processing but not in syllabus. The capacity can be judge by the way or quality of efforts put, the simulation developed to verify the algorithm and brain storming done and presented to evaluator in form of presentation.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Make a report by literature review on new Embedded boards
- b) Identify the possible reduction in mathematical burden required in particular AI/ML algorithm, modify the code and test verify the speed for new algorithm.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
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Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit 1: Background and Framework for Industry 4.0	1	1a. <u>Understand</u> the origin and key benefits of deploying Industry 4.0 1b. <u>Understand</u> design principles and Pillars for Industry 4.0. 1c. <u>Transform</u> a small and medium enterprises (SMEs) into Industry 4.0. Do cost benefit and ROI analysis for deployment of industry 4.0.	Lecture Only	Mixture of black board and ICT, Videos from Internet
	2	Pro 1: <u>Make</u> 3 D model in various freely available software. Pro 2: <u>Simulate</u> Mechanical and Electrical Systems in industries, using software like ANSYS and MATLAB. 1d. <u>Develop</u> mathematical model/ simulation/Digital twin for the given physical process or object. 1e. <u>Select</u> appropriate additive manufacturing process/3D printer for given application.	Lab only	Mixture of black board and ICT
	3	1f. <u>Understand</u> Environmental Management and Technology potential for Environmental Management 4.0. and hence, <u>Apply</u> design principles looking to it. 1g. <u>Understand</u> Technical and Human Resource view of Industry 4.0 and <u>incorporate</u> it in deployment and management strategies.	Lecture Only	Mixture of black board and ICT
Unit 2: Industrial IoT and Industrial Communication	4	2a. <u>Understand</u> Various Industrial Sensor and smart sensors. 2b. <u>Understand</u> various industrial communication protocols.	Lecture only.	Mixture of black board and ICT
	5	Pro 3: <u>Develop</u> IoT solution for Industries. a) <u>Integrate</u> sensor and <u>Write</u> code to collect the data using M2M communication. 2b. <u>Understand</u> various industrial communication protocols. (continued from last week).	Lab and Lecture both.	Mixture of black board and ICT
	6	Pro 3: <u>Develop</u> IoT solution for Industries. b) <u>Write</u> code to interconnect and interact among smart devices/PLC/Robots with IOT systems.	Lab and Lecture both.	Mixture of black board and ICT

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		2c. <u>Understand</u> difference between IoT and M2M communication. Understand basics of M2M communication. 2d. <u>Understand</u> structural of IIoT and key features. Accordingly, <u>select</u> appropriate cloud and software offering for IIoT.		
Unit 3: AI-ML application in Industry 4.0	7	3a. <u>Understand</u> role and various application of AI and ML in Industry 4.0. 3b. <u>Understand</u> various statistical feature extraction criteria and code it.	Lecture Only	Mixture of black board and ICT
	8	Pro 4: <u>Apply</u> Fourier, Wavelet and similar transform to Mechanical Vibration data set, <u>generate</u> features and <u>Write</u> code for Predictive maintenance of machines, using AI ML applications. 3c. <u>Write</u> code to Apply Fourier and Wavelet on given data set.	Lab and Lecture both.	Mixture of black board and ICT
	9	3d. <u>Understand</u> LSTM, Random forests, Decision Trees, and logistic regression and <u>Write</u> code for it using ready libraries.	Lecture Only	Mixture of black board and ICT
	10	3e. <u>Understand</u> classification algorithms like SVM, and code it. <u>Write</u> code using ready libraries, for Predicting machines health using classification algorithm.	Lecture Only	Mixture of black board and ICT
Unit 4: Industrial Robotics and Programming	11	4a. <u>Understand</u> various types of Industrial Robot and their expected role and Functions in various industrial processes. 4b. <u>Understand</u> various parts of robot: Robot Manipulator and Robot Controller, Sensors and actuators in Robot, Other mechanical arrangement. 4c. <u>Understand</u> various Specification of Industrial Robot. Learn key features of Various Robot (Model) from leading Industrial Robot Manufacturer	Lecture Only	Mixture of black board and ICT
	12	4d. <u>Understand</u> Mathematical back-ground behind various types of motion and develop the simulation for evaluating the response.	Lecture Only	Mixture of black board and ICT
	13			

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
		<i>Pro7. <u>Design</u> Mechanical System for various types of industrial Robots and make 3 D model, simulate the response of robot.</i> <i>Pro8. <u>Program</u> Industrial Robot for various types of kinematics and movements.</i> <i>4e. <u>Program</u> Industrial robot for various types of motion using Rapid, ROS and PLC ladder logic.</i>	Laboratory Only	Mixture of black board and ICT
	14	<i>Pro9. <u>Develop</u> Digital Twin(DT) and Cyber Physical Systems (CPS) for various industrial task and process.</i> <i>a) Develop Digital Twin for industrial processes with ANSYS, MATLAB and Amazon AWS/PTC Thingwork/or suitable cloud offering.</i>	Laboratory Only	Mixture of black board and ICT
	15	<i>Pro9. Develop Digital Twin(DT) and Cyber Physical Systems (CPS) for various industrial task and process.</i> <i>b) Develop CPS and use Library support available in Amazon AWS for implementing CPS.</i> <i>Pro 10. <u>Write</u> MATLAB code for parameter tuning of digital twin using Least Square Estimation (LSQ), Weighted Least Square Estimation (WLS) and Maximum Likelihood(ML) methods.</i>	Laboratory Only	Mixture of black board and ICT

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	<i>Industry 4.0: Current Status and Future Trends,</i>	<i>Edited by Jesús Hamilton Ortiz, William Gutierrez Marroquin, Leonardo Zambrano Cifuentes</i>	<i>An IntechOpen Publication, Published in London, United Kingdom DOI: 10.5772/intechopen.90396</i>

S. No.	Title of Book	Author	Publication including ISBN
2	Industry 4.0 The Industrial Internet of Things	Alasdair Gilchrist,	Apress Publication ISBN: 978-1-4842-2047-4
3	Industrial Robots Programming: Building Applications for the Factories of the Future.	J. Norberto Pires, Mechanical Engineering Department, University of Coimbra, Portugal	SpringerLink, ISBN 0-387-23325-3 ISBN 9780387233253
4	Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing,	Ian Gibson, David Rosen and Brent Stucker	SpringerLink, Second Edition ISBN: 978-1-4939-2113-3
5	Machine Learning using Python	Manaranjan Pradhan, U Dinesh Kumar	Wiley, ISBN: 978-81-265-7990-7 ISBN: 978-265-8855-8(ebk)

XV SOFTWARE/LEARNING WEBSITES

- NPTEL Courses by Prof. Sudip Misra, IIT Kharagpur
https://onlinecourses.nptel.ac.in/noc20_cs69/preview
- OpenSim/OpenSimulator, FreeScad, Tinkercad
- MATLAB, ANSYS, RobotStudio 2.2

XVI PO-COMPETENCY-CO MAPPING

Course Code	Competency and COs	POs and PSOs													PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning			
EE 481	Semester VI	Industry 4.0 Mark 'H'=3 for high, 'M'=2 for medium or 'L'=1' for the relevant correlation of each PO or PSO, Competency or CO														
481.1	Find opportunity of applying industry 4.0	2	2	2	2	2	1	1	1	1	1	2	--	--	--	

Course Code	Competency and COs													PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
		PO 1 Engineering knowledge	PO 2 Problem analysis	PO 3 Design/development of solutions.	PO 4 Conduct investigations of complex problems	PO 5 Modern tool usage	PO 6 The engineer and society	PO 7 Environment and sustainability	PO 8 Ethics	PO 9 Individual and teamwork	PO 10 Communication	PO 11 Project management and finance	PO 12 Life-long learning		
	<i>Solutions for various industrial process, <u>advise</u> deployment on basis of the cost involved in deployment and possible return on investment.</i>														
481.2	<i>Design the 3D model of an object to be 3D printed.</i>	3	2	2	2	3	--	--	--	1	--	--	--	--	--
481.3	<i>Write Python and C Code for smart sensor, Industrial and IIoT communications.</i>	2	1	2	1	2	--	--	--	1	--	--	--	---	--
481.4	<i>Develop dynamic Simulation for a physical system and test it. Develop Digital twin and Cyber Physical System(CPS).</i>	3	3	3	2	2	--	--	--	--	1	1	2	--	2
481.5	<i>Write code for preventive/predictive maintenance.</i>	3	3	2	3	2	--	--	--	--	--	--	--	---	2
481.6	<i>Program Industrial robot for required action and motion.</i>	3	3	3	2	3	--	--	1	1	1	--	--	--	2

XVII NAME OF RESOURCE PERSONS

a) Jivanadhar A. Joshi

Course Code: **EE482**
Course Title: **Smart Grid Technologies and Applications**

UG Engineering Programme	Semester
Electrical Engineering	Seventh

I RATIONALE

The smart grid can help measure and reduce energy consumption and costs. One of the visions for the smart grid is that consumers will be able to monitor their energy consumption in real time, or at least near real time, via the Web. This curriculum course structure is developed such that the student is able to understand the basic concepts, components, architecture, various measurement technologies of smart grid and the importance of renewable energy in smart grid.

II COMPETENCY

The basic purpose of this course is to help the student to acquire the following competency which is a **macro-level industry-oriented task** for this course:

- **Evaluate the performance of various renewable energy sources and energy storage equipment which are used in industry and power system.**

III COURSE OUTCOMES (COs)

Each of the following course outcomes (COs) are the **meso-level industry-oriented sub-tasks** of the above-referred competency. Each CO is an integrated performance of the cognitive domain outcomes, practical outcomes and social skills. The industry expects the students to display them individually and collectively:

- 1) **Understand** the basic concepts, components and architecture of smart grid.
- 2) **Compare** the various measurement technologies in smart grid.
- 3) **Design** renewable energy sources which are used in smart grid.
- 4) **Illustrate** about battery technology and energy.
- 5) **Interpret** about role of Electric Vehicles in smart grid.

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)			Total Credits (L+T+P)	Examination Scheme				
				Theory Marks		Practical Marks		Total Marks
L	T	P	C	ESE	PA	ESE	PA	150
3	0	2	4	70	30*	25	25	

Legends: L-Lecture; T – Tutorial/Teacher Guided Theory Practice; P - Practical; C – Credit, ESE - End Semester Examination; PA - Progressive Assessment

(*): Under the **Theory PA**, out of 30 marks, **10 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is converted from 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V SUGGESTED PRACTICAL OUTCOMES/ EXERCISES

The following practical outcomes (PrOs) predominantly related to the *Psychomotor Domain* are related to the industry-oriented COs and the competency. The acquisition of these PrOs to the **relevant level of Dave's Psychomotor Domain taxonomy** by the students through the **practicum** will facilitate in integrating them with other LOs in order to display the COs to ultimately attain the industry-required competency.

S. No.	CO No.	Practical Outcomes (Practicum) (PrOs in Psychomotor Domain)	Approx. Hrs. required
1	1	Create the hybrid AC/DC microgrid test system in grid-connected mode using MATLAB software.	04
2	2	Computes the PMU benchmark model.	02
3	3	Test the MPPT analysis of PV and wind generator using P & O method.	02
4	3	Analyze the effect of partial shading on a PV generation.	02
5	3	Design the energy management system of microgrid using MATLAB software.	04
6	3	Design the deep neural network for MPPT analysis using MATLAB software.	04
7	4	Create test scenarios for battery management system using Simulink Test	02
8	5	Design vehicle-to-grid system to control microgrid system parameters using MATLAB software.	04
		Total	24

VI MAJOR EQUIPMENT/ INSTRUMENTS/SOFTWARES REQUIRED

S. No.	Equipment Name with Broad Specifications	PrO No.
1	MATLAB Simulation software (version 2021a)	1,5,6,7,8
2	Test rig of doubly fed induction generator based wind energy conversion system	2
3	Test rig of MPPT boost converter for solar PV	3,4

VII AFFECTIVE DOMAIN OUTCOMES (ADOs)

The following Affective Domain Outcomes (ADOs) are not specific to any of the above listed PrOs, but they are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he performs the series of practical exercises and other T-L strategies throughout the semesters.

- 1) Follow safe practices.
- 2) Practice energy conservation.
- 3) Function as a team member.
- 4) Function as a team leader.
- 5) Follow ethics.

The level of achievement of the ADOs according to **Krathwohl's 'Affective Domain Taxonomy'** should gradually increase as planned below:

- a) 'Responding Level' in 1st year
- b) 'Valuing Level' in 2nd year

- c) 'Organising Level' in 3rd year
- d) 'Characterising Level' in 4th year.

VIII UNDERPINNING THEORY COMPONENTS

The following topics are to be learned by the students and assessed by the teachers in order to develop the '**Apply**' and above level sample **Cognitive Domain** learning outcomes denoted by unit outcomes (UOs) related to **Revised Bloom's Taxonomy** (RBT) given below for achieving the related COs to be integrated to perform the identified competency. More UOs could be added.

<i>Unit No. & Title</i>	<i>Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related ProOs required by industry)</i>	<i>Topics and Sub-topics</i>
Unit – I Introduction	CO 1. <u>Understand</u> the basic concepts, components and architecture of smart grid. <i>Pro 24. <u>Create</u> the hybrid AC/DC microgrid test system in grid-connected mode using MATLAB software.</i> 1f. <u>Prepare</u> the drawing of smart grid for the given situation 1g. <u>Design</u> the smart grid system for the given area 1h. <u>Determine</u> the components used for the specified smart grid application	1.6 Today's Grid Versus Smart Grid, Rationale for Smart Grid, 1.7 Computational Intelligence, Power System Enhancement, 1.8 Communication and Standards. 1.9 Environment and Economics, Shareholders Roles and Function, 1.10 Architecture of Smart Grid 1.11 Smart Grid components functions.
Unit– II Sensors and Measurement	CO 2. <u>Compare</u> the various measurement technologies in smart grid. <i>Pro 2. <u>Computes</u> the PMU benchmark model.</i> 2p. <u>Determine</u> the sensors for the specified smart grid application. 2q. <u>Choose</u> the correct sensor for the specified application. 2r. <u>Design</u> the PMU for the given rating of smart grid.	2.28 Sensors for Smart Grid 2.29 Monitoring and Measurement Technologies 2.30 PMU, Smart meters, Smart Appliances 2.31 Multi Agent Systems (MAS) Technology 2.32 Micro grid and Smart grid comparison 2.33 Wide Area Measurement
Unit– III Distributed Generation	CO 3. <u>Design</u> renewable energy sources which are used in smart grid. <i>Pro 3. <u>Test</u> the MPPT analysis of PV and wind generator using P & O method.</i> <i>Pro 4. <u>Analyze</u> the effect of partial shading on a PV generation.</i> <i>Pro 5. <u>Design</u> the energy management system of microgrid using MATLAB</i>	3.6 Solar Energy, PV Systems 3.7 Wind turbine Systems 3.8 Biomass, Small and Micro Hydro Power 3.9 Fuel Cell 3.10 Geothermal heat pumps

Unit No. & Title	Unit Outcomes (UOs) in Cognitive Domain (to develop COs and related PROs required by industry)	Topics and Sub-topics
	software. Pro 6. <u>Design</u> the deep neural network for MPPT analysis using MATLAB software. 3e. <u>Determine</u> the maximum power generation point from <u>the given</u> type of PV array. 3f. <u>Evaluate</u> the effect of partial shading on the PV array. 3g. <u>Design</u> the energy management system for <u>the given</u> smart grid/microgrid system. 3h. <u>Design</u> the deep neural network for the given application.	
Unit-IV Energy Storage	CO 4. <u>Illustrate</u> about battery technology and energy. Pro 6. <u>Create</u> test scenarios for battery management system using Simulink Test 4l. <u>Select</u> the relevant battery for <u>the specified</u> application. 4m. <u>Evaluate</u> the performance of <u>the given</u> type of battery for <u>the specified</u> condition. 4n. <u>Design</u> the battery management system for <u>the specified</u> application.	4.9 Batteries, Flow Batteries 4.10 Fuel Cell and hydrogen electrolytes 4.11 Flywheel 4.12 Super conduction magnetic energy storage systems, super capacitors, Simulation and case studies
Unit – V Electric Vehicles	CO 5. <u>Interpret</u> about role of Electric Vehicles in smart grid. Pro 7. <u>Design</u> vehicle-to-grid system to control microgrid system parameters using MATLAB software. 5c. <u>Evaluate</u> the performance of the plugin electric vehicle for the specified condition. 5d. <u>Evaluate</u> the performance of the hybrid electric vehicle for the specified condition. 5e. <u>Design</u> the vehicle to grid system to control the parameters of microgrid.	5.6 Plugin Electric Vehicles and hybrid, 5.7 Vehicle classes, 5.8 Vehicle Architecture, 5.9 Grid to Vehicle (G2V) Charging, Grid Impacts, 5.10 Vehicle to Grid (V2G).

IX TABLE OF SPECIFICATIONS (TOS) FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks

Unit No.	Unit Title	Teaching Hours	Distribution of Theory Marks						
			R Level	U Level	A Level	An Level	E Level	C Level	Total Marks
I	Introduction	09	4	4	4	2	0	0	14
II	Sensors and Measurement	09	4	4	4	2	0	0	14
III	Distributed Generation	09	0	4	2	2	2	4	14
IV	Energy Storage	09	2	2	4	2	0	4	14
V	Electric Vehicles	09	0	4	2	2	4	2	14
Total		45	10	18	16	10	6	10	70

Legends: R= Remember; U= Understand; A= Apply; An= Analyse; E= Evaluate; C= Create; (Revised Bloom's taxonomy).

Note: This TOS shall be treated as a general guideline for students and teachers for distribution of marks. The actual distribution of marks in the question paper may vary slightly to fulfil the MiLOs.

X MICRO-PROJECTS

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. The student ought to submit the micro-project by the end of the semester to develop the industry-oriented COs. In the first four semesters, the micro-projects are group-based. However, in the seventh and eight semesters, it should be preferably be **individually** undertaken to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry. In special situations where groups have to be formed for micro-projects, the number of students in the group should **not exceed three**.

The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of PrOs, UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than **15 (fifteen) student engagement hours** during the course.

A suggestive list of micro-projects **for a minimum Learning effort of 15 hours per semester** are given here. Similar micro-projects could be added by the concerned faculty of the course:

- 1) **Electric Vehicle:** Collect the specifications of different types of electric vehicles manufacturing by different types of industries.
- 2) **Battery:** Collect the specifications of different types of battery used in different Electric vehicle by different types of industries.
- 3) **Renewable generator:** Prepare case studies on control issues in renewable energy integration and microgrid.
- 4) Undertake the survey in reputed industry and submit a report on different type of sensors and standard used to implement smart grid.

XI SPECIAL INSTRUCTIONAL STRATEGIES (if any)

- a) Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.

- b) '**L**' in **item No. 4** does not mean only the traditional lecture method, but different types of teaching methods and media that are to be employed to develop the outcomes.
- c) About **15-20% of the topics/sub-topics** which are relatively simpler or descriptive in nature that can be understood without the teacher assistance are to be given to the students for **self-directed learning**, but to be progressively assessed for their integration into the COs.

XII SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related **co-curricular** activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a) Library survey regarding different protection schemes used in industries.
- b) Prepare 15 min. power point presentation on how to choose relay for the particular application.

XIII COURSE PLAN

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
Unit – I Introduction	1	Pro 1. <u>Create</u> the hybrid AC/DC microgrid test system in grid-connected mode using MATLAB software. 1c. <u>Prepare</u> the drawing of smart grid for the given situation	Lecture with media	PPTs, black board
	2	1d. <u>Design</u> the smart grid system for the given area	Lecture With media	PPTs, black board
	3	1e. <u>Determine</u> the components used for the specified smart grid application	Lecture with media,	PPTs, black board
Unit– II Sensors and Measurement	4	2a. <u>Determine</u> the sensors for the specified smart grid application.	Lecture with media, Tutorial	PPTs, black board
	5	2b. <u>Choose</u> the correct sensor for <u>the specified</u> application.	Lecture with media, Tutorial	PPTs, black board
	6	Pro 2. <u>Computes</u> the PMU benchmark model. 2c. <u>Design</u> the PMU for the given rating of smart grid.	Lecture with media	PPTs, black board
Unit– III Distributed Generation	7	Pro 3. <u>Test</u> the MPPT analysis of PV and wind generator using P & O method. Pro 4. <u>Analyze</u> the effect of partial shading on a PV generation.	Lecture with media,	PPTs, black board

Unit No. & Title	Week No.	Learning Outcomes	Teaching method	Teaching media
on		3a. <u>Determine</u> the maximum power generation point from <u>the given</u> type of PV array. 3b. <u>Evaluate</u> the effect of partial shading on the PV array.		
	8	Pro 5. <u>Design</u> the energy management system of microgrid using MATLAB software. 3c. <u>Design</u> the energy management system for <u>the given</u> smart grid/ microgrid system.	Lecture with media	PPTs, black board
	9	Pro 6. <u>Design</u> the deep neural network for MPPT analysis using MATLAB software. 3d. <u>Design</u> the deep neural network for the given application.	Lecture with media, + PPT	PPTs, black board
Unit-IV Energy Storage	10	4a. <u>Select</u> the relevant battery for <u>the specified</u> application.	Lecture with media, Tutorial	PPT, black board
	11	4b. <u>Evaluate</u> the performance of <u>the given</u> type of battery for <u>the specified</u> condition.	Lecture with media	PPTs, black board
	12	Pro 6. <u>Create</u> test scenarios for battery management system using Simulink Test 4c. <u>Design</u> the battery management system for <u>the specified</u> application	Lecture with media, case study	PPT, black board
Unit – V Electric Vehicles	13	5a. <u>Evaluate</u> the performance of the plugin electric vehicle for the specified condition.	Lecture with media	PPTs, black board
	14	5b. <u>Evaluate</u> the performance of the hybrid electric vehicle for the specified condition.	Lecture with media, Tutorial	PPTs, black board
	15	Pro 7. <u>Design</u> vehicle-to-grid system to control microgrid system parameters using MATLAB software. 5c. <u>Design</u> the vehicle to grid system to control the parameters of microgrid	Lecture with media, Tutorial	PPTs, black board, Video

XIV SUGGESTED LEARNING RESOURCES

S. No.	Title of Book	Author	Publication including ISBN
1	Smart Grid: Fundamentals of design and analysis	James Momoh	John Wiley & sons, 2012. ISBN: 9780070671188, 0070671184
2	Smart Grid: Technology and Applications	Janaka Ekanayake, Kithsiri Liyanage, Jianzhong Wu, Akihiko Yokoyama, Nick Jenkins	John Wiley. ISBN: 9788126557356
3	Smart Grid Applications,	Lars T. Berger, Krzysztof Iniewski	Wiley, 2012 ISBN: 978-1-118-00439-5

S. No.	Title of Book	Author	Publication including ISBN
	Communications, and Security		
4	Smart Grid: Integrating Renewable, Distributed and Efficient Energy	Fereidoon Sioshansi	Academic Press, 2012, ISBN: 9780071077743
5	Innovative Testing and Measurement Solutions for Smart Grid	Qi Huang, Shi Jing, Jianbo Yi, Wei Zhen	Wiley, 2016 ISBN: 978-1-118-88992-3

XV SOFTWARE/LEARNING WEBSITES

1. Introduction to Smart Grid: - <https://nptel.ac.in/courses/108107113>
2. Fundamentals of Electric Vehicles: Technology & Economics: - <https://nptel.ac.in/courses/108106170>

XVI PO-COMPETENCY-CO MAPPING

Cour se Code	Competency and COs	PO 1 Engin eerin g know ledge	PO 2 Probl em analys is	PO 3 Desig n/dev elop ment of soluti ons.	PO 4 Condu ct invest igation s of comp lex probl ems	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Envir onme nt and sustai nabilit y	PO 8 Ethics	PO 9 Indivi dual and team work	PO 10 Com muni cation	PO 11 Proje ct mana gement and finan ce	PO 12 Life-long learni ng	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE482	Semester VII														
482	Competency Evaluate the performance of various renewable energy sources and energy storage equipment which are used in industry and power system	3	2	2	2	1	-	1	-	-	-	-	1	3	-
482.1	Understand the basic concepts, components and architecture of smart grid.	3	3	3	2	1	-	2	-	-	-	1	1	3	2
482.2	Compare the various measurement technologies in smart grid	3	3	1	1	3	-	1	-	-	-	2	-	3	1
482.3	Design renewable energy sources which are used in smart grid	3	3	1	2	-	-	-	-	-	-	2	-	3	2

Cour se Code	Competency and COs	PO 1 Engin eerin g know ledge	PO 2 Probl em analy sis	PO 3 Desig n/dev elop ment of soluti ons.	PO 4 Condu ct invest igatio ns of comp lex probl ems	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Envir onme nt and sustai nabili ty	PO 8 Ethics	PO 9 Indivi dual and team work	PO 10 Com muni cation	PO 11 Proje ct mana geme nt and finan ce	PO 12 Life- long learni ng	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
482. 4	Illustrate about battery technology and energy	3	2	2	-	1	-	-	-	-	-	-	-	2	-
482. 5	Interpret about role of Electric Vehicles in smart grid	3	2	2	2	-	1	2	-	1	-	1	-	2	1

XVII NAMES OF RESOURCE PERSONS

a) Dr. Jigar Sarda

OCEE4001: Computer-Aided Design of Electrical Machines

Description:

This course OCEE4001 – Computer-Aided Design of Electrical Machines is offered from SWAYAM

URL: https://onlinecourses.nptel.ac.in/noc24_ee50/preview

Credit and Week:

Teaching Scheme	Week	Theory	Practical	Total
Marks	12	100	-	100
Credit	-	3	1	4

About this course:

This course deals with the design of electrical machines using an analytical and a computer-aided approach. The course covers the detailed part by part design of various machines derived using fundamental electromechanical equations. Various conventional electromechanical devices/machines including transformers, induction, synchronous and dc machine design is covered along with design approaches presenting more recent special machines such as brushless permanent magnet (PMSM and BLDC), switched reluctance, and synchronous reluctance machines along with numerical problems. A computer-aided approach to design the aforementioned machines is presented in each chapter, and the course also contains exercises and tutorial to enable the reader to understand the design process and test out the design problems on their own.

Industry Support:

All Electrical Machines production Industry

Course Layout:

Week 1:

Introduction - I

Introduction – II : Fundamentals and Principles of Design

Equivalent Circuit approach to Design

Week 2:

Transformer Design - I

Transformer Design - II

Transformer Design - III
Transformer Design - IV

Week 3:
Windings in Electrical Machines
Design of DC Machines – I
Design of DC Machines – II
Design of DC Machines – III

Week 4:
Design of Three-phase Induction Motors - I
Design of Three-phase Induction Motors - II
Design of Three-phase Induction Motors - III
Design of Three-phase Induction Motors - IV

Week 5:
Design of Single-Phase Induction Machines – I
Design of Single-Phase Induction Machines – II
Design of Single-Phase Induction Machines – III

Week 6:
Design of Three-phase Synchronous Machines - I
Design of Three-phase Synchronous Machines - II
Design of Three-phase Synchronous Machines - III
Design of Three-phase Synchronous Machines - IV

Week 7:
Design of Synchronous Reluctance Machines – I
Design of Synchronous Reluctance Machines – II
Design of Synchronous Reluctance Machines – III

Week 8:
Design of Brushless Permanent Magnet Machines – I
Design of Brushless Permanent Magnet Machines – II
Design of Brushless Permanent Magnet Machines – III
Design of Brushless Permanent Magnet Machines – IV
Design of Brushless Permanent Magnet Machines – V

Week 9:
Design of Salient Reluctance Machines – I
Design of Salient Reluctance Machines – II
Design of Salient Reluctance Machines – III

Week 10:
Design of Stepper Machines – I
Design of Stepper Machines – II
Design of Axial Flux Machines – I
Design of Axial Flux Machines – II

Week 11:

Computer-aided Design and Analysis Methods - I

Computer-aided Design and Analysis Methods - II

Week 12:

Case studies and tutorials – I and II

Case studies and tutorials –III

Case studies and tutorials –IV

Books and references:

1. A. K. Sawhney, “A Course in Electrical Machine Design”, Dhanpat Rai and Co., 2016.
2. T. A. Lipo, “Introduction to AC Machine Design”, IEEE Press – Wiley Publications, 2017.
3. J. R. Hendershot and T. J. E. Miller, “Design of Brushless Permanent – Magnet Motors”, Motor Design Books LLC, 2nd edition, 2010.
4. J. Pyrhonen, T. Jokinen, and V. Hrabovcova, “Design of Rotating Electrical Machines”, John Wiley and Sons Inc., 2nd edition, 2013.
5. R. Krishnan, “Switched Reluctance Motor Drives”, CRC Press LLC, USA, 2001.

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course .

Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Madras. It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

CE491: DEEP LEARNING APPLICATIONS AND AI

B.TECH 7TH SEMESTER (Electrical)

INSTITUTE ELECTIVE – IV

(For Minor Specialization in “Artificial Intelligence and Machine Learning”)

Credits and Hours:

Teaching Scheme	Theory	Tutorial/Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Objective of the Course:

The main objectives of the course are:

- To make students aware about Deep Learning and Artificial Intelligence applications
- To learn about the most effective AI techniques, and gain practice implementing them
- To learn how to quickly and powerfully apply these techniques to new problems
- To make students aware about field where AI can be applied in electrical engineering problems

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	12
2	Deep neural network	12
3	Improving deep neural network	12
4	Convolution Neural Network	12
5	Sequence models	12

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

C. Detailed Syllabus:

1	Introduction	12 Hours	20%
1.1	Introduction: deep learning		
1.2	Tools for deep learning		
1.3	Shallow and deep neural network		
1.4	Tensor flow		
2	Deep Neural Network	12 Hours	20%
2.1	Forward propagation		
2.2	Backpropagation		
2.3	Application: Load Forecasting		
3	Improving Deep Neural Network	12 Hours	20%
3.1	Learning rate, Activation Function		
3.2	Number of Layers and Neurons		
3.3	Overfitting and Underfitting Test		
4	Convolution Neural Network	12 Hours	20%
4.1	Convolution operation		
4.2	Padding and strides		
4.3	Activation functions		
4.4	Pooling and dropout layers		
5	Sequence models	12 Hours	20%
5.1	Time series and sequence data		
5.2	Modelling data for sequence learning		
5.3	Recurrent neural network		
5.4	Long short-term memory network		

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and practical session which carries 10 Marks and 5 marks weightage respectively.

- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a practical session where students have an opportunity to build an appreciation for the concepts being taught in lectures.

E. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To understand fundamental of Deep Learning and Artificial Intelligence (Understand).
CO2	To solve difficult and complex problem of computer science using AI techniques (Understand, Apply).
CO3	To identify fields where AI can be applied (Understand, Apply).

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	1	1	2	-	-	-	-	-	-	1	1	1
CO2	2	2	2	2	1	1	-	-	-	-	-	1	1	1
CO3	2	1	1	2	1	1	-	-	-	-	-	1	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Reference Books:

1. Stuart Russell, Artificial Intelligence – A Modern Approach (3 Edition), Pearson, U.K., 2016.
2. Tariq Rashid, Make your own neural network, CreateSpace Independent Publishing Platform, U.S., 2016.

3. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, The MIT Press, Cambridge, England, 2017.
4. Seth Weidman, Deep Learning from scratch, O'Reilly media, U.S.A., 2019.

Web Materials:

1. <https://www.youtube.com/watch?v=zu6VKOtI0PY>

LIST OF PRACTICALS

Practical No.	Name of Practical
1	Application of hybrid artificial intelligence approaches for hydraulic problems
2	Using artificial neural networks for estimation of the energy consumption obtained by dams and hydropower plants
3	Swarm intelligence for optimum weight of the structures
4	Metaheuristic optimization algorithm in design optimization of 2D and 3D steel or concrete structure
5	Application of fuzzy logic in electrical engineering problems
6	Optimization model for single- and multi-objective solutions for construction management

EE491: ELECTRIC VEHICLES IN SMART GRID
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)
Minor Specialization Subject-4
Institute Elective

(For Minor Specialization in “Electric Vehicle Systems”)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

As electrical power generation and economy is essential gradient for the industrial and all around development of any country, the objectives of the course are:

1. To understand the economics and effects on grid during charging.
2. To identify the impacts on system demand and on distribution system during different penetration level of EV charging.
3. To design control strategies for EVs to support frequency control of power system
4. To develop solutions based on ICT to support EV deployment.
5. To understand different protocol for grid interfacing and communication related to EVs.

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to EVs in Smart Grid	10
2	Influence Of EVs On Power System	15
3	Frequency Control Reserves & Voltage Support From EVs	17
4	ICT Solutions To Support EVs Deployment	12

5	Vehicular communication	06
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Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

E. Detailed Syllabus:

- 1 Introduction to EVs in Smart Grid 10 Hours 16.67%
Introduction, Impact of charging strategies, EV charging options and infrastructure, energy, economic and environmental considerations, V2G ,G2V and V2V Technology ,PEVs Charging Infrastructures ,Impact of EV charging on power grid, effect of EV charging on generation and load profile, Smart charging technologies, Impact on investment
- 2 INFLUENCE OF EVS ON POWER SYSTEM 15 Hours 25%
Introduction, identification of EV demand, EV penetration level for different scenarios, classification based on penetration level, EV impacts on system demand: dumb charging, multiple tariff charging, smart charging, case studies, Effect of large scale EV charging on Distribution Systems.
- 3 FREQUENCY CONTROL RESERVES & VOLTAGE SUPPORT FROM EVS 17 Hours 28.33%
Introduction, power system ancillary services, electric vehicles to support wind power integration, electric vehicle as frequency control reserves and tertiary reserves, voltage support and electric vehicle integration, properties of frequency regulation reserves, control strategies for EVs to support frequency regulation.
- 4 ICT SOLUTIONS TO SUPPORT EV DEPLOYMENT 12 Hours 20%
Introduction, Architecture and model for smart grid & EV, ICT players in smart grid, smart metering, information & communication models, functional and logical models, technology and solution for smart grid: interoperability, communication technologies.
- 5 Vehicular communication 06 Hours 10%
Communication within vehicle, with grid, digital communication systems, vehicle network theory, embedded control, actuators, data analysis and importance of data in maintainability, concept of driverless car , NETWORKS AND PROTOCOLS : Overview of

general-purpose networks and protocols -Ethernet, TCP, UDP, IP,ARP,RARP - LIN standard overview -workflow concept-applications -LIN protocol specification -signals - Frame transfer -Frame types -Schedule tables -Task behaviour model -Network management -status management - overview of CAN -fundamentals -Message transfer - frame types-Error handling -fault confinement-Bit time requirements.

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	25	20	15	15	5

G. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand the economics and effects on grid during charging.
CO2:	Identify the impacts on system demand and on distribution system during different penetration level of EV charging.
CO3:	Design control strategies for EVs to support frequency control of power system
CO4:	Develop solutions based on ICT to support EV deployment.
CO5:	Understand different protocol for grid interfacing and communication related to EVs.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2:	3	2	2	1	1	-	-	-	-	-	-	-	-	-
CO3:	3	2	3	1	2	-	2	-	1	-	1	2	2	-
CO4:	3	3	3	2	2	-	2	-	1	-	1	2	2	-
CO5:	3	1	-	-	-	1	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realize the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, and realize the operation of relay under pre-determined conditions.

I. Recommended Study Material:

Text Books:	
1.	SumedhaRajakaruna, FarhadShahnia and Arindam Ghosh, “Plug In Electric Vehicles in Smart Grids-Integration Techniques”, Springer Science + Business Media Singapore PvtLtd., 2015.
2.	Canbing Li, Yijia Cao, YonghongKuang and Bin Zhou, “Influences of Electric Vehicles on Power System and Key Technologies of Vehicle-to-Grid”, Springer-Verlag Berlin Heidelberg, 2016.
3.	James Larminie and John Lory, “Electric Vehicle Technology – Explained”, John Wiley & Sons Ltd, 2003

Reference Books:	
1.	Qiuwei Wu, "GRID INTEGRATION OF ELECTRIC VEHICLES IN OPEN ELECTRICITY MARKETS", John Wiley & Sons, Ltd, 2013.
Web Material:	
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Other Material:	
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B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VIII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Charotar University of Science and Technology, Gujarat

(use as template)

Re-designed Course Structure of UG in Electrical Engineering

Course Code: **EE458**

Course Title: **Major Project**

UG Engineering Programme	Semester
Electrical Engineering	Eight

I RATIONALE

This course enables the students to apply some of the knowledge and/or skills developed during the programme to new problem for which there are number of engineering solutions. This course include planning of the tasks which are to be completed within the time allocated, and in turn, helps to develop ability to plan, , use, monitor and control resources optimally and economically. By studying this course abilities like creativity, imitateness and performance qualities are also developed in students. Leadership development and supervision skills are also integrated objectives of learning this course.

II COMPETENCY

The course should be taught and implemented with the aim to develop the required course outcomes (COs) so that students will acquire following competency needed by the industry:

- **Find the solution of a real-life problem.**

III COURSE OUTCOMES (COs)

Depending upon the nature of the projects undertaken, the following could be some of the major course outcomes that could be attained, although, in case of some projects few of the following course outcomes may not be applicable.

- 1) Apply the engineering knowledge to analyze, provide solution and carried out investigation of the problems by individual or team.
- 2) Identify the engineering problem; solve, report and present it by teamwork and work with other discipline.
- 3) Use experience related to professional and ethical principles and responsibility during their project.
- 4) Evaluate the problem in commercial, societal, health, safety, legal, cultural and environment aspects before reach to final solution.
- 5) Apply the engineering skills to contemporary issues related to engineering in general with advanced tools and technologies

IV TEACHING AND EXAMINATION SCHEME

Teaching Scheme (In Hours)	Total Credits (L+T+P)	Examination Scheme		
		Theory Marks	Practical Marks	Total Marks

L	T	P	C	ESE	PA	ESE	PA	800
0	0	36	20	00	00	400	400	

Legends: *L*-Lecture; *T* – Tutorial/Teacher Guided Theory Practice; *P* - Practical; *C* – Credit, *ESE* - End Semester Examination; *PA* - Progressive Assessment

(*): Under the **Theory PA**, out of 40 marks, **20 marks** are for **micro-project assessment** to facilitate integration of COs and the remaining 20 marks is the average of 2 tests to be taken during the semester for the assessment of the cognitive domain UOs required for the attainment of the COs.

V COURSE DETAILS

- i. Major Project showcases that, a student during the final semester will be able to apply the engineering knowledge to a practical problem.
- ii. Students can carry out their project work in-house at institute/ industry/ research organization/govt. organization.
- iii. Project should be related to the Product/prototype design and development or Experimentation, Simulations and evaluation of any system etc.
- iv. A student is required to carry out the project work related to Electrical Engineering or some interdisciplinary work under the guidance of a faculty member and the supervisor of the concerned industry/institute/organization.
- v. The major project work provides student an opportunity to develop something on their own or group of maximum 3 students. Final decision related to the number of students in one group is based on internal supervisor recommendation.
- vi. Project will be evaluated at least twice during the semester by internal faculty as a part of continuous evaluation and final evaluation will be carried out by external examiner at the end of semester.
- vii. Student should submit the project report at the end of the semester duly signed by project supervisor and head of department. Report comprising literature review, objective, methodology, scope of the project work undertaken, outcome of project and conclusions.
- viii. The students are required to identify their problem and they are required to follow all the rules and instruction issued by department/industry.
- ix. Students should finalize the project at the end of seven semester (before end of November)

VI PROJECT REPORT

At the end of the semester student should prepare project report. Guideline of the Project report is as follow

- a) Certificate
- b) Acknowledgements
- c) Abstract (in one paragraph not more than 150 words)
- d) Content Page
- e) Chapter-1 Introduction and background of the Industry or User based Problem/Task
- f) Chapter-2 Literature Survey for Problem/Task Identification and Definition,

- g) Chapter-3 Proposed Detailed Methodology of solving the problem/completing the task
- h) Chapter-4 Methodology actually followed to solve the problem/complete the task
- i) Chapter-5 The solution, its development and its implementation/execution (with details including drawings, software, used for or developed for the solution etc. in detail). In case of survey/experiment type of project this chapter will include data analysis, information presentation and analysis.
- j) Chapter-6 Final conclusions and recommendations if any for future projects.

Note:

- a) The report should contain as many diagrams and figures, charts etc. as relevant for the project.
- b) Originality of the report (written in own words) would be given more importance rather than quality of printing and use of glossy paper or multi-colour printing

VII ASSESSMENT OF PROJECT WORK

Assessment of Project work also has two components, first is Progressive Assessment, while another is End of the Term Assessment.

iii. Progressive Assessment Criteria

- a) Project guide or the group of internal faculty are supposed to carry out this assessment. It is a continuous process, during which for developing desired qualities in the students, faculty should orally give informal feedback to students about their performance and interpersonal behaviour while guiding them on their project work every week.

S. No.	Criteria	Marks
First Progressive Assessment at the end of 4th week		
9	Project Proposal for work to be completed at the end of semester: Including problem definition and methodology adopted, literature review adopted (in case of working models prepared then quality of workman ship, appropriateness of raw materials/components/ technology being used, functioning of the prototype etc. has to be also assessed)	40
	General Behaviour (initiative, resourcefulness, reasoning ability, imagination/creativity, self-reliance)	Only written feedback/suggestions
Second Progressive Assessment at the end of 12th week		
10	Execution of Plan during semester (marks to be also given based on ability to collect relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also	120

S. No.	Criteria	Marks
	assessed	
11	Log book (for work done during semester- detailed and regular entry would be basis of marks)	24
12	General Behaviour (persistence, interest, confidence, problem solving ability, decision making ability, initiative to act, team spirit, sharing of material etc., participation in discussions, completion of individual responsibilities, leadership)	Only written feedback./ suggestions
Third Progressive Assessment at the end of 14th week		
13	Portfolio for Self-learning and reflection (for work done in semester- marks based on amount of reflection and completion of the portfolio)	48
14	Project Report including documentation - for work done in semester (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	
15	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation for work done during semester)	48
16	Defence (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce for work done during semester)	56
Total		400

Note: Oral feedback on general behaviour may also be given whenever relevant/ required during day to day guidance and supervision.

iv. END SEMESTER EXAMINATION

a) Criteria of Marks for ESE for Capstone Project Execution.

S. No.	Description	Marks
1	Problem Identification/Project Title (Innovation /Utility of the Project for industry/ User/Academia) marks to be also given based on (i) Accuracy or specificity of the scope and (ii) Appropriateness of the work with reference to desired course outcomes.	16
2	Industrial Survey and Literature Review (marks to be given based on extent/volume and quality of the survey of Industry / Society / Institutes/Literature/Internet for Problem Identification and possible solutions)	24
3	Project Proposal (In case of working models, detailed design and planning of fabrication/assembly of the prototype has to be also assessed). Marks to be given also based on appropriateness, flexibility, detail and clarity in methods/planning.	24
4	Execution of Plan (marks to be also given based on ability to collect	120

S. No.	Description	Marks
	relevant information, ability to follow correct procedure, manipulative skills, ability to observe, record & interpret, ingenuity in the use of material and equipment, target achievement) In case of working models, quality of workman ship (including accuracy in dimensions, shape, tolerance limits), appropriateness of raw materials/components/ technology being used, functioning of the prototype, cost effectiveness, marketability, modernity etc. has to be also assessed.	
5	Log book (detailed and regular entry would be basis of marks)	16
6	Portfolio for Self learning and reflection (marks based on amount of reflection and completion of the portfolio)	40
7	Final Report including documentation. (marks based on: clarity in presentation and organization; styles and language; quality of diagrams, drawings and graphs; accuracy of conclusion drawn; citing of cross references; suggestion for further research/project work)	80
8	Presentation (presentation skills including communication skills to be assessed by observing presentations and asking questions during presentation)	40
9	Defence (ability to defend the methods/materials used and technical knowledge, and involvement of individual to be assessed by asking questions during presentation and viva/voce)	40
Total		400

XVI Guidelines for Portfolio Preparation and assessment

The main purpose of portfolio preparation is learning based on self-assessment and portfolio is not to be used for assessment in traditional sense.

- Each student has to prepare his/her portfolio separately. However, he/she can discuss with the group members about certain issues on which he/she wants to write in the portfolio.
- Whatever is written inside the portfolio is never to be used for assessment, because if teachers start giving marks based on whatever is written in the portfolio, then students would hesitate in true self-assessment and would not openly describe their own mistakes or shortcomings.
- Some marks are allocated for portfolio, these marks are to be given based on how sincerely portfolio has been prepared and not based on what strengths and weaknesses of the students are mentioned in the portfolio.
- Portfolio has to be returned back to the students after assessing it (assessment is only to see that whether portfolio is completed properly or not) by teachers. Because student is the real owner of the portfolio.
- Students mainly learn during portfolio preparation, but they can further learn if they read it after a gap. And hence they are supposed to keep the portfolios with them even after completion of the diploma because it is record of their own experiences (it is like diary some people write about their personal experiences), because they can read it again after some time and can revise their learning (about their own qualities)

XVII PO-COMPETENCY-CO MAPPING

Cour se Cod e	Competency and COs	PO 1 Engin eerin g know ledge	PO 2 Probl em analy sis	PO 3 Desig n/dev elop ment of soluti ons.	PO 4 Condu ct invest igatio ns of compl ex probl ems	PO 5 Mode rn tool usage	PO 6 The engin eer and societ y	PO 7 Envir onme nt and sustai nabili ty	PO 8 Ethics	PO 9 Indivi dual and team work	PO 10 Com muni cation	PO 11 Proje ct mana gemen t and finan ce	PO 12 Life- long learnin g	PSO1 Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.	PSO2 Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.
EE45 8	Semester VIII														
458	Competency <i>Find the solution of a real-life problem.</i>	3	2	2	2	2	2	1	1	2	2	2	1	2	2
458. 1	Apply the engineering knowledge to analyze, provide solution and carried out investigation of the problems by individual or team.	3	-	1	-	-	1	-	1	-	1	1	1	-	1
458. 2	Identify the engineering problem; solve, report and present it by teamwork and work with other discipline.	3	3	2	2	1	2	-	-	3	2	1	2	2	-
458. 3	Use experience related to professional and ethical principles and responsibility during their project.	3	2	1	-	-	-	-	3	1	-	2	-	-	1
458. 4	Evaluate the problem in commercial, societal, health, safety, legal, cultural and environment aspects before reach to final solution.	3	-	-	3	3	3	1	-	2	1	3	2	3	3
458. 5	Apply the engineering skills to contemporary issues related to engineering in general with advanced tools and technologies	3	1	3	2	3	3	-	-	-	2	2	3	3	2

XVIII NAMES OF RESOURCE PERSONS

- a) Dr. Nilay Patel
- b) Prof. Soaib Saiyad